Titel der Arbeit:

Calibration of Standard Stars

D i s s e r t a t i o n

zur Erlangung des akademischen Grades

d o c t o r r e r u m n a t u r a l i u m

(Dr. rer. nat.)

im Fach Physik

Spezialisierung:
Experimentalphysik

eingereicht an der
Mathematisch-Naturwissenschaftlichen Fakultät
der Humboldt-Universität zu Berlin

von
M.Sc. **Benjamin Bastian-Querner**

Präsidentin der Humboldt-Universität zu Berlin:
Prof. Dr. Julia von Blumenthal

Dekanin der Mathematisch-Naturwissenschaftlichen Fakultät:
Prof. Dr. Caren Tischendorf

Gutachter/innen:	1. Prof. Dr. Marek Kowalski 2. Dr. Susana Deustua 3. Prof. Dr. Nicolas Regnault
Tag der mündlichen Prüfung:	24.05.2024

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Für Lea

Abstract

The measurement of dark energy Ω_Λ and its equation of state parameter w plays an important role in cosmology. The Hubble diagram of a Type Ia supernova (SNe-Ia) constrains these parameters. Supernova cosmology compares the light curves of SNe-Ia at different redshifts and filter bands. The accuracy of the Hubble diagram, and hence of the above parameters, depends on the accuracy of currently available spectrophotometric standards. This requires advances to improve the connection between current astrophysical flux standards and those established in laboratories.

CALSPEC is a standard stellar network with an internal consistency of 0.5 %, frequently validated with STIS at the Hubble Space Telescope. Recent studies achieve a DETF Figure-of-Merit (FoM) of the order of 45. New instruments such as the Vera C. Rubin Observatory require flux calibration uncertainties of the order of 0.1 % to exceed the 600. Lombardo, Greg Aldering, et al. 2014 proposed SCALA, a flux calibration device that allows traceable calibration of SNIFS. Using SNIFS as a reference system, the calibration of SCALA can be transferred to CALSPEC. SCALA aims to transfer the calibration of the NIST laboratory standard with uncertainties dominated by the NIST calibration uncertainties. SCALA uses two sequential monochromators to simultaneously illuminate the SNIFS + telescope system and the calibrated photodiodes with traceable calibration for the range between 3000 Å to 10 000 Å.

Lombardo, Küsters, et al. 2017 reported the first results of SCALA and Daniel Küsters 2019 identified several sources of systematic uncertainty that affected SCALA's first measurements. This work aims to upgrade SCALA to address these systematic uncertainties as suggested by the previous study. As part of the SCALA upgrade, the VASC, a spectral comparator setup was developed that achieves a flux accuracy of 0.1 % and a spectral uncertainty of <1 Å.

Twenty photosensors were calibrated, and SCALA was upgraded with the proposed improvements in June 2022. At the end of the upgrade, standard stars from the standard star network used by the Supernova Factory were observed for four nights between June 19 and 22, 2022. During the day, SNIFS was calibrated against the previously calibrated photosensors, allowing the calibration to be traced back to the NIST laboratory standard.

It was shown that the calibration transfer from SCALA contributes in the order of 0.1 % to the total uncertainty budget. The SNIFS calibrations against CALSPEC and SCALA show several artifacts caused by the SNIFS analysis pipeline using SCALA-exposed data. The causes are discussed, and corrections are proposed.

Improvements to the VASC that were made by the end of this work but were not part of this ensure that the calibration transfer from the NIST laboratory standard to SNIFS is subordinate to the NIST contribution. An adjustment of the SNIFS analysis pipeline will allow comparison with CALSPEC with uncertainties less than 0.5 %.

Die Messung der dunklen Energie Ω_Λ und des Equation-of-State Parameters w spielt eine wichtige Rolle in der Kosmologie. Das Hubble-Diagramm von Typ Ia Supernova (SNe-Ia) gibt Aufschluss über diese Parameter. Die Supernova-Kosmologie vergleicht die Lichtkurven von SNe-Ia bei verschiedenen Rotverschiebungen und Filterbändern. Die Genauigkeit des Hubble-Diagramms und damit der oben genannten Parameter hängt von der Genauigkeit des derzeit verfügbaren spektrophotometrischen Standards ab. Dies erfordert Fortschritte bei der Verbesserung der Verbindung zwischen den derzeitigen astrophysikalischen Flussstandards und den in den Labors erstellten Standards.

CALSPEC ist ein stellares Standardnetzwerk mit einer internen Konsistenz von etwa 0,5 Prozent, das regelmäßig mit STIS am Hubble-Weltraumteleskop validiert wird. Jüngste Studien erreichen ein DETF Figure-of-Merit (FoM) in der Größenordnung von 45. Neue Instrumente wie das Vera C. Rubin-Observatorium erfordern Flusskalibrierungsunsicherheiten in der Größenordnung von etwa 0,1 Prozent, um ein FoM 600 zu überschreiten. Lombardo, Greg Aldering, et al. 2014 schlug SCALA vor, ein Flusskalibrierungsgerät, das eine rückführbare Kalibrierung von SNIFS ermöglicht. Unter Verwendung von SNIFS als Komparator kann die Kalibrierung von SCALA auf CALSPEC übertragen werden. SCALA zielt darauf ab, die Kalibrierung des NIST-Laborstandards mit Unsicherheiten zu übertragen, die von den NIST-Kalibrierungsunsicherheiten dominiert werden. SCALA verwendet zwei sequentielle Monochromatoren zur gleichzeitigen Beleuchtung des SNIFS + Teleskopsystems und der kalibrierten Photodioden mit rückführbarer Kalibrierung für den Bereich zwischen 3000 Å und 10 000 Å.

Lombardo, Küsters, et al. 2017 berichtete über die ersten Ergebnisse von SCALA und Daniel Küsters 2019 identifizierte mehrere Quellen systematischer Unsicherheit, die die ersten Messungen mit SCALA beeinträchtigten. Diese Arbeit zielt darauf ab, SCALA zu verbessern, um diese systematischen Unsicherheiten zu beseitigen, wie in der vorherigen Studie vorgeschlagen. Als Teil des SCALA-Upgrades wurde der VASC, ein spektraler Komparator entwickelt, der eine Flussgenauigkeit von 0,1 Prozent und eine spektrale Unsicherheit von 1 Å erreicht.

20 Fotosensoren wurden kalibriert und SCALA wurde im Juni 2022 mit den vorgeschlagenen Verbesserungen aufgerüstet. Nach Abschluss der Aufrüstung wurden in vier Nächten zwischen dem 19. und 22. Juni 2022 Standardsterne aus dem von der Supernova Factory verwendeten Standardsternnetz beobachtet. Tagsüber wurde SNIFS anhand der zuvor kalibrierten Fotosensoren kalibriert, so dass die Kalibrierung auf den NIST-Laborstandard zurückgeführt werden kann.

Es wurde gezeigt, dass die Kalibrierungsübertragung von SCALA in der Größenordnung von etwa 0,1 Prozent zum gesamten Unsicherheitsbudget beiträgt. Die SNIFS-Kalibrierungen gegen CALSPEC und SCALA zeigen mehrere Artefakte, die durch die SNIFS-Analysepipeline unter Verwendung von SCALA generierten Daten verursacht werden. Die Ursachen werden diskutiert und Korrekturen vorgeschlagen.

Verbesserungen am VASC, die bis zum Ende dieser Arbeit vorgenommen wurden, jedoch nicht teil dieser Arbeit waren, stellen sicher, dass der Kalibrierungstransfer vom NIST-Labornormal zu SNIFS dem NIST-Beitrag untergeordnet ist. Eine Anpassung der SNIFS-Analysepipeline wird den Vergleich mit CALSPEC mit Unsicherheiten von weniger als 0,5 Prozent ermöglichen.

Brief Summary

The measurement of dark energy Ω_Λ and its equation of state parameters w is a critical task in cosmology. These parameters are constrained by the Hubble diagram of Type Ia supernovae (SNe-Ia). In the field of supernova cosmology, the analysis involves comparing the light curves of SNe-Ia at different redshifts and filter bands. The precision of the Hubble diagram, and thus the accuracy of these parameters, depends on the reliability of current spectrophotometric standards. It is therefore imperative to improve the agreement between modern astrophysical flux standards and those established in laboratories.

After a motivational introduction in Chapter 1, the reader is introduced to supernova (SN) cosmology in Chapter 2. The cosmological standard model is derived, photometric methods are explained, and supernova cosmology is introduced, emphasizing the need for accurate calibration in the photometric system. The need for the zero-point calibration uncertainty $\sigma(ZP)$ over several bands that capture SNIa light curves to exceed a precision finer than 0.1 % in relative flux is motivated. The photometric standards are discussed, in particular the laboratory standard, the Electrical Substitution Radiometer, and the astrophysical standard CALSPEC. The chapter concludes with a discussion of the bottom-up calibration approach.

The thesis continues with the details of the laboratory equipment for photometric calibrations covered in this thesis in Chapter 3. It focuses on the Variant Angle Spectral Comparator (VASC), the comparator setup used to calibrate between photometers by comparing their responsivities. The content includes a look at the components of the VASC, the light sources used, the double monochromator, and the model for predicting the output wavelength. The method of wavelength calibration using low-pressure gas lamps is discussed. The chapter also examines the test chamber and gantry robot for precise positioning and measurements, and the relay optics for light direction and focusing. Additional sections describe the LArge Resistance Amperemeter (LARA), including the working principles of the photodiode and the design of the amperemeter. The chapter also covers the SNFactory and the SuperNova Integrating Field Spectrograph (SNIFS), detailing their operation and functionality. The “SNIFS CAlibration Apparatus” (SCALA) is also examined, which aims at traceable calibrations of SNIFS in the range of 3000 Å to 10 000 Å with uncertainties below 0.5 %. Initial challenges and improvements made in the 2022 SCALA upgrade are also outlined.

The characterization of the CMOS test sensors for the ULTRASAT is discussed in Chapter 4 as an example of the use of the VASC.

Chapter 5 describes the analysis of the calibration data. It starts with a discussion of the data reduction pipeline, including the data reduction steps and the calibration of the LARAs for SCALA, and ends with the performance comparison of the VASC against the NIST calibration.

The thesis concludes with the standard star observations in June 2022. Chapter 6 describes the calibration of SNIFS using its pipeline against SuperNova Factory standard stars and against SCALA, which allows us to trace the calibration back to the NIST laboratory standard. The chapter concludes with a discussion of the remaining artifacts in the comparison of the calibrations caused by the SNIFS pipeline and a look at future work.

Die Messung der dunklen Energie Ω_Λ und ihres Zustandsgleichungsparameter w ist eine wichtige Aufgabe in der Kosmologie. Diese Parameter werden durch das Hubble-Diagramm von Supernovae vom Typ Ia (SNe-Ia) eingeschränkt. Im Bereich der Supernova-Kosmologie beinhaltet die Analyse den Vergleich der Lichtkurven von SNe-Ia bei verschiedenen Rotverschiebungen und Filterbändern. Die Präzision des Hubble-Diagramms und damit die Genauigkeit dieser Parameter hängt von der Zuverlässigkeit der aktuellen spektrophotometrischen Standards ab. Es ist daher zwingend notwendig, die Übereinstimmung zwischen modernen astrophysikalischen Flussstandards und den in Labors etablierten Standards zu verbessern. Nach einer motivierenden Einführung in Kapitel 1 wird der Leser in Kapitel 2 in die Kosmologie der Supernova (SN) eingeführt. Das kosmologische Standardmodell wird hergeleitet, photometrische Methoden werden erklärt und die Supernova-Kosmologie wird vorgestellt, wobei die Notwendigkeit einer genauen Kalibrierung des photometrischen Systems betont wird. Die Notwendigkeit, dass die Nullpunkt-Kalibrierungsunsicherheit $\sigma(ZP)$ über mehrere Bänder, die SNIa-Lichtkurven erfassen, eine Genauigkeit die nicht mehr als 0.1 % im relativen Fluss überschreitet, wird begründet. Die photometrischen Standards werden besprochen, insbesondere der Laborstandard, das Elektrische Substitutions Radiometer, und der astrophysikalische Standard CALSPEC. Das Kapitel schließt mit einer Diskussion über den Bottom-up-Kalibrierungsansatz.

In Kapitel 3 wird die Laborausrüstung für photometrische Kalibrierungen im Detail vorgestellt. Der Schwerpunkt liegt dabei auf dem "Variant Angle Spectral Comparator" (VASC), dem Komparator, der für die Kalibrierung von Photometern durch den Vergleich ihrer Sensitivitäten verwendet wird. Der Inhalt umfasst einen Blick auf die Komponenten des VASC, die verwendeten Lichtquellen, den Doppelmonochromator und das Modell zur Vorhersage der Wellenlänge des austretenden monochromatischen Lichtes. Die Methode der Wellenlängenkalibrierung unter Verwendung von Niedrigdruckgaslampen wird besprochen. Das Kapitel befasst sich auch mit der Prüfkammer und dem Portalroboter zur präzisen Positionierung und Messung sowie mit der Relaisoptik zur Lichtlenkung und Fokussierung. Weitere Abschnitte beschreiben das "LArge Resistance Amperemeter" (LARA), einschließlich der Funktionsprinzipien der Fotodiode und des Aufbaus des Amperemeters. Das Kapitel befasst sich auch mit der SNFactory und dem SuperNova Integrating Field Spectrograph (SNIFS) und beschreibt deren Arbeitsweise und Funktionalität. Auch der "SNIFS CAlibration Apparatus" (SCALA) wird untersucht, der auf rückführbare Kalibrierungen des SNIFS im Bereich von 3000 Å bis 10 000 Å mit Unsicherheiten unter 0.5 % abzielt. Außerdem werden die ersten Herausforderungen und Verbesserungen des SCALA-Upgrades 2022 beschrieben. Die Charakterisierung der CMOS-Testsensoren für den ULTRASAT wird in Kapitel 4 als Beispiel für den Einsatz des VASC erörtert. Kapitel 5 beschreibt die Analyse der Kalibrierungsdaten. Es beginnt mit einer Diskussion der Datenreduktionspipeline, einschließlich der Datenreduktionsschritte und der Kalibrierung der LARAs für SCALA, und endet mit dem Leistungsvergleich des VASC mit der NIST-Kalibrierung.

Die Arbeit schließt mit den Standardsternbeobachtungen im Juni 2022 ab. Kapitel 6 beschreibt die Kalibrierung von SNIFS mit Hilfe seiner Pipeline gegen die Standardsterne der SuperNova Factory und gegen SCALA, was es uns ermöglicht, die Kalibrierung bis zum NIST-Laborstandard zurückzuverfolgen. Das Kapitel schließt mit einer Diskussion der verbleibenden Artefakte im Vergleich der beiden Kalibrationen ab, die durch die SNIFS-Pipeline verursacht werden, und einem Ausblick auf zukünftige Arbeiten.

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1 Introduction

Our current scientific worldview is shaped by our understanding of the universe's evolution over time. This understanding is solely based on the general relativity and the cosmological principle. The discovery of the acceleration of the universe's expansion in 1998 Perlmutter et al. 1999; Adam G Riess et al. 1998 revealed the existence of "dark energy". This energy makes 70 % of all substances contained by the universe within the standard model of cosmology. Yet the dynamic of the universe is dominated by this component today, there is "no theoretical explanation for its existence or magnitude". This compelling problem requires accurate probing of the universe's evolution, "demanding an ambitious observational program to determine the dark energy properties as well as possible" (Albrecht et al. 2006). The method of Supernova cosmology compares apparent fluxes of supernovae at different bands and redshifts. Flaws in the differential flux calibration between the different bands lead to errors in the Hubble Diagram, necessary for the determination of the cosmological parameters. For recent studies, the contribution of the calibration uncertainties is comparable to the statistical uncertainty (Abbott et al. 2019; MEA Betoule et al. 2014; Jones et al. 2019; Daniel Moshe Scolnic et al. 2018; Sullivan et al. 2011) Next-generation surveys such as the Vera C. Rubin Observatory require improvements in photometric calibrations to make use of all its statistical information (Marc Betoule et al. 2023; Hazenberg et al. 2018). Chapter 2 starts with the derivation of the standard model and introduces the concept of supernova cosmology (see 2.1.3).

In the second half, it introduces the concept of photometric calibrations and introduces the current standard for photometric calibrations, which is given by "Calibration Spectra" (CALSPEC), a databank of stellar spectra (R. C. Bohlin, Hubeny, and Rauch 2020) (see Sec. 2.4). These spectra ground on white dwarf models of three carefully selected stars (Vennes et al. 1994). The calibration is transformed into a network of 121 stars via STIS spectroscopy. The last bottom-up calibrations of standard stars were done in the 80s (Hayes 1985). It further discusses the impact of calibration uncertainties on the cosmological parameters (see Sec. 2.2).

Lombardo, Greg Aldering, et al. 2014 proposed a flux calibration device (SCALA) to calibrate the supernova integral field spectrograph (SNIFS) permanently mounted on the UH88 telescope of Hawaii University. This calibration is traceable to the photometric laboratory standard of the National Institute of Standards and Technology (NIST). SCALA had its first commission in 2015 with the first calibration results published in Lombardo, Küsters, et al. 2017. Daniel Küsters 2019 examined the systematic effects that arise in the usage of SCALA and made suggestions on how to address those, to ensure the calibration transfer from the NIST measurement to SNIFS adds uncertainties of below 0.1 % to the overall budget.
One notable conclusion is the necessity of the development of a spectral comparator setup to perform responsivity calibrations of photosensors

in the laboratory, which is part of this work. Chapter 3 introduces the "Variant Angle Spectral Comparator" (VASC), SCALA, and the SNIFS instrument and summarizes the SCALA upgrade in 2022. It ends with the implementation of the in Daniel Küsters 2019 suggested improvements into SCALA.

As an application example of the VASC, Chapter 4 introduces the characterization of the ULTRASAT test sensors.

Chapter 5 explains the methodology of the calibration with the VASC, and the calibration of the photodetectors that serve as monitors for SCALA.

Chapter 6 discusses the observation of standard stars with SNIFS and the calibration of SNIFS with SCALA and compares the results with CALSPEC.

The work ends with a summary in chapter 7.

Scientific Motivation 2

This chapter lays the foundation for understanding the connection between calibration methodologies and exploring cosmological measurements with a focus on dark energy.

In 2.1, we dive into the field of cosmology. It derives the cosmological standard model and provides all necessary for supernovae cosmology.

Sec. 2.2 sheds light on the pivotal role of calibration in cosmological measurements. It emphasizes its profound influence on the accuracy and reliability of our cosmological findings.

In Sec. 2.3, we go into the concept of photometric standards. The section describes various calibration methods, traces the calibration chain, and explains the Electrical Substitution Radiometer. Additionally, we examine the light sources employed in calibrations at the National Institute of Standards and Technology (NIST), providing insights into the tools and techniques used to establish photometric standards.

Sec. 2.4 introduces the concept of Standard Stars and their modeled spectral energy densities (SEDs), known as CALSPEC. These reference spectra are essential for ensuring the accuracy of astronomical measurements.

The last Sec. 2.5 describes a new approach for a bottom-up calibration of the standard stars of CALSPEC and others to underly the modeled SEDs with new experimental data, which has not been done since the early '80s (Hayes 1970; Landolt 1983; Oke and Gunn 1983; Oke and Schild 1970).

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2.1 Cosmology

The study of the cosmos, or cosmology, delves into the fundamental questions about the universe's origin, evolution, and fate. It is a field that is continuously evolving and expanding our understanding of the cosmos. This chapter explores several key aspects within the realm of cosmology, shedding light on the structure and dynamics of the universe.

Sec. 2.1.1 begins with the foundational framework of cosmology, the cosmological model. This model forms the bedrock of our comprehension of the universe, describing its expansion, geometry, and composition. It will explain the standard Big Bang model, focusing on different models for the contribution of dark energy.

Vital photometry and spectro radiometry techniques are explored in Sec. 2.1.2. These techniques are necessary for measuring celestial objects' brightness and spectral properties. It will delve into the principles behind these methods and their applications in determining the physical instrumental characteristics and their calibration.

In Sec. 2.1.3, the attention shifts to the notable phenomenon of supernovae within cosmology. Supernovae of the Type Ia (SNIa) have significantly contributed to our efforts to understand the universe's expansion better.

The section describes the role of SNIa as a standardizable candle and will derive observables that can probe the cosmological standard model. It gives a practical description from measuring light curves over supernovae standardization to a model fit. Ultimately, it will explain the figure of merit, the current uncertainty measure.

2.1.1 Cosmological Model

Only the following assumptions are necessary to derive a mathematical model for the universe's evolution: the cosmological principle holds and a model for an action that leads to the field equation of general relativity (GR). Under *large scales* (>100 Mpc), the universe appears homogenous and isotropic. *Isotropic* means there is no preferred direction in the universe; regardless of where one points, it looks the same. On the contrary, *Homogenouity* means no preferred location in the universe exists. A translation in position does not change its appearance. The easiest action for GR is the Einstein-Hilbert action (Carroll 2019)[1]:

$$S_H = \int \mathbf{dx^4}\sqrt{-\mathfrak{g}}\left[\frac{1}{2\kappa}(R - 2\Lambda) + \mathscr{L}_\mathrm{M}\right] \tag{2.1}$$

With R and g being the Ricci Scalar and the determinant of the metric tensor $\mathfrak{g} := \det g_{\mu\nu}$. The field $\mathscr{L}_\mathrm{M}$ describes any matter or substance content in the universe. Λ can be added to the action to introduce the cosmological constant, which is interpreted as a part of the source term. See side note for further motivation. $\kappa = 8\pi G/c^4$ is a constant determined by taking the non-relativistic limit to regain the Newtonian law of gravity, with G and c the Gravitational constant and the speed of light. Variation of this action $\delta S = 0$ is a textbook exercise (Carroll 2019) and leads to the common Einstein field equations:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = T_{\mu\nu} \tag{2.2a}$$

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \tag{2.2b}$$

By defining the Stress-Energy tensor as $T_{\mu\nu} := \frac{-2}{\sqrt{-\mathfrak{g}}}\frac{\delta(\sqrt{-\mathfrak{g}}\mathscr{L}_\mathrm{M})}{\delta g^{\mu\nu}}$. Next, we need the general metric that follows from the cosmological principles and describes the universe's space-time at all times. We start with the Minkowski metric that describes the infinitesimal element ds^2 of flat space-time:

$$ds^2 = c^2dt^2 - \mathbf{dx^2} \tag{2.3}$$

Whereas $\mathbf{dx^2}$ are the spatial coordinates and dt^2 the time dimension. Isotropy in space allows the change of the spatial coordinates from cartesian to spherical:

$$\mathbf{dx^2} \rightarrow \frac{dr^2}{1-kr^2} + r^2(d\Theta^2 + \sin^2\Theta d\Phi^2) \tag{2.4}$$

The resulting metric (Eq. 2.5) is the Friedmann-Lemaître-Robertson-

Note that the cosmological constant Λ in the Einstein field equations can be motivated from the properties of energy stress tensor **T**, which is identified as the source of the gravitational field. **T** must be equal to a differential operator **O** acting on the metric **g**, as the field equations are the generalization of the Newtonian Poisson equation. Because **T** is a rank-two tensor, and it must vanish under the covariant derivative to ensure energy conservation ($T^{\alpha\beta}{}_{;\beta} = 0$), the left-hand-side of the field equations 2.2a has to be a combination of $g_{\mu\nu,\lambda\sigma}$, $g_{\mu\nu,\lambda}$, or $g_{\mu\nu}$ (Schutz 2022). The last term can be added with the arbitrary constant Λ but is unnecessary to describe gravity. Einstein published his field equations in 1915 without the Λ term (Einstein 1915).

k represents the curvature of the space. If $k = +1$, the space is closed and is finite. Suppose $k = 0$ then space is flat and follows Eq. 2.3 but infinite. For the last case, $k = -1$, it is open and infinite.

[1] Another action is the Goldstone action $S_G = \int \mathbf{d^4x}\sqrt{-\mathfrak{g}}\mathscr{L}[U, (\partial_\mu U)^2, \Box U, \ldots]$ with $S = S_G + S_H$ (Baumann and McAllister 2015) or a minimally coupled single scalar field ϕ; $S_\phi = \int \mathbf{d^4x}\sqrt{-\mathfrak{g}}[R/2 - 1/2(\partial\phi)^2 - V(\phi)]$ with $S = S_\phi + \int \mathbf{d^4x}\sqrt{-\mathfrak{g}}\mathscr{L}_\mathrm{M}$ (Lehners 2015), leading to inflationary and ekpyrosis theories that allow solving the Horizon Problem (see App. A.5.2).

Walker-Metric describing a universe that is homogenous and isotropic at all times but allows distances to expand. To parameterize this expansion, the spatial components of the metric are multiplied with a time-dependent scale factor $a(t)$ (Carroll 2019).

$$ds^2 = c^2dt^2 - a^2(t)\left(\frac{dr^2}{1-kr^2} + r^2(d\Theta^2 + \sin^2\Theta d\Phi^2)\right) \tag{2.5}$$

For describing the universe's evolution, the task is to find the function $a(t)$. Next to the metric, a description of the stress-energy tensor is needed that expresses the distribution of the entities that interact with gravity the universe is filled with. To fulfill the homogeneity requirement, these substances can be assumed to be a liquid with a certain energy density ρ underlying a pressure p. Isotropy implies that all off-axis elements are non-zero. Therefore, the energy stress tensor has the form:

$$T_{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & p & 0 & 0 \\ 0 & 0 & p & 0 \\ 0 & 0 & 0 & p \end{pmatrix} \tag{2.6}$$

The field equations 2.2a allows the derivation of the Friedmann equations (Eq. 2.7), which are the necessary differential equations that describe the scale factor's behavior.

$$\begin{aligned} H^2(a) &= \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3} \end{aligned} \tag{2.7}$$

The first equation follows from the time component of the Einstein equation, and the second from the spatial part. The second is also called the acceleration equation, which describes how the universe's expansion rate changes over time. The scale factor is, per definition, $a(t_0) = 1$ for present times. The local expansion rate is scaled by a and defined as the Hubble constant $H(t) = \dot{a}/a$, whereas H_0 refers to the Hubble constant[2][3] at present times $H(t_0) = 67.4(5)\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ Both Friedmann equations connect the scale factor's evolution $a(t)$ with the time dependend energy density $\rho(t)$ and the pressure $p(t)$. For cosmologically important substances, the equation of state can be written in a simple linear form of $p = w\rho$. Only one of ρ and p are independent in that form. From the Energy conservation $\dot{E} + p\dot{V} = 0$ within a spherical volume scaled with a, the fluid equation that connects the scale factor a and the energy density ρ can be derived.

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0 \tag{2.8}$$

Solving Eq. 2.8 leads to a simple relation between a substance's density ρ_x and the scale factor a depending on the equation of state (EoS) parameter w (Eq 2.9). w depends on the substance it describes.

$$\rho_x(a) = \rho_{x,0}a^{-3(1+w)} \tag{2.9}$$

Please note that the exact value of H_0 is under debate. The value $67.4(5)\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ comes from predictions of the ΛCDM-model based on measurements of the Planck satellite from the "early" universe (for redshift $z > 1000$). A second approach is to build up a distance ladder from parallax over Cepheids to Supernovae Ia (SNIa) measurements to compare redshifts with corresponding distances. This local value is estimated to be $73.2(13)\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from the SH0ES collaboration. This 4σ discrepancy is called the Hubble tension. A detailed study can be found in Valentino et al. 2021.

[2] From Planck observations (Lemos and Shah 2023), see side note.
[3] The numbers in parenthesis are the uncertainty of the last significant digits.

$$w = \begin{cases} \frac{\langle v^2 \rangle}{3c^2} \approx 0 & \text{Matter} \\ \frac{1}{3} & \text{Radiation} \\ -1 & \Lambda \\ \omega_0 + \omega_a(1-a) & \text{Varying EoS} \end{cases} \tag{2.10}$$

Eq. 2.10 lists the interesting cosmological liquids. Matter can be assumed to be a non-relativistic gas from which the pressure follows the perfect gas law. Together with the equipartition theorem and the energy density $\rho_M \approx \mu c^2/V$, with μ the mean mass of the particles, it follows that the pressure, compared to the mean squared velocity of the particles, is negligible against the square of the speed of light. Its EoS parameter is $w_M \approx 0$. In the case of photons, the mean velocity is c and, therefore, $w_R = 1/3$. As the cosmological constant in Eq. 2.7 does not scale with a, $w_\Lambda = -1$ (Ryden and Partridge 2004). In the case of the Hilbert-Einstein action Eq. 2.1, the energy conservation and the relativistic limits confine the EoS parameter within $w \in [-1, 1/3]$ for all substances between the cosmological constant and radiation. It is shown that scalar fields ϕ introduced in the action that has negative kinetic energy[4] can have $w_\phi < -1$ (Lucchin and Matarrese 1985). It is convenient to call all liquids 'dark energy' if $w_x < -1/3$. Generally, the EoS parameter does not have to be constant over time. A practical approximation is to assume w linear in a, with $w_x(a) = w_{x,0} + w_{x,a}(1-a)$ (Amendola et al. 2018). The energy density in the first equation 2.7 is the sum of all substances the universe contains $\rho = \sum_x \rho_x$. Usually, they are parametrized by their value today in units of the current critical density $\rho_c = 3H_0^2/8\pi G$ at which a matter-dominated universe has a flat curvature (Pierre Astier 2012). The first Friedmann equation gets the form of

$$H^2(a) = H_0\left(\frac{\Omega_{R,0}}{a^4} + \frac{\Omega_{M,0}}{a^3} + \frac{\Omega_k}{a^2} + \sum_x \Omega_x(a)\right) \tag{2.11}$$

with Ω_x describing substances contained by the universe next to radiation, non-relativistic matter, and the influence of the universe's curvature k. Ω_x can have every form of:

$$\Omega_x(a) = \begin{cases} \Omega_\Lambda \\ \Omega_{x,0} a^{-3(1+w)} \\ \Omega_{x,0} \exp\left(3\int_a^1 (1+w(a'))a' da'\right) \end{cases} \tag{2.12}$$

Depending on whether it is the cosmological constant Λ, has an arbitrary constant EoS parameter, or varying one. Choosing concrete values for these Ω_x means choosing a concrete cosmological model, which might or might not describe the one we live in. For example, an Einstein-De-Sitter universe is a model with $\Omega_k = \Omega_R = \Omega_\Lambda = 0$ and $\Omega_M > 0$. As

[4] Measurements of the cosmical microwave background (CMB) radiation show a uniformity of the order of $< 10^{-4}$ which implies that the complete observable universe must have been in casual contact to reach that kind of thermal equilibrium which can not be explained with standard Big Bang evolution (Eq. 2.11) where the comoving Hubble radius $(aH)^{-1}$ grows with time (Horizon Problem, see App. A.5.2). That can be solved by postulating a shrinking Hubble radius from negative conformal time ($\tau = dt/a$) so that the integral $\Delta\tau = \int_{-\infty}^{a_{CMB}} da/H(a)$ is dominated from early times and allows all content in the universe to come into causal contact. $\tau = 0$ the spacelike singularity is replaced by the transition from inflation to standard Big Bang evolution. These inflation theories can be achieved by introducing a flat scalar field ϕ into the action 2.1. If the field's change is negligible over time (slow-roll model), it can have an EoS parameter of $w_\phi \approx -1$ (Baumann and McAllister 2015) and behave as a cosmological constant.

observational data shows that the universe's curvature is flat over large scales, and the radiation is diluted by the scale factor to the power of four, one can assume that both components are negligible at present times:

$$\left.\begin{array}{l}\Omega_{R,0} \approx 10^{-4} \\ \Omega_k\end{array}\right\} \approx 0 \tag{2.13}$$

That brings the first Friedmann equation to the following practical representation:

$$H^2(a) = H_0\left(\frac{\Omega_{M,0}}{a^3} + \Omega_x(a, \omega_0, \omega_a)\right) \tag{2.14}$$

Meanwhile, the correct form of Ω_x is a matter of observation. The established representation of Ω_x to describe the available data is:

$$\Omega_x(a, \omega_0, \omega_a) \xrightarrow{w_0=-1,\, w_a=0} \Omega_\Lambda \tag{2.15}$$

A non-evolving EoS parameter $w = -1$ with a substance of constant energy density being represented by the cosmological constant. This model is the ΛCDM[5] model and is called the standard model of cosmology. It is noteworthy that it is just a coincidence that the cosmological constant, which finds its way into the field equations because of a mathematical reason (see note 1) but has no fundamental grounds, seems to be a sufficient description of the dark energy[6].
According to the dark energy task force (DETF) (Albrecht et al. 2006), there are four approaches to probing the model's parameters: Baryon Acoustic Oscillations, Galaxy Cluster Counting, Weak Gravitational Lensing, and Type Ia Supernovae cosmology. This work deals with the last approach.

For further reading on cosmology, the reader is referred to Amendola et al. 2018; Pierre Astier 2012; Baumann and McAllister 2015; Carroll 2019; Lemos and Shah 2023; Ryden and Partridge 2004.

2.1.2 Photometry and Spectrophotometry

Precise photometry is crucial for accurately measuring cosmological parameters, as it involves comparing fluxes in different bands of supernovae at various redshifts. The flux of a source can be measured either spectroscopically, which gives the spectral energy distribution (SED) of an object, or photometrically, which gives the integrated flux F_b over a certain bandpass b. The integral of the whole restframe SED of an object is called bolometric flux F. The bolometric apparent magnitude m is defined as $\mathbf{m} := -\mathbf{2.5}\log_{\mathbf{10}}(\mathbf{F}/\mathbf{F_x})$ with the reference flux of $\boldsymbol{F_x = 2.518\,021\,002 \times 10^{-8}\ \mathrm{W\,m^{-2}}}$. In contrast, the absolute magnitude M is defined as $\mathbf{M} := -\mathbf{2.5}\log_{\mathbf{10}}(\mathbf{L}/\mathbf{L_x})$ with $\mathbf{L}$ the object's luminosity and the reference luminosity $\boldsymbol{L_x = 3.0128 \times 10^{28}\ \mathrm{W}}$[7].

[5] CDM stands for "Cold Dark Matter", such that the dark matter contributes to the Ω_M term with the same EoS parameter as the baryonic matter.
[6] Not to be confused with the cosmological "Coincidence Problem" (Velten, Marttens, and Zimdahl 2014)
[7] The flux F_x correspond to the measured flux of an object with zero absolute bolometric magnitude at 10 pc distance. L_x is chosen such that the sun's bolometric magnitude is 4.74 mag (Mamajek et al. 2015).

Usually, photometry is done with a set of pass-band filters. A photometric system is defined by the used filters τ_B and the telescope's throughput $\tau_{\text{tel.}}$; the fractional throughput $R_\lambda = \tau_b \tau_{\text{tel.}}$[8], and a zero-point the fluxes are compared with. In the reality of ground-based observatories, R_λ also includes the atmospheric throughput $\tau_{\text{atm.}}$, which must be corrected for. For example, the Landolt photometric system consists of a set of filters called UBVRI, with α-Lyrae (Vega) as the zero-point of all filters[9] (Landolt 1983). The magnitude in bands is often labeled with the filter names, or the filter's name is given as a subscript such as $m_V := V$ (Pierre Astier 2012). The magnitude m_b in Landolt's system can be expressed as

$$m_b = m_b(\text{Vega}) - 2.5 \log\left(\frac{\langle F_\lambda(Star)\rangle}{\langle F_\lambda(Vega)\rangle}\right) \tag{2.16}$$

with $m_b(Vega)$ the zero point for the given band b, which is shown above for the visual band V and the integrated physical fluxes in the band $\langle F_\lambda(\text{Star})\rangle$ of the target star and $\langle F_\lambda(\text{Vega})\rangle$ Vega as reference (Colina and R. C. Bohlin 1994). Factor 2.5 is chosen historically to match the magnitudes of Ptolemey's Algmagenst that categorized the intensity of stars into six categories, with the brightest star having a magnitude of one and the faintest visible star with the human eye having six (Pogson 1856). A magnitude change of factor five corresponds to a change in physical flux in factor 100. The conversion between two filter systems is also called *K-corrections*. Blanton and Roweis 2007 gives an unexhausted list of various common conversions.
The wavelength weighted mean flux $\langle F_\lambda \rangle$ can be expressed as

$$\langle F_\lambda \rangle = \frac{\int F_\lambda R_\lambda \lambda d\lambda}{\int R_\lambda \lambda d\lambda} \tag{2.17}$$

The collected photoelectrons N_e are:

$$N_e = \frac{A}{hc} \int F_\lambda R_\lambda \lambda d\lambda \tag{2.18}$$

with the telescope's collection area or aperture A, the speed of light c, and the plancks constant h. By defining the sensitivity S_λ of the instrument, which includes the filter, where $\langle S_\lambda \rangle$ converts the flux $\langle F_\lambda \rangle$ into the measured photoelectrons:

$$\langle F_\lambda \rangle = \langle S_\lambda \rangle N_e \tag{2.19}$$

One can express the sensitivity source independently:

$$\langle S_\lambda \rangle = \frac{hc}{\int \lambda R_\lambda d\lambda} \tag{2.20}$$

In practice, $\langle S_\lambda \rangle$ is calibrated by measuring the instrument's response in a band to a standard star with a known SED (R. Bohlin, Collins, and Gonnella 1997; R. C. Bohlin, Gordon, and Tremblay 2014; Colina and R. C. Bohlin 1994). The difference between two filter band magnitudes is called a color and is often designated as $E(b_1 - b_2) = b_1 - b_2$, with

[8] The subscript λ indicates the wavelength dependence of that variable.
[9] Later on the zero point of Landolt was estimated to be V = 0.029 mag to fit newer comparison with laboratory standard (Colina and R. C. Bohlin 1994)

b_1 being the bluer filter. Objects with higher color values are redder than those with smaller colors. Because of the logarithmic nature of the magnitudes, the color is a flux ratio between the filter bands.

In the case of spectrophotometry, the integral in Eq. 2.20 goes away, and the Flux can be expressed with:

$$F_\lambda = \frac{hc}{A\lambda R_\lambda d_\lambda} = S_\lambda N_{e,\lambda} \tag{2.21}$$

with d_λ being the spectral dispersion in angstroms per pixel. The same applies to the calibration; the spectral sensitivity is calibrated by comparing the response concerning a standard star (R. Bohlin, Collins, and Gonnella 1997; R. C. Bohlin, Gordon, and Tremblay 2014). For completeness, the monochromatic magnitude AB is

$$AB = -2.5 \log \left(F_\lambda \left[\frac{\text{erg}}{\text{cm s}^2 \text{ Å}} \right] \frac{c}{\lambda^2} \right) + 48.6 \tag{2.22}$$

The constant +48.6 is selected such that $AB = V$ for objects with flat spectra and V the visual magnitude of Vega $V = 0.029$ mag (Oke and Gunn 1983).

2.1.3 Supernovae Cosmology

Type Ia Supernovae cosmology uses supernovae as 'standard candles.' They are "objects of reproducible luminosity and enable one to measure distances through their observed energy flux: $f = L/4\pi d_L^2$" (Pierre Astier 2012). The luminosity L can be expressed in terms of luminosity distance d_L, which is the corresponding distance of an object in a flat space according to the measured flux f. The SNIa allows probing of the relation between d_L and the scale factor a at the time of the event. d_L can be predicted from assumptions about the cosmological model.

Supernovea Type Ia as a Standard Candle

Supernovae are dynamical objects that display their internal structure via multiple spectral lines that appear and evolve. This allows a classification of these events to filter and identify those with reproducible luminosity. The appearance of spectral lines of various elements classifies supernovae. The first classification is the appearance or absence of hydrogen. The latter is the Type I Supernovae (SNIa). With the appearance of silicon, the supernovae are classified as Type Ia, commonly accepted to originate from the same progenitor event. Until the '90s, SNIa were supposed to be identical (Cappellaro and Turatto 2001). A carbon-oxygen white dwarf (WD) is assumed to be the progenitor of these explosions. It accretes mass from a companion star until it reaches the Chandrasekhar mass ($M_{\text{Chan}} = 1.38\,\text{M}_\odot$ (Chandrasekhar 1931)) at which the carbon is triggered to start a fusion reaction of the whole star within seconds. There are two consensus models for this binary system. The single-degenerated model assumes a single sub-Chandrasekhar white dwarf accreting hydrogen and helium from a companion star of low-surface-gravity such as a late main-sequence star (Whelan and Iben Jr 1973). The second double-degenerated model considers a second WD as a

companion. Both stars merge after losing orbital angular momentum during gravitational radiation (Iben Jr and Tutukov 1984; Webbink 1984). A short discussion of these models can be found in Astier 2012 (Pierre Astier 2012). The maximal absolute magnitude in the B band M_B and the host galaxy's redshift z gives a redshift-luminosity relation that can be predicted from the cosmological model. Nevertheless, measurements from supernovae in 1991 and 1992 (Filippenko et al. 1992; Leibundgut et al. 1993) have questioned reproducibility. Phillips concluded in his 1993 paper (Mark M Phillips 1993) that the apparent maximal magnitude must be standardized to use the SNIa as standard candles and proposed the "**brighter-slower**" relation that describes the correlation of the maximal absolute magnitude in the B-band and its decay rate $\Delta m_{15}(B)$ the averaged decline over the first 15 days after the novae's maximum. The magnitude also correlates with the width of the light curve of the rest-frame timescale, which can be parametrized by the stretch factor s[10] first suggested in Goldhaber et al. 1996. The decay rate $\Delta m_{15}(B)$ can be translated into the stretch factor and vice-versa (Perlmutter et al. 1999), so correcting for both is redundant. The magnitudes must be compared in their rest frame and the same band to compare the supernovae. As the universe's expansion reddens more distant novae, the photometric map of a spectrum differs with varying redshift, or the setup itself differs used to measure the light curves because the spectrum of the redshifted supernovae does not overlap with the B band filter and must be measured with a redder one. Secondly, the supernovae are measured with different telescopes using an arbitrary set of filters that must be transformed into a standard set. In literature, this is known as "**K-corrections**" that transfers the flux $f(T_1, T_2, z)$ measured with the filter with a transmission curve of $T_2(\lambda)$ at redshift z to the standard filter $T_1(\lambda)$ without redshift.

$$f(T_1, T_2, z) = f(T_2)\frac{\int S(\lambda)T_1(\lambda)d\lambda}{\int S(\lambda(1+z))T_2(\lambda)d\lambda} \tag{2.23}$$

The accuracy depends merely on the knowledge of the supernovae's spectrum $S(\lambda)$, which must be taken from a template and can only be an approximation. If known, the rest frame magnitude in the B band could be extracted directly from $S(\lambda)$ (Pierre Astier 2012). Another correlation is the "**brighter-bluer**" relation, which reflects the extinction by dust and the intrinsic relation between the maximal magnitude M_B and the color $E(B-V) = B-V$ (Tripp 1998). B and V are the maximum apparent magnitude at their respective bands. This relation is similar to the CCM "extinction law" $A(V) = E(B-V)R_V$, mainly describing the dust extinction in the Milky Way with the extinction parameter R_V (Cardelli, Clayton, and Mathis 1989). The common ansatz is to extract an averaged value for R_V, more recently referred to as β, for all supernovae in the dataset, treating it as a free parameter. In contrast, newer investigations show evidence that each supernova has an individual extinction parameter depending on the host galaxy (Brout and Daniel Scolnic 2021). Tripp 1997 (Tripp 1998) used the **brighter-slower** and the **brighter-bluer** relation to parameterize a modified or standardized absolute magnitude $M_B = -19.48 + b(\Delta m_{15} - 1.05) + R(B -$

[10] More recent literature uses the parameter $x_1 = s - 1$ (Pierre Astier 2012). This parameterization comes from the lightcurve model fitter SALT that uses x_0 for scaling the amplitude, x_1 to scale the median variability of the light curve template, and c the color relative color offset (Julien Guy et al. 2007).

V)[11] while holding R and b as nuisance parameters in the fit of H_0 and the deceleration parameter q_0 describing the universe's kinematics without any physics[12]. In modern times, empirical light curve models are used to determine the color c and stretch factor x_1. Different models are available to parameterize the interpolation between the empirical light curve data. A non-exhausted list of models is *Multi-Light Curve Shape (MLCS)* (Jha, Adam G Riess, and Kirshner 2007), *Spectral Adaptive Light curve Template (SALT)* (Guy, J. et al. 2005), *SALT2* (Julien Guy et al. 2007), & *SALT3* (Taylor et al. 2023), and *SiFTO* (Alex Conley et al. 2008). After fitting the cosmology described above, or even the cosmological parameter Ω, the remaining residual in luminosity is called intrinsic scattering σ_{int}. A wide range of supernovae in nearby galaxies allows a quantification of the dispersion σ_{int}. A nonexhausting list of historical samples that investigate the properties of SNIa from low to high redshifts can be found in Jha, Kirshner, et al. 2006, Hicken et al. 2012, Contreras et al. 2010 (low redshift sample), Holtzman et al. 2008 (intermediate redshift), and Pierre Astier et al. 2006 (high redshift).

Oberservables to Probe Cosmology

To probe the parameters $\Omega_{M,0}$, and $\Omega_{X,0}(w_0, w_a)$, they must be connected with a set of observables. The first step is to express the scale factor a concerning the corresponding redshift z[13] and rewrite the Friedmann equation.

$$a = \frac{1}{1+z} \rightarrow H^2(z) = H_0\Big(\Omega_{M,0}(1+z)^3 + \Omega_x(z, \omega_0, \omega_a)\Big) \tag{2.24}$$

One helpful measure is the luminosity distance d_L, which connects the observed flux f and luminosity L of an object with the distance in flat and static space with $f = L(4\pi d_L)^{-2}$. In the case of a curved and expanding universe, the flux through the shell that the d_L spans must be scaled with a^{-2} as the energy is scaled by the expansion with one factor of a^{-1}, and secondly, the arrival time between two photons is stretched by a further factor of a^{-1}. By comparing the flux in static Minkowski with an expanding FLRW metric, one can define the luminosity distance to be:

$$d_L = S_k(r)(1+z) \tag{2.25}$$

With $S_k(r)$ the integral over the radial part of the FLRW metric, which can be one of:

$$S_k(r) = \begin{cases} a(t)\sinh^{-1}\left(\frac{r}{a(t)}\right) & k = -1 \\ r & k = 0 \\ a(t)\sin^{-1}\left(\frac{r}{a(t)}\right) & k = +1 \end{cases} \tag{2.26}$$

[11] The −19.48 mag are the mean absolute magnitude from SNIa based on the distance modulus measurements of the Large Magellanic Cloud. Further explanation can be found in the footnote of Tripp 1997 (Tripp 1998).
[12] For further explanation of the physics free kinematics, see (Ryden 2017).
[13] For a derivation of the redshift, see App. A.5.1

Where r is also the proper distance. With $0 = ds^2 = c^2dt^2 - a(t)^2dr^2/(1 - kr^2)$ it follows that:

$$S_k(r) = c\int_0^t \frac{dt'}{a(t')} = \int_0^z \frac{dz'}{H(z')} \tag{2.27}$$

From Eq. 2.25 and Eq. 2.27 and flat curvature $k = 0$ the luminosity distance d_L is:

$$\Rightarrow d_L(z) = \frac{1+z}{H_0}\int_0^z \frac{dz'}{\sqrt{\Omega_{M,0}(1+z')^3 + \Omega_x(z', \omega_0, \omega_a)}} \tag{2.28}$$

As described in Sec. 2.1.2, fluxes are measured in magnitudes. The relation between absolute M and apparent magnitude m follows as $M = m - 5\log_{10}(d_L/10\,\mathrm{pc})$[14]. As a measure of distance, it is established to take the *distance modulus* μ, the difference between apparent and absolute magnitude:

$$\mu = m_{app} - M = 5\log_{10}(d_L[\mathrm{Mpc}]) + 25 \tag{2.29}$$

Measurement of the Cosmological Parameters

To probe a cosmological model, one must measure m_{app} of standard candles at different redshifts z and compare it with the prediction of Equation 2.28 and 2.29. The absolute magnitude M only scales the Hubble constant H_0. It does not play a role in probing the cosmological parameters Ω_M and Ω_X as those are not sensitive to the standard candle's absolute luminosity. From above, the goal is to get a distance modulus from a flux proportional to a supernova's bolometric luminosity. The absolute magnitude M determines the Hubble Constant only but not the evolution of the scale factor, as M is only an additive value in the Hubble diagram. A precise comparison of the Supernovae's fluxes is important to probe the cosmological evolution. The established observable is to derive a standardized distance modulus compared to 10 pc from a calibrated observed peak magnitude in the restframe B band m_B^* in the UBVRI filter system. The techniques to measure the cosmological parameters from SNIa lightcurve measurements have a long history. Adam G Riess et al. 1998 and Perlmutter et al. 1999 were the most noticeable first attempts to determine the cosmological parameters. In Perlmutter et al. 1999, the peak magnitude in the observed R-band m_R is corrected with the stretch factor s fitted from a template and the "**bluer-brighter**" relation by fitting an individual reddening factor $R_B = A_R/(B - V)$ to match the template's color. Further, a **K-correction** to the standard B-band was done with K_{BR}:

$$\mu = m_R + \alpha(s-1) - K_{\mathrm{BR}} - A_R - M_B \tag{2.30}$$

With M_B, the absolute magnitude that scales the Hubble Constant only. M_B and α are nuisance parameters in the cosmological-parameter fit at a dataset of 42 SNIa. The result for a ΛCDM fit is $\Omega_M = 0.28^{+0.09}_{-0.08}$ with $\Omega_\Lambda = 1 - \Omega_M > 0$ and an intrinsic scattering 0.154(40) mag to 0.157(25) mag.
In recent literature, the standardization is done with light curve model fitters. The description here is restricted to the SALT model used in the

[14] f_x is the expected flux of a light source placed at a standard distance of 10 pc with the defined reference luminosity of L_x.

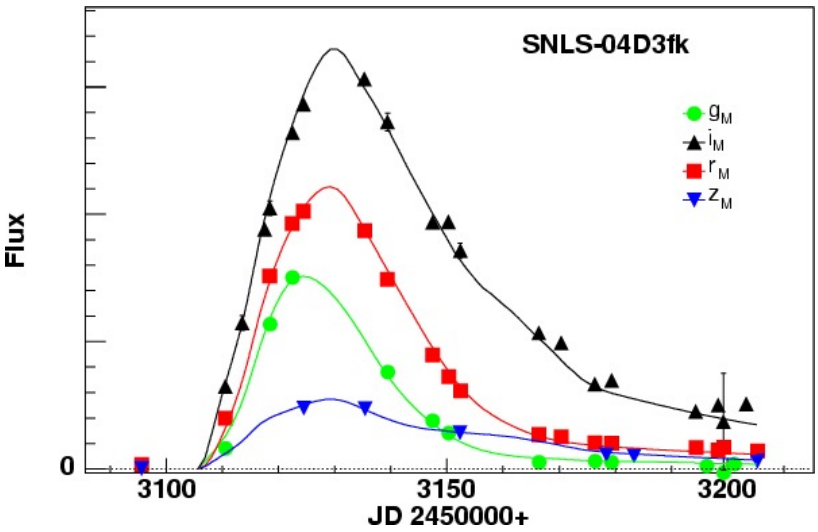


Figure 2.1: SNIa SNLS-04D3fk measured with g_M, r_M, i_M and z_M bands with the MegaPrime/MegaCam instrument at the Canada-France-Hawaii Telescope (CFHT) for the Supernova Legacy Survey (SNLS) taken from Pierre Astier et al. 2006.

most recent measurements of the cosmological parameters. The model consists of a template of the spectral energy density (SED) of an SNIa that evolves. The template has the form of

$$S(\lambda, p) = x_0[S_0(\lambda, p) + x_1 S_1(\lambda, p) + \ldots] \exp\left(-cCL(\lambda)\right) \tag{2.31}$$

S_0 is the averaged spectral sequence depending on the time p after the rest frame peak maximum in the B band. The wavelength λ and S_k express the variability, whereby x_1 correlates with the stretch parameter (Julien Guy et al. 2007). x_0 can be converted into the maximal B-band magnitude as it scales the spectrum amplitude with $m_B^*(x_0) = -2.5\log_{10}(x_0)$. In contrast to the measured B-band maximum, the value $m_B^*(x_0)$ is independent of the band's shape and does not require a K-correction (Marriner et al. 2011). The SN's color is reflected by c, and CL corresponds to a wavelength-dependent reddening parameter similar to R_V. Figure 2.1 shows the measured light curve of the SNLS-04D3fk at a redshift $z = 0.358$. The solid lines show the results of the SALT2 light curve fitter that varied the above parameters at the template SED to reproduce the data with the MegaPrime/MegaCam filters[15].

$$\mu(z) = \overbrace{-2.5\log_{10}(x_0)}^{m_B^*} - M_0(z) - \alpha(z)x_1 - \beta(z)c + \Delta\mu_{\text{bias}} \tag{2.32}$$

The standardized distance modulus μ can be described with Eq. 2.32. The absolute magnitude M_B correlates with the host galaxy (Johansson et al. 2013; Sullivan et al. 2011) and can be addressed with a simple step function 2.33.

$$M_0 = \begin{cases} M_1^B & \text{if } M_{\text{stellar}} < 10^{10} M_\odot \\ M_1^B + \Delta_M & \text{else} \end{cases} \tag{2.33}$$

MEA Betoule et al. 2014 show the step in the Hubble residuals $\mu - \mu_{\text{model}}$ if no step function is considered in the Hubble fit. The step in Fig. 2.2 is $\Delta_M = -61(12)$ mmag and is of 5σ significance. In the case of the

[15] One example implementation of the SALT-fitter can be found in the SNCosmo python package DOI:10.5281/zenodo.8091892 (Team 2023).

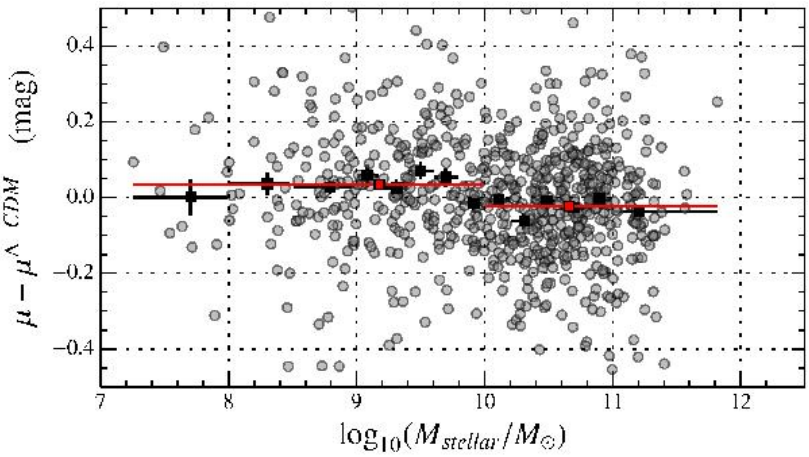


Figure 2.2: Hubble residuals from MEA Betoule et al. 2014 without the mass step function 2.33 included in the Hubble parameter fit of a ΛCDM model. The red line shows the averaged residual below and above $10^{10} M_{\odot}$.

uncertainty estimate, it is necessary to point out that M_1^B must be a nuisance parameter, too. Otherwise, it would introduce constraints on the Hubble constant, leading to underestimating the uncertainties. On the contrary, recent studies suggest that the mass split function "is an artifact of improper fitting and arises from significantly different R_V distributions for different types of galaxies" and has to be addressed by a more complex dust model (Brout and Daniel Scolnic 2021). $\Delta\mu_{\text{bias}}$ corrects for distance biases arising from survey and spectroscopic selection efficiencies and the analysis and light curve fitting and is determined with lightcurve simulations (Abbott et al. 2019). The χ^2 that must be minimized can be defined as

$$\chi^2 = (\mu - \mu_{\text{model}})^T C^{-1} (\mu - \mu_{\text{model}}) \tag{2.34}$$

with the standardized distance modulus $\mu = \mu(z; \alpha, \beta, M_1^B, \Delta_M)$, the predicted $\mu_{\text{model}} = \mu_{\text{model}}(z; \Omega_M, \Omega_X(w_0, w_a))$, and the covariance matrix that reflects all systematic and statistical uncertainties (Abbott et al. 2019; MEA Betoule et al. 2014; Marriner et al. 2011). Daniel Moshe Scolnic et al. 2018 provide a detailed explanation of the matrix. Figure 2.3 shows the resulting Hubble plot of the analysis from the Dark Energy Survey (DES) Collaboration of the first three years of the Supernova Program (DES-SN). The distance modulus of the supernova is plotted with the fit results of the nuisance parameter α, β, M_1^B, Δ_M and Eq. 2.32. The dashed black line shows the results of the flat wCDM fit. In comparison, the plot also shows two Einstein-de Sitter models with $\Omega_M = 0.3$ (dotted green line) and $\Omega_M = 1$ (dashed-dotted blue line). The best fit here gives $w = 0.978 \pm 0.059$ and $\Omega_M = 0.321 \pm 0.018$, which is compatible with the cosmological constant hypothesis.

To investigate the physical origin of the dark energy substance, probing a possible variation of the equation-of-state parameter is of special interest. Combining the SNIa constraints with those from cosmic microwave background (CMB) radiation and the baryon acoustic oscillations (BAO), one can get results for a flat $w_0 w_a$CDM model fit. To compare the quality of the fit. Albrecht et al. 2006 suggests the figure-of-merit (FoM), which is defined to be the reciprocal of the area in the w_0-w_a plane that encloses the 95 % confidence level region, which is proportional to $[\Delta w_0 \Delta w_a]^{-1}$. Figure 2.4 shows the confidence contours of the $w_0 = -0.957 \pm 0.124$ and $w_a = -0.336 \pm 0.552$ cosmological parameters from the Joint Light curve Approach (JLA) from MEA Betoule et al. 2014 resulting in a FoM of 31.3. This result is comparable with the ΛCDM model and does not resolve

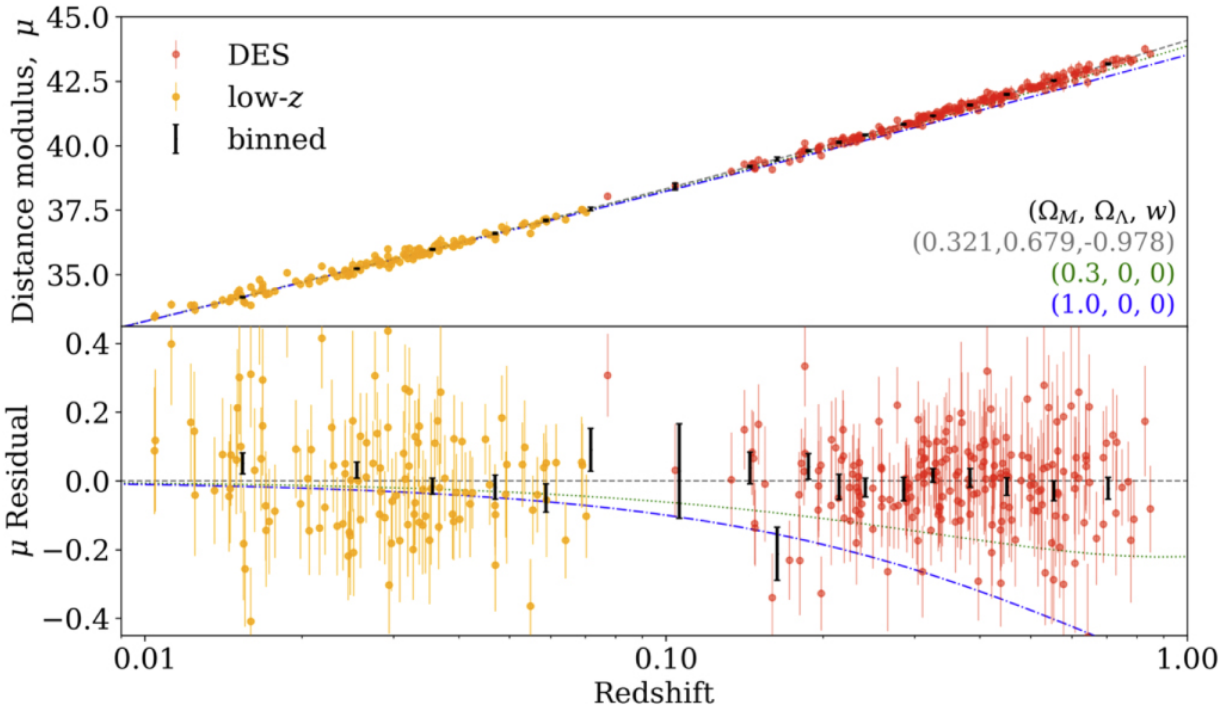


Figure 2.3: Hubble diagram for the DES-SN3YR sample. The low redshift subset $z < 0.1$ is taken from Harvard-Smithsonian Center for Astrophysics surveys. Top: distance modulus (μ) from fit (black bars, which are used for cosmology χ^2-fits) and for each SN (red, orange circles). The dashed gray line shows our best-fit model, while the green and blue dotted lines show models with no dark energy and matter densities $0.3\,\Omega_M$ and 1.0, respectively. Bottom: residuals to the best-fit model; 1σ error bars show 68 % confidence (Abbott et al. 2019).

any evolution of the EoS parameter. This result is close to Abbott et al. 2019 with a FoM of 45.5. The latter also showed that the DES SN sample reduces the uncertainties on w_0 and w_a by a factor of 1.5 to 2 compared to the results using BAO and CMB alone (Abbott et al. 2019).

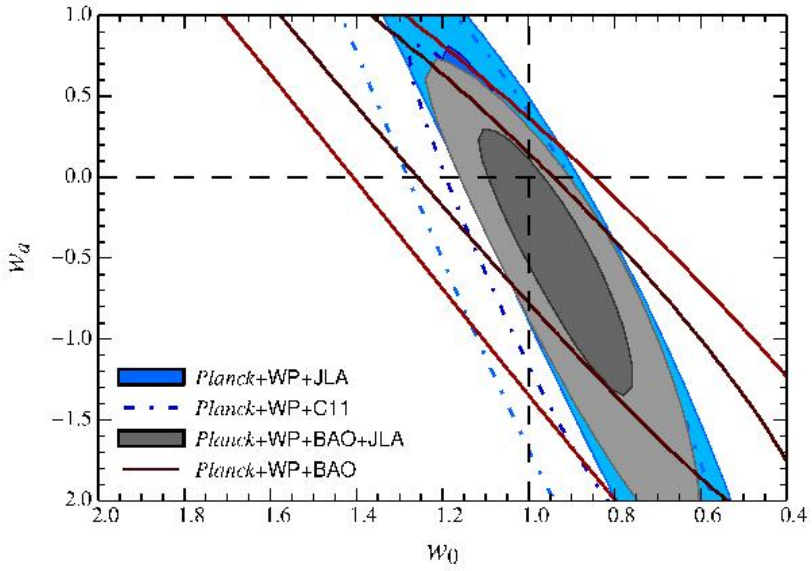


Figure 2.4: Confidence contours at 68 % and 95 % (including systematic uncertainty) for the w_0 and w_a cosmological parameters for the flat w_0w_aCDM model. The point $w_0 = -1$, $w_a = 0$ equals to the ΛCDM model. Taken from MEA Betoule et al. 2014

2.2 Calibration and its Impact on Cosmological Measurements

Measuring the flux from a source requires measuring a standard source of known spectral flux. This involves comparing the measured flux with the standard and target sources through sequential and repeated measurements. In the case of a telescope and stars, the resulting flux measurement is influenced by various factors, including the throughput of the atmosphere (τ_{atm}), instrument (τ_{inst}), filters (τ_{filt}), and the external quantum efficiency (QE_{ext}) of the detector (Stubbs and Tonry 2006). Importantly, all these quantities exhibit wavelength dependencies. The calibration process, therefore, addresses the product $\tau_{\text{inst}} \cdot \tau_{\text{filt}} \cdot \tau_{\text{atm}} \cdot QE_{\text{ext}}$, which collectively accounts for the influence of these variables on the measured flux.

In scenarios where atmospheric properties remain stable throughout a night, with negligible changes in τ_{atm} between measurements of standard stars, the night is considered photometric. In cases where τ_{atm} varies ($\dot{\tau}_{\text{atm}} > 0$), standard stars at the same frame as the target are employed for calibration.

Notably, while relevant for determining the Hubble constant (H_0), the absolute magnitude is considered a nuisance parameter that does not directly impact the cosmological parameter fit. Consequently, an absolute calibration does not contribute to the uncertainty budget of the cosmological parameters.

In the context of cosmological measurements, it is crucial to compare light curves across different bands, especially given that supernovas are observed at varying redshifts. Any systematic biases arising from the spectral calibration of the detector can lead to distortions in the ratios between light curves, consequently resulting in an erroneous color $c + \delta c$ for individual supernovae. This bias δc directly influences the parameter fit, as it introduces a bias in the standardization of the supernova.

Hazenberg et al. 2018 conducted a comprehensive investigation into

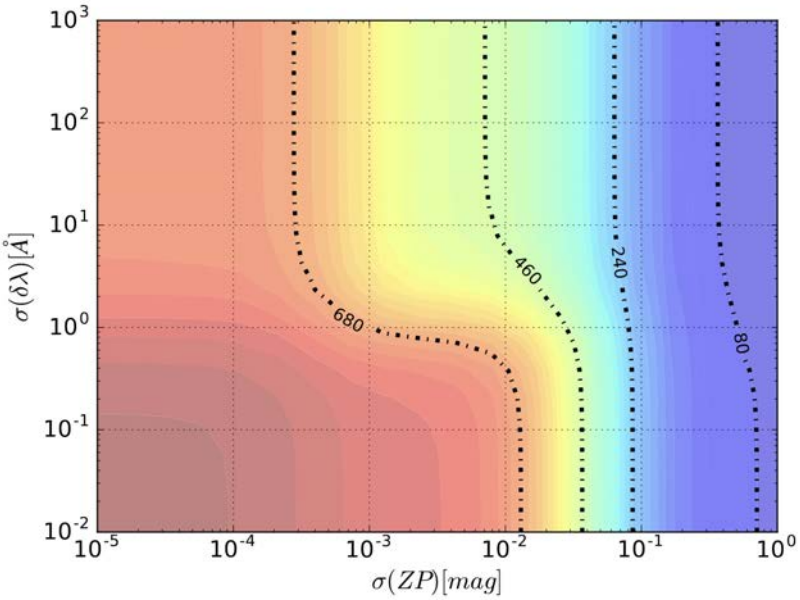


Figure 2.5: Evolution of the Figure of Merit (FoM) concerning the filter position uncertainty and the filter zero-point (zp), from investigation of Hazenberg et al. 2018 The transition area occurs at $\sigma_{zp} = 1\,\mathrm{mmag}$.

the dependence of the Figure of Merit (FoM)[16] on the uncertainties associated with the calibration of the zero-point (σ_{zp}) and the mean filter position (σ_λ) of a certain band. Their study involved simulating a typical Vera C. Rubin Observatory[17] survey spanning ten years. As depicted in Fig. 2.5, a plateau in the FoM is observed for uncertainties of $\sigma_{zp} < 10\,\mathrm{mmag} \approx 1\,\%^{\mathrm{rel\ unc}}$ and $\sigma_\lambda < 1\,\text{Å}$. Importantly, they underscored the nearly degenerate nature of the cosmological parameter and calibration zero points, emphasizing that this degeneracy can only be resolved through precise external constraints on the flux scale. Moreover, their findings advocate for a relative calibration of the zero points between filters, surpassing 1 mmag, to extract the majority of the statistical information from a ten-year Rubin Observatory supernova sample.

2.3 Photometric Standard

To express a quantity, a unit is multiplied by a number. This unit must be defined and properly described. The ancient way was to create an artifact that realizes that unit. The latest example was the **kg**. Before the redefinition of the SI[18] unit system in 2019, a platinum-iridium cylinder defined the kg, the International Prototype of the Kilogram (IPK). The artifact has the disadvantage that it can be lost or damaged. The modern method is an abstract description of the unit by defining the values of natural constants that fix the unit's quantity. That allows the realization of a unit by experimenting. This ansatz lacks any drift of an artifact and makes the task a purely metrological problem. The uncertainties rely on the experimentalist competence and the available technologies. Any advance in metrology will result in a better determination of the unit. In the case of the IPK, lower uncertainties uncovered that its copies, the

[16] The DETF figure of merit is the reciprocal of the area of the error ellipse enclosing the 95% confidence limit in the w_0–w_a plane (Albrecht et al. 2006).
[17] Formerly Large Synoptic Survey Telescope LSST
[18] Système international d'unités or Internation System of units

national prototypes of different countries, drifted away from each other, leading to the conclusion that the IPK itself must drift, too. Next to the scientific question, this is also an all-day matter, as small changes in a unit can cause economic damage.
With the reform in 2019, all SI base units are defined abstractly via a fixed value for seven natural constants (the lowest hyperfine transition frequency of Cs $\boldsymbol{\Delta\nu_{\mathrm{Cs}}}$, the speed of light in vacuum **c**, the Planck constant **h**, the elementary charge **e**, the Boltzmann constant **k**, the Avogadro constant $\boldsymbol{N_A}$ and, the luminous efficacy $\boldsymbol{K_{\mathrm{cd}}}$). To realize a unit, a user is free to choose any given physics equation that links the defining constant to the quantity to be measured in the experiment (Brochure 2019b).
As this work is about photometric measurements, the unit that is of interest is the optical Watt. Even though the watt is the more fundamental unit, the SI unit that is realized by an experiment is the candela (Blevin and Steiner 1975). It quantifies the electromagnetic power integrated over the emitter's spectrum in a given direction. The physical quantity $\mathrm{W\,sr^{-1}}$ is an equivalent quantification of a candela and must be realized equally. One candela (**cd**) is roughly the luminosity a burning candle emits per steradians. The candela is defined in the following:

> **The candela, symbol cd, is the SI unit of luminous intensity in a given direction. It is defined by taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency** $540 \times 10^{12}\,\mathrm{Hz}$, $K_{\mathbf{cd}}$, **to be 683 when expressed in the unit** $\mathrm{lm\,W^{-1}}$, **which is equal to** $\mathrm{cd\,sr\,W^{-1}}$, **or** $\mathrm{cd\,sr\,kg^{-1}\,m^{-2}\,s^{3}}$, **where meter and second are defined in terms of h, c and** $\Delta\nu_{\mathrm{Cs}}$ **(Brochure 2019b).**

In other terms, a monochromatic source of $540 \times 10^{12}\,\mathrm{Hz}$ and one cd emits in a given direction $1/683\,\mathrm{W\,sr^{-1}}$[19]. Although it is realized from the second (a base unit) and the Watt (a derived unit), it is, per definition, a base unit. The Consultative Committee for Photometry and Radiometry (CCPR) of the International Committee for Weights and Measures (CIPM) suggested a list of approaches to realize the candela.

These methods are the easiest to implement or have the smallest officially recognized uncertainties and are referred to as "detector-based" or "source-based[20]". Under certain conditions, a Planckian radiator[21] and an electron storage ring producing synchrotron radiation[22] can be seen as absolute sources by the CCPR. The National Institut of Standards and Technology (NIST) (Larason and Jeanne M. Houston 2008), as well as the Physikalisch Technische Bundesanstalt (PTB) (Meindl et al. 2006), realize

[19] The arbitrary value of 683 is set such that the efficacy of a candela matches the sensation of the human eye to that luminosity of a candle. It is necessary to point out that in the case of other wavelengths, the candela and lumen are weighted with the sensation of a human eye.
[20] In the case of the candela, a source-based method is an experiment which realizes a photometric source of known flux, whereas a detector-based approach provides a primary that measures a flux traceable to the other base units.
[21] A black body whose spectral radiance can be predicted from thermodynamics traceable to the Kelvin.
[22] Synchroton radiation from circular moving electrons is traceable to electrical and length units.

the radiometric units with the Electrical Substitution Radiometer (ESR).

2.3.1 Methods of Calibration

To transfer the responsivity from one photodetector to another, there are two methods of calibration: the *photodetector substitution* and the *substitution with a monitor*.
For the first method, the signal Y_t of a device under test (DUT) to a monochromatic input X is compared with the signal Y_r from a detector of known response under the same input. The detector's signal Y depends linearly on the input X with the response s: $Y = s * X$. Under the circumstances that the input for each subsequential measurement is the same, the response of the device under test S_t follows as

$$S_t = \frac{Y_t}{Y_r} * S_r \tag{2.35}$$

from the response of the reference detector of known response S_r.
The comparison can be done in two ways. The first is to substitute the detectors at each wavelength. Secondly, one detector is scanned over the whole wavelength range before substitution. This approach assumes that the light source is stable over the comparison time. The former way is less affected by a light source drift but consumes more measurement time. The latter is faster but requires a highly stable light source. To encounter the drift of the light source, a beamsplitter between the source and detector is placed, and a monitoring detector receives the reflected beam. Under the assumption that the reflectance of the beamsplitter or the properties of the following optics and the monitor's response does not change between the comparison times, variations of the light source can be effectively eliminated on a time scale greater than the signal sampling rate, with simultaneous measurements of the monitor. The transmittance and the response of the monitor detector cancel out. So the response S_t is scaled with the ratio of the monitor's signals during measurements with the DUT Y_{mt} and with the reference detector Y_{ms}:

$$S_t = \frac{Y_t}{Y_r} * \frac{Y_{mt}}{Y_{mr}} * \frac{G_r}{G_t} * S_r \tag{2.36}$$

Where G_r and G_t are the different gains of the compared detectors. The NIST special publication (Larason and Jeanne M. Houston 2008) Sec. 3.2.4 provides a detailed calculation description.

2.3.2 Calibration Chain

The photosensor's response to a known flux must be quantified to calibrate it. This flux could be emitted by a source-based photometric primary or a source calibrated from a detector-based primary. As the primary is often hard to handle and for reasons of practicability, it is convenient to transfer the calibration to Working Standard (WS) detectors whose response to a given source is compared with the detector under test (DUT). Figure 2.6 shows the calibration chain for spectral power responsivity as it is realized at NIST. The primary detector is the Primary

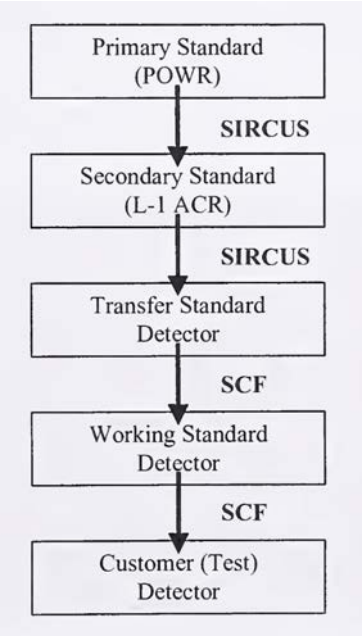


Figure 2.6: Calibration chain for spectral power responsivity measurements (Larason and Jeanne M. Houston 2008).

Optical Watt Radiometer (POWR). This absolute radiometer connects the received electromagnetic power with electrical power units and is part of the facility for spectral irradiance and radiance responsivity calibrations using uniform sources (SIRCUS). In the first place, the absolute scale is transferred via photodetector substitution (see Sec. 2.3.1) to the L-1 Absolute Responsivity Radiometer (ARC) from L-1 Standard and Technology, Inc., which is a more compact and mobile detector but has a verified low uncertainty (Jeanne M Houston and Rice 2006) serving as a secondary standard. The laser-based light sources of SIRCUS are used to compare the two absolute radiometers. In the next step, the responsivity of silicon trap detectors is measured by comparing their response to the received flux from SIRCUS quantified by a sequential measurement with the L-1 ARC with the same method as before. As the trap detectors do not provide the necessary long-term stability to serve as WS detectors, their response calibration is again transferred to S1337 Hamamatsu silicon photodiodes[23]using the Spectral Comparator Facility (SCF). The SCF uses a monitor to compensate for the flux drifts of its light source. The responsivity of a DUT is compared with the SCF from these WS detectors. The WS are calibrated annually (Larason and Jeanne M. Houston 2008).

2.3.3 Electrical Substitution Radiometer

The principle of an Electrical Substitution Radiometer (ESR) is the comparison of the heating power of electromagnetic radiation with the heating power from a substituted electrical power, causing the exact temperature change in a receiver cavity. The CCPR also suggests this ansatz in the appendix of the SI Brochure (Brochure 2019a). The NIST developed the POWR, previously known as HACR 2, to replace the high-accuracy cryogenic radiometer (HACR). The radiometer can be separated into three parts: the cryostat, the detector module, and the baffle section. The cryostat has an outer liquid nitrogen shield at 77 K and a liquid Helium reservoir working as the heat bath for the cold plate, the reservoir's bottom. The cold plate holds the detector module and baffles section shown in Fig. 2.7. The baffle section measures scattered light with an off-axis parabolic mirror and a photodiode along the optical path. The detector module has a copper-made cavity with a 30° slant closing off the back of the cylinder. The cavity is coated with specular black paint with an absorptance of 0.999 995 at 633 nm. For most measurements, POWR is operated at the boiling point (4.2 K) of Helium[24]. Cryogenic temperatures reduce the copper's heat capacity, increase its responsivity on heating, lower its emitted thermal radiation, and allow the use of superconducting wires. Two germanium resistance thermometers (GRTs) measure the cavity's temperature. The cavity is surrounded by heating wires that hold its temperature ≈1 K above those of the heat bath (Daniel Küsters 2023a). The radiation to be measured is heating the receiver cavity after entering the vacuum isolation of the cryostat and passing the baffle (Jeanne M Houston and Rice 2006). The rise in temperature in the cavity caused by the absorbed radiation is compensated by a reduction in the resistive ohmic heating to hold the temperature. A high-temperature sensitivity

[23] S1337 photodiodes are found to be best suited for high accuracy transfer measurements for the visible and UV spectral regions (Lei and Fischer 1993).
[24] If a lower thermal noise is required, the operating temperature can be reduced to 2 K.

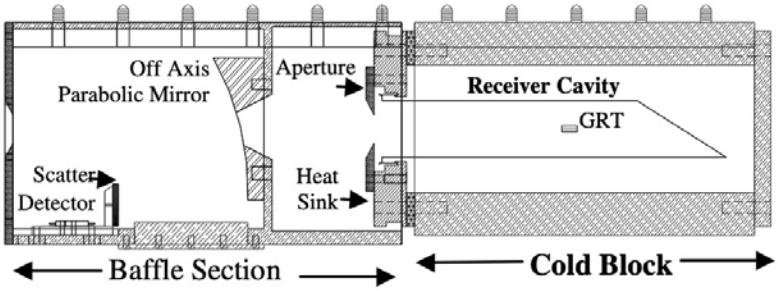


Figure 2.7: Side view of the detector module from (Jeanne M Houston and Rice 2006). The left half shows the baffle section with the parabolic mirror collecting stray light from the entering beam. The mirror has a hole to let the beam pass. The right half shows the copper-made receiver cavity with the germanium resistance thermometer.

of the thermistors is crucial. An operation on the critical temperature of a superconductor is also an option. The measured electrical power is maintained via voltage and resistive standards from NIST (Electrical Engineering Laboratory) (Larason and Jeanne M. Houston 2008). The receiver can measure microwatt to milliwatt optical power with an uncertainty of 0.01 % within a spectral window of 200 nm to 12 µm (Jeanne M Houston and Rice 2006).

2.3.4 Light Sources used in calibrations at NIST

To perform a calibration of the secondary standard or to transfer the calibration to a photodetector, a light source that is tunable in wavelength is needed. For the photodetector substitution method to transfer the standard to the trap detectors, the source must also be stable between the subsequential measurements with the absolute radiometer and the photodetector. The NIST uses these measurements in the *facility for spectral irradiance and radiance responsivity calibrations using uniform sources* (SIRCUS), which also contains the POWR radiometer. For substitution measurements with a monitor, NIST uses the Spectral Comparator Facility. Both are described in the following paragraph.

SIRCUS The NIST provides two facilities for their calibrations. The first is the *facility for spectral irradiance and radiance responsivity calibrations using uniform sources* (SIRCUS). It constitutes high-power tunable lasers whose emissions are introduced into an integrating sphere, producing a quasi-Lambertion flux source. The wavelength coverage depends on the available tunable lasers. It consists of the POWR radiometer and two light sources, SIRCUS (UV-VIS-IR) covers the range from 210 nm to 960 nm and IR SIRCUS from 700 nm. SIRCUS light source consists of several key components for precise measurement and control of the optical properties. A set of tunable lasers covers a wide spectrum of wavelengths, from 210 nm to 1050 nm. Specifically, the following lasers are utilized:

- Dye laser (Johnston, Brady, and Proffitt 1982)
- Ti:sapphire laser
- Nd:YAG laser

The Ti:sapphire laser is pumped by an argon-ion laser[25]. To cover the entire wavelength range, optical frequency multipliers are employed to double or triple the frequency of the laser's emitted

[25] This also provides strong lines in the visible range.

light[26]. The built-in etalon is used to tune the laser's output. An intensity stabilizer with a precision of 0.1 % is integrated into the setup to ensure consistent laser output. A portion of the laser output is directed to a Michelson interferometer with a sensitivity of 0.01 Å. Additionally, a Fabry-Perot interferometer is utilized to measure the bandwidth of the optical signals. Optical fibers are used for efficient light transmission, and a speckle-removing system, either an ultrasonic bath or a rotating mirror, is employed to minimize speckle effects (Mielenz, Saunders Jr, and Shumaker 1990). An integrating sphere is used as a diffusor, and a monitoring diode is placed within the sphere to monitor the incident light. To ensure minimal external interference, a light-tight box encloses the entire setup. The box is coated with Ultrapol, which has a reflectance of 0.1 % to 0.3 % over the wavelength range of 300 nm to 2.5 µm. Baffles are placed between the light source and the detector to reduce stray light and enhance measurement accuracy.

SCF The *Spectral Comparator Facility* (SCF) is a monochromator-based system that covers a wavelength range from 350 nm to 1800 nm. Two different photodiodes are employed as WS in the SCF:

- Hamamatsu S1337-1010BQ silicon photodiode (350 nm to 1100 nm)
- GPD Optoelectronics Corporation indium gallium arsenide (InGaAs) GAP5000 photodiode (1100 nm to 1800 nm)

These WS photodiodes are calibrated annually for accurate measurements. Fig. 2.8 shows the schematic of the SCF. A Helium-Neon laser allows the alignment of the detectors under test, guaranteeing that the illuminating beam hits the photosensitive area vertically. The typical light source is a 100 W quartz-halogen lamp. The optics within the SCF are confined in a light-tight box. A flat mirror mounted on an automated stage is used for source collection, and a spherical mirror pictures the source's image onto the monochromator's entrance slit. The Applied Physics Corporation Cary-14 monochromator is modified as a double monochromator with one featuring a prism and the second a grating with a blaze angle of 30° and 600 lines per mm density. The full-width at half-maximum (FWHM) bandwidth is set to 4 nm, achieved with 1.1 mm slit widths. The monochromator has a stray light rejection, with a rejection ratio of 10^{-8}. The optical power on the device under test (DUT) typically varies between 0.05 µW to 1 µW. A flat fused silica quartz plate is used as a beam splitter to monitor variations in the source during the comparison with a photodiode. The SCF retains a scanning carriage, enabling spatial uniformity measurements of the detector. A precision transimpedance amplifier circuit is employed to read out the photodiodes.

The facilities' detailed description and schematics can be seen in S. W. Brown, Eppeldauer, and Lykke 2006.

[26] For doubling, a lithium barium oxide crystal is typically used.

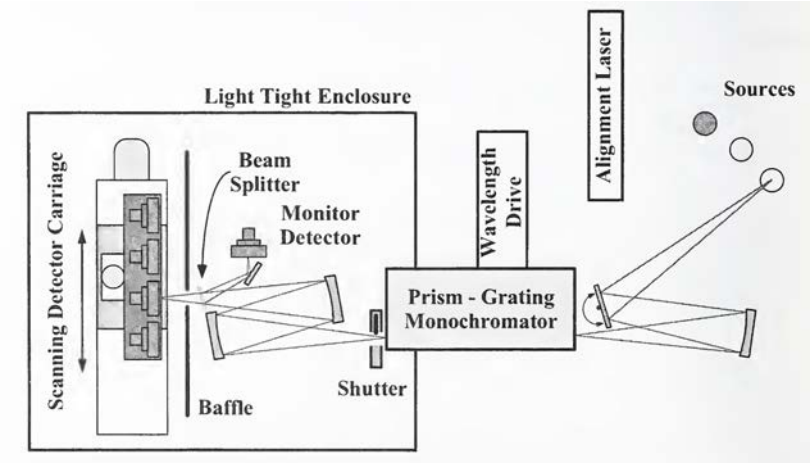


Figure 2.8: Schematic of the Spectral Comparator Facility for the VIS/NIR. Taken from (S. W. Brown, Eppeldauer, and Lykke 2006)

2.4 Standard Stars "Calibration Spectra" (CALSPEC)

CALSPEC, short for "Calibration Spectra", is a database of high-resolution stellar spectra for calibrating and characterizing astronomical instruments, particularly spectrographs. These spectra are obtained from well-known stars and are reference standards for astronomical observations. The CALSPEC database includes the spectra of stars with a wide range of spectral types and luminosities and cover all hour angles, making it a valuable resource for astronomers. These spectra are often used to calibrate and verify the performance of telescopes and spectrographs, ensuring that the data collected by these instruments is accurate and reliable. These spectra are essential for astronomical research applications, such as studying stellar properties, analyzing the interstellar medium, and determining distances to astronomical objects, especially supernova cosmology, as a bias in color will directly bias the Hubble diagram.

2.4.1 CALSPEC History Review

Up to the late '60s, the standard stars were not satisfactorily calibrated. Since 1960 all published energy distributions for α-Lyrae (Vega) disperse between 50 mmag and 100 mmag in the range from the ultraviolett to the infrared. Several groups (e.g. (Hayes 1970; Oke and Schild 1970)) made attempts to measure the absolute flux of Vega by comparison with a laboratory standard. The laboratory primary standard at that time was source-based. A tungsten lamp held at the melting point of copper or platinum, serving as a black body radiator (see Sec. 2.3). Bottom-up measurements had an uncertainty of <20 mmag in the visual and a monochromatic flux at 5556 Å of 3.48×10^{-9} erg s^{-1} cm^{-1} Å^{-1} (Oke and Schild 1970).

D. Finley, Basri, and Bowyer 1984 and R. C. Bohlin, Colina, and David S Finley 1995 suggested the usage of pure hydrogen white dwarf (WD) models to calibrate the International Ultraviolet Explorer satellite. R. C. Bohlin, Colina, and David S Finley 1995 calculated the spectral energy distributions (SED) for the four stars GD153, GD71, G191B2B, and HZ43. The WDs are selected to cover a wide range of effective temperatures

with high flux to minimize HST STIS[27] observing time, with negligible interstellar extinction (Vennes et al. 1994) and zero metallicity (R. C. Bohlin, Colina, and David S Finley 1995). Because FOS[28] photometry of these stars agrees with Landolt spectrography to be better than 1 %, these stars are suggested as primary reference standards for calibrations between 1000 Å to 10 000 Å (R. C. Bohlin, Colina, and David S Finley 1995). HZ43 is a hot, hydrogen-rich white dwarf (DA). It has an abundance of metal considerably lower than the measured abundance in white dwarfs exhibiting similar atmospheric parameters (Dupuis et al. 1998). However, as it has a companion M-star, it was rejected from the list of primary stars because the companion contaminates measurements with STIS.

2.4.2 CALSPEC Description

The hydrogen models depend on the effective surface temperature T_{eff} and the surface gravity $\log g$, which are determined by χ^2-fits of the models onto the observed Balmer lines in STIS data (R. C. Bohlin, Gordon, and Tremblay 2014) taken in Gianninas, Bergeron, and Ruiz 2011. Even though the interstellar reddening is small, a correction is applied to the models that correspond to a change in color of $E(B-V) = 0.5\,\text{mmag}$. The models are differentiated between Local Thermodynamic Equilibrium (LTE) and Non-Local Thermodynamic Equilibrium (NLTE). The newest models used in the CALSPEC database are the NLTE TLUSTY207 (Hubeny and Lanz 2017) and TMAP2019 models[29]. GB191-B2B is modeled with TLUSTLY, GD71, and GD153 with TMAP (R. C. Bohlin, Hubeny, and Rauch 2020). Fig. 2.9 shows the resulting models in the CALSPEC database, which are available as FITS files. The model uncertainty is also given and estimated by the quadratic sum of the variation of T_{eff} and $\log g$ with its uncertainties and the differences between the TMAP and TLUSTLY models. To quantify the goodness of the models, a calibration with STIS is done, as described in Sec. 2.1.2, and the residuals STIS/Model for all standard stars are estimated. Whereby the instrument's sensitivity $\hat{S}_\lambda$(GD71, GD153, GB191-B2B) for the different bands of the spectrograph is calculated as an equally weighted average from the results of all three stars (R. C. Bohlin 2000). Fig. 2.10 shows the above residual between $\hat{S}_\lambda$-calibrated STIS measurements and the models. With except the narrow hard-to-model Balmer lines at 3800 Å, the models show self-consistency with the STIS data within <0.5 % over the wavelength range of 1750 Å to 8000 Å (R. C. Bohlin, Hubeny, and Rauch 2020).

This consistency check assumes that the parameter coverage of the effective temperature and the surface gravity is large enough to probe STIS's sensitivity at a wide enough range of flux levels and that the model's parameters do not correlate with the resulting sensitivity. Assuming STIS to be linear and neglecting its noise contribution from Eq. 2.21, the residual res_i of the star i can be written as:

$$\text{res}_i = \frac{N_i^{\text{obs}}}{3F_i^m}\left(\frac{F_1^m}{N_1^{\text{obs}}} + \frac{F_2^m}{N_2^{\text{obs}}} + \frac{F_3^m}{N_3^{\text{obs}}}\right) \tag{2.37}$$

[27] Space Telescope Imaging Spectrograph at the Hubble Space Telescope.
[28] The Faint Object Spectrograph is aboard the Hubble Space Telescope. The calibrations are based on measurements from Oke, Schild, and Hayes (Hayes 1970; Oke and Schild 1970).
[29] Available at the Theoretical Stellar Spectra Access (TheoSSA) `www.dc.g-vo.org/theossa`

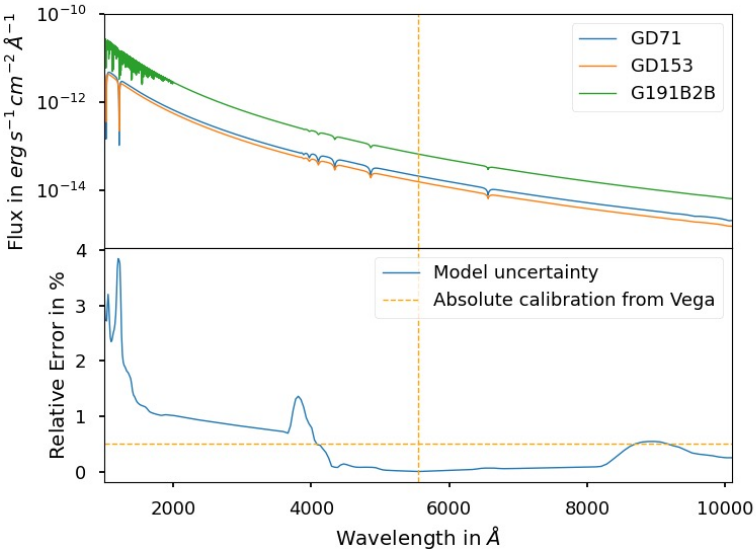


Figure 2.9: Upper panel: Blue, orange, and green pure hydrogen flux models of the stars GD71, GD153 (TMAP2019), and GB191-B2B (TLUSTLY207). In the lower panel, the square root of the covariance matrix's trace is in blue. The yellow dashed line shows the anchor point of the absolute flux scale at 5556 Å of $F(5556) = 3.44(17) \times 10^{-9}\,\mathrm{erg\,cm^{-2}\,s^{-1}\,\AA^{-1}}$ which gives another 0.5 % overall uncertainty. The latter does not affect the color (R. C. Bohlin, Gordon, and Tremblay 2014).

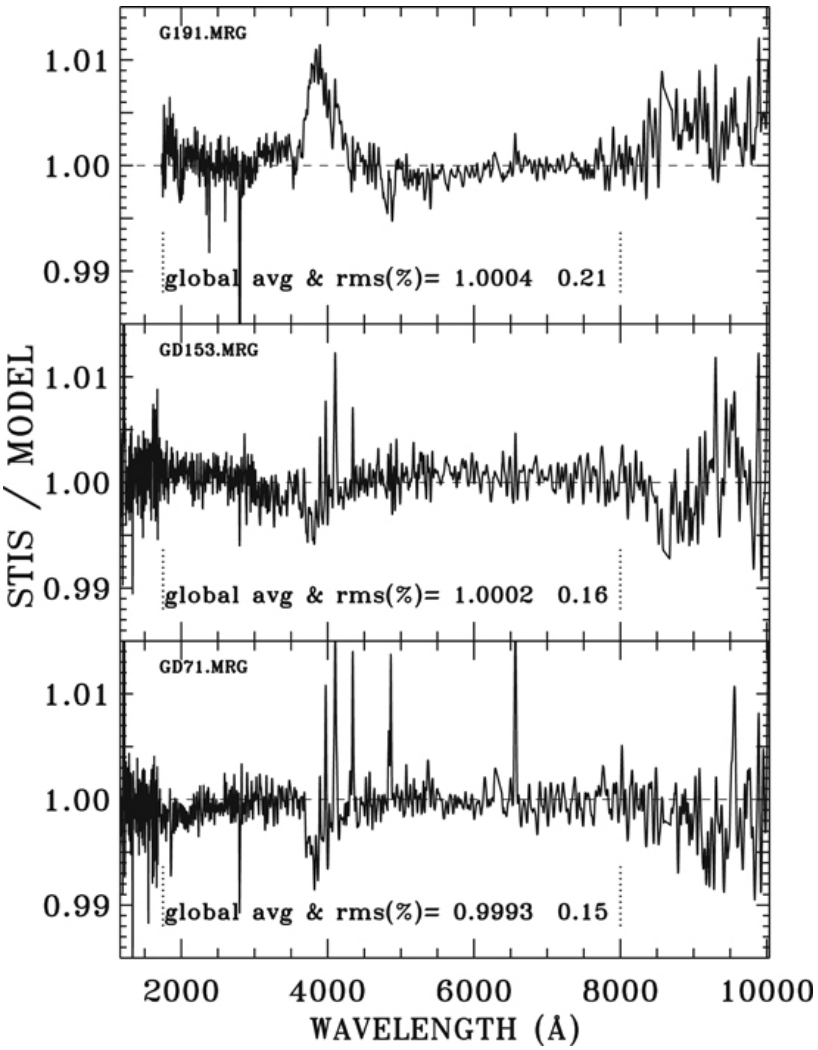


Figure 2.10: Residuals for the new set of primary standard WDs. The numerator flux distribution is created by combining STIS SEDs from the five low-dispersion observing modes in the 52×2 slit. At the same time, the denominators consist of the new model SED calculations obtained from TMAP for GD153 and GD71 and from TLUSTLY for G191-B2B. The average and RMS values, indicated on each panel, are computed within the range defined by vertical dotted lines at 1750 Å and 8000 Å. Taken from R. C. Bohlin, Hubeny, and Rauch 2020.

With N_i^{obs} the observed spectral counts and F_i^m the modeled SED. As the differences in the models F_i^m only differ in the used parameters (T_{eff} and log g, ignoring the differences between TLUSTLY and TMAP) that are determined by fitting the same models on STIS data which was calibrated with the former, less accurate models, it can not be excluded that the non-trivial terms in Eq. 2.37 cancel out per construction of the self-consistency between models and STIS data.

As a further note, R. C. Bohlin, Hubeny, and Rauch 2020 updated the models of the standard stars, and the models changed in the range from 3000 Å to 10 000 Å by 1 %. In the whole range of the models (0.1 μm to 20 μm) the changes are >2 %.

Based on the upper discussion, flux measurements with instruments that have a calibration chain to the laboratory standard are needed to verify the above approach. To enhance the spectral calibration, bottom-up calibrations are needed that are more accurate than 0.5 % in the flux scale.

2.5 The Bottom-up Calibration

As described in Sec. 2.2, the zero-point calibration uncertainty $\sigma(ZP)$ of the different bands that measure SNIa light curves must surpass 0.1 % level in absolute flux to extract all statistical information in the example of 10-years Vera C. Rubin Observatory data. Stubbs & Tonry 2006 (Stubbs and Tonry 2006) reviewed the calibration of ground-based CCD photometric measurements and identified the two components to break the 1 % "precision barrier":

1. The full characterization of the atmospheric transmission.
2. The identification, measurement, and removal of instrumental artifacts.

The content of this work will concentrate on the second point as this affects the measured color of an object, while the first only affects the absolute flux scale, in the case of a photometric night, if a standard source is measured at different air mass (Pierre Astier 2012).

Based on the upper review, Lombardo, Greg Aldering, et al. 2014 suggested a device to calibrate the Integral Field Spectrograph (SNIFS) on the University of Hawaii 2.2-meter telescope. The SNIFS Calibration Apparatus (SCALA) comprises 18 mirrors, each with a diameter of 16 cm, and these mirrors are exposed to light from a tunable monochromatic source through six integrating spheres. Positioned at the telescope's pupil, SCALA produces narrow beams with a 1-degree width, ensuring uniform focal plane illumination. SCALA is a spectral comparator that allows calibration of the standard stars with the substitution method with a monitor, where the monitoring photodiodes are calibrated at the relative scale.

This allows the first bottom-up calibration of stars since Hayes 1970, Oke and Schild 1970, and Landolt 1983 in the '70s and early '80s. Part of this work is the development of a spectral comparator that allows the transfer of relative responsivity calibrations of photodetectors. This comparator follows the description in Sec. 2.3.1 for the substitution method with a

monitor (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020; Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022). Its uncertainty is estimated to be below the target of 0.5 % over the spectral coverage of SCALA 3000 Å to 9500 Å. Fig. 2.11 shows the calibration chain from the NIST laboratory standard to the stars' SEDs. The NIST calibrated and provided three Hamamatsu S2281 photodiodes, which are used to calibrate 18 further photodiodes with their amperemeters[30]. The comparator setup is also used to test the amperemeter's linearity and quantify the reproducibility of those measurements. Another advantage of the new comparator system is the gantry robot that allows illumination of the DUT under different angle-of-incidence (AoI) that is unique under similar setups (Larason and Jeanne M. Houston 2008; Meindl et al. 2006).

SCALA is then equipped with calibrated photodiodes that monitor the incident flux on the telescope. The calibration of SNIFS is done during the daytime, and the observation of the stars is performed on the same night, allowing direct transfer of the relative calibration from the photodiodes to the stars' SEDs.

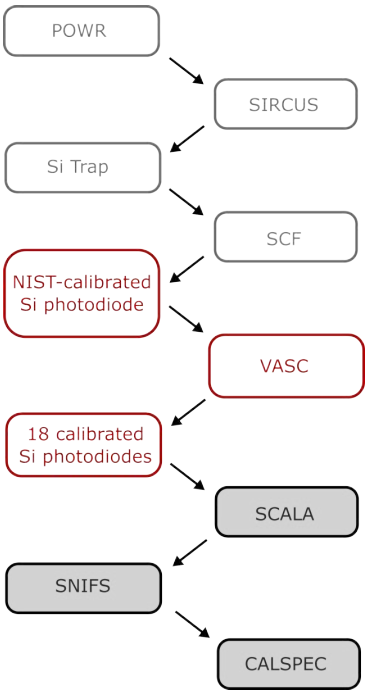


Figure 2.11: Proposed calibration chain for the standard stars. The chain follows the same structure as in Marc Betoule et al. 2023. The left blocks show the detectors, while the right ones display the sources. In grey, it starts as explained in Sec. 2.3.2 with the measurements conducted at NIST (Larason and Jeanne M. Houston 2008). Contributions are made with the VASC, developed in this work, and ends (grey background) with observing the stars.

[30] Self-developed amperemeters based on the transimpedanz amplifier described in Larason and Jeanne M. Houston 2008.

Experimental Setup 3

This chapter presents the laboratory equipment and apparatus for photometric calibrations explored in this thesis. It describes the Variant Angle Spectral Comparator (VASC) as the device used for transferring the calibration from one photometer to another by comparing their responsivities.

The discussion commences in Sec. 3.1, detailing the multiple components of the VASC. The light sources are explained in Sec. 3.1.1. The double monochromator and wavelength model are explained in Sec. 3.1.2 and 3.1.2, and 3.1.2, respectively. Sec. 3.1.3 describes using a holmium-didymium filter for wavelength calibration, and the setup for linearity testing is presented in Sec. 3.1.4. The section on optical fibers is discussed in Sec. 3.1.5, followed by a depiction of the test chamber and gantry robot employed for positioning and measurements in Sec. 3.1.6. Sec. 3.1.7 centers on the relay optics responsible for directing and focusing the light. The chapter then provides an overview of the control software in Sec. 3.1.8, emphasizing its function in managing the operations of the VASC.

Sec. 3.2 moves on to the LArge Resistance Amperemeter (LARA), starting with a discussion of the photodiode's working principle in Sec. 3.2.1. Subsequently, Sec. 3.2.2 covers the amperemeter's design and function.

The SNFactory and the SuperNova Integrating Field Spectrograph "SNIFS" are discussed in Sec. 3.3. An overview of the SNFactory is provided in Sec. 3.3.1, while the operational details of SNIFS are explained in Sec. 3.3.2. SCALA is discussed in Sec. 3.4, which summarizes the initial issues and subsequent improvements from Küsters' 2019 study in Sec. 3.4.1. The SCALA-Upgrade 2022 enhancements are described in Sec. 3.4.2.

This chapter provides an up-to-date description of the laboratory setup, including the latest modifications. Any differences from the information presented in the SPIE proceedings (Daniel Küsters 2019) and (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020) are due to these updates. This chapter presents a thorough explanation of the photometric calibration setup and procedures used in the laboratory.

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3.1 Variant Angle Spectral Comparator (VASC)

The Variant Angle Spectral Comparator (**VASC**), formerly known as the Photometric Calibration Setup (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020; Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022), enables fully automated sub-percent calibration of photosensors with wavelength uncertainties in the angstrom range and performs tests at various angles of incidence (AoI) and positions. In addition to SCALA, the setup is designed to meet the requirements of the IceCube and ULTRASAT projects (refer to Table 3.1). The setup can be divided into the light source and the movable relay optics. The light

Table 3.1: Requirements from the different research groups. The most constraining parameters are bold. Cited from (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020).

Project	Wavelength Range in Å	unc. in %	Flux in γ/s	unc. in %	Detector Size in cm
IceCube	2500–6000	~ 1	**< 1k**	1	⌀**40**
SCALA	3000–10000	**< 0.1**	**> 10M**	**< 0.1**	1×1
Ultrasat	**2000–11000**	< 0.5	< 3	< 0.5	10×10

source includes two broadband light sources and low-pressure gas lamps, which enable the calibration of the wavelength scale using spectral lines. It also has LASERs for alignment purposes and uses an LED for linearity tests. Two asymmetrical in-plane Czerny-Turner monochromators are combined to suppress out-of-band emission. Encoders measure the position of the grating turret to calculate the expected output wavelength, calibrated with low-pressure gas lamps. The goal of illuminating the device under test (DUT) is achieved by a relay optic mounted on a gantry robot with three spatial and two rotational degrees of freedom. The optic includes a monitoring diode, which records the current flux and allows the transfer of the NIST photometric laboratory standard to the DUT by determination of the momentary splitting ratio of the beam splitter. This ensures sub-percentage calibration of the DUT's responsivity.

3.1.1 Lightsources

Several light sources are needed for wavelength, photometric calibration, and alignment. Fig. 3.1 shows a drawing of the light source setup that produces the monochromatic light. The setup is equipped with two broadband sources a 150 W Xenon arc lamp (S1)[1] and a Laser-Driven Light Source (LDLS), a lamp socket for several low-pressure gas lamps (see Fig. 3.2 and 3.4), and a 532 nm class two LASER (<1 mW). Each light source can be selected separately with a motorized flat mirror M3. The default light source for photometric calibrations is the LDLS, which has the highest ultraviolet (UV) flux output. Two parabolic mirrors, M1 and M2 ($\varnothing$ = 2″), change the focal ratio of the output beam from f/1 to fit the monochromator's focal ratio of f/4. The plasma point's image is then pictured by a spherical mirror (M4) over an intermediate flat mirror (M3) to the monochromator's entrance slit. M3 has a distance of ≈ 500 mm to M4 and is mounted on a rotation stage with a stepper motor to change between the images of the different light sources (Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022). M4 and the focus of M3 form a circle of focus (CoF) around the mirror's rotation axis with a radius of ≈ 300 mm. Depending on the position of M3, M4 can depict an image of a light source on the monochromator's entrance slit sharply if it is on CoF.

Spectral lines of several Low-Pressure Gas (LPG) lamps (see Fig. 3.2)[2] are used to calibrate the absorption lines of the Holmium-Didymium Filter (HoDi). A broadband light source with the same image as the

[1] For an overview of all necessary components, see Tab. A.1 in the Appendix.

[2] There are lamps from NARVA with main component Na (Fig. A.1a), Cd (Fig. A.1b), Ne (Fig. A.2a) and Hg (Fig. A.2b) are used for calibration (see technical specification for Hg Fig. A.8).

low-pressure lamps is needed to transfer the calibration from the spectral lines to the HoDi filter. To achieve this, a hollow cylinder (integrating cylinder, IC)[3] was fabricated (Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022), which can be selectively illuminated by a Xenon lamp (XeLamp) via IC_{in} and one LPG lamp enclosed by the IC. The lamp is aligned off the cylinder's axis to ensure only scattered light exits IC_{out}.

[3] Costum made from SphereOptics GmbH; Gewerbestrasse 13; 82211 Herrsching; Germany for technical drawing see appendix (see Sec. A.4) (Daniel Küsters 2023b).

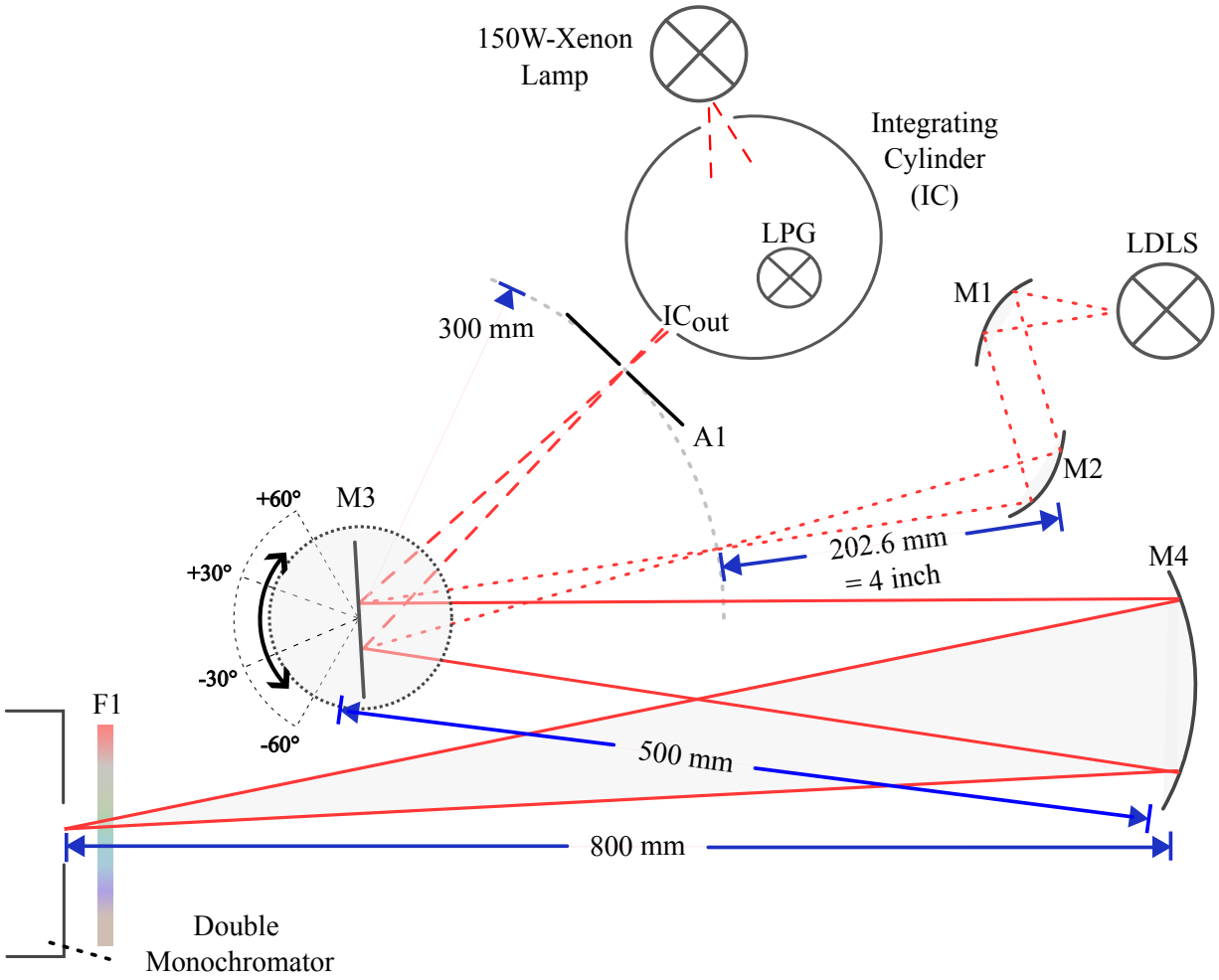


Figure 3.1: Schematic of the light source. The rotational mirror **M3** allows the selection of different light sources. The Laser-Driven Light Source is the default broadband source (**LDLS**) used for ordinary measurements. The second broadband light source, a 150 W-Xenon Lamp (**XeLamp**), and the Low-Pressure Gas (**LPG**) lamp socket are used for wavelength calibrations. The integrating cylinder (**IC**), a hollow PTFE cylinder, ensures that the light path for the XeLamp and the LPG lamp are the same and allows a transfer of the wavelength scale from the LPG lamps to the broadband XeLamp. The mirrors **M1/2** change the focal ratio from f/1 to f/4, as the monochromators require. The spherical mirror **M4** projects the lamps' images at the 300 mm-circular plane around **M3** onto the monochromator's entrance slit. A detailed description of the components see Tab. A.1.

To suppress stray light, an aperture of two blades forming a 25.4 mm in height times ≈ 3 mm slit and aluminum foil are placed in CoF formed by M3 in front of IC_{out} (A1). The LASER is deflected into the monochromator plane with flat mirrors so that M3 can depict it. Optionally, an LED powered by a stable power source can be replaced by the LASER to operate as a steady light source, as all other non-coherent lamps vary with 2 Hz to 100 Hz fluctuations in the % range.

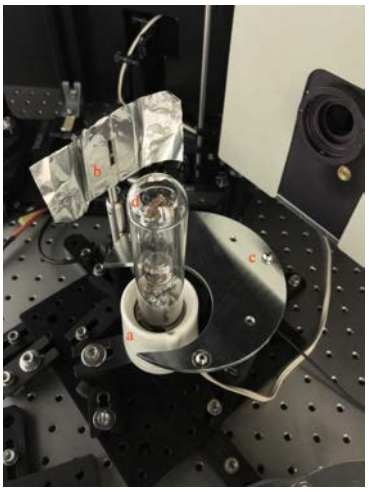


Figure 3.2: Low-pressure gas lamp mounting without housing with a: being the lamp socket, b: aperture to suppress stray light, c: alignment plate of the integrating cylinder, and d: a mounted low-pressure gas lamp. The metal balls at c: and indentations at the bottom plate of the integrating cylinder's housing ensure a reproducible alignment of the cylinder.

3.1.2 Monochromator Setup and Wavelength Model

To fulfill the requirements for SCALA, there are three requirements for the monochromator system. First, it should provide monochromatic light with negligible out-of-band rejection. Secondly, the wavelength shall be determined below 1 Å to fulfill the requirement in Tab. 3.1 for SCALA. One Å wavelength uncertainty leads to a relative uncertainty contribution to the responsivity calibration of a photodiode below 0.1 % (see Fig. 3.6) and the band-pass shape must be known.

Out-off-band Rejection and double monochromator

A monochromator system can be seen as a bandpass filter with a certain suppression level. Internal scattering within the monochromator leads to out-of-band light, which produces a floor of broadband light. To quantify this floor, one measures the throughput of a bandpass filter within its rejection floor using a photodiode and a broadband light source. To get the transmittance, the photocurrent is recorded[4] with and without the high-pass filter. The monochromator is tuned from 2000 Å to 6000 Å over the rejection band. The scattered light bypasses the monochromator and is integrated from the cut-off 4950 Å to the end of the photodiode's sensitivity range at ≈11 500 Å. Fig. 3.7 shows the measured transmittance of a Schott GG495 high-pass filter with a cut-off wavelength of 4950 Å and a suppression of $< 10^{-5}$. The measurement is performed two times with two different Cornerstone 260 Monochromators. Within the rejection range, a clear signal is still left, for both monochromators (green and red graphs). The gray dashed line shows the same measurement performed with the SCALA monochromator used in its 2015 commission (SCALA will be described in Sec. 3.4).

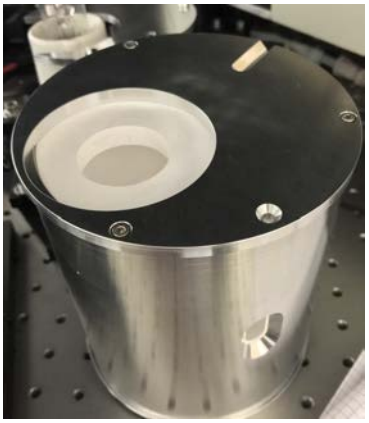


Figure 3.3: The bottom view of the integrating cylinder.

The solution is the sequential usage of a second Cornerstone 260 Monochromator. This relais will be called a double monochromator in the following, and Mono1 and Mono2 if referred to as the first and the second monochromator, respectively (see Tab. A.1). The NIST uses for UV calibrations in its spectral comparator facility a similar approach with a monochromator with two gratings (Larason and Jeanne M. Houston 2008) and the *Physikalisch-Technische Bundesanstalt* uses a Prism-grating double monochromator (Meindl et al. 2006). Fig. 3.8 shows a drawing of the monochromator setup together with the optics to correct for astigmatism, beam alignment (described in Sec. 3.1.2), HoDi transmittance imprint (see Sec. 3.1.3), intensity reduction and linearity tests (see Sec. 3.1.4), and the coupling into the optical fiber (see Sec. 3.1.5). Cornerstone 260 Monochromators from Newport green and red curve in Fig. 3.7. The

[4] Using a Keithley 6482 with a detection limit of 0.1 pA.

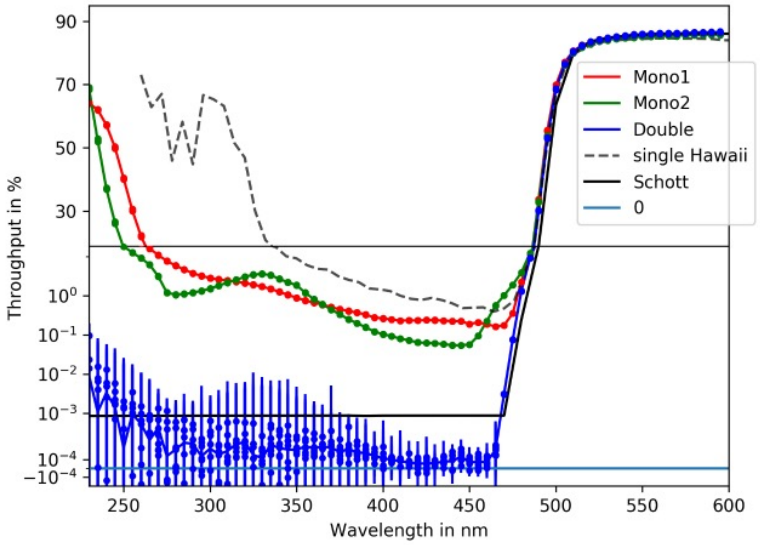


Figure 3.7: Measurements of the transmission of Schott GG495 filter with Mono1, Mono2 and in series from (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020).



Figure 3.4: Low-pressure gas lamp with the integrating cylinder with the 150 W-WXenon lamp on the left upper corner to illuminate into the integrating cylinder.

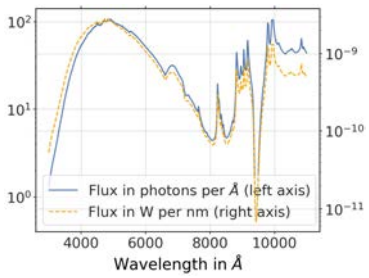


Figure 3.5: Available flux with the LDLS in photons per Å (left axis, solid blue line) and in W per nm (right axis, orange dashed line) at the DUT. The throughput of all optical components limits the maximal Flux in the UV and IR. In contrast, the limitation on the flux in the VIS is due to the saturation level of the reference photodiode's amperemeter ≈250 pA.

former transmittance measurement is repeated with the resulting double monochromator system, and the blue graph in Fig. 3.7 shows that it is capable of suppressing the out-off-band light below the detection limit (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020).

Synchronisation

The manufacturer of the Cornerstone 260 Monochromators refers to an accuracy of 3.5 Å and a precision of 0.8 Å for a single monochromator if calibrated with a two-point calibration. To suit SCALA accuracy of one Å, a calibration of the monochromator's wavelength is necessary with multiple points over the whole range for the complete double monochromator. As a reference, a set of line spectra is recorded with a LARA with different low-pressure gas lamps (LPG lamps see Tab. A.1) as a light source. The lines are assigned to the Atomic Spectra Database from NIST (Standards and Technology 2009). Before calibration, both monochromators must be synchronized, as their internal calibration is insufficiently accurate. These relative wavelength differences $\Delta\lambda_{\text{Mono}}(\lambda_1)$[5] need to be determined by maximizing transmission. The first monochromator is iterated from 300 nm to 1000 nm and the second is tuned to maximize the double monochromator's throughput $F(\lambda_1, \lambda_2)$ that is proportional to the recorded photocurrent $I(\lambda_1, \lambda_2)$[6]. The Xenon lamp is used as a light source. Each wavelength sweep of the second gives a top-hat profile, whereas the difference between the first monochromator's wavelength λ_1 and the center of mass of the profile $I(\lambda_1, \lambda_2)$ is calculated with:

$$\Delta\lambda_{\text{Mono}}(\lambda_1) = \frac{\sum_{\lambda_2} I(\lambda_1, \lambda_2) * \lambda_2}{\sum_{\lambda_2} I(\lambda_1, \lambda_2)} \tag{3.1}$$

[5] Wavelength tuning of the first/second monochromator is labeled as λ_1, resp. λ_2.
[6] Measured with a Keithley 6485 and a photodiode illuminated by the output beam.

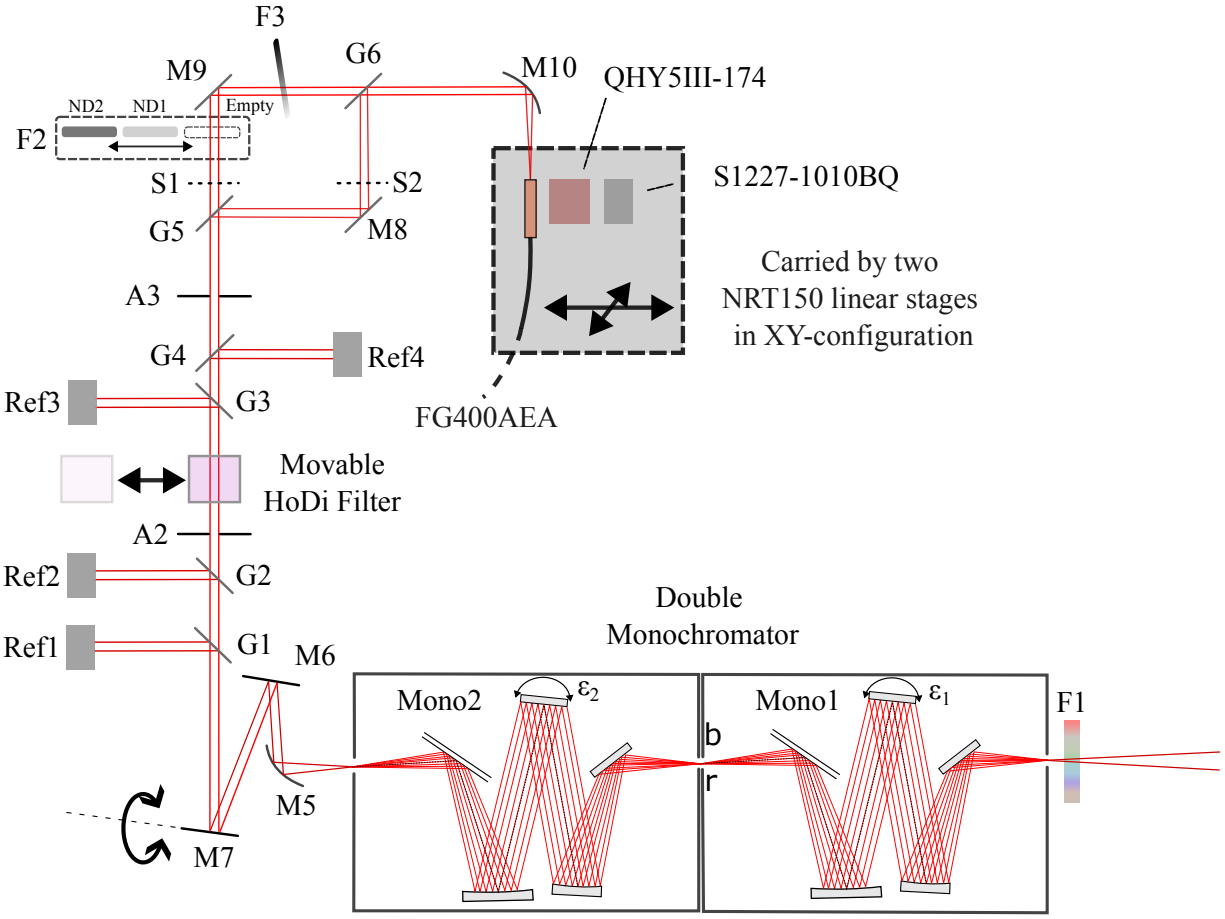


Figure 3.8: Schematic of the double monochromator setup. The light source is depicted in Fig. 3.1. The monochromator setup consists of two Cornerstone 260 Monochromators (Mono1 and Mono2) with a focal length of 260 mm. The first monochromator is illuminated with the output of the light source. The second monochromator is illuminated with the output of the first monochromator. Both monochromators are equipped with three gratings with different blaze angles to ensure optimal thought for different wavelength ranges (see Tab. 3.2). The gratings are mounted on a turret, whose position is measured with an encoder. The monochromator's output is collimated with a parabolic mirror (**M5**), and corrected for astigmatism with a cylindrical lens (**M6**). The HoDi filter is mounted on a linear stage for movements in and out of the beam. The goniometer-mounted flat mirror (**M7**) corrects for the beam's tilt changes due to imperfect alignment of the gratings. The iris aperture (**A3**) defines the beam's diameter. The two neutral density filters (OD:1 **ND1** and OD:2 **ND2**) on the linearity stage and the variable neutral density filter allow continuous reduction of the intensity to avoid saturation of the photosensors. The linearity setup (described in Sec. 3.1.4) consists of two glass plates (**G5/6**), two plane mirrors (**M8/9**) and two shutters (**S1/2**). The last part is the parabolic mirror (**M10**) that focuses the beam onto a linear fiber FG4000AEA or two detectors all mounted on NRT150 linear stages in XY-configuration (described in Sec: 3.1.5).

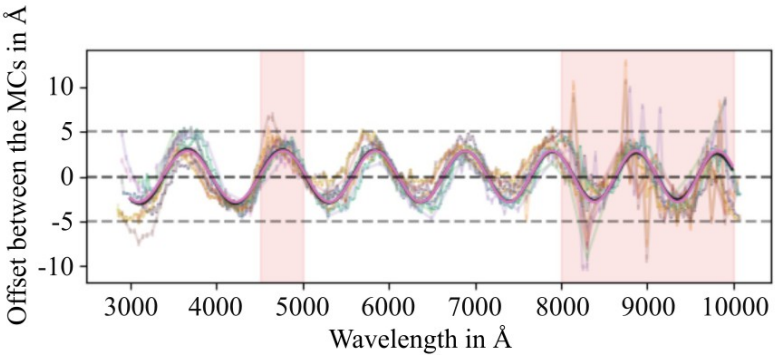


Figure 3.9: Offset between the monochromators to get maximal throughput of the double monochromators from (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020). The different repititions are different 1200 lines mm^{-1}-grating. The red-shaded regions are those with strong emission lines of the used XeLamp which affects the CoM of Eq. 3.1 making that data useless for the fit. The red solid line showcases a sinusoidal fit with all data included. The masked data results in the black solid line with a periodicity of 1000 Å describing the data.

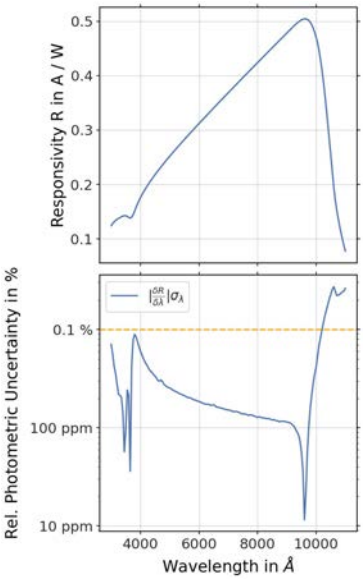


Figure 3.6: Upper plot, responsivity from a Hamamatsu S2281 photodiode calibrated at NIST (Larason and Jeanne M. Houston 2008). The lower plot, the resulting relative uncertainty on incident flux for $\sigma_\lambda = 1$ Å The scatter at 3600 Å are due to the finite resolution of the responsivity data around the local maximum.

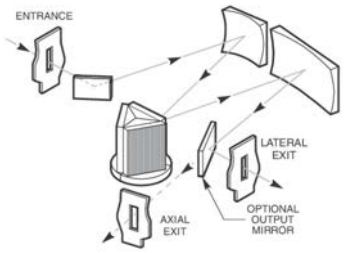


Figure 3.10: Optical configuration of the Cornerstone 260 Monochromators (Newport 2011).

Fig. 3.9 shows $\Delta\lambda_{\text{Mono}}$ as a function of λ_1. The different repititions are different 1200 lines mm^{-1}-grating. The sinusoidal modulation comes from a mechanical error in the worm gear for both monochromators. The used XeLamp has emission lines in the red-shaded region, in which Eq. 3.1 gives insufficient values. A sinusoidal function $A * \sin 2\pi \frac{\lambda_1 - \varphi}{p}$ is fitted to the data to describe the offset with a periodicity p of ≈1000 Å and an amplitude A of ≈2.5 Å with the shaded region being excluded and used to determine the tune (Eq. 3.2) for monochromator 2 concerning the other.

$$\lambda_2(\lambda_1) = \Delta\lambda_{\text{Mono}}(\lambda_1) + \lambda_1 \tag{3.2}$$

Wavelength Model

The Cornerstone 260 Monochromators works with an asymmetrical in-plane version of a Czerny-Turner monochromator (see Fig. 3.10). The input light must have a focal ratio of F/3.9. A motor rotates the grating turret holding three gratings (Tab. 3.2) to the angle on which the desired wavelength fulfills constructive interference. The focal distance for the output monochromatic light is ≈260 mm, and the shape of the bandpass is a triangle (Newport 2011).

After synchronization (Eq. 3.2) the requested wavelength $\lambda_{\text{Target}} := \lambda_1$ must be mapped to a traceable wavelength calibrated scale λ_{NIST} to achieve the needed accuracy <1 Å. The worm gear error introduces a sinusoidal modulation. The values between the spectral lines must be interpolated, for which no model is available.
Instead, the expected wavelength is calculated by modeling the double monochromator. The actual position of the grating turret is measured, from which the expected model wavelength λ_{model} can be derived. In the last step, λ_{model} is calibrated with a linear model against spectral lines of the low-pressure gas lamps with reference values from the NIST Atomic Database (Standards and Technology 2009).

Table 3.2: Table of Mounted Plane Ruled Gratings Acquired from Richardson Gratings, a Division of Newport Corporation. Note *: Absence of Catalogue Number and incomplete Master Number. Note #: Blaze Angles for the 1200 $\frac{\text{lines}}{\text{mm}}$ gratings are extrapolated based on data from similar gratings, as these specific models are no longer available from the manufacturer, precluding data recovery.

Position		Catalogue No.	Master No.	Nominal Wavelength in Å	Blaze Angle in °	Grooves in $\frac{\text{lines}}{\text{mm}}$
	1	53-∗-330R	1624	4000	14	1200
Mono1	2	53-∗-360R	MR136	7500	26.7	1200
	3		02L0*	2250	11.3	600
	1	53-∗-200H	5233	2500	8.8#	1200
Mono2	2	53-∗-280R	1597	5000	17.5	1200
	3		02L0*	2250	11.3	600

Fig. 3.11 shows both monochromators fixed on a 800 mm × 450 mm aluminum bottom plate with 10 mm thickness. The plate has holes to allow access to the turret's axis (see Fig. 3.12) for mounting the encoder adapter (see Fig. 3.13). The entrance slit of the first and the exit slip of the second monochromator are set to 1 mm, giving a bandwidth of 18 Å for a combination of two gratings with 1200 lines mm^{-1} and 36 Å for a combination with two times 600 lines mm^{-1} (the calculation is shown in Daniel Küsters 2019). Figure 3.14 shows the mounted Heidenhain encoder for Mono2. The encoders are connected to a controller unit (see Fig. 3.15), which a self-written Python API reads out. The retrieved information about the turret's orientation for both monochromators $\epsilon_{1/2}$ can be translated into an expected wavelength λ_{model}, which is derived in the following.



Figure 3.11: Two Cornerstone 260 Monochromators mounted on a bottom plate to allow access to the turret's worm gear from below. The plate also fixes the alignment between the monochromators.

Derivation of the Model Wavelength

For constructive interference, the path length difference Δs of two neighboring beams separated with the groove distance d at the gratings must be a multiple m[7] of the wavelength λ (compare Fig. 3.16). For the incident angle α and the diffraction angle β that means:

$$\Delta s = m\lambda_i n_{\text{air}} = d(\sin\alpha_i - \sin\beta_i) \tag{3.3}$$

and

$$\lambda_i(\alpha,\beta) = \frac{d}{m n_{\text{air}}}(\sin\alpha_i - \sin\beta_i) \tag{3.4}$$

With the incidence α_i and diffraction angle β_i within the monochromator $i \in [1,2]$ onto their gratings and n_{air} the air's refraction index[8]. Within the monochromators, the sum of both $\gamma_i = \alpha_i + \beta_i$ is fixed at ≈24° (see Tab. 3.3)[9]. Whereas α_i and β_i are functions of the orientation of the grating measured from the Heidenhain encoders ϵ_i. In the case of the expected wavelength after Mono2, the interference condition must be fulfilled for both monochromators simultaneously. Mono2 is illuminated with a spectrum $\lambda_c(x,\epsilon_1)$ from Mono1 with $x \in W_{\text{mid}} = [-w_{\text{mid}}/2, w_{\text{mid}}/2]$ the position at the entrance/exit slit of width w_{mid} of monochromator two/one. For the light at the output of Mono2 x fulfills



Figure 3.12: Axis of the grating turret of a Cornerstone 260 Monochromators. The axis is a worm gear that is countered by a feather to increase the mechanic's reproducibility if rotated by the stepper motor.

$$\lambda_{\text{model}} := \lambda_2(x,\epsilon_2) = \lambda_c(x,\epsilon_1). \tag{3.5}$$

The spectrum $\lambda_c(x,\epsilon_1)$ can be described with a linear displacement x from the center of W_{mid} with the wavelength λ_1 from Eq. 3.4. The linear displacement in wavelength can be described with the reciprocal dispersion $R_d = |\frac{\delta\lambda}{\delta x}|_{x=0}$. From that $\lambda_c(x,\epsilon_1)$ follows:



Figure 3.13: Axis extension on the monochromator's turret axis to allow the attachment of the Heidenhain encoder.

$$\lambda_c(x,\epsilon_1) = \frac{\lambda_1(\epsilon_1)}{n_{\text{air}}} + R_{d,1}x \tag{3.6}$$

The orientation $\epsilon_{1,2}$ for the gratings of the monochromator one and two



Figure 3.14: Mounted Heidenhain encoder at monochromator two.

[7] Which is also the order of diffraction.

[8] To calculate the refractive index the air properties are measured with two BME280 sensors that log air pressure temperature and humidity in the lab at the moment of each photometer-read-out. The formulas follow the method of Edlen published from NIST (Stone and Zimmerman 2001)

[9] To estimate the correct values of γ_i for each monochromator and grating the emission line of a low-pressure gas lamp was scanned. Together with the measured angle from the Heidenhain encoders, the wavelength provided by NIST and the current air's refractive index γ can be calculated from Eq. 3.7



Figure 3.15: Heidenhain controller to read out the ND287 encoders.

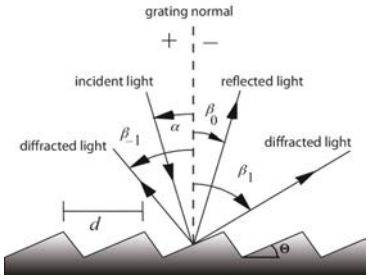


Figure 3.16: Schematic of plane ruled diffraction gratings. With α as the incident angle and β_m the diffraction light for different orders of diffraction $m \in [-1, 0, 1]$. In this work $m = +1$. The blaze angle Θ shifts the intensity maximum of the enveloping distribution of the interference pattern from 0th-order to a desired wavelength to the 1st. The shape of the reflectivity depends, therefore on Θ and the overall throughput of the double monochromator system on the choice of the different gratings from Tab. 3.2.

Gratings	γ_1 [°]	γ_2 [°]
1 & 1	23.23681	21.66943
1 & 2	23.90396	22.41484
2 & 2	23.80299	22.26506
3 & 3	24.40978	23.19181

Table 3.3: Indivdual values for γ_i for different grating configurations. As the correct geometry depends on the alignment of the individual grating, γ_1 depends on the choice of the grating for Mono2 and vice versa. The errors on those values lead to a systematic shift in the model wavelength, which is assumed to be small enough to be treated as linear and will be included in the calibration of λ_{model}. The encoder system sets the resolution.

are defined as zero if the grating's zeroth-order illuminates the slit. That means the monochromator is bypassed with white light ($\alpha = \beta$). From this, it follows that $\epsilon = \alpha - \frac{\gamma}{2} = \frac{\gamma}{2} - \beta$ together with Eq. 3.4 one gets:

$$\lambda_1(\epsilon_1) = \frac{2dn_{\text{air}}}{m}\cos(\gamma_1/2)\sin(\gamma_1/2 - \beta(\epsilon_1)) = \frac{2dn_{\text{air}}}{m}\cos(\gamma_1/2)\sin(\epsilon_1) \tag{3.7}$$

For the dispersion relation R_d one allows the displacement x at W_{mid} with $\beta_1 \rightarrow \beta_1 + \frac{x}{f_{\text{obj}}}$ in Eq. 3.4 and evaluates the first derivative at $x = 0$:

$$R_{d,1} = \left|\frac{\delta}{\delta x}\left(\frac{dn_{\text{air}}}{m}\left(\sin(\alpha_1) - \sin\left(\beta_1 - x/f_{\text{obj.}}\right)\right)\right)\right|_{x=0} = \frac{dn_{\text{air}}}{m f_{\text{obj.}}}\cos(\beta_0) \tag{3.8}$$

with the objective's focal length $f_{\text{obj}} = 256.9\,\text{mm}$ and the line constant d. For the lefthand side of Eq. 3.5 one allows the displacement x for $\alpha_2 \rightarrow \alpha_2 - \frac{x}{f_{\text{col}}}$ with the collimator's focal length $f_{\text{col}} = 219.15\,\text{mm}$ of Mono2 in Eq. 3.4:

$$\lambda_2(\alpha, \beta) = \frac{d}{m}(\sin\left(\alpha_2 - \frac{x}{f_{\text{col}}}\right) - \sin\beta_2) \tag{3.9}$$

Whereas, the angles $\alpha_2 = \epsilon_2 + \frac{\gamma_2}{2}$ and $\beta_2 = \frac{\gamma_2}{2} - \epsilon_2$ come from the measurement with the Heidenhain encoder at Mono2. All together Eq. 3.5 leads to

$$\frac{m}{d}\left(\frac{\lambda_1}{n} + x\frac{dn}{m f_{obj}}\cos(\beta_1)\right) = \sin\left(\alpha_2 - \frac{x}{f_{col}}\right) - \sin(\beta_2) \tag{3.10}$$

Which must be solved concerning the displacement x. To solve 3.10 analytically, it is necessary to approximate $\sin(\alpha_2 - x/f_{\text{col}}) \approx \sin\alpha_2 - \frac{x}{f_{\text{col}}}\cos\alpha_2 + O(x^2)$ with respect to x/f_{col}. With only the linear term, the solution is:

$$x = \frac{\sin\alpha_2 - \sin\beta_2 - \frac{m\lambda_1}{dn}}{\frac{n}{f_{obj}}\cos(\beta_1) + \frac{\cos(\alpha_2)}{f_{col}}} \tag{3.11}$$

The model wavelength $\lambda_{\text{model}} = \lambda_2 = \lambda_c$ can now be derived from Eq. 3.6. The SPIE proceeding (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020) shows a second derivation in which Eq. 3.10 has been solved numerically. In this work, it is solved analytically with up to the quadratic term of $\sin(\alpha_2 - x/f_{\text{col}})$.

Collimator optics after the monochromator

It is necessary to have collimated light for the parts of the setup after the double monochromator. An off-axis parabolic mirror from Thorlabs (M5, see Fig. 3.8 and Tab. A.1) collimates the light from the monochromator's exit slit in one direction. A concave cylindrical mirror (M6) collimates in the other. With perfectly aligned gratings, the light beam could be assumed to follow one optical plane. Instead, the grating turret's mechanics of the Cornerstone 260 Monochromator do not allow a sufficient alignment of the gratings to yield 90° alignment concerning the optical plane. Each grating is held by four countering screws, two for tip and tilt. Fig. 3.17 depicts the degrees of freedom of the gratings. The tip and

tilt correspond to the β angle, while changes in α correspond to changes in wavelength. γ rotates the spectrum within the monochromator. The changes in β from 90° deflect the beam off the optical plane. With two monochromators, there are eight degrees of freedom to be adjusted to align the grating vertically with respect to the optical plane. All gratings have been adjusted so that the light beam leaves the second monochromator following the optical plane over a lever of ≈1.5 m at 8000 Å. To correct the residual rotation in β, the flat mirror M7 is mounted on a GNL18-Z8 goniometer from Thorlabs that can be moved in tilt automatically. For each grating configuration, a look-up table over 2000 Å to 12 000 Å allows the goniometer to hold the beam in the optical plane while changing the wavelength.

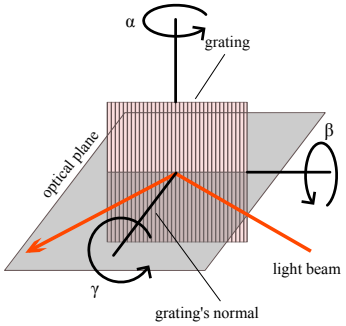


Figure 3.17: A schematic of the degrees of freedom for the gratings. The angles α, β, and γ illustrate the rotational degrees of freedom of the gratings.

3.1.3 Holmium Didymium Filter

The light from the monochromator is first collimated and then focused with an objective (parabolic mirror M10) onto an optical fiber to transport it to the testing chamber. As the light beam suffers from optical aberrations, the exact bandwidth differs spatially in the collimated beam. The last aperture that defines the bandwidth for the photometric calibrations is the circular entrance into the fiber with 400 μm in diameter. As the exact adjustment of all components varies over time (e.g., thermal expansions), the wavelength calibration of λ_{model} changes too. Instead of calibrating λ_{model}, the absorption lines of a spectral absorption filter are calibrated, and the photometric calibration is done twice with and without the filter to imprint its calibrated lines into the signal of the device under test.
The liquid filter (see Fig. 3.18) is called Holmium Didymium (HoDi) consisting out of holmium oxide, praseodymium, and neodymium in solution, giving a set of absorption lines in the range from 2410 Å to 8640 Å (Fig. 3.20a).[10] To calibrate the filter, the spectral lines of the low-pressure gas lamps have been scanned to calibrate λ_{model} for the specific light path from the integrating cylinder. A subsequential measurement of the filter's absorption spectrum using the broadband Xenon lamp gives the position of the line with a traceable wavelength scale to NIST atomic database (Standards and Technology 2009). The exact procedure is outlined in Ch. 5.



Figure 3.18: Picture of the HoDi liquid absorption filter UV45 from Hellma (KG 2023).

3.1.4 Linearity Setup

A necessary condition to reach sub-percent calibration is the linearity of the device under test (DUT) and all reference devices used in the setup. A part of the photometric comparator setup allows us to measure two different light beams separately and their sum, which can differ in intensity by two orders of magnitude.
The collimated monochromatic beam passes an iris diaphragms[11] **A3** and is split by a UV fused silica window (G5) which reflects ≈1 % to the light path **A**. The transmitted path (**B**) can be attenuated by a set of neutral density filters, which allows attenuation of four orders of magnitude. F2

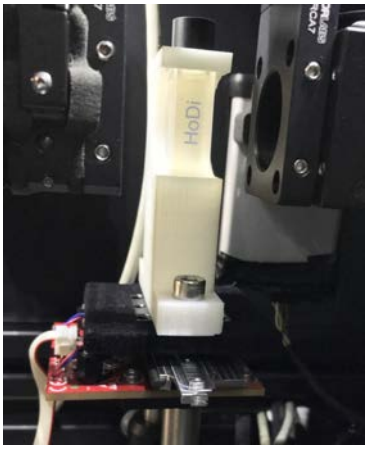
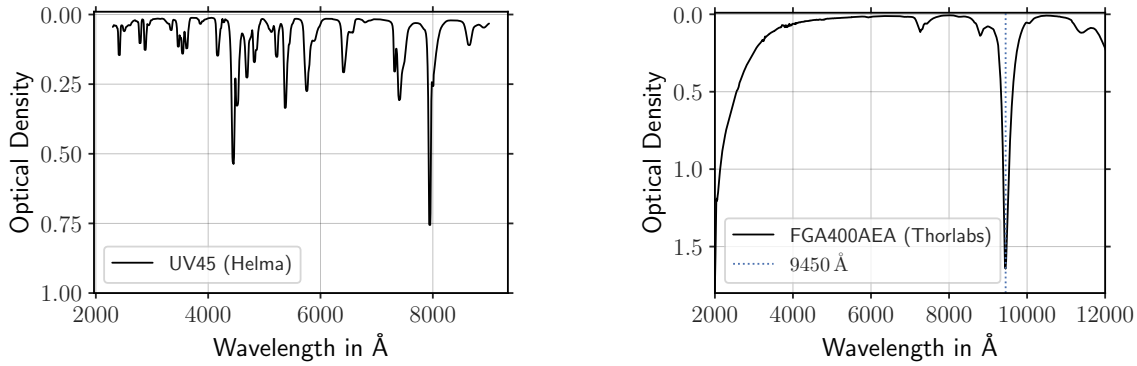


Figure 3.19: Picture of the HoDi filter in the setup mounted on a motorized linear stage (ELL20/M) from Thorlabs. The filter can be moved into the beam and out by demand. The holding structure is 3D-printed.

[10] An absorption filter does not depend on the AoI or any other geometrical shifts, in contrast to an interference filter.
[11] Thorlabs



(a) A typical absorption spectrum of the HoDi filter. Data from Hellma (KG 2023).

(b) Attenuation of an 18 m-FG400AEA optical fiber given by the manufacturer (black solid line). Notable is the absorption line around 9450 Å that reduces signal-to-noise significantly for that region (blue dotted line). Data from Thorlabs Thorlabs n.d.

Figure 3.20: Absorption spectra of the HoDi filter (a) and the optical fiber (b).

in path B (see Fig. 3.8) is a Multi-Position Slider[12] allowing to insert 1″ neutral density filters with an optical density of one or two. A variable neutral density filter[13] (F3) allows step-less attenuation of beam B. F2 and F3 filters are adjusted at a slight angle to the beam to avoid back reflections. Two plane mirrors (M9 & M10) redirect the beams **A** and **B** to an intersection point at another UV-fused silica window **G6** to merge the transmitted light from path **B** with the reflection of **A**. The filters F2 and F3 are used to limit the flux at the DUT and the reference photodiode such that the last one's amperemeter does not exceed its saturation limit of ≈250 pA (compare Fig. 3.5).

3.1.5 Optical Fiber

The Off-axis parabolic mirror (M10, see Fig. 3.8, and Tab. A.1) from Thorlabs works as an objective and focuses the collimated merged light from the linearity part into the optical fiber. The fiber FG400AEA from Thorlabs has a costume length of 18 m. It is a UV Solarization-Resistant, Step-Index Multimode Fiber that allows broadband transmission (1800 Å to 12 000 Å) and has a core diameter of 400 μm. The numerical aperture is NA = 0.22, which corresponds to a focal ratio of F/2.3 in the case of $n_{\text{air}} \approx 1$. At 9450 Å, the fiber has a strong absorption line that reduces the signal-to-noise significantly in that region (see Fig. 3.20b), which will be discussed in Ch. 5 in more detail. The fiber is mounted on two NRT150 stepper motor-driven linear translation stages from Thorlabs in an XY Configuration (see Fig. 3.22a) to allow a systematic search of the optimal coupling position. Next to the fiber, an imaging camera (QHY5III-174) and a photodiode for a remote test of the light intensity and focus are also mounted on the translation stages. As the light suffers from aberrations, the shape of the monochromator's slit image is distorted.

[12] ELL9K from Thorlabs with four positions for 1″ optical components. **The piezoelectric-driven sliders are powered only for actual movement to avoid stray infrared light from its optical limit switch.**

[13] NDC-100C-2M round continuously variable neutral density filter (maximal attenuation is OD2) from Thorlabs mounted on a motorized rotation stage.

Fig. 3.21 depicts a CCD image of the slit image at the optical fiber. It shows multiple maxima. The global maximum is found by performing a two-dimensional scan of the fiber's position with the linear stages, maximizing the monitoring diode's signal at the robot after each beam alignment before the fiber.

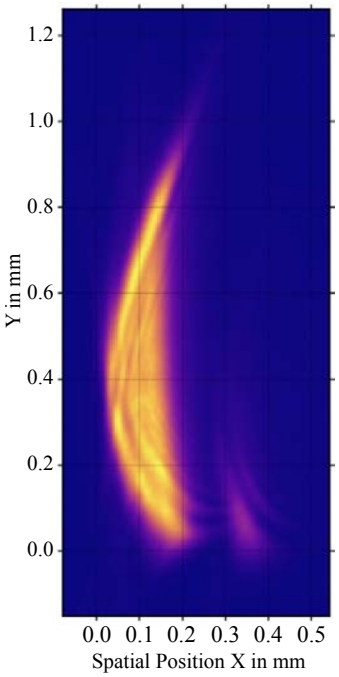


Figure 3.21: Focussed light after **M10**, imaged with the QHY5III-174 showing the optical abberations.

3.1.6 Test chamber and Gantry Robot

The devices under test must be screened from environmental light and are enclosed in a box of $2 \times 1.5 \times 2\,\mathrm{m}^3$ (see Fig. 3.22b). The box consists of black anodized aluminum plates. To ensure reproducibility and automatize spatial scans, a gantry robot moves the relay optic freely in three spatial and two rotational dimensions. The robot system (see Fig. 3.23) has an available volume of $1070 \times 680 \times 580\,\mathrm{mm}^3$ and the maximal load to guarantee the spatial repucibility of 50 μm in height and 15 μm[14] in the axes of the horiontal plane is 15 kg. Two Heavy-Duty Rotation Stages (HDR50) from Thorlabs are mounted on the gantry robot to allow automatic rotation in two axes (see Fig. 3.24 left). These stages have an accuracy of 0.05° and a bi-directional reproducibility of 0.02°. The control algorithm assumes targets as coordinate systems. It performs a passive coordinate transformation $\varphi : (\phi,\ \vartheta) \in \mathbb{R}^2 \rightarrow (x,\ y,\ z,\ \alpha_{\mathrm{polar}} = \varphi,\ \alpha_{\mathrm{azimuth}} = \theta) \in \mathbb{R}^3 \times [70, 110] \times [0, 360)$ to map the surface coordinates from the DUT to real coordinates of the gantry robot and the rotation stages[15]. This allows an easy iteration over even curved surface grids and coordinate systems or DUTs without explicitly programming a trajectory. For a more detailed algorithm description, see (Bastian-Querner 2019).

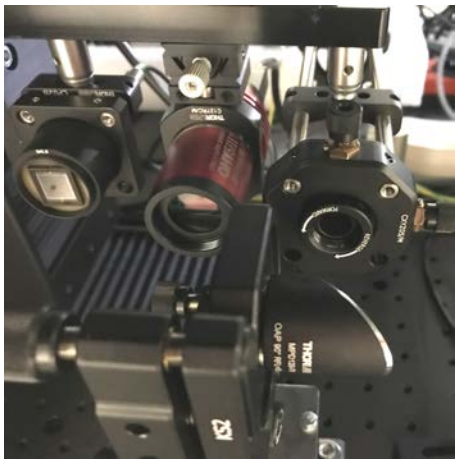
3.1.7 Objective and Relais Optics

The gantry robot optics must fulfill some requirements to illuminate a device under test (DUT) with known flux. A reference photometer is needed that measures a known fraction R of the original flux, and the light that exposes the DUT must be focused with negligible stray light or interference pattern. The product of the fraction R and the reference photometer's responsivity S must be constant over at least two subsequential measurements to transfer the calibration of one photometer to a second with RS as the transfer variable. It can depend on wavelength λ but must be stable concerning ambient conditions like temperature.
During this work, there have been two different configurations used to transfer photometric calibrations traceable to NIST[16] and PTB[17] laboratory standards.

Optic configuration used for ULTRASAT scout measurements:

Fig. 3.24 shows the robotic relay optic used to characterize ULTRASAT test CMOS sensors (scouts) with a different anti-reflection coating (ARC)

[14] Given by the manufacturer Schneider.
[15] The algorithm tests for each movement different kinds of trajectories by calculating the minimal distance between the components of the robot and the different surfaces. In that way, the system did not perform any collisions during >1000 h of operation for the last > 4 years without any proximity sensor.
[16] National Institut of Standards and Technology U.S. Department of Commerce
[17] Physikalisch Technischen Bundesanstalt, Nationales Metrologieinstitut, in Germany



(a) The objective **M10** focussing the monochromatic light onto the optical fiber. From left: A S1337-1010BQ photodiode from Hamamatsu for remote light intensity control, a QHY5III-174 camera with CMOS sensor to image the focused light, and the FG400AEA multimode fiber.



(b) Test chamber to screen the devices under test from the environment.

Figure 3.22: Image of the optical fiber (a) and the test chamber (b).

in Bastian-Querner, Kaipachery, et al. 2021a (For further motivation, see Sec. 4). For references these optic is called "scout-optic". The light from the FG400AEA fiber is collimated with an off-axis parabolic mirror with a RFL = 2"[18] (see Fig. 3.24) and reflected along the horizontal rotational axis. A continuously variable iris aperture limits the diameter of the collimated beam. A reflective neutral density filter NDUV05 with optical density OD = 0.5 splits the beam into two parts of nearly the same intensity. The transmitted beam is focused with a UV fused silica lens with 35 mm focal length onto an S1337-1010BQ photodiode[19] working as a reference, and the reflected beam passes a second lens of the same kind. The image of the fiber on the CMOS sensor has, therefore, a diameter of approximately 400 μm $*$ 35 mm/50.8 mm = 276 μm.

Optic configuration used for SCALA measurements

As the photometer used in SCALA measures unpolarized light, the light used to characterize the angle dependency must also be unpolarised. Using a neutral density filter with 45° inclination introduces a nonnegligible polarisation[20]. Fig. 3.25 shows a picture of the improved configuration, and Fig. 3.26 a schematic. A continuously variable iris aperture allows for control of the diameter of the collimated beam. The neutral density filter[21] is inclined with only 5° to the optical axis instead of 45° like the ULTRASAT configuration. Using the TMM, the mixing between s and p-polarisation mixing can be expected to be 50.8 % to 49.2 % from personal communication (Daniel Küsters 2021). The reflected beam is redirected three times with flat folding mirrors[22] (see schematic 3.26).

[18] MPD129-F01, Thorlabs
[19] From Hamamatsu
[20] The splitting between p and s-polarisation can be expected as 70 % to 30 % for a nickel coated ND-filter calculated with the Transfer Matrix Method (TMM) (Daniel Küsters 2021).
[21] Same as above.
[22] MRA25-F01 from Thorlabs

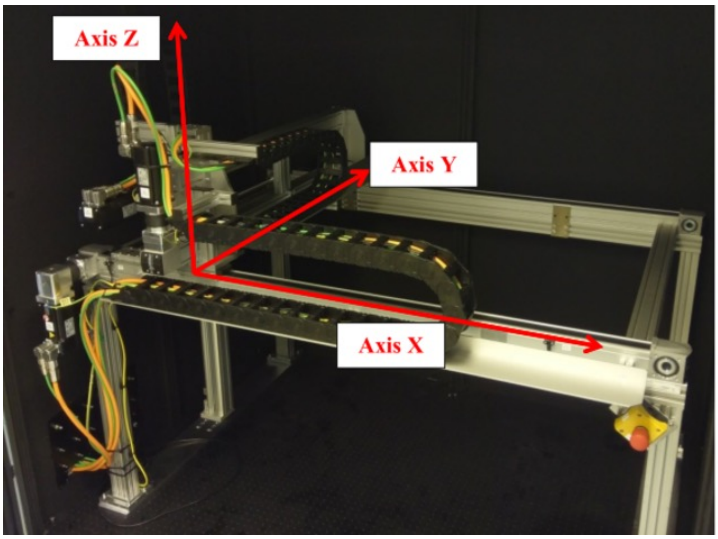


Figure 3.23: Gantry robot delivered from Schneider Electric and build-up by the DESY internal workshop.

An off-axis parabolic mirror replaces the lens on the DUT side with a RFL = 4″[23] to avoid chromatic aberration. The resulting image diameter of the fiber at the DUT is approximately 800 μm calculated as above. This optic is called "SCALA-optic" in the following.

3.1.8 Control Software

To control the VASC, a Python package was written called **LabDAQ** (Bastian-Querner 2023a). Every robotic component of the setup has its Python class inherited into one module. All necessary parts can be automatically power cycled. The software is resilient against serial connection errors and catches up and fixes common errors of the individual devices that occur during operation of more than 48 h automatically. This resilience is needed to operate this setup with more than 20 devices generating more than 100 observables[24] throughout >10 d without any human intervention. For each measurement, the status of the setup, the reference photometers, and the DUT must be read out simultaneously. For this, a data acquisition (DAQ) class (included in LabDAQ) is written that generalizes the readout for an arbitrary amount and type of devices. Correctly set up, if a Python software API for a device is installed (either manually or with Pip), the DAQ imports the API on its own and reads the device out synchronized with all other devices without the necessity of hard coding it into the measurement scripts. A command line tool allows one to toggle the installed devices for the next measurement. That gives a standard that allows a controller with minimal knowledge about the software to implement a new DUT into the setup for calibration purposes. The setup generates twenty to fifty individual measurements per week if running automatically. To overcome the amount of data that must be analyzed with different analysis routines depending on the measurement's kind, each measurement gets a unique running hex number ID, and the program chooses the data location automatically. A data butler[25] is written that logs all necessary parameters and descriptions

[23] MPD149-F01, Thorlabs
[24] Giving a full view over the exact setup state.
[25] Motivated by the Vera C. Rubin Observatory Science Pipelines on AWS (Bektesevic et al. 2020)

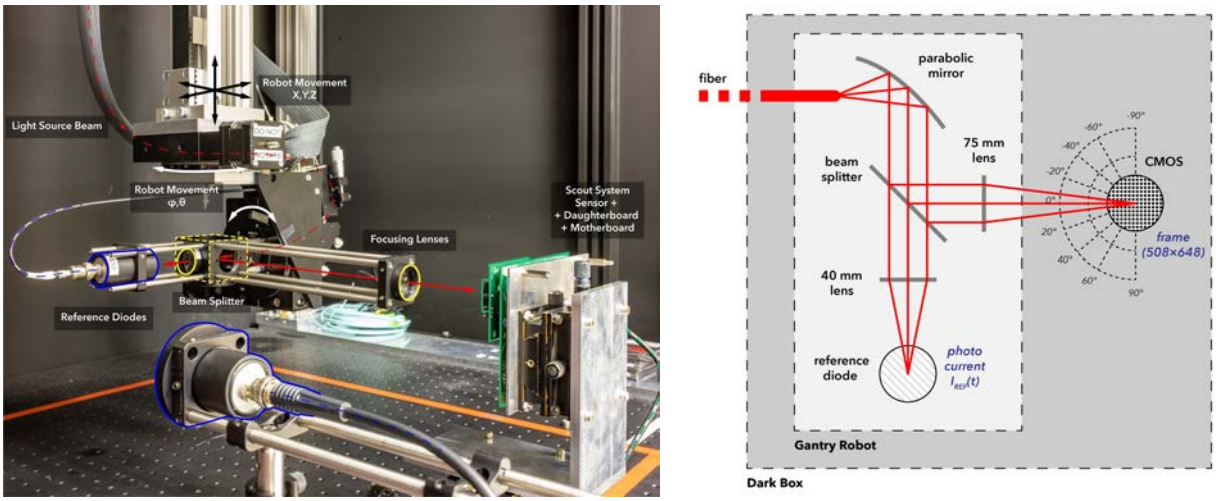


Figure 3.24: On the left, the Gantry robot and the intermediate relay optic with 45° beamsplitter. Used to measure the scout devices in ULTRASAT. Right: A schematic of the robotic "scout-optic" to expose the CMOS sensor. Both are taken from (Bastian-Querner, Kaipachery, et al. 2021a).

that characterize the $O(10^3)$ measurements and allow a command line search function for all measurements done with the setup and maps the ID to the data's location. The butler can also be used in scripts, allowing a loop over measurements without explicitly knowing the data's locations. The documentation can be found in the repository (Bastian-Querner 2023a) and DESY confluence (Bastian-Querner 2023b).

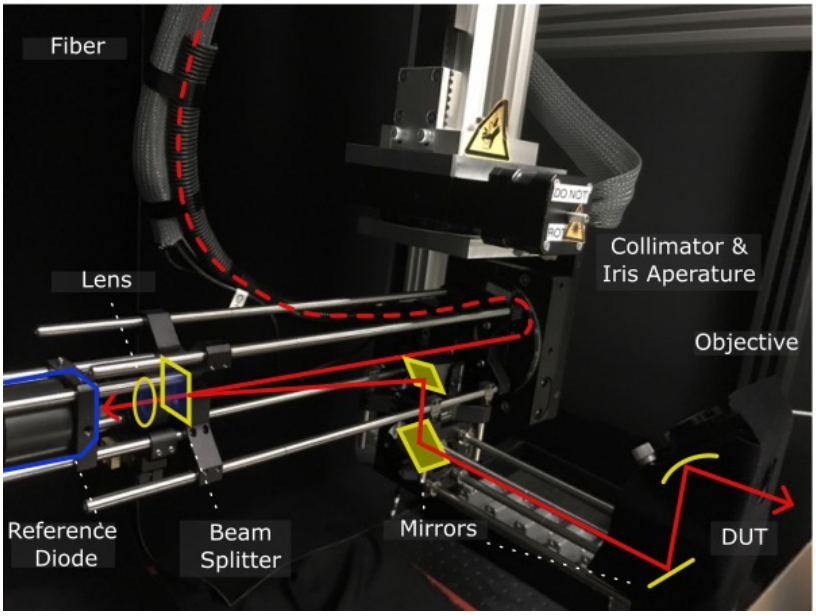


Figure 3.25: Gantry robot and the improved "SCALA-optic" relay optic with ≈5° inclinated beamsplitter to reduce polarization. Used to characterize the photometer used in SCALA (Bastian-Querner, Kaipachery, et al. 2021a)

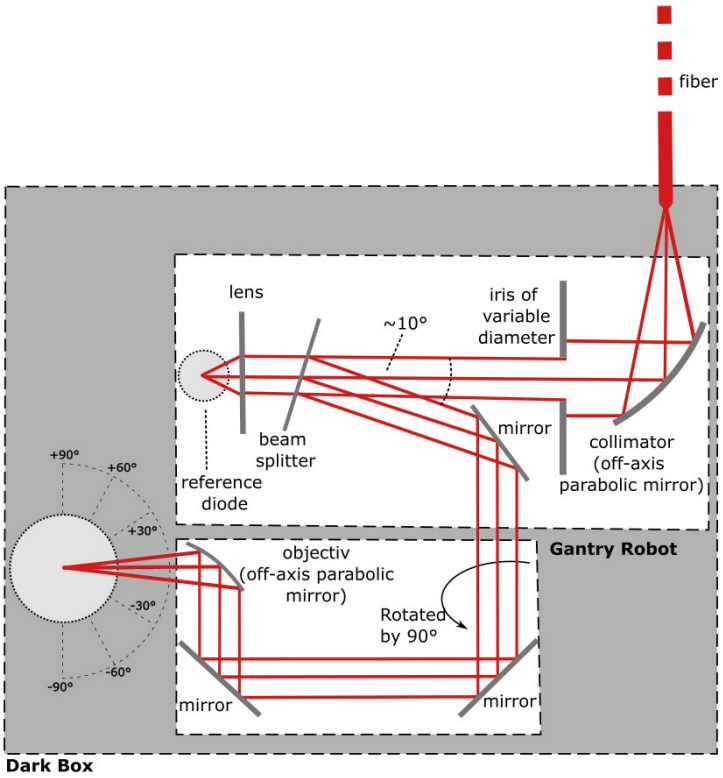


Figure 3.26: Schematic of the optical configuartion with ≈5° inclinated beamsplitter to reduce polarization. Used to characterize the photometer used in SCALA. The schematic is not in scale.

3.2 The LArge Resistance Amplifier (LARA)

The LArge Resistance Amplifier (LARA) is an extremely sensitive ampermeter developed by Dr. Daniel Küsters (affiliated as Post Doc., Department of Physics. University of California, Berkeley) and Marko Kossatz (Electronic Development Engineer, Deutsches Elektronen-Synchrotron (DESY). It is used to measure a photocurrent of a photodiode in zero-biased mode. Optionally, the LARA can be equipped directly with a photodiode (in this work S1337-1010BQ, Hamamatsu) making it to a photodetector. The LARAs have in combination with the S1337 diode a noise floor of 10 fA in 4 s at 25 °C and saturate around 400 pA. This section describes the working principle of the device and its characterization. Before describing the LARA, the working principle for a photodiode is sketched.

3.2.1 Principal of Photoconductor and Photodiode

Silicon is a solid crystal that can be seen as a vast molecule formed by periodically repeating atoms and bonds. The distinct energy levels of the individual atoms are perturbed by the presence of neighboring atoms, causing new energy levels to emerge. Because of the sheer number of atoms involved (approximately 10^{20}), the energy levels become a continuum called bands. Distinct electron energy levels evolve into different bands, if bands don't overlap, they produce 'band gaps'.

In Silicon, interference between the 3p and 3s states prompts a crossover degeneracy, forming sp3-hybrid orbitals. These orbitals constitute the valence or bonding bands of lower energy and the higher-energy conduction band emerging from the bonding and anti-bonding states. The energy differential between the valence band E_v and the conduction band E_c is termed the band gap energy, represented as $E_G = E_c - E_v$.

Materials can be differentiated based on this band gap energy. The gap is so vast for insulators that populating the conduction band becomes infeasible. Conversely, metals exhibit an overlap between these bands. Semi-conductors strike a balance: the band gap is sufficiently small to overcome through the photoelectrical effect, temperature-induced excitations, or phonon interactions. When electrons are excited enough to cross the band gap, holes are formed in the valence band. This 'hole' behaves oppositely to an external electrical field. When an electron reverts to its original state, filling the hole, it is termed 'recombination'. In the context of Silicon, which has a band gap $E_g = 1.2eV$, its conductivity augments with temperature.

A photodiode is constructed by bringing p-doped and n-doped materials into contact. As a reminder, doped materials are semiconductors intentionally contaminated with specific elements to introduce holes or supply extra valence electrons. A notable example of n-type doping employs phosphorus, whereas p-type doping utilizes boron.

Upon establishing contact between these doped materials, charge carriers undergo an exchange, which culminates in forming the 'depletion region'. This region has an intrinsic electric field. When a photon is absorbed in this region, an electron in the conduction band and a hole in the valence

band are created, as long as the photon's energy $E_\gamma \geq E_G$. The fraction between the fraction of absorbed photons, denoted as N_d, and the number of incident photons, represented as N_γ, defines the intrinsic quantum efficiency ($QE_i = N_d/N_\gamma$). In practice, the external QE is of interest which also depends on the probability of the photon's transmittance into the material seized by the Fresnel Equations.

The generated photocurrent I_p is proportional to the incident photon flux F and depends on wavelength. In the case of photons near the band gap energy, phonon interactions can lever the electron excited by a photon with $E_\gamma \lesssim E_g$ and excite it into the conduction band. This process is called the phonon-assisted photo effect. As the energy density of the phonons depends on the temperature, the QE depends on the temperature in the regime where the photon energy is comparable to the band gap energy, which is for silicon near-infrared.

A photodiode can be operated in three modes:

- Photoconductive mode: The photodiode is reverse biased, and the photocurrent that is proportional to the incident flux is measured. Increased sensitivity is achieved by increased depletion region width. Dark noise also increases with the reverse bias voltage.
- Power-cell mode: The photodiode is connected to an electrical load. The voltage drop across the load is proportional to the incident flux. This operation is used in a solar cell. The response is slow.
- Photovoltaic mode or zero bias mode: The photodiode is operated without an external bias voltage. The photocurrent is held at zero. This can be achieved by e.g. connecting it to a voltmeter with high impedance. This mode has the advantage of low noise.

For more details, see Chromey 2016.

3.2.2 Design

The LARA's amperemeter is a current-to-voltage converter using a transimpedance amplifier based on (Larason and Jeanne M. Houston 2008). Fig. 3.27 shows the read-out principle of the LARA. The generated photocurrent of the p-n diode is converted by an operational amplifier (OPA) with a feedback resistance[26] of 10 GΩ working as a transimpedance amplifier. The second OPA is set up as a voltage follower that decreases the load at the output of the first one, matches impedance, enhances stability and bandwidth, facilitates multiple output connections, and improves driving capability.

Furthermore, it acts as a low-pass filter and converts the single-ended output to a differential. A sigma-delta analog-to-digital converter (ADC) digitalizes the differential signal with a high sampling frequency. Another advantage of this ADC is that it shifts the noise frequency above the signal's bandwidth[27] which a low-pass filter can remove. It promises higher linearity than other ADCs. Fig. 3.28 shows the circuit board of version 4.6.

[26] The used resistor is the UM030Z1008JE from Ohmite a high voltage resistor. This type has the advantage that the thicker isolation seals its core from pollution that can change its resistance.

[27] The ADC performs an oversampling (two to four times the Nyquist-Theorem requires), which reduces quantization noise.

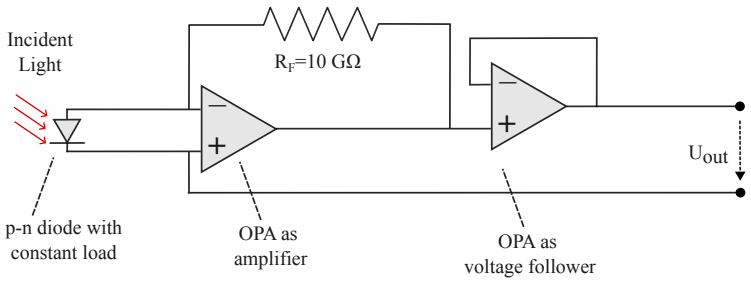


Figure 3.27: Schematic of the readout principle for the LARAs. Motivated from (Larason and Jeanne M. Houston 2008).

The ADS1264 32-bit analog-to-digital converter (ADC) includes an on-board temperature sensor that performs self-monitoring with the PT1000 attached to the photodiode for temperature measurement. Furthermore, the LARA system has a DS18B20 ome-wire thermometer with a resolution of 0.5 K to ensure measurement consistency. Although the PT1000 was designed to record the detector's temperature, during the laboratory calibration, the gantry robot vibration caused damage to the contacts of the PT1000, causing the PT1000 data to be untrustworthy. As a result, the temperature data provided by the ADC's internal sensor had to be relied upon, and the PT1000 readings had to be disregarded.

Küsters investigated the noise of the amperemeters by comparing the observed noise spectrum with a physical model. The LARAs noise is limited by the thermal noise of the feedback resistor, which is at 25 °C 10 fA in 4 s (Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022).

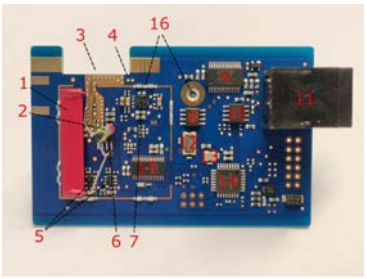


Figure 3.28: Picture of the final circuit board of the LARA version 4.6 designed by Marko Kossatz, DESY Electronics. See Tab. 3.4 for the legend.

1	Feedback Resistance
2	Capacitance
3	Contacts for p-n diode
4	Contacts for PT1000
5	Filter for power supply
6	OPA as transimpedanz amplifier
7	OPA as voltage follower
8	One Wire bus Thermometer
9	FTDI UART to USB
10	Optocoupler for shutter state
11	Ethernet Port
12	Oscillator for ADC
13	Oscillator for Processor
14	Microprocessor
15	Sigma-Delta ADC
16	Contacts for shielding

Table 3.4: Labels for Fig. 3.28.

3.2.3 Temperature Dependence

The feedback resistor[28] has a nominal value of $R_{\text{nominal}} = 10\,\text{G}\Omega$, but underlies a temperature dependence α. The dependency can be described linearly in the temperature drift ΔT from nominal temperature:

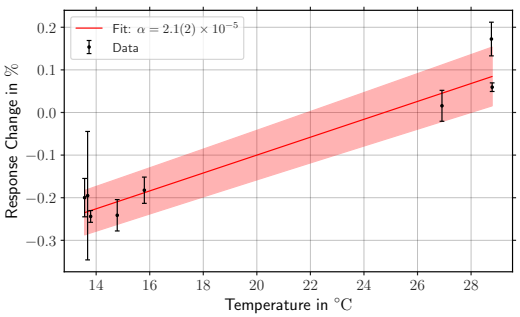
$$R(\Delta T) = R_{\text{nominal}}(1 + \alpha \Delta T) \tag{3.12}$$

To estimate α for the used resistance, one LARA is enclosed in a Thermal Vacuum Chamber (TVaC) as shown in Fig. 3.29. Its responsivity is compared with the reference diode at the robot optics (see Sec. 3.1.7). Multiple wavelength scans are done around temperatures of 15 °C and 27 °C. Light in the visual was used to ensure the only temperature dependence in the responsivity due to the feedback resistance. Fig. 3.30a shows the change in responsivity concerning 24 °C dependent on the TVaC's temperature. A linear χ^2 fit (solid orange line) results for $\alpha = 2.1(2) \times 10^{-5}$.

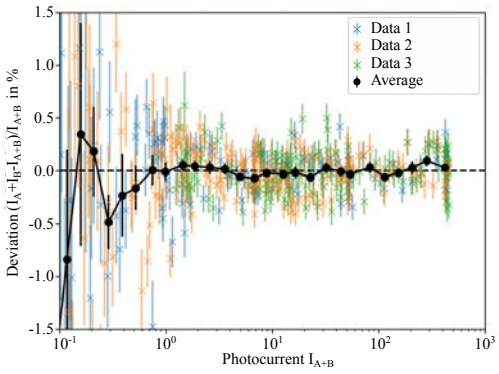
3.2.4 Linearity

For all flux calibrations, the linearity of the detectors is crucial, especially if the flux throughput from the light source varies over the wavelength range. Any non-linearity would lead to an unwanted bias in the calibration.

[28] UM030Z1008JE of Ohmite



(a) Relative change in the LARA's responsivity on the temperature concerning 24 °C. The solid line shows a linear χ^2-fit with the value of $\alpha = 2.1(2) \times 10^{-5}$. Data from personal communication with Dr. Daniel Küsters.



(b) The deviation from linearity for measurements performed with the linearity setup in a current range of 0.1 pA to 400 pA (both beams open). The data sets one to three are three subsequential measurements. The black points and curve represent the averaged values of the individual measurements, which were binned. Throughout the range of 1 pA to 400 pA, no deviation greater than 0.1 % was observed. Taken from (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020).

Figure 3.30: Temperature dependence and non-linearity of the LARA prototype.

Sec. 3.1.4 describes the linearity part of the setup. Fig. 3.30b shows the deviation from one of the ratios between the algebraic sum $I_A + I_B$ of the beams measured individually with the LARA prototype and the current I_{A+B} of the summed flux against I_{A+B}. The flux of beam **B** is varied such that the summed signal of **A** and **B** result in a photocurrent from 0.1 pA to 400 pA. The different colors indicate different measurements. The black graph shows the binned average of the individual measurements. Below 1 pA, the scatter is due to a low signal. In the rest of the range, the maximal significant deviation is below 0.1 %. It can be concluded that the LARA with the transimpedance amplifier is sufficiently linear over its range for the requirements of SCALA.

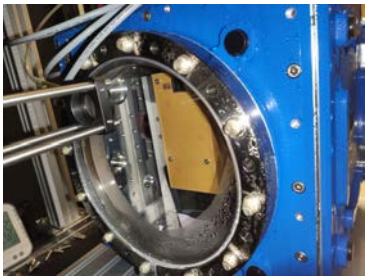


Figure 3.29: The LARA prototype in the TVaC provided by the DESY ULTRASAT group is visible through a UV-enhanced quartz glass window.

3.3 The Supernova Factory and SNIFS

The Supernova Factory (SNFactory) is an ambitious astronomical project that primarily aims to study Type Ia supernovae (SNIa) and collect spectroscopic data. The goal is to improve our understanding of their physics and use them as a reliable distance indicator. This program involves a systematic and well-coordinated approach to observing and analyzing SN. The SNFactory operates its own developed spectrograph, the "SuperNova Integral Field Spectrograph" (SNIFS).

3.3.1 The SNFactory

The factory focuses on obtaining time series data for SNIa, with a particular emphasis on tracking the supernova within a range of -15 to +45 days from its maximum brightness with a frequency of 2 to 3 days (Yannick Copin 2013). The basic program of the SNfactory, as described in Greg Aldering et al. 2002, focused on producing spectrophotometric light



Figure 3.31: SNFactory Logo from snfactory.lbl.gov.

curves covering a broad spectral range from 3200 Å to 10 000 Å for at least 300 nearby Type Ia supernovae. These supernovae are selected to cover a wide range of parameter space. The redshift covers $0.03 < z < 0.08$ to minimize the contribution of the host galaxy's velocities to the error budget. The intent is to detect supernovae shortly after their explosion, typically 10-15 days before maximum light incidence, with 3–5 days for spectrophotometry between -15 and +45 days after maximum light incidence. This results in around 15 spectra per supernova.

The primary purpose of the Supernova Factory program are as follows (Greg Aldering et al. 2002):

1. **Study Type Ia Supernovae (SNe Ia)**: The intention is to systematically study and collect data on Type Ia supernovae. This includes observing their properties, behavior, and spectral characteristics.
2. **Improve Understanding of SNe Ia Physics**: The program aims to enhance our understanding of the physics governing Type Ia supernovae. This involves investigating their formation, evolution, and underlying mechanisms.
3. **Utilize SNe Ia as Distance Indicators**: Type Ia supernovae are known for their consistent luminosity, making them valuable as distance indicators in cosmology. The program seeks to refine their use as standard candles for measuring cosmic distances.

The SNFactory uses ultra-wide-angle CCD mosaic imagers for blind searches instead of targeted searches. This approach is similar to that used to detect high redshift supernovae and is more efficient because each image covers a large area of the sky containing multiple galaxies. This method ensures that important subsets of supernovae, such as superluminous Type Ia supernovae, are not missed (Greg Aldering et al. 2002). Furthermore, the factory coordinates discovery and follow-up observations, ensuring minimal delays in acquiring previous light curves. Follow-up observations can begin as early as three hours after the initial discovery observations.

The international SNfactory collaboration is composed of several founding members, including:

- **Lawrence Berkeley National Laboratory (LBNL)**: LBNL serves as the principal institution responsible for leading the SNfactory project, with G. Aldering as the principal investigator (PI).
- **Yale University**: Yale University manages the SNfactory research programs, initially including Palomar-quest and later LSQ.
- **Centre de Recherche Astrophysique de Lyon (CRAL)**: CRAL is responsible for constructing and maintaining the SNIFS instrument, a crucial project component.
- **Institut de Physique Nucléaire de Lyon (IPNL)**: IPNL plays a key role in processing and producing spectro-photometric data, a vital aspect of the research.
- **Laboratoire de Physique Nucléaire et des Hautes Énergies**: This laboratory is responsible for data acquisition and transfer operations.

These founding members comprise around thirty researchers and students involved in the collaboration.

In 2010, as part of negotiations for the extension of observations, the SNfactory collaboration, referred to as SNfactory-II, expanded to include new partners, including the Tsinghua Center for Astrophysics, Physikalisches Institut Universität Bonn, Max-Planck-Institut für Astrophysik, and the Centre de Physique des Particules de Marseille. This expansion reflects the growing international scope of the project and its collaboration with additional research institutions.

The collaboration has developed a unique tool called the Supernova Integrating Field Spectrograph "SNIFS" (Yannick Copin 2013).

3.3.2 The SuperNova Integrating Field Spectrograph "SNIFS"



Figure 3.32: Photograph of SNIFS at the UH88 telescope.

The Supernova Integral Field Spectrograph (SNIFS, see Fig. 3.32) is a spectrograph with spatial resolution developed by the SNFactory. In contrast to a slit-based spectrograph, which produces only a spatially integrated spectrum, SNIFS depicts the spatial information together with the spectral. The so-called data cubes are three-dimensional data packages with 15×15 pixels in spatial dimension along the spectral axis. One spectrum at one spatial position is called spaxel (x, y, & λ). This allows a spectrum correction for wavelength-dependent PSF and placement, as the PSF can be fitted wavelength-wise. SNIFS is permanently mounted at the University of Hawaii 2.2-meter telescope (UH88, see Fig. 3.33) on the summit of Maunakea, making it continuously available for use, typically up to 20 % of the time.



Figure 3.33: UH88 at the summit of Maunakea.

The primary purpose of SNIFS is to allow astronomers to capture spectral and spatial information from astronomical objects simultaneously. It was developed in collaboration between the Centre de Recherche Astrophysique de Lyon (CRAL) and the Institut de Physique Nucléaire de Lyon (IPNL), and it has been used for a wide range of research projects, including the study of supernovae and other celestial phenomena (Yannick Copin 2013).

Technical Characteristics

SNIFS is characterized by technical specifications that make it a powerful instrument for astronomical observations. Tab. 3.5 to 3.7 from (Greg Aldering et al. 2002; Yannick Copin 2013) show some key technical characteristics:

Table 3.5: Integral Field Unit: Specifications of the spatial properties of the spectrograph.

Microlens frame	15×15
Spatial sampling	0.4251(3)″
Field of view	6.38″×6.38″
Rotation/CCD	3.8°
Interspectrum separation	8.5 pixel

Table 3.6: Spectrograph: Spectroscopic properties of Blue and Red channel.

	Channel B	**Channel R**
Total spectral range [Å]	3200 – 5300	5000 – 10200
Useful spectral range [Å]	3300 – 5150	5100 – 9700
Spectral Sampling [Å pixel^{-1}]	2.38	2.93
Spectral resolution (FWHM) [Å]	5.23	7.24
Detector	Marconi 2k × 4k	E2V-DD 2k × 4k
Model	CCD44-82	CCD44-82-0-A72
Series/Class	9252-20-2/0	01502-14-0170
Pixel [μm]	15	15
Dark current [e^-/h]	0.09	0.144
Reading noise [e^-]	1.9	2.2
Gain	0.758	0.764

Scale	$0.14''$ pixel^{-1}	
Field of View	two $4.7' \times 9.4'$ regions	
	Channel P	**Channel G**
Detector	LBNL E2V 2k × 4k	LBNL E2V 2k × 4k
Model	CCD44-82-1-B23	CCD44-82-2-B23
Series/Class	02263-15-0171	02111-12-02/2
Filters	Multi-Filter: F1-5	Bessel V

Table 3.7: Properties of Auxillary Camera P-channel and Guider Camera G-channel. Both channels are mounted side by side. For the filter F1 to F5 see Fig. 3.37.

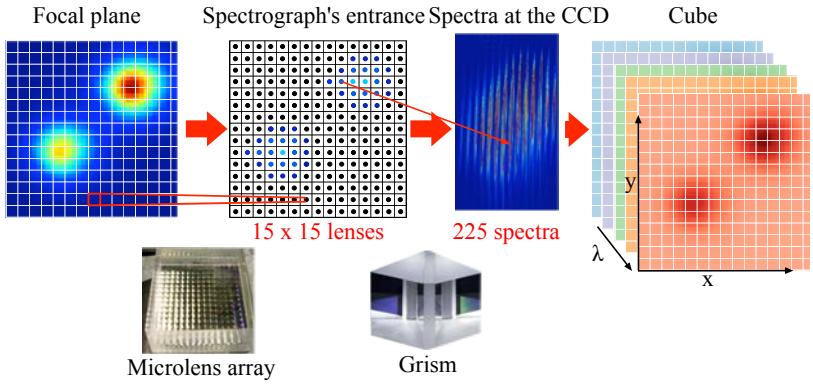


Figure 3.34: Schematic illustrating the principle of the integral field spectrograph. Taken from Clément Buton 2009. Translated from French and modified in its labels.

Working Principle

Fig. 3.34 visualizes the principle of the integral field spectrograph. The telescope's image in the focal plane is split into 225 spaxels by the Micro-Lens Arrays (MLA) resulting in 225 individual light beams. Each beam from the MLA is refracted by the grism (TIRG). The grism is designed such that the first-order of its diffraction pattern has the highest intensity. The relais optic, therefore, reflects the first-order of the pattern to the CCD detector. This results in a set of spatially separated spectra on the detector, with each spectrum corresponding to a specific spatial element or spaxel in the field of view. The data collected by SNIFS is processed to create data cubes $C(x, y, \lambda)$ containing two spatial and one spectral dimension. Each data cube element represents a specific spatial element within the field of view and contains spectral information for that region.

Fig. 3.35 shows a schematic of SNIFS. The f/10.14[29] beam from the telescope enters SNIFS by striking the pick-off prism (POP; exaggerated in the figure). The POP reflects a fraction of the entering beam to the spectroscopic channel via total internal reflection. The first optical element is the enlarger (EL), to ensure coverage of the MLAs that allow spatial resolution. SNIFS is divided into blue and red channels. A dichroic filter splits the light after the EL for the red (transmitted beam) and blue (reflected beam) channels. The spatial extent of the beam in the focal plane is $(0.7\,\text{mm})^2$. The MLAs continuously cover a 6" x 6" field of view with a dense grid of 7 mm focal length micro-lenses. Each microlens captures light from a specific spatial region within the field. To cover the $\approx 20\,\text{mm} \times 20\,\text{mm}$ MLAs, the enlarger has an enlargement factor of $e_f = 32$. The image of the telescope has a size of $7\,\text{mm} \cdot 2.2\,\text{m}/(22.6\,\text{m}\, e_f) \approx 22\,\mu\text{m}$. This size is negligible in comparison with the pitch of the microlenses (1.3 mm), so the result is a good separation of the 225 diffraction patterns called spaxels. Noteworthy, these lenses are made of two glass plates with fifteen cylindrical cuts, lying twisted 90 degrees to each other (see lower

[29] A focal length of 22.6 m compared to an entrance pupil of 2.2 m.

left photograph in Fig. 3.34), generating lenses of squared shape. This shape adds a second parasitic diffraction pattern to the spaxels beside those from the grism. In the case of monochromatic light, this leads to a cross-shaped pattern on the CCD (discussed later in Sec. 6.2.3). In the nominal use case of broadband sources, this additional pattern can be treated as a homogeneous background and is subtracted from the data (Daniel Küsters 2019).

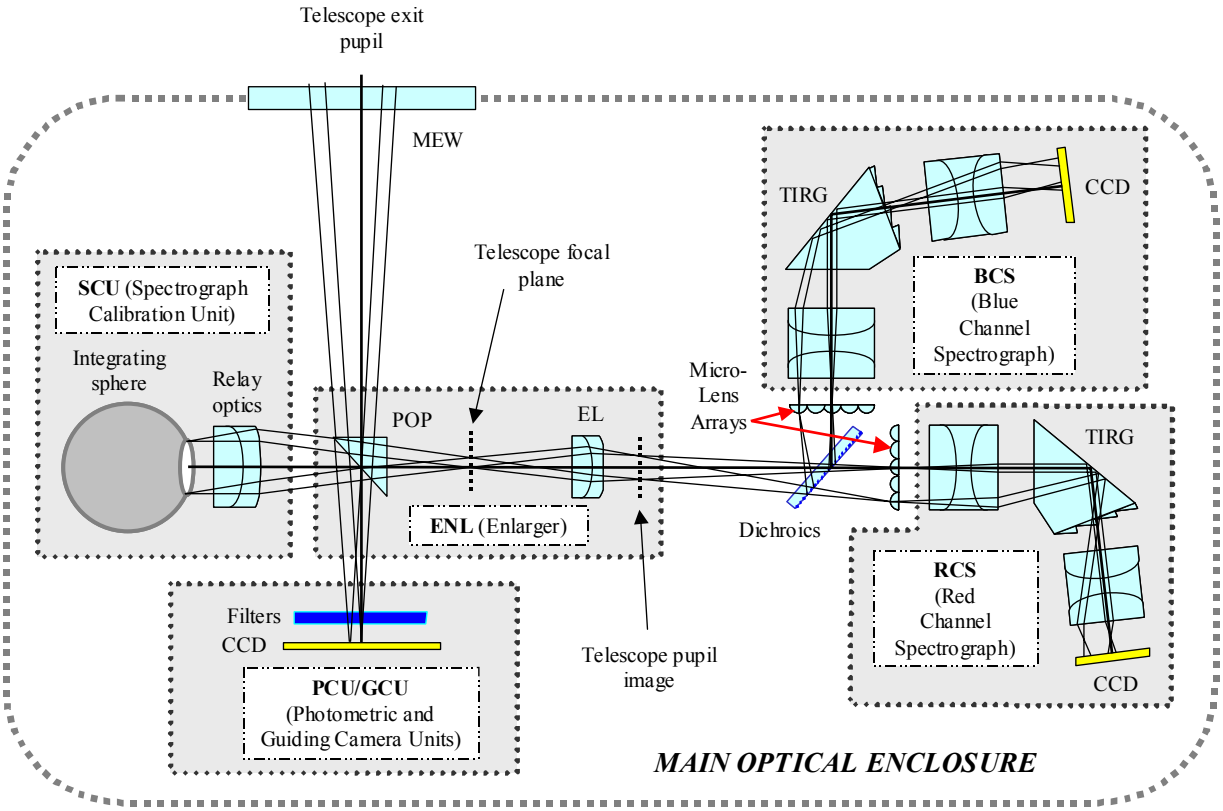


Figure 3.35: Schematic of SNIFS components: The telescope's f/10 beam hits a pick of prism (POP) that redirects a fraction of the light to an enlarging lens. It is noteworthy, that the size of the POP is exaggerated. Its actual size is ≈6 mm and it is extended into the telescope's beam which has an approximate diameter of 100 mm. This picked of light is divided by a dichroic mirror and projected onto the blue and red MLA. It then travels through specific collimators, grisms, cameras, and CCDs for both blue (BCS) and red (RCS) channels. The majority of light goes through a filter wheel to the photometry camera (PGC), with a unique multi-band filter for atmospheric monitoring during spectroscopic observations. Off-center light also hits the guiding camera for precise guidance during spectroscopy and focus or offset adjustments between shots. On the left side, the Spectrograph Calibration Unit (SCU) provides a continuous halogen lamp and arc lamps for spectrograph calibration (Greg Aldering et al. 2002; Henault 2002).

For flat fielding and wavelength calibration, SNIFS uses its Spectrograph Calibration Unit (SCU). The SCU allows frequent calibrations of the optics to provide each data cube with individual calibrations. It contains for each channel one broadband lamp for three-dimensional flat field corrections and arc lamps for wavelength calibriations (Greg Aldering et al. 2002; Daniel Küsters 2019). Fig. 3.36 shows a schematic of the SCU. The SCU is equipped with the following lamps:

- **Xenon Arc Lamp**: The Xenon arc lamp is a broadband light source that provides a continuous spectrum in the UV range (3200 Å to 5400 Å). It is used for flat field measurements for the blue channel.
- **Quartz Tungsten Halogen (QTH) Lamp**: The QTH lamp is a broadband light source that provides a continuous spectrum in the IR range (5400 Å to 10 000 Å). It is used for flat field measurements in the red channel.

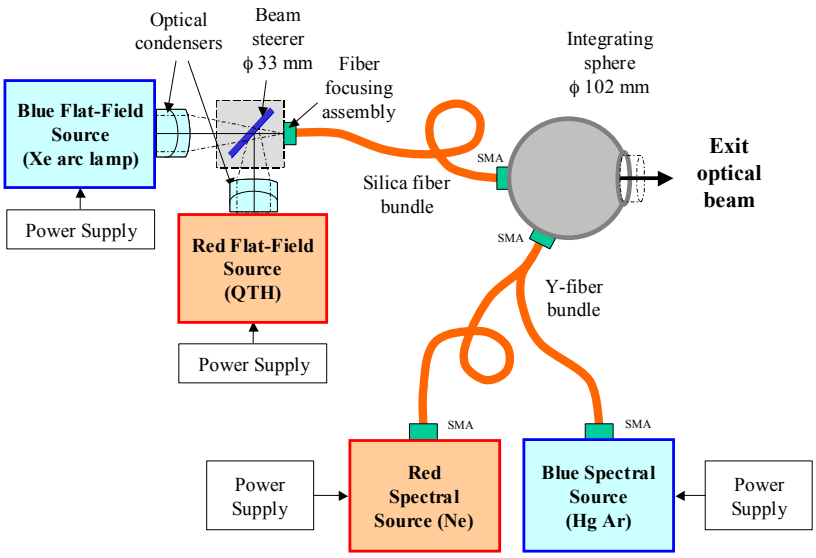


Figure 3.36: Schematic of the SNIFS Spectrograph Calibration Unit showcasing the four different lamps for flat field measurements and wavelength calibrations for each channel (Henault 2002).

- **Hg Ar Arc Lamp**: The Hg Ar lamp provides spectral lines in the UV range. It is used for wavelength calibrations of the blue channel.
- **Ne Arc Lamp**: The Ne lamp provides spectral lines in the IR range. It is used for wavelength calibrations of the red channel.

The light is fed into an integrating sphere (IS). The homogeneous light from the IS is focused by the calibration optics (see Relay Optics in Fig. 3.35) at the telescope's focal plane. When the POP is removed from its nominal position, SNIFS can be exposed by the SCU's lamps.

The telescope's light, not reflected by the POP illuminates two *LBNL E2V* CCD detectors with 2 kpixel × 4 kpixel each. The detectors refer to the photometric (P) and guiding or tracking (G) channels. A filter wheel between the P-channel and the CCDs provides SDSS-"ugriz"-like filters, a B filter plus a pinhole aperture with a periodic pattern of holes, and a Multi-Filter (MF) that separates the imaged sky into five distinct filter bands (see the left regions F1–5 in Fig. 3.37).

The MF was originally designed to measure atmospheric extinction's intensity and chromatic variability, with filters chosen explicitly for this task. Due to issues such as poor photometric quality in some bands and limited visible secondary sources, this purpose could not be achieved. However, detailed modeling revealed that variations in atmospheric extinction are color-neutral, and their chromatic aspects can be measured with standard stars. Thus, the chromatic function of MF is not used to track atmospheric extinction, and all filter sources are concurrently processed (Yannick Copin 2013).

Nevertheless, the MF proved valuable in determining the chromatic variance of the spatial point distribution function, which is essential for subtracting the galactic background. Thus, the chromatic function of MF is not used to track atmospheric extinction, and all filter sources are concurrently processed. A wide-field exposure with the G-channel in a few seconds in the V band allows confirmation of the presence of SNIFS's target and allows accurate positioning of the spectrograph's field-of-view (see MLA square in the Bessel V image on the right of Fig. 3.37). The G-channel is then used in video mode to guide the telescope in real-time by observing a star's apparent path (Yannick Copin 2013; Daniel Küsters

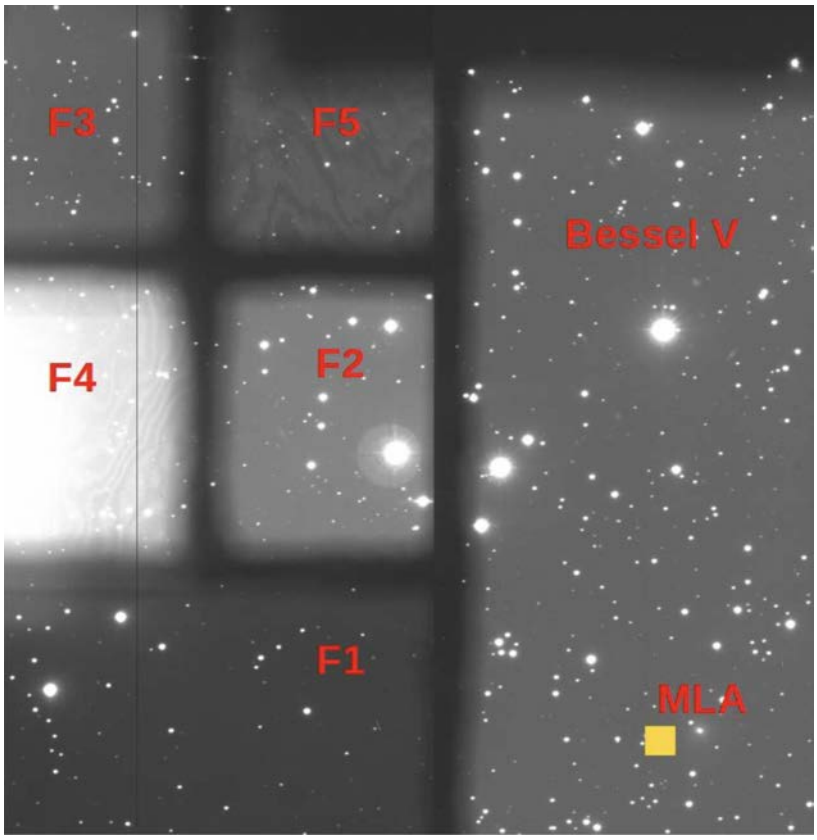


Figure 3.37: Observing using the P and the G imaging channel of SNIFS. The Multi-Filter array and the Bessel V separate the sky into six adjacent filters with varied passbands, chosen to be sensitive to specific properties of the atmosphere such as air-glowing in the IR (see F4). The filter names are highlighted in red. The yellow square, marked as MLA, denotes the potential placement area for the field of view of the spectroscopic channels. Taken from (Yannick Copin 2013; Daniel Küsters 2019).

Label	λ_{eff} in Å
F1	3900
F2	5900
F3	7250
F4	8500
F5	9500
Ch. G	5450 (Bessel V)

Table 3.8: Filter names and effective wavelengths of the Multi-Filter array and the Bessel V filter. Taken from (Yannick Copin 2013; Daniel Küsters 2019).

2019). An overview of the filters' labels and their effective wavelengths can be found in Tab. 3.8.

In summary, SNIFS captures spectral and spatial information from astronomical objects within its field of view using a Micro-Lens Array and two spectroscopic channels. Data processing and calibration steps are applied to create calibrated spectra and data cubes for detailed analysis of astronomical objects. The data processing is outlined in Sec. 6.1.1.

3.4 SCALA

The SNIFS Calibration Apparatus (SCALA, as mentioned in 2.5) is an in situ calibration device developed for SNIFS. The goal is to provide a bottom-up calibration. It is permanently located within the dome of the UH88 telescope. The following section gives an overview of SCALA's working principle and its capabilities until its upgrade in 2022. The results of the pilot observation are summarized too.

Fig. 3.38 shows a schematic of SCALA. It consists of 18 f/4 mirrors with a diameter of 16 cm to illuminate the telescope's entrance pupil. To feed the mirrors, SCALA utilizes broadband light sources combined with a monochromator to produce a tunable bandpass that illuminates the telescope aperture. For the UV and blue light spectrum, a Xenon lamp is employed. However, a halogen lamp is preferred to circumvent the strong emission lines in the Xenon spectrum for red light.

The lower schematic in Fig. 3.38 shows the design of the complete setup. A flip mirror is used to switch between the different light sources. The

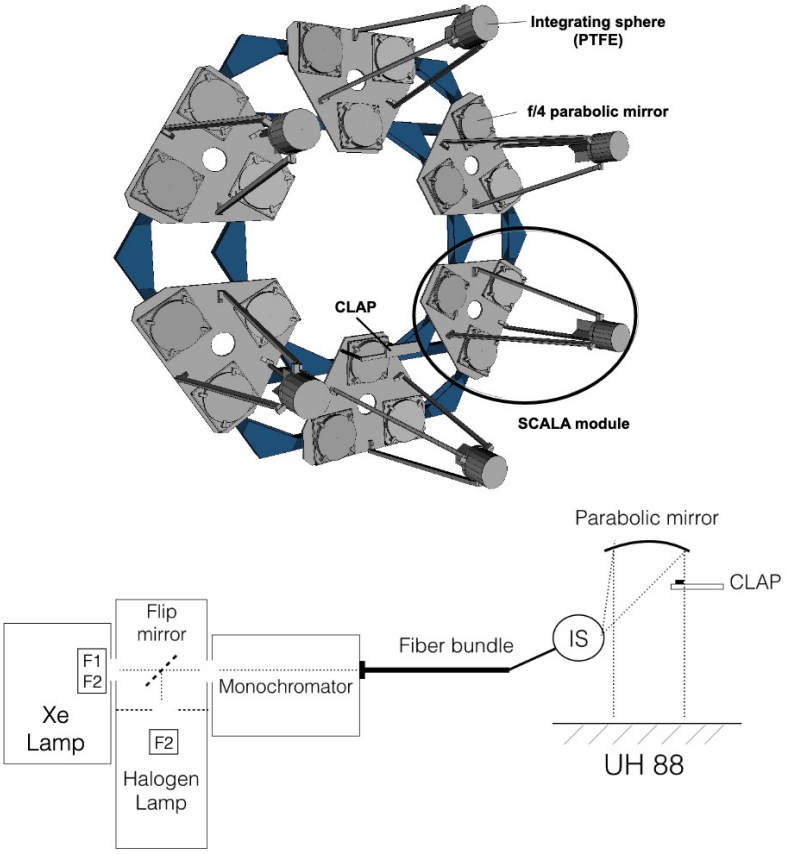


Figure 3.38: A schematic of SCALA before update until summer 2022. The six submodules of SCALA are arranged in a hexagonal configuration and mounted directly in front of the telescope's entrance pupil. The reference photodiode's (CLAP) location can be found on the lower submodule. The SCALA design is showcased below, featuring the lamp system and flip mirror that permits the selection of lamp output shining on the monochromator entrance. This setup includes a fiber optic cable leading to the integrating sphere (IS) and a calibrated photodiode (CLAP) oriented towards the collimated beam bouncing off the mirror. A dotted line shows the light's path. The flipping mirror is enclosed within the halogen lamp to avoid any light leakage. Two small boxes inside the lamp compartments denote where the order sorting filters (F1 and F2) are located (Lombardo, Küsters, et al. 2017).

monochromator is a Newport Cornerstone 260 using 1200 lines mm^{-1}-grating with blaze wavelengths of 3500 Å and 7500 Å, respectively. The slit widths are set up to provide light with a FWHM of 35 Å.

Two order sorting filters, F1/2 with a cut-on wavelength of 3090 Å and 4950 Å, are supposed to prevent second-order light from the gratings. Tab. 3.9 lists the chosen configuration between lamp, grating, and filter depending on the wavelength (Lombardo, Küsters, et al. 2017).

A bundle of fibers is used to transport the light to the separated modules. This is designed so that its six branches have fibers spread along the exit slit of the monochromator (see Fig. 3.39). This ensures a consistent sampling of the entire slit across the integrating spheres and averages the inhomogeneities of the monochromator's outcoming light (Daniel Küsters 2019; Lombardo, Küsters, et al. 2017).

Table 3.9: Table representing wavelength ranges, lamps, grating blaze, and filter cut-ons (Lombardo, Küsters, et al. 2017).

Wavelength range in Å	Lamp	Grating blaze in Å	Filter cut-on in Å
3200–4500	Xe	3500	None
4500–5220	Xe	3500	3090 (F1)
5220–6240	Xe	7500	3090 (F1)
6240–7020	Xe	7500	4950 (F2)
7020–10000	Halogen	7500	4950 (F2)



Figure 3.39: Fiber-bundle set in front of the monochromator's exit slit. Each color indicates a different Integrating Sphere. The $33 \times 6 = 198$ fibers are equally distributed over the slit height to average the spatial inhomogeneities (Daniel Küsters 2019).

Thirty-three fibers from the bundle guide the light into one of the six integrating spheres (IS)[30]. The spheres are positioned such the exit holes are in the mirror's focal point of 80 cm to illuminate the telescope's entrance pupil with collimated light (Daniel Küsters 2019). Fig. 3.38 depicts an angle between the IS and mirror's optical axes that is exaggerated and does not accurately reflect reality. Additionally, the output of the IS is illustrated as emanating from a single point, which oversimplifies the system. Baffle plates are strategically placed under the IS to minimize cross-illumination risk. These plates are arranged to be evenly spaced and form an angle of 120 degrees between each plate, as shown in Fig. 3.40. The color of the baffle is black, and its hue is specifically chosen for the absorption of extraneous light and the minimization of reflection.

The reference flux standard is a NIST-calibrated S3477-04 photodiode from Hamamatsu combined with a two-stage Peltier cooler. This combination, named the "Cooled Large Area Photodiode" (CLAP), integrates the above photodiode with a low-noise amplifier and a digitization unit. The diode tracks the light reflected off the mirrors to measure the SCALA light output. As shown in the lower Fig. 3.38, it is positioned within the beam of the upper mirror of the lowest module. To thoroughly understand the device and possible systematics, please refer to chapter 5 of Küsters 2019 (Daniel Küsters 2019).

An effective beam number has been introduced to take the wavelength-dependent background stray light and the different throughputs of the different beams into account.

The telescope mirror has inhomogeneities. The throughput is, therefore, spatially dependent on where the plane wave enters the entrance pupil of the telescope. To account for the fact that SCALA can only provide



Figure 3.40: One of the integrating spheres of SCALA is visible. The cylindrical aluminum casing and the internal white PTFE are evident through an exit port (Lombardo, Greg Aldering, et al. 2014).

[30] The integrating spheres are hollow spheres made out of sintered Teflon. The material provides a diffuse scattering to ensure the light is reflected multiple times before it leaves the cavity. The exiting beam can be seen as homogeneous.

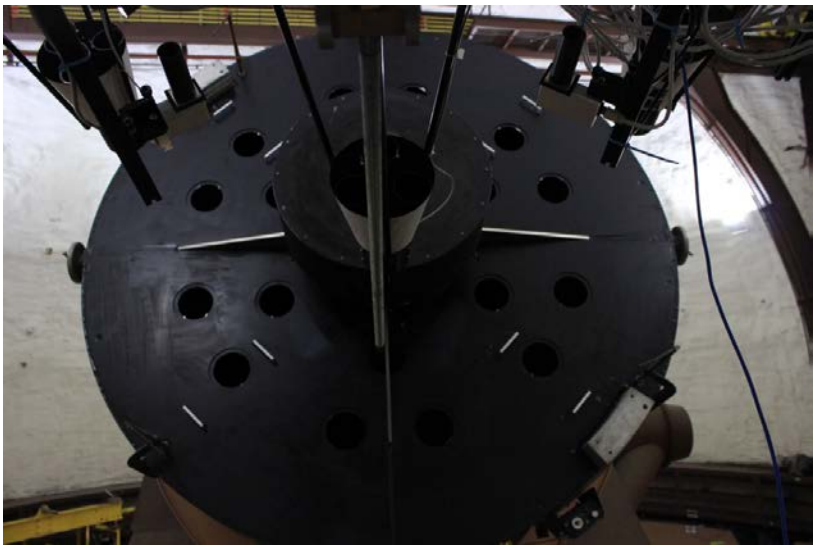


Figure 3.41: Image of the mask mounted on the telescope pupil. The mask is made of 6 mm ALUCORE, divided into four sections for easier handling. Each hole for the rays has a diameter of 16 cm. A 20 cm opening was made to remove the front layer and honeycomb structure of the ALUCORE, after which the 16 cm hole was cut into the back layer. The mask ensures consistent instrument throughput for both standard star observations and SCALA illumination and is finished in matte black.

collimated light that illuminates $18 \cdot 20\,\mathrm{cm}^2/2.2\,\mathrm{m}^2 = 14.9\,\%$ of the primary mirror, a mask is mounted on the pupil as shown in Fig. 3.41. The mask comprises 6 mm ALUCORE. It is divided into four sections for ease of use[31]. Each hole designed for the beams spans 16 cm in diameter. To create a clear edge around these holes, a 20 cm opening was created to eliminate the front layer of the ALUCORE panel and its honeycomb structure, leaving only the back layer intact. The 16 cm opening was then cut out of the remaining back layer. The panels were then coated with a matte black finish (Daniel Küsters, Lombardo, et al. 2016). The mask ensures that the instrumental throughput is the same when observing the standard stars and illuminating with SCALA.

The telescope and instrument calibration process simultaneously uses measurements from the calibrated photodiode and readout from SNIFS. In a single SNIFS exposure, SCALA is observed at multiple distinct wavelengths where the monochromator shutter controls the exposure time for each wavelength. The line profile of SCALA is triangular with a full width at half maximum (FWHM) of 35 Å. It has been confirmed that a 500 Å separation between observed wavelengths prevents overlap (Daniel Küsters 2019). The method restricts the wavelengths per exposure to four for the blue channel and ten for the red. A SNIFS exposure of SCALA's observation in the blue channel is shown in Fig. 3.42. Multiple exposures with four or ten monochromatic lines, based on the SNIFS channel, are required to calibrate the entire SNIFS wavelength range. The goal is to attain a signal-to-noise ratio (S/N) greater than 100 per spaxel. Exposure duration ranges from 30s to 180s. The calibration lasts 8 hours, but the software's flexibility can shorten this time. Using larger wavelength steps as an option can lessen calibration time. This leads to quicker throughput measurements, particularly useful in tests and daily checks.

During the June 2015 commissioning, nighttime SCALA measurements were conducted using the new aperture mask with SNIFS. SCALA observations were alternated with standard star observations to evaluate the "telescope + SNIFS" calibration's stability throughout the night. Only observations from the final night, with an average seeing of 1.35 arcsec,

[31] The chosen material, ALUCORE, offers the advantage of being light while providing the necessary rigidity to withstand wind pressure.

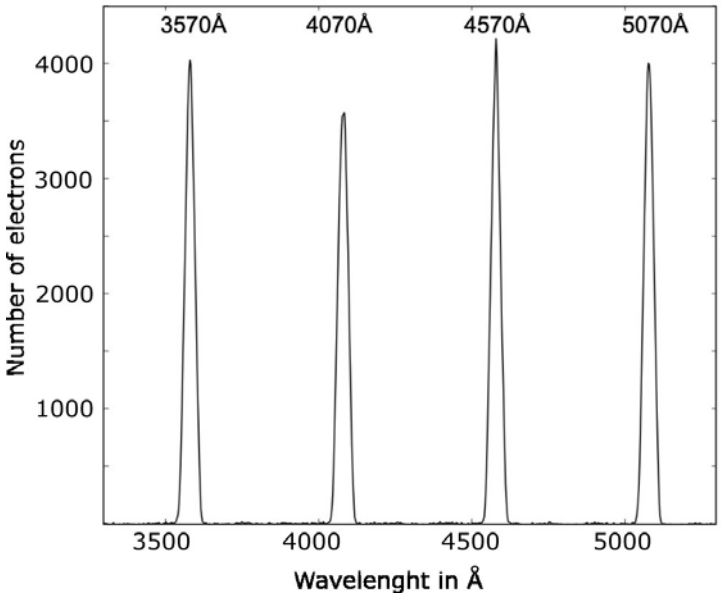


Figure 3.42: A spectrum from one SNIFS cube (B13, see Sec. 6.2) in the blue channel provides an example of a SCALA spectrum. This representative exposure demonstrates a 500 Å disparity between the observed wavelengths taken during the pilot observation after first commissioning in 2015. The electron count for each wavelength is almost uniform due to the careful choice of the SCALA exposure duration for every wavelength. The unique triangular shape and 35 Å FWHM of the SCALA line profile are also easily discernible (Lombardo, Küsters, et al. 2017).

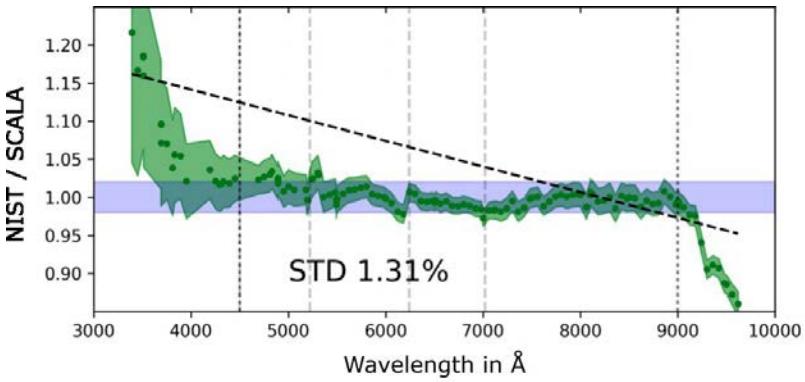
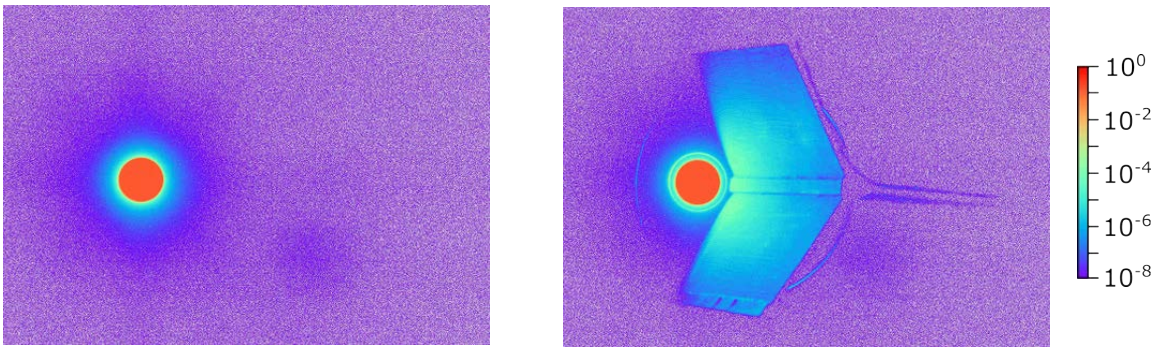


Figure 3.43: The ratio is calculated based on SCALA and CALSPEC calibrations. The diagonal dashed line shows that it is adjusted for out-of-band emissions from the monochromator and wavelength-dependent background. To highlight the wavelengths where the SCALA configuration was altered, vertical dashed lines correspond to modifications in gratings, order-sorting filters, and lamps as specified in Tab. 3.9. Additionally, the ratio is normalized by the average over 4500 Å to 9000 Å (Daniel Küsters 2019).

were considered appropriate for analysis due to several problems. Details are provided in Lombardo, Küsters, et al. 2017.

Fig. 3.43 shows the resulting ratio of the "telescope + SNIFS" responsivity calibrated with SCALA (traceable to NIST laboratory standard) and CALSPEC spectra from Daniel Küsters 2019. The primary standard star GD153 and several secondary stars are observed (sourced from the CALSPEC archive). After corrections for out-of-band emissions and the background, the residual scatter is $\sigma = 1.31\,\%$ over the complete range 3000 Å to 9500 Å. The residual structure is supposed to be due to a change in the out-of-band emissions. Nevertheless, both calibrations agree with the uncertainties.

It is concluded that the photodiode's precision does not constrain the precision of the results. The SCALA system's enhancement in minimizing backgrounds (external, internal, and out-of-band) will enable calibrations with even finer precision, potentially at the sub-percent level (Daniel Küsters 2019).



(a) The exit holes for the other beams are covered.

(b) Same measurement with the baffle from 2015 commissioning.

Figure 3.44: Broadband measurement with a CMOS camera. Stacked images show the light distribution from the integrating sphere in log-scale and compare an optimal configuration with the 2015 baffle design. The color bar beside it indicates light intensity, where red represents the highest intensity and violet is the lowest. The maximal intensity is normalized to one. From personal communication with Küsters 2023 (Daniel Küsters 2023c).

3.4.1 Issues and Suggested Improvements Identified from Küsters 2019 Study

Based on the results presented in Küsters 2019, several problems were identified in that study to achieve sub-percentage uncertainties (Daniel Küsters 2019):

1. **Crosstalk between SCALA beams:** Light from an integrating sphere port fails to reach the corresponding collimating mirror and instead shines on the SCALA structure. The scattered light then reaches the photodiode after reflecting off the collimating mirror. Crosstalk amplitude stays consistent up to 7000 Å but increases for longer wavelengths, which mirrors the reflection coefficient of black anodized aluminum. Over time, dust settling on the relevant surfaces causes the amplitude to decrease. The light shields are also incapable of blocking all light from the other beams. Fig. 3.44 compares the light outcome from the IS between an ideal light shield and the shield from the 2015 commissioning. Light from the other beams is reflected from the IS housing, visible as a ring.
2. **Reflection off IS light shield:** The light shield attached to the integrating sphere reflects light (see Fig. 3.44b). However, due to the black anodized surface of the shield, it exhibits the same wavelength dependence as the crosstalk. The amplitude varies depending on the amount of dust present on the shield.
3. **Reflection off IS port and photodiode housing:** The vicinity of the integrating sphere port and the metal casing of the photodiode can serve as reflective surfaces, thereby generating a background.

In addition, the standard photodiode was calibrated by NIST in 2008. The pilot observations were performed after the first commissioning (Lombardo, Küsters, et al. 2017) in 2015. Fig. 3.45) depicts the uncertainty contributions in the SNIFS calibration with SCALA from Daniel Küsters 2019. A significant contribution to the systematic uncertainty is the error in the adequate number of beams, the background from the effects mentioned above, and the out-of-band emissions of the monochromator.

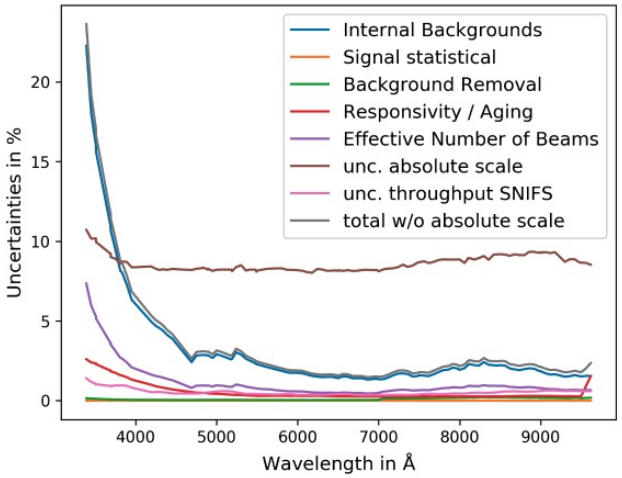


Figure 3.45: Uncertainties of the individual identified systematic effects. The main factor influencing the absolute scale is the reflection from the photodiode's housing. The wavelength-specific variation of the SCALA calibration is primarily limited by the uncertainties linked to the out-of-band emission and the measurements regarding the effective number of beams. Subsequent uncertainties arise from the aging of the photodiode and the uncertainty associated with SNIFS throughput on our specific observational night.

Several notable ones arise from the suggested improvements. Firstly, regarding the SCALA shutter, extended exposure durations can impact photodiode readings due to statistical uncertainty related to external background measurements. A potential solution involves replacing the monochromator shutter with a chopper. The chopper can operate in the 1 Hz to 3 Hz frequency range, enabling alternating background and signal measurements for equal amounts of data collection. To mitigate internal reflections, integrating sphere ports can be equipped with a baffle system to reduce the emitted beam's angular breadth and, thus, diminish reflected light. It is vital to reduce the time gap between calibrating the reference diodes and the actual observation to achieve accurate results while calibrating the reference photodiode. This eliminates any aging effect. Additionally, to account for the effective number of beams, avoiding a model for variances between individual beams is recommended. Instead, equipping each beam with a calibrated photodiode and an amperemeter module is the ideal approach for consistent and precise measurements.

3.4.2 SCALA-Upgrade 2022

Since its initial commissioning, SCALA has been mounted on the dome of the UH88. It was removed in 2021, and the mirrors were removed for recoating in the Canada France Hawaii Telescope (CHFT) vacuum chamber[32], which is located on the summit of Maunakea 250 m next to the UH88, allowing a low-cost process. In June 2022, the abovementioned improvements were implemented in the SCALA facility in the UH88 dome. The advances include mirror recoating, implementation of baffles for the integrating spheres, monitoring diodes for each beam, beam alignment, and implementation of the second monochromator plus a chopper.

The upgrade was done in collaboration with Dr. Daniel Küsters, with the support of Dr. Greg Aldering, Senior Scientist at Lawrence Berkeley National Laboratory. The author primarily focused on the software implementation and adjustments in the SCALA software. The following sections describe the upgrade.

[32] The CFHT has its vacuum chambers for mirror recoating for low-cost maintenance (Magrath 1997; McGrath and Nahrstedt 1996).

Mirror Recoating

Fig. 3.47 shows SCALA mirrors before and after recoating in 2021. The degraded state of one of the mirrors is shown in Figure 3.47a, where the scattered light indicates its degradation. The reflectance of such a degraded mirror is estimated to decrease by about 0.34 % per month in the visual range (Magrath 1997), resulting in a total degradation of nearly 25 % over six years. While cleaning of these mirrors is possible, recoating is often preferred. Direct contact with the coating can cause damage (McGrath and Nahrstedt 1996), necessitating recoating, which presents operational risks and scheduling challenges. Before 2022, these mirrors were removed from the SCALA setup and recoated one month before the upgrade in the CFHT vacuum chamber showcased in Fig. 3.46.

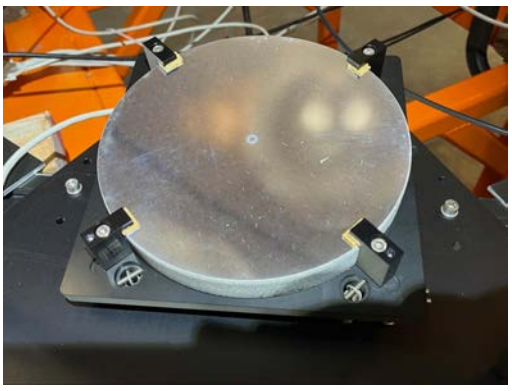
No scratches were found on any of the mirrors, only dust particles. Since all the particles can be removed with bellows, their size is probably larger than 20 μm (McGrath and Nahrstedt 1996). Conversely, the degraded mirror shows the blueish scattering characteristic of Rayleigh scattering. Such scattering is associated with particles smaller than about 50 nm, which cannot be removed by an air stream.



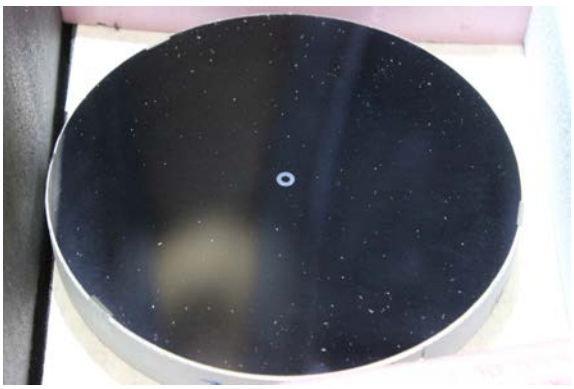
Figure 3.46: Figure shows the vacuum chamber used to recoat the SCALA mirrors in the CFH observatory. Private communication Dr. Greg Aldering, Senior Scientist at Lawrence Berkeley National Laboratory.

Baffle

The Küsters 2019 study (Daniel Küsters 2019) showed that stray light is present between various light paths, leading to a rise in background levels. Additionally, the reflected light from the light shield, as depicted in Fig. 3.44b, also contributes to the intensified background. See Fig. 3.48 for the upgraded IS housing baffle. This new design utilizes five circular PCB plates, each with three elliptical holes that intersect with the ideal path of the light beams. The plates are positioned roughly 0.8 cm, 4 cm, 7.5 cm, 12 cm, and 19 cm from the IS exit holes. A visual representation of the beam's path through the exit hole is shown in Fig. 3.49a. One design consideration was that only surfaces parallel or inclined by less than the cone angle of the beam towards the optical axis in the direction of the light could reflect light to the mirror and raise the background. To reduce these surfaces, the edges of the holes are sharpened as shown in



(a) A degraded SCALA mirror from 2021. Scattered light emphasizes degradation.



(b) Recoated SCALA mirror from 2021 showing some dust particles on its surface.

Figure 3.47: SCALA mirrors: (left) pre-recoating and (right) post-recoating in 2021.

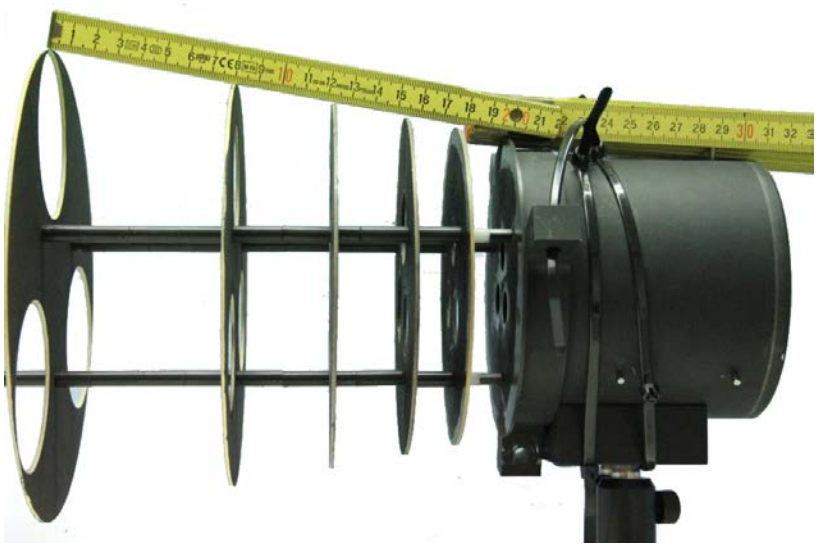
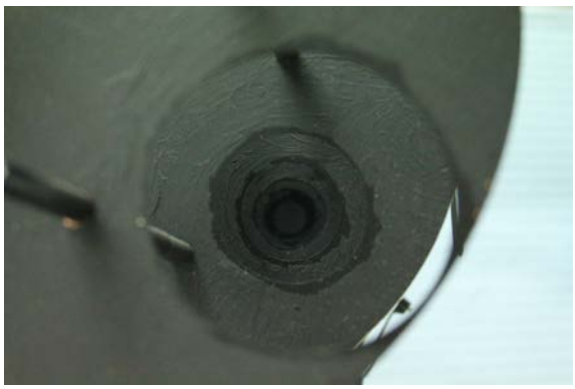
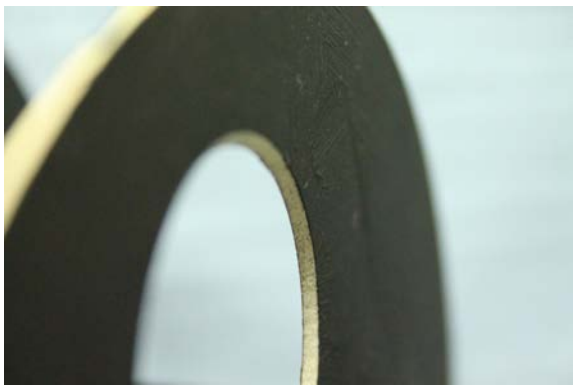


Figure 3.48: Prototype of the new baffle with integrating sphere on the right.



(a) Baffle prototype: view into the integrating sphere to visualize the light path



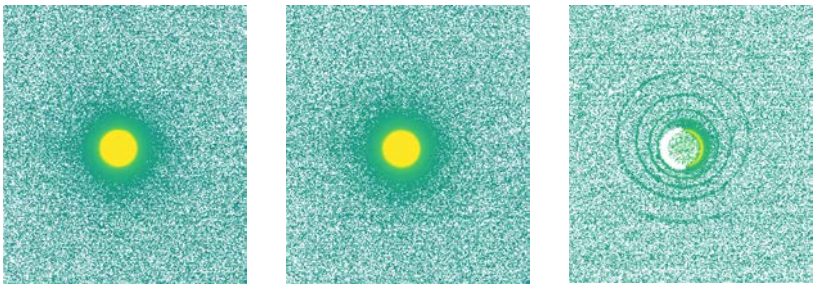
(b) Showing the ground edge of one of the PCB plates. The edge has a 45° angle, and the sharp edge is the side with the copper plate side.

Figure 3.49: Pictuers of the new baffle. Picture (a): The mirror's perspective into the IS exit hole along the beam path. Picture (b): The sharped 45° edge of a hole.

Fig. 3.49b, where a 45° edge is inclined away from the direction of the light to minimize reflection to the mirror. To evaluate the efficiency of the baffle, measurements are conducted similar to those shown in Fig. 3.44. The difference in the sum of the broadband CMOS measurements is an indicator of the baffle's quality, with only a 0.02 % increase in light with the baffle compared to the ideal beam. Note that only the sharp contour of the copper side has the potential to enhance the background, visible as rings in Fig. 3.50c. To prevent the scattering of light from the baffle that could cause the background in the dome to appear brighter, Acktar's Black Foil is used to encircle the conical baffle structure as depicted in Fig. 3.51.

Beam Monitoring

As shown in Fig. 3.45, the throughput variation of different beams and its modeling by the effective beam number N_e contributes as the second major uncertainty. Monitoring each beam, along with the light background in the dome, is essential to improve accuracy.



(a) CMOS measurement with an ideal beam.

(b) CMOS measurement with the new baffle.

(c) Difference between the two measurements.

Figure 3.50: Broadband CMOS measurements demonstrate the quality of the new baffle on a logarithmic scale. Bright yellow indicates high intensity, while cyan and white represent low intensities.

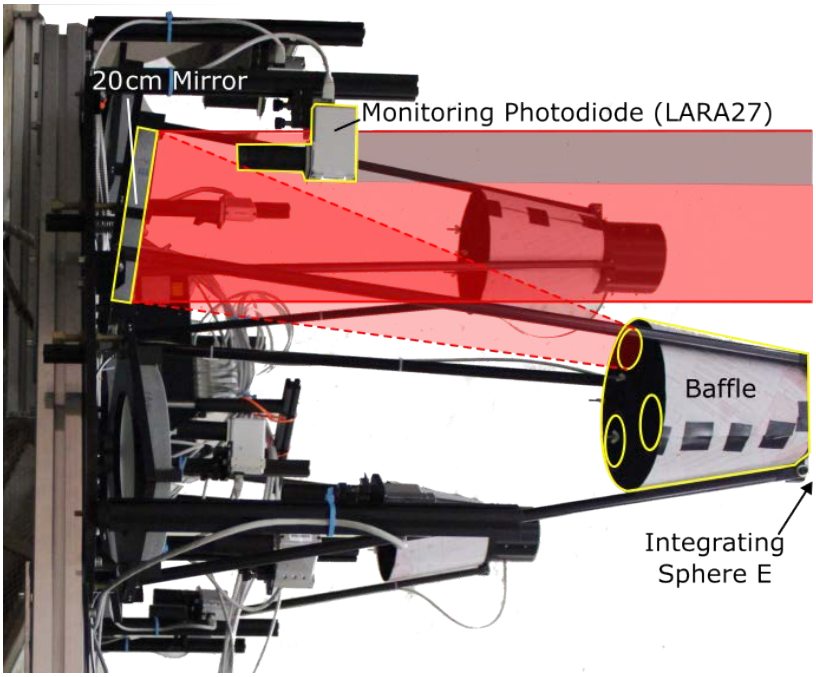


Figure 3.51: Module E Visualization: Components of Beam E Mirror 2, including the baffle, the F/4 16 cm mirror, and the monitoring LARA No. 27, are delineated with yellow outlines. A schematic representation of the integrating sphere, absent in the original photo, is also depicted. The beam path is traced in red, with the shaded grey segment representing shadowing from the LARA.

As discussed in Sec. 3.2, 21 LARAs were calibrated using the **VASC**. Eighteen of these LARAs are installed on the SCALA structure to monitor individual beam intensities. Each LARA is equipped with a lens tube of 1″ and 60 mm length. This design minimizes background interference from stray light by narrowing the field of view. These tubes are also equipped with an aperture at the end. This further limits the angle of view. As a result, LARA can only be illuminated by its corresponding mirror. Fig. 3.51 shows a module illustrating the beam path. LARA is positioned so that the pupil of the lens tube is about 80 mm above the mirror, and it extends 1 cm into the beam.

Fig. 3.52 shows the labeling of modules A to E and their mirrors 1-3, while Tab. 3.10 lists the LARAs associated with each mirror. In particular, LARA 18 and LARA 21 are for background measurements and monitoring of the shutter and chopper status of the monochromator. LARA 21 is mounted on the light source mount about ≈1 m below SCALA oriented sideways and only records the chopper's state. In contrast, LARA 18 is mounted on module D, facing the telescope, to measure the light background.

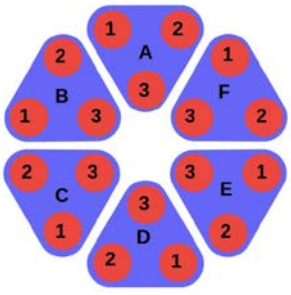


Figure 3.52: The labeling convention for SCALA modules. Modules range from A to F, with mirrors within each module numbered 1 to 3. Convention taken from Lombardo 2017.

Table 3.10: LARAs assigned to the SCALA modules, as visualized in Fig. 3.52.

Mirror	Module					
	A	B	C	D	E	F
1	08	11	15	19	24	31
2	09	12	16	22	27	30
3	10	14	17	23	28	29

Beam Allignment

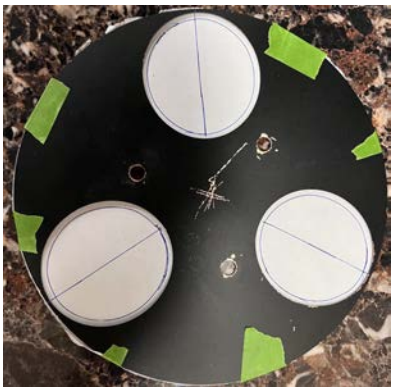
The alignment procedure is similar to that in Lombardo, Küsters, et al. 2017. An LED light source is placed directly at the telescope's Cassegrain focal plane to ensure proper alignment of the SCALA optics with the telescope. By that, SCALA is illuminated by a collimated light beam that aids each mirror's adjustment. In contrast, to Lombardo, Küsters, et al. 2017, a guide mask, as in Fig. 3.53a, is mounted on the module's baffle exit shown in Fig. 3.53b. The mirrors are fine-tuned to direct the light cone to the beam's respective mask's center. Fig. 3.53d shows the telescope directed to the SCALA structure during calibration using SCALA. For precise positioning, SCALA illuminates the telescope with white light. The P-channel and pinhole grid allow accurate correction of the telescope's CCD pointing, producing a regular pattern of images from the SCALA beams depicted in Fig. 3.53c.

Double Monochromator and Chopper

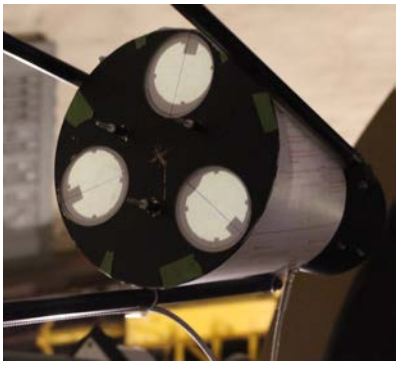
It was shown (see Sec. 3.4.1) that out-of-band emissions incline the background. As discussed in Sec. 3.1.2, a single Cornerstone[33] 260 Monochromators (see Fig. 3.54 No. **2**) has a significant amount of out-of-band light and a second monochromator can reduce this background to a non-detectable level. Like the double monochromator setup described in the named section, a second monochromator (**3**) is put in series with the first.

When measuring with the monitoring reference diodes and the SNIFS spectrograph, all devices must be illuminated with the same exposure time. Secondly, the background of the amperemeters must be measured sequentially. The pilot measurements in 2015 (Lombardo, Küsters, et al. 2017) synchronized the exposure of the devices by using the monochromator's shutter.

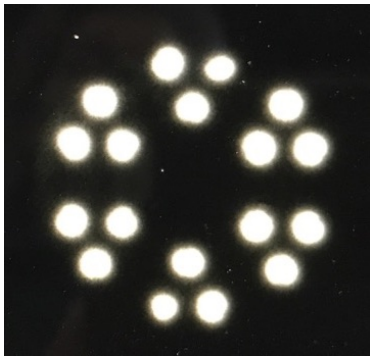
[33] Formerly Oriel



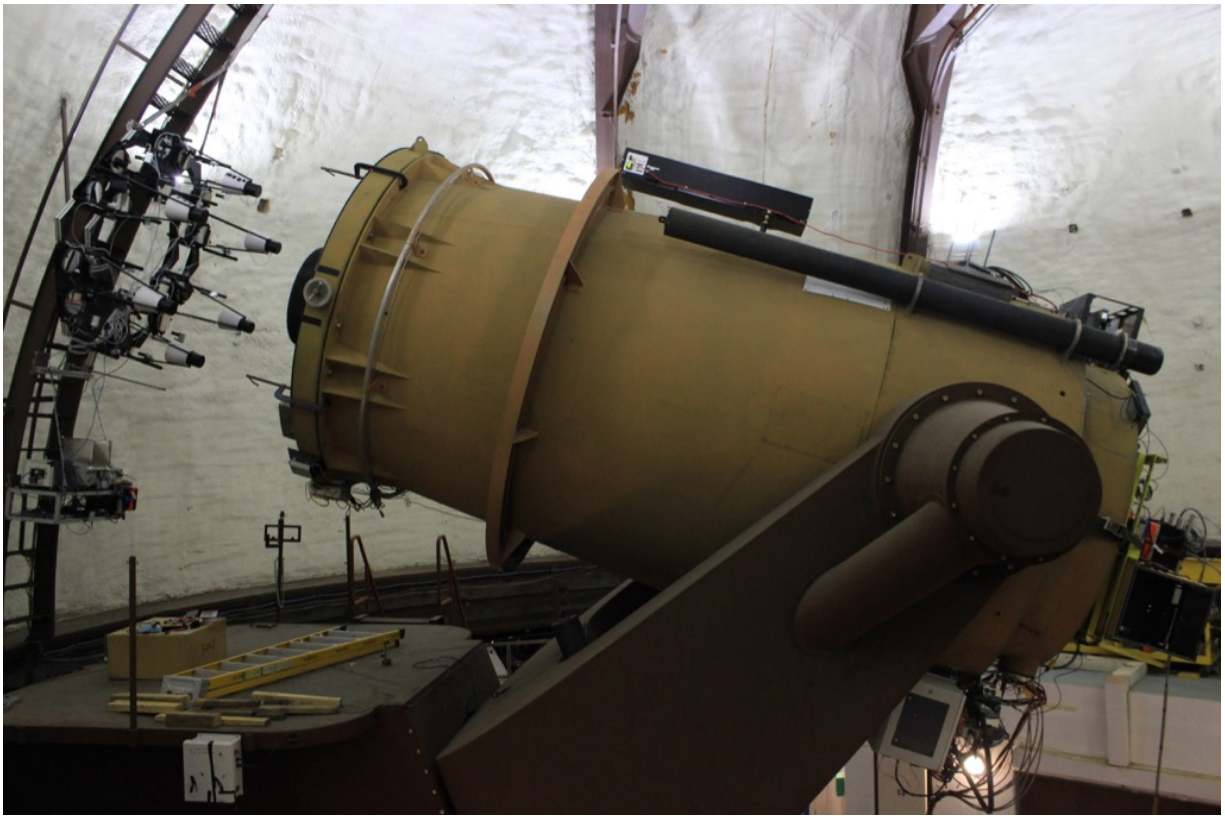
(a) Guide mask used to align the mirrors. The elliptical drawing on the paper indicates the optimal light path for the beam.



(b) Guide mounted on the baffle of an integrating sphere illuminated by the telescope. Pictures show the light beams before mirror fine-tuning.



(c) CCD image from the P-channel with the B filter pin-hole grid. A picture of the SCALA beams allowing precise positioning of the telescope.



(d) The figure shows the UH88 telescope directed to SCALA.

Figure 3.53: A collection of images illustrating various components and stages of the SCALA setup and alignment process.

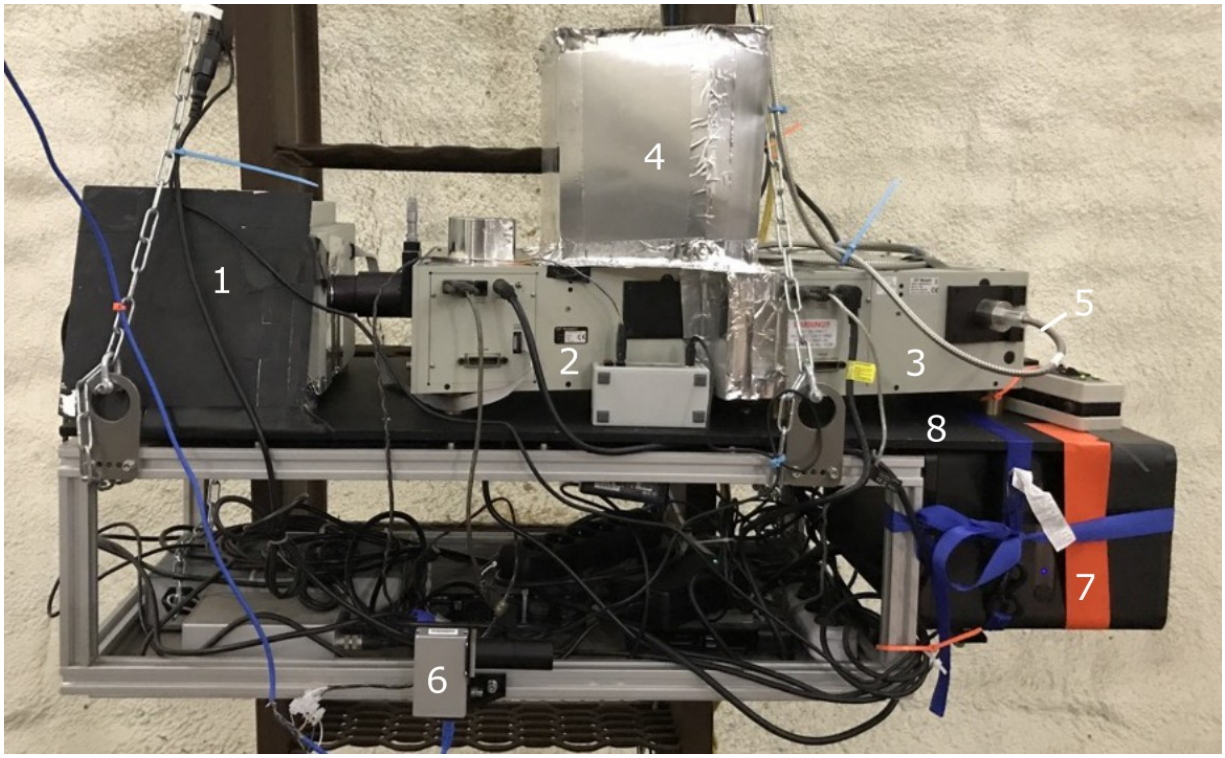


Figure 3.54: The photograph shows the light source setup of SCALA after the upgrade. At position **1**, a Xenon arc lamp named Newport's Oriel Apex Monochromator Illuminator, MODEL 70812. Adjacent to it, the Cornerstone 260 Monochromators from Newport sits at **2**, with a counterpart monochromator at **3**. The chopper in the housing at **4** modulates the light beam. The fiber bundle is seen at **5**, directing the light to the integrating spheres. LARA No. 21 monitors the chopper and shutter states at **6**. A PC controls the setup at **7**, operating on a Linux Ubuntu OS, and manages the system functionalities. Lastly, an aluminum plate at **8** provides structural support to the monochromator.

A chopper is placed between the monochromators to take over the shutter's function for synchronization, as shown in Fig. 3.55. To record the present operational condition of the chopper blade, a mechanical micro switch, furnished with a momentary roller lever, is placed above the blade's wheel. When the wheel rotates, the blade's notches come into contact with the roller lever, activating the switch and indicating the chopper's current state. As shown in the figure 3.54, the chopper, denoted as **4**, is encased in an aluminum housing to minimize light scatter. This casing ensures that extraneous light from the chopper's operations does not inadvertently reach the telescope, enhancing the SNIFS background.

In summary, the SCALA upgrade in June 2022 includes the following improvements:

- Recoating of the mirrors
- Baffle for the integrating spheres
- Monitoring diodes for each beam
- Beam alignment
- Implementation of the second monochromator plus a chopper



Figure 3.55: The chopper wheel between the two monochromators.

ULTRASAT Test Sensor Characterization

4

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4.3 Results and Discussion . . 75

The Ultraviolet Transient Astronomical Satellite (ULTRASAT) is a wide-field (204 degree2) UV transient explorer that will be operated in geostationary orbit. The launch is scheduled for 2026. It is sensitive to 22.5 mag with 900 s exposure in the near UV (230 nm to 290 nm) (Shvartzvald et al. 2023). ULTRASAT stands for time domain astronomy and multi-messenger astronomy with two key science goals: EM-counterparts to binary neutron star mergers (BNS) and supernovae (SNe). ULTRASAT is lead by Weizmann Institut of Science (WIS) and the Israel Space Agency (ISA). Launch by NASA, telescope, and CMOS developed by industry, FPA provided by DESY.

Massive star explosions are powerfully studied with early UV observations. EM signals are constituted by the shock breakout, which is expected to be in the X-ray/UV for supergiant stars. Due to ULTRASAT's sensitivity, it can be expected that up to 20 afterglows of untriggered and ten orphan GRBs will be detected. The estimated detection rate for tidal disruption events (TDEs) is $\approx$ 35 events per year leading to > 10 TDEs per year at 90 % confidence level. NUV-band observations will chart the unexplored territory of physical parameters of the emission from massive BHs at the center of galaxies (AGNs) and explore the potential for high redshift quasars. The sensitivity in the NUV band allows the exploration of variable stars. With accounted stellar variability, there are ≈1000 stars with sufficient accuracy at each frame that allows the detection of Jupiter-size planets. "The survey will find hundreds of such planets within the mission lifetime" (Sagiv et al. 2014). The wide field of view will detect thousands of asteroids and near-Earth objects (NEOs), even if these objects are relatively faint in the NUV. Furthermore, it can detect EM counterparts for gravitational waves as it allows a rapid search of large areas (Sagiv et al. 2014).

Intermediate sensor characterizations of test sensors for the camera's final sensor array are vital for design decisions. These evaluations use the VASC and demonstrate its practical application and synergistic benefits.

4.1 ULTRASAT Satallite

The spacecraft (see Fig. 4.1) system can be parted into a telescope and a bi-propellant propulsion system and weights ≈1100 kg in total. The propulsion system utilizes approximately half the total payload. The system has been constructed based on the long-term experience of the Israel Aerospace Industries (IAI). One of the critical features of this spacecraft is its antenna, which enables real-time data download at any given time. This capability allows immediate access to the acquired observations and facilitates quick response times for capturing new targets. The spacecraft can begin observations within 15 minutes of receiving the command. The spacecraft has been designed to withstand three hours of negative power balance. This scenario occurs in approximately 33 percent of the potential targets.

The accessible sky is restricted to 66 percent due to stray light originating from sources such as the Sun, Earth, and Moon. The system has been designed to ensure that the 3σ pointing jitter remains below 3″ over 300 seconds. This level of stability is crucial for capturing high-quality images, particularly for faint celestial sources located more than 5 degrees away from the field of view (FOV) center.

DESY develops the focal plane array (FPA). Furthermore, the quantum efficiency (QE) of the complementary metal-oxide-semiconductor (CMOS) sensor (see Fig. 4.2) is enhanced through the implementation of a high-K dielectric layer and a UV-optimized anti-reflection coating (ARC) (Shvartzvald et al. 2023). Extensive testing of various ARC designs has been conducted as part of this work at DESY to determine the most efficient configuration for optimizing the sensor's performance.

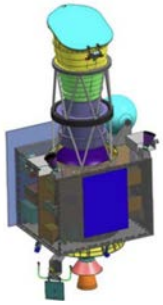


Figure 4.1: The ULTRASAT satalite from (Particle Physics ULTRASAT Project and Science 2023).

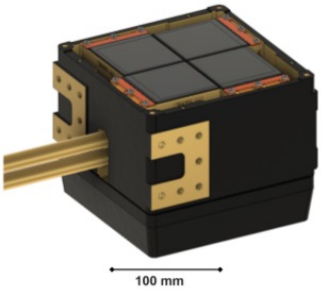


Figure 4.2: Digital renderings of the Detector Assembly of the ULTRASAT camera from (Asif et al. 2021).

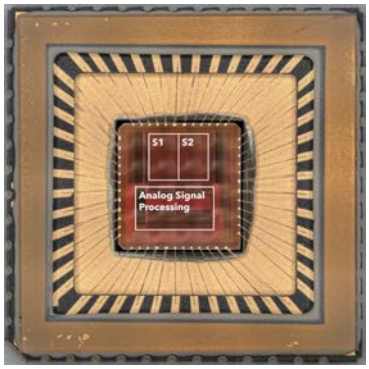


Figure 4.3: The test sensor. The sensor consists of two active pixel arrays S1/S2, and the analog signal processing region. Taken from (Bastian-Querner, Kaipachery, et al. 2021b).

4.2 Quantum Efficiency of the Test Sensors

Test sensors have been produced to carry out selection studies to determine the coating used in the final sensor. These sensors share a similar pixel design with the final sensor. Three ARCs were under study. The test sensor (see Fig. 4.3) has two active pixel arrays with 120×80 pixels of 20 μm (S1) and 15 μm (S2) pitch, respectively. The pixels on S1 have guard rings to reduce charge leakage (blooming) but a reduced fill factor of one-third (Kaipachery 2023). The quantum efficiency measurements are done on S2 (Bastian-Querner, Kaipachery, et al. 2021b).

Among the quantum efficiency, the test parameters include gain, linearity, dark current, and quantum yield to make the design decision on the ARC variant. The SPIE Proceeding (Bastian-Querner, Kaipachery, et al. 2021a) explains the results and the complete methodology. The measurements and the analysis of these tests are the achievements of the authors of this proceeding.

The test sensors are read out by a development board provided by Tower Semiconductors and must be operated by a Windows machine with LabView software. Tower provides no API to embed the software into a larger framework. The software must be controlled as a graphical user interface (GUI) application with human input devices (HID).

The quantum efficiency measurement consists of multiple exposures between five and ten frames per nm-wavelength over a range of 220 nm to 1100 nm what results in $\approx$ 2000 to 8000 individual operations per measurement with a different set of exposure times to compensate the variation of the light source's throughput over wavelength per sensor. Dark current frames are taken at different exposure times for a gain measurement. Granular measurements are required over large exposure times to understand the gain's behavior, leading to hundreds of single operations with changed sensor parameters. The quantum yield measurement consists of gain measurements but with the illumination of monochromatic light and spans a parameter space of different wavelengths and exposure times.

An API is required to ensure the needed quality and reproducibility that no human controller can do. This work focuses on developing the above API to embed the test sensor into the data acquisition of the photometric

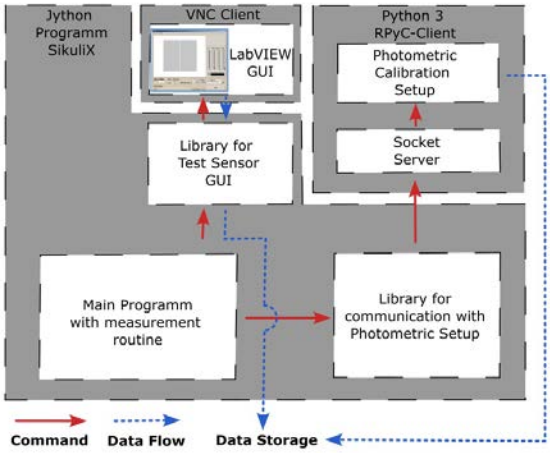


Figure 4.4: Schematic of the readout framework developed in this work for the ULTRASAT test sensor characterization. The software is accessible from the DESY hosted GitLab: UC Sensor Test Measurements repository (Bastian-Querner 2021).

setup to realize the quantum efficiency measurements.

To embed the readout of the test sensor into the calibration setup's framework, an API was written that allows automated control of the sensor's parameters, such as exposure time and the trigger. The software uses SikuliX, a Jython-based framework (Hocke 2017b) running on the host Linux machine that controls a remote Windows machine via Virtual Network Computing (VNC) with image recognition from OpenCV (Hocke 2017a).

Fig. 4.4 shows a schematic of the framework to read out the test sensors developed in this work. The Jython script works as the main program controlling the LabVIEW software from the test sensor's manufacturer on the remote machine. Two libraries are written to control the LabVIEW GUI remotely on a Windows machine, as well as the photometric setup that includes the readout of the reference diodes. The software framework of the photometric setup (see Sec. 3.1.8) runs in an RPyC (Remote Python Call) session, which is controlled via a socket from the main program.

4.3 Results and Discussion

We measured the gain, the quantum yield, and the quantum efficiency of twelve test sensors with different ARCs for the test sensor characterization. An average gain for all sensors has been calculated, and the quantum yield data are comparable to other publications. Fig. 4.5 shows the granular, spectral QE scans for three test sensors exemplary for each coating published in Bastian-Querner, Kaipachery, et al. 2021a. The ARC T2 is concluded to enhance the QE in the ULTRASAT band of 220 nm to 280 nm most and is recommended by the publication. The uncertainties of the spectral QE's shape are dominated by the gain measurement of ≈2 %. The overall scale has an additional uncertainty of 1 % by the used Keithley 6482 amperemeter. A detailed discussion of the methods and conclusion was discussed in Bastian-Querner, Kaipachery, et al. 2021a. The curve of T2 peaks up to 80 % in the ULTRASAT band decreases to a minimum of ≈300 nm and shows a double peak in the visual, after

Methodology of the Quantum Efficiency measurement

The measurement of the quantum efficiency is separated into two parts. First, the determination of the splitting ratio of the beam splitter by measuring two calibrated photodiodes a Hamamatsu S1337-1010BQ (calibrated by PTB between 2000 Å to 4000 Å, σ_{rel} = 0.3 %) and an S2281 (calibrated by NIST between 3000 Å to 11 000 Å, σ_{rel} = 0.15 %) photodiodes are transferred to the reference diode on the "scout-optic" described in 3.1.7. Secondly, the test sensor is illuminated while the flux is calculated from the former calibrated responsivity. The splitting ratio of the beamsplitter is included in the former calibration. The double monochromator uses grating number three for both monochromators (see Tab. 3.2) for wavelengths below 365 nm to increase flux which causes a bandwidth of 3.4 nm instead of 1.7 nm as it is in the VIS and IR. The used relay optic is the ULTRASAT configuration (see Sec. 3.1.7). The reference photodiode on the robotic relay optic and the primary calibrated photodiode S1337-1010BQ photodiode[a] are read out with a Dual-Channel Picoammeter 6482 from Keithley. To compensate for the variation of the flux (compare Fig. 3.5) over the wavelength, the exposure time is chosen to be 100 ms for UV and 20 ms in the VIS/IR. The variable flux in Fig. 3.5 is damped over that the signal varies between 1 pA to 200 pA. The statistical uncertainty varies between 0.1 % for 1 nA and 2 % for 1 nA for the relative responsivity. The absolute scale and non-linearity have an additional 1 % from the manufacturer's calibration.

[a] From Hamamatsu

which the QE decreases in the infrared IR with a long tail. The QE shows oscillations in the IR from interference caused by back reflections after a finite sensor thickness. The observed period $\delta\lambda$ at $\lambda = 8000$ Å is ≈80 Å. The sensitivity outside the ULTRASAT band is also vital as it sets an upper limit for the rejection performance of the filters and the stray light baffles. For sufficient characterization, it is necessary to resolve that interference pattern, which has an amplitude of up to 20 % in the spectral QE with a minimal wavelength ≈80 Å. Each oscillation would have been sampled with at least three points[1]. That makes (11 000 Å − 2000 Å)/80 Å/3 ≈ 340 individual samples per QE measurement. To get a complete picture of the reproducibility of the coatings, several sensors of the same ARC have to be measured. The uncertainty of the gain of 2 % and the statement for the upper limit of exposure time and collected charge caused by non-linearity could only be achieved with the above-discussed granularity in exposure time and measurements from all twelve sensors. There have been 158 gains, 47 quantum yields, and 261 spectral quantum efficiency measurements over three months. Even for the QE measurements alone and a reduced granularity in wavelength, that would have been ≈ 89.000 operations. A human operator would have been able to perform ≈ 15.000 operations if errors are not counted[2]. That leads to the conclusion that the development of the API based on OpenCV image recognition via VNC from this work was crucial for the quality of the data and the findings taken in the SPIE proceeding (Bastian-Querner, Kaipachery, et al. 2021a).

[1] The Nyquist-Shannon sampling theorem requires two samples per period, while the period in this oscillations vary.
[2] Assuming 2 min per operation and 160 h of working time per month.

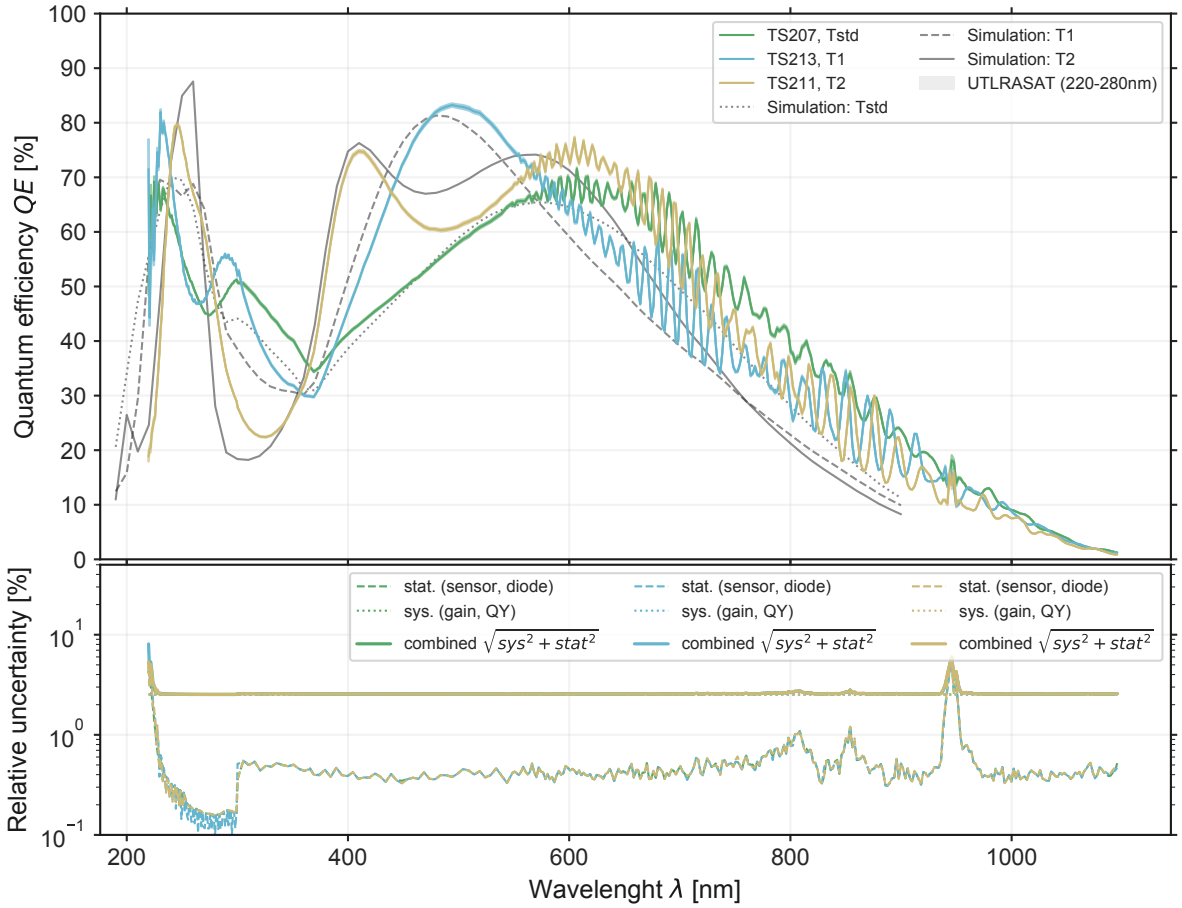


Figure 4.5: The results of spectral quantum efficiency for all three ARC options compared with Simulations. Taken from (Bastian-Querner, Kaipachery, et al. 2021a)

Calibration of LARAs 5

This chapter describes the methods and algorithms used to calibrate the LARAs with the VASC and analyze the data. It starts with the mathematical formulation of the calibration of photosensors using the comparison method with monitor (Sec. 5.1) and a description of the data acquisition of the LARAs (see Sec.5.2). Next, it deeps into the methodology of the calibration of the LARAs (see Sec. 5.3), which includes the calibration of the wavelength scale (see Sec. 5.3.2), the alignment of the LARAs (see Sec. 5.3.1), and the determination of the responsivity (see Sec. 5.3.3).

At the end of this chapter, the results of the calibration are presented by a comparison of all LARAs (see Sec. 5.4), a full discussion of all uncertainties (see Sec. 5.4.1), and a comparison of the VASC's transfer capabilities with the NIST calibration (see Sec. 5.5).

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5.1 Mathematical Description of Photometric Calibration for Photodiodes

A standard is needed to perform a calibration. That standard must be traceable down to the laboratory realization of the observable of interest. In the case of the photometric measurement, the laboratory standard is the POWR operated by NIST (see Sec. 2.3) (Larason and Jeanne M. Houston 2008). The NIST provides three calibrated S2281 photodiodes[1] manufactured by Hamamatsu (see Fig. 5.1a). Fig. 5.1b shows the responsivity from the calibration sheet of one representative diode (7J051). The orange curve shows the quadratic sum of all systematical and statistical relative uncertainties. These diodes are now referred to as NIST diodes.

The NIST diodes are read out with two LARAs. The calibration transfer from the NIST photodiodes to the DUT can be separated into two parts: the measurement of the NIST diodes to calibrate the reference diode mounted on the robotic optics (see Sec. 3.1.7) and the measurement of the photodiode or Device Under Test (DUT). Calibration is transferred by comparing the photocurrents:

$$r_i = \frac{I_i}{I_{\text{Ref}}}, \tag{5.1}$$

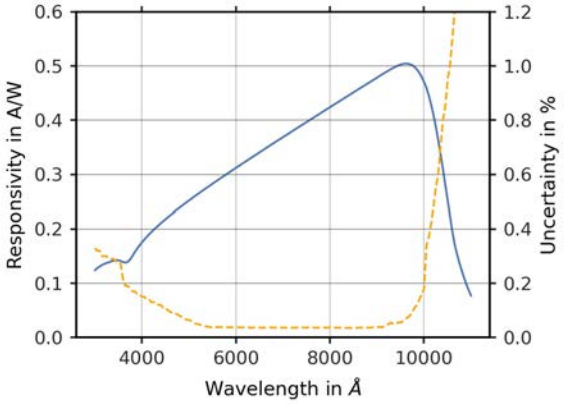
Where $i \in \{\text{NIST}, \text{DUT}\}$, represents the NIST, reference, and DUT diodes. The final step involves eliminating the responsivity of the reference diode:

$$R_{\text{DUT}} := \frac{r_{\text{NIST}}}{r_{\text{DUT}}} = \frac{I_{\text{NIST}}}{I_{\text{DUT}}}. \tag{5.2}$$

[1] Serialnumbers: 7J048, 7J052, & 7J054



(a) Top: SN 7J052; Bottom: SN 7J054.



(b) Responsivity S of one of the NIST calibrated Hamamatsu S2281 photodiodes. Blue solid line: The responsivity in A/W and dashed orange line: the relative uncertainty on that value.

Figure 5.1: NIST calibrated S2281 photodiodes and their responsivity.

With R_{DUT} being the calibration, that is the responsivity ratio between the DUT and the NIST diode. Since the product of photocurrent and responsivity is equal for DUT and NIST diodes, the flux $F = S_{\mathrm{NIST}} I_{\mathrm{NIST}} = S_{\mathrm{DUT}} I_{\mathrm{DUT}}$, the responsivity of the DUT is given by:

$$S_{\mathrm{DUT}} = S_{\mathrm{NIST}} R_{\mathrm{DUT}} \tag{5.3}$$

Considering the spectral quantum efficiency as:

$$QE = \frac{hc}{\lambda e} \cdot S \left[\frac{A}{W} \right], \tag{5.4}$$

The argument leads to a similar result:

$$QE_{\mathrm{DUT}} = QE_{\mathrm{NIST}} R_{\mathrm{DUT}} \tag{5.5}$$

Therefore, the ratio r_i (Eq. 5.1) is the observable to be measured for making a calibration transfer between two photodiodes with the VASC.

The following sections describe the detailed recording of the photocurrents, their analyses, and the necessary corrections on r_i.

5.2 Data Acquisition of the LARA Photo Diodes

The LARA records discrete values of the voltage $U_i(t_j)$ over their feedback resistance over time t_j with a sampling rate of 1200 Hz. The feedback resistance has a nominal value of 10 GΩ, which has a temperature dependence α as described in Sec. 3.2.2. To convert the voltage $U_i(t_j)$ into a current $I_i(t_j)$ the voltage is devided with the temperature corrected resistance $R_{\mathrm{nominal}} = 10\,\mathrm{G\Omega} \rightarrow 10\,\mathrm{GV\,A^{-1}}[1+\alpha(20\,^{\circ}\mathrm{C} - \langle T \rangle_t)] \times k$ instead of the nominal. An individual multiplication factor k reflects the error from the feedback resistor's nominal value, and $\langle T \rangle_t$ is the averaged

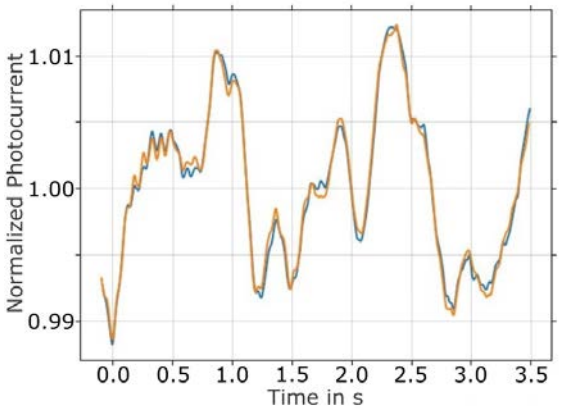


Figure 5.2: Measured relative photocurrent after synchronization at a reference photodiode (Blue) and a DUT LARA (Orange) illuminated with light from the LDLS. The current is normalized by its mean value (≈200 pA) to visualize the flux output variability.

calibrated temperature value from the ADU sensor[2]. The module's gain G is then the inverse of the temperature-dependent feedback resistance, and the nominal gain $G_{\text{nominal}} = 10^{-10}\,\text{A}\,\text{V}^{-1}/k$. The current is, therefore, calculated as

$$I = \frac{U}{10\,\text{GV}\,\text{A}^{-1}}[1 + \alpha(20\,^\circ\text{C} - \langle T\rangle_t)] \tag{5.6}$$

If photocurrents of two LARAs are compared like in Eq. 5.1, the individual error on the nominal gain does not cancel out, as it differs from LARA to LAR. Eq. 5.1, 5.3, and 5.5 become:

$$r_i = \frac{I_i}{I_{\text{Ref}}}\frac{k_{\text{Ref}}}{k_i}, \tag{5.7}$$

$$S_{\text{DUT}} = S_{\text{NIST}} R_{\text{DUT}} \frac{k_{\text{DUT}}}{k_{\text{NIST}}}, \tag{5.8}$$

$$QE_{\text{DUT}} = QE_{\text{NIST}} R_{\text{DUT}} \frac{k_{\text{DUT}}}{k_{\text{NIST}}}, \tag{5.9}$$

To determine the factors k, an absolute calibration of the LARA's feedback resistors would be necessary. However, it is omitted, as it varies only on the absolute scale, which does not affect the calibration of the color. In all further argumentations, the ratio $k_{\text{DUT}}/k_{\text{NIST}}$ is overlooked but should be held in mind.

To compare the responsivities of the photodiodes, variability in flux must cancel out. The Laser-Driven lightsource varies in its flux up to 1 % with 2 Hz to 25 Hz (see Fig. 5.2). The read-out of the devices must be synchronized to the smallest scales of variabilities of significant amplitude. Fig. 5.2 shows small scale variation at 25 Hz with an amplitude of 0.4 %. Depending on the LDLS's bulb lifetime, there are 1 % jumps in intensity observed. To allow a comparison, a synchronization of better than $25\,\text{Hz}^{-1} = 40\,\text{ms}$ of all devices must be done. In the case of the LARA for every read-out, a timestamp t_{start} is recorded from software

[2] The build-in ADC temperature sensor records 24 samples in parallel of the waveform read-out. The sensors are calibrated by comparing a GTH 175 / Pt 1000 Digital thermometer from Greisinger Electronic at room temperature. The thermometer's accuracy is given as 0.1 °C from its manufacturer. This is assumed to give a relative temperature calibration of the above uncertainty.

that allows relative comparison between the waveforms down to 5 ms[3]. The synchronization is done in two steps. The first is a read-out delay time t_{delay} set for each amperemeter individually to synchronize the t_{start}. These parameters are found manually after each new device is connected to the read-out system, which changes the computer's load. The devices are read out multiple times, and the offsets between t_{start} are averaged and used as the delays to synchronize for read-out. As the DAQ computer's processor load changes over time, it is insufficient to reproduce the exact multitask every time down to the 40 ms. At every read-out, the offsets between t_{start} are calculated, and the measurement is repeated if the largest offset concerning the reference diode at the robot optics is larger than 100 ms. The second part of the synchronization concerning the recorded t_{start} is the removal of samples in the waveform during the data reduction to bring the waveforms to an exact overlay before being compared. After the synchronization, the waveforms overlap with jitter less than the sampling time (compare Fig. 5.2).

5.2.1 Ionization Radiation

During read-out, the photodiodes are hit by ionizing particles from natural radioactivity and cosmic radiation. If such a particle hits the diode, a minimum amount of energy is released in the silicon depending on the sensor thickness, adding additional current to the photocurrent. Fig. 5.3 shows a waveform of one LARA hit by an ionizing particle. The subfigure illustrates a zoom into the deposited charge and is referred to as cosmic from now on. A mask is estimated in three steps to remove the cosmics from all waveforms. First, the ratio r_i is calculated after synchronization sample-wise without background correction. Secondly, a sigma clipping is done on the ratio r, assuming the ratio between the responses of diodes is constant, and the waveforms are sufficiently synchronized. Last, ten[4] samples ($\approx$ 8 ms) are removed at the peak's beginning and end to remove the cosmic's residual tail.

5.2.2 Averaging and Background Subtraction

A signal s waveform $I_i^{s,b}(t_j)$ with background b and background waveform $I_i^b(t_j)$ are recorded for the devices $i \in \{\text{NIST}, \text{DUT}\}$. Each waveform of the length t_{exp} consists of $N_{\text{wave}} = t_{\text{exp}} f_s$ samples j at the time t_j, recorded with a sampling rate f_s =1200 Hz. The background measurement is subsequently done with a closed monochromator shutter and is assumed to have no variations. The cosmic removal is done with $I_i^b(t_j)$ as explained above but without taking the ratio. The standard deviation $\sigma(I_i^b)$ and the average $\langle I_i^b(t_j)\rangle$ are calculated from the masked background and applied to $I_i^{s,b}(t_j)$ to get $I_i^s(t_j) = I^{s,b}(t_j) - \langle\, I_i^b(t_j)\rangle$ the background corrected waveforms $I_i^s(t_j)$.

[3] This has been verified by measured waveforms from different LARA with the monochromator's shutter being closed during recording. The left jitter of 5 ms is from the computer's ability to reproduce the same processes in parallel depending on the system's processor load.

[4] Empirical value.

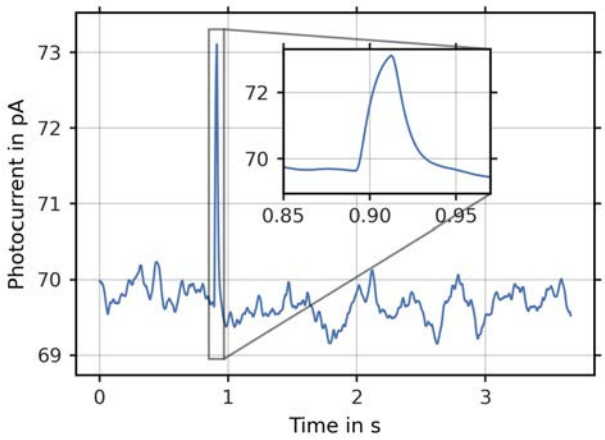


Figure 5.3: The photocurrent, measured using a calibrated LARA for SCALA, exhibits an additional charge resulting from the impact of an ionizing particle on the diode during read-out.

For each Device i, the ratio $r_i(t_j)$ (Eq. 5.1) is calculated from $I_i^s(t_j)$ and $I_{\text{ref}}^s(t_j)$ and averaged $\langle r(t_i)\rangle$ over all samples. The quadratic sum of the relative uncertainties $\sigma^{\text{rel}}(I_i^b)$, and $\sigma^{\text{rel}}(r_i)$ is taken as the statistical error for r_i.

5.3 Methodology

SCALA needs 18 calibrated photodetectors with one additional detector for recording the background variation outside a beam. During that work, 21 LARA modules have been calibrated concerning their relative spectral sensitivity S_λ. Fig. 5.4 shows twelve LARAs ready for shipment after calibration, and Fig. 5.5 shows the S1337-BQ1010 photodiode with a 9.5 mm × 9.5 mm aperture. The method used here is the substitution method with the monitor as described in Sec. 2.3.

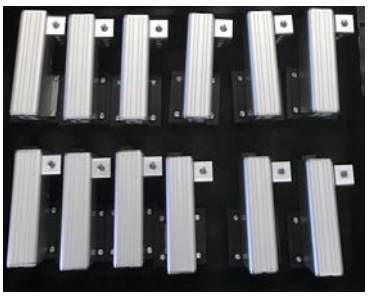


Figure 5.4: Twelve of 21 LARAs are ready for shipment to the UH88 telescope after calibration.

Fig. 5.19 shows eight LARAs in the dark box for calibration with the **VASC**. The gantry robot has the SCALA relay optic described in Sec. 3.1.7. It automatically moves between the photodiodes.

For the calibration process, the responsivity of the LARA is compared with the NIST-calibrated photodiodes by illuminating the LARA and NIST subsequently. One measurement is conducted from 3000 Å to 12 000 Å with 8 Å and 16 Å stepsizes. LARA module's spectral sensitivity S_λ is derived from Eq. 5.3 as described in Sec. 5.3.3. As the final bandpass is determined by the coupling into the optical fiber (compare Sec. 3.1.3), this scan is also conducted to measure the transmittance of the calibrated Holmium-Dydimium filter that allows a wavelength calibration for the bandpass after the optical fiber during a line fit, described in Sec. 5.3.2.



Figure 5.5: S1337-BQ1010 photodiode of 10 mm × 10 mm size integrated into a representative LARA with its aperture.

5.3.1 Alignment

Because of imperfections in the optics (described in Sec. 3.1.7), the image of the fiber is not perfect and exhibits stray light, which decreases radially following a power law. Fig. 5.6 shows the point spread function (PSF) of the image in the objective's focal point as the spatial decay constant is at the same scale of the photodiodes' active area (10 mm),

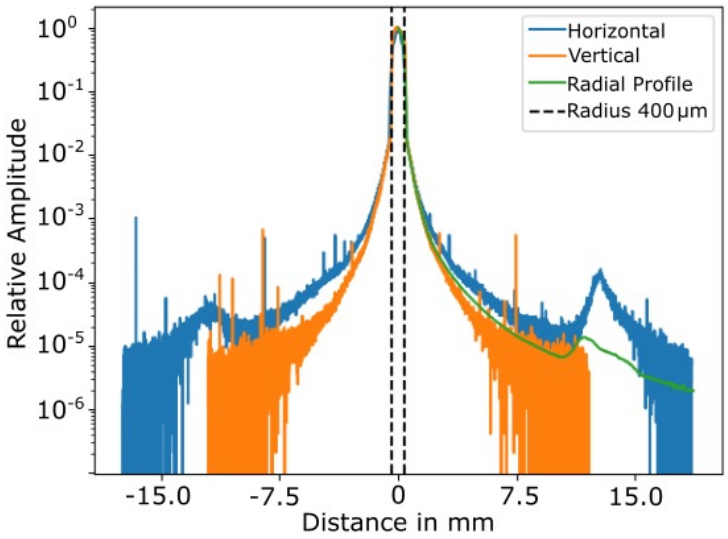


Figure 5.6: Slices from a High Dynamic Range (HDR) image in the objective's focal point from monochromatic light of 5000 Å. The image was taken with a QHY 600 camera, which has an active area of 36 mm × 24 mm. Blue: The relative amplitude horizontal to the ground or parallel to the robot's X-axis. Orange: Amplitude vertically (parallel to the robot's Z-axis) sliced. Green: Radially averaged profile of the relative amplitude. Black dashed: The theoretical size of the image in the focal point (see Sec. 3.1.7). The spikes are due to hot pixels. Notable are the interference modes in the horizontal line.

a significant amount of light (> 1 %, see Sec. 5.3.3) is not collected by the photodiode. Furthermore, the PSF is not radially symmetrical and shows interference modes at horizontal lines that move from the image with increasing wavelength. The surface of the off-axis parabolic mirror that works as the objective is processed with diamond turning, which leaves behind periodic scratches with an RMS-roughness of 2 Å to 5 Å (Personal communication with Thorlabs customer service 01–17–2022). These scratches work as a diffraction grid. That makes the alignment of the photodiodes crucial for substitution measurements and reproducibility. Any misalignment leads to a change in collected flux and, therefore, to a systematic error. The external quantum efficiency of a photodiode depends on the transmission of the incident light of the optical properties of silicon described by the Fresnel Equations. It also depends on the angle of incidence (AoI). In Summary, before the responsivity measurements, the relative position and rotation of the diode to be probed must be estimated.

For the calibration measurements, the DUT is positioned approximately 50 mm outside the robot's accessible volume, with its normal parallel to the depth (Y) axis (compare Fig. 5.19). It lies within the width (X) and height (Z) plane. To determine the DUT's position, three scans are conducted.

- XZ-slices scan
- Triangulation scan
- Angle of Incidence scan

These scans together give the exact orientation and position of the probed sensor.

XZ-slices

In the alignment process, the diodes are manually adjusted to ensure the light spot is positioned at the center of each diode within the robot's coordinate system. Fine-tuning of their orientation is achieved through

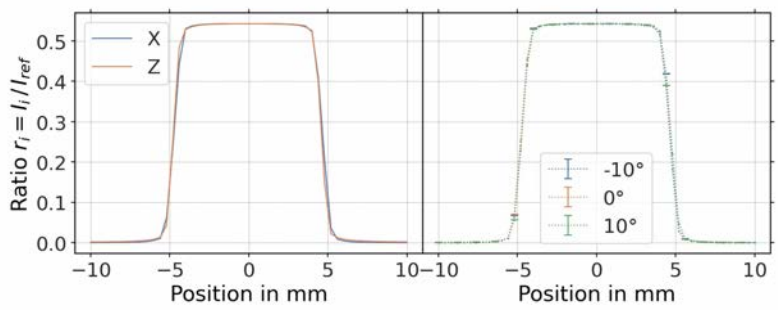


Figure 5.7: Results of the second XZ-slices and the triangulation scans after positional calibration for LARA No. 27. Left: Relative response r_{27} concerning the reference diode in dependence of the light spot's position relative to the diode's center. The blue line shows the response with $Z = 0$ and varying X (X-slice), and vice versa for the orange line. Right: Response of an X-slice with $\theta \in \{-10° : \text{Blue}, 0° : \text{Orange}, 10° : \text{Green}\}$. The repeated scans after calibration give a residual offset of $|\delta_X| = (39 \pm 15)$ µm, $|\delta_Z| = (78 \pm 50)$ µm, and $|\delta_Y| = (82 \pm 15)$ µm. The uncertainties come from the robot's reproducibility given by its manufacturer, which is assumed to be the limiting error.

precise scans of the spatial responses. During these scans, the relative responsivity ($r_i = I_i/I_{\text{ref}}$, as in Eq. 5.1) is measured as the robot moves within a range of ±10° along both the X and Z-axes. The resulting response curve, illustrated in the left plot of Figure 5.7, represents a convolution of the optics' Point Spread Function (PSF) and the diode's Top-Hat spatial response function. To determine the offset of the diode's center relative to the robot's coordinate system, the Center-of-Mass (CoM) is calculated using the formula $\text{CoM} = \frac{\sum_j r_j X_j}{\sum_j X_j}$, where X_j represents the position of the j-th relative response r_j. Once the spatial calibration of the diode is completed, the scan is repeated to assess the alignment error.

Triangulation

To determine the deviation in distance, the spatial shift of relative responses between two X-slices with different angles of incidence is measured. For verification, measurements are conducted on three X-slices with angles of incidence in the horizontal plane ($\{-10°, 0°, 10°\}$). The averaged displacement (δ_X) of the Centers of Mass (CoMs) from the ±10° scans to the 0° scan is used to calculate the deviation ($\delta_Y = \delta_X \tan(\delta_\theta)$) between the expected distance and the actual distance.

The right plot in Figure 5.7 illustrates the repeated scan for LARA No. 27 after distance calibration. All CoMs of the responses align within 25 μm, resulting in a residual distance of (82 ± 65) μm. The difference between the −10° and 10° scans serves as the limiting uncertainty for δ_X.

Angle of Incidence

The internal quantum efficiency of a photodiode depends in first-order on the transmittance of the light flux into the silicon, which is described as the transition from one optical medium (usually air) to another (in this case, silicon with $n_{\text{Si}} > n_{\text{Air}}$). The Fresnel equations (Eq. A.10) describe the transmittance or reflectivity dependent on the incidence angle and the light's polarisation. The maximal transmittance for unpolarised light, and therefore the maximal response, lies on normal incidence. The correct orientation of the photodiode in space concerning the incident pointing vector is essential to avoid underestimated responses. Figure 5.8 shows a heatmap of the two-dimensional AoI-scan of one example S1337

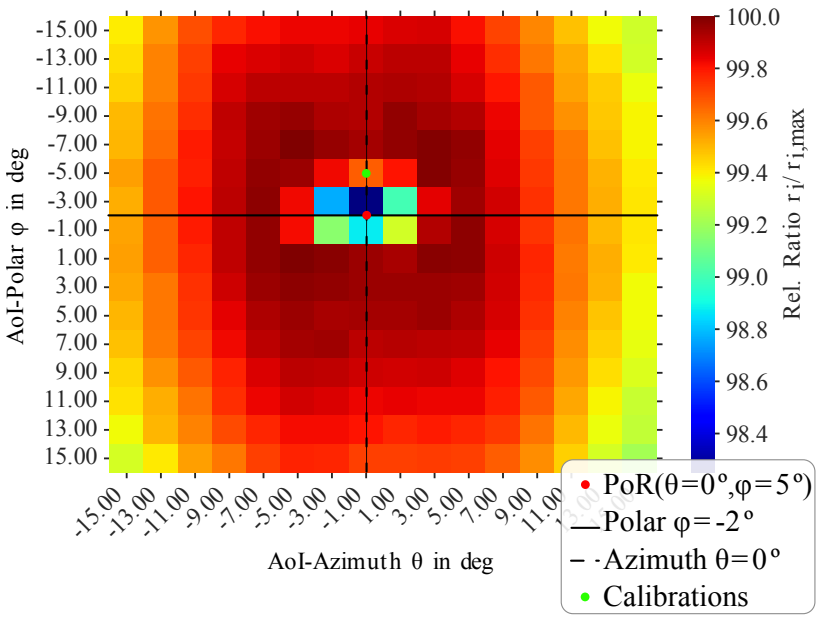


Figure 5.8: Measured relative response r_i of a photodiode depending on the angle of incidence (AoI) in two dimensions ($\alpha_{\text{Polar}} = \varphi$ and $\alpha_{\text{Azimuth}} = \theta$). In the first-order, r_i is inversely proportional to $a\varphi^2 + b\theta^2$, which gives elliptical contour lines. $a = b$, only in the case of unpolarized light, which is not generally given in this work. Note the different axis ranges, which increases the apparent ellipticity.

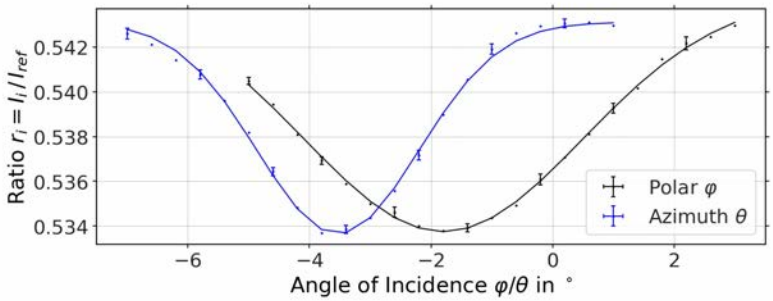


Figure 5.9: Results of the angle of incidence (AoI) scan for LARA No. 27. Blue: Scan with fixed polar angle ($\varphi = 0°$) and varying azimuth angle θ. Black: Vice-versa. Solid lines show fits with a Gaussian function to estimate the minima. Solid lines show The displacement of the AoI-angles from perpendicular incidence is $|\delta_\varphi| = 1.80(5)°$ and $|\delta_\theta| = 3.53(5)°$. Using the HDR50 accuracy as the limiting uncertainty.

photodiode. The diode's response is elliptical, with a minimum in its center. As the light from the relay optics described in Sec. 3.1.7 has a partial polarization from the monochromator gratings, the contour lines of the response r_i are elliptical instead of circular. The minimum (red point) indicates the normal incidence. The back-reflected beam propagates back through the relais optic and causes a relative signal increase on the monitoring diode. The exact light way is not investigated. This peak of reflection (PoR) is visible with all probes measured with this optics. A measurement at this particular angle must be avoided, which makes its determination before each calibration essential.

After spatial calibration at the X, Y, and Z positions, the light is pointed at the diode's center; two scans are performed. One with constant azimuth angle $\theta = 0°$ and varying polar angle φ, and vice-versa. Fig. 5.9 shows the results of an AoI-scan for LARA No. 27. To determine the minimum, a Gaussian function is fitted on the data, visible as solid lines. Under the assumption that the PoR is an ellipse with its axes in parallel with φ and θ or, at best, rotationally symmetric, the resulting minima represent the global minimum.

The AoI for each calibration is chosen to be $\Delta\varphi = 5°$ away from the PoR (see the green point in Fig. 5.8). This point is close enough to the normal incidence to either allow a simple correction or find consideration in the uncertainty budget but far enough from the PoR.

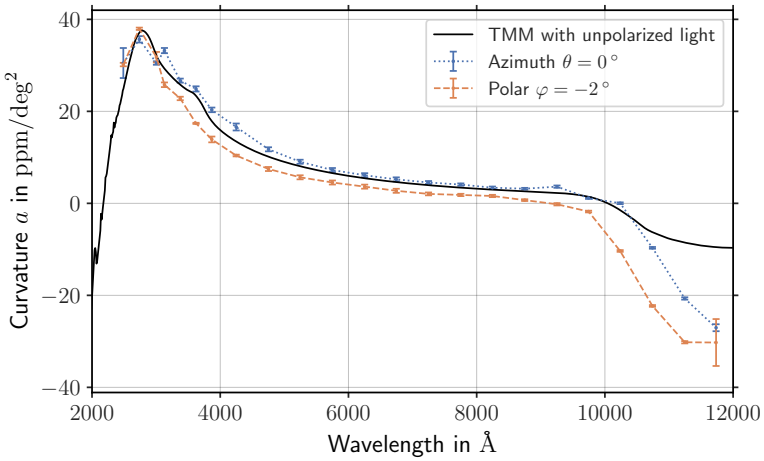


Figure 5.10: Fitted curvature a in dependence on the wavelength for a S1337 diode. The blue dotted line shows the measurements with constant azimuth angle $\phi = -2°$ (see Fig. 5.8 dashed black line) and the dashed orange line depicts the curvatures for measurements with constant polar angle $\theta = 0°$. The black solid line shows the resulting curvature from a TMM (from Daniel Küsters, personal communication (Daniel Küsters 2021)) from the S1337 diode for unpolarized light. The model lies between the data points from the two scans. The curvature is used to estimate the uncertainty from the AoI in the calibrations of the LARAs' photodiodes and those done with SCALA.

Uncertainty caused by the AoI

When a photodiode is adjusted concerning a source, its orientation remains a certain error. For unpolarized light, it is sufficient to describe the orientation error by one variable Φ, which is the angle between the normal incidence and the normal vector of the detector surface. To quantify the change in responsivity, multiple scans at different wavelengths λ were performed with varying azimuth θ and constant polar angle $\varphi = -2°$ and varying φ but constant $\theta = 0°$ (see Fig. 5.8 dashed and solid black lines). The relative response ratio $r_i/r_{i,\max}$ normalized with the value of maximal response $r_{i,\max}$, is fitted with a second order polynomial $1 - a_\Phi \Phi^2$ with $\Phi \in [\varphi, \theta]$ (see Eq. A.15) [1] to estimate the curvature a that describes the angle dependence of the responsivity. The curvature a is plotted in Fig. 5.10 in dependence on the wavelength. A positive curvature corresponds to a declination in responsivity for increasing incidence angle Φ, and vice versa. As a comparison, the black solid line shows the curvatures derived by a TMM model of the S1337 photodiode from personal communication with Daniel Küsters (Daniel Küsters 2021) for unpolarised light. In the range from 3000 Å to 10 000 Å the theoretical curvature is between both data curves. The VASC emits polarized light of unknown mixing. The deviation on the model is less than 3 ppm deg^{-2} between 2500 Å and 10 000 Å. As the curvature determination is distorted by the unknown polarization mixing of the VASC's light, the absolute value of the maximum of both curves $\max(|a_\theta|, |a_\phi|)$. The actual relative error in responsivity with an AoI Φ follows as:

$$\Delta(\Phi, \lambda)_{\text{rel}} = \left| \frac{S(\Phi, \lambda)}{S_\perp(\lambda)} - 1 \right| = \max(|a_\theta|, |a_\phi|) \cdot \Phi^2 \tag{5.10}$$

1: The polynomial can be motivated by calculating the second order Taylor expansion of the normalized transmission $T_\mu(\theta_i, \lambda)/T_\mu(\theta_i = 0, \lambda)$ with the angle of incidence θ_i given by the Fresnel equations for polarized light $\mu \in$ [s-polarization, p-polarization]. This corresponds to the relative sensitivity of the detector $S(\theta_i, \lambda)/S_\perp(\lambda)$, where the λ dependence comes from the wavelength dependence of the refractive indices and $S_\perp(\lambda) = S(\theta_i = 0, \lambda)$. To approximate the dependence on θ, it is sufficient to assume a single refractive index for the detector. A complete derivation can be found in Appendix A.5.3.

5.3.2 Spectral Calibration

The wavelength calibration can be separated into two parts: The first part is the calibration of the wavelength model λ_{Model} and the second is the transfer to the Holmium-Didymium Filter (compare Sec. 3.1.3). By measuring the transmission curve of the filter with the DUT, the wavelength calibration is later imprinted in its signal.

Model Wavelength λ_{Model}

To achieve < Å uncertainty in the wavelength calibration, the wavelength is independent and traceable to data published by a national metrology institute. The scale is calibrated against the spectral lines database from NIST (Standards and Technology 2009). There are four low-pressure gas lamps available (see Tab. A.1) that provide spectral lines from Cadmium, Mercury, Sodium, Neon, and Argon; the last serves as ignition gas in two of the lamps. In the first step, the spectral lines emitted by the gas lamps must be identified. This is necessary to obtain a subset of lines from the database. These identified lines will enable the assignment of the measured lines obtained through the double monochromator using visual inspection. Spectral measurements from the lamps are done with proprietary high accuracy spectrograph OSA201C[5]. The OSA is a scanning Michelson interferometer. Depending on the fiber's core refractive index, the light is fed into the spectrograph via an optical fiber with focal ratios between 2.0 and 2.2. Figure 5.11 shows the working principle of the interference spectrometer.

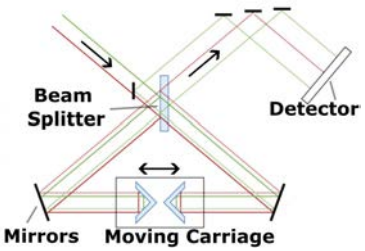


Figure 5.11: Schematic of the Optical Spectrum Analyzer from Thorlabs. Taken from (Thorlabs 2023). Both retroreflectors share the same carriage and are moved simultaneously. This optical configuration translates the optical movement into four times the movement of the carriage. The optical path of the probed light is green, while the reference laser is red. Solid lines indicate incoming beams and dotted lines show reflected beams ending in the detector assembly.

After passing a collimator and a beamsplitter, the light (red line in Fig. 5.11) of the two beams is reflected by retroreflectors. The two beams generate an interferogram on the beamsplitter shifted to the entering beamline. A photometer records the interferograms. The spectrum is scanned by moving the carriage of the retroreflector, resulting in a phase shift between the beams in dependence on the carriage's position. The spectrum is calculated from a Fourier transform of the varying interference pattern. Next to the probed light, the interferogram of a reference laser of a known wavelength is recorded. As the physical path of the light does not change over the wavelength, a one-point calibration of the spectrum is sufficient to scale the spectrum. This makes the spectrograph self-calibrating and allows, in the case of the OSA201C, a spectral accuracy of 2 ppm without external calibration. The OSA is sensitive in a range from 3500 Å to 11 000 Å. A 6329.918 Å Helium-Neon Laser is used as a reference. The light from the probe is coupled to the instrument via an optical fiber. A fiber of 125 μm core diameter is positioned in front of the low-pressure gas lamps and connected with the spectrometer to increase the signal to noise at the expense of alignment. All four low-pressure gas lamps and the background are scanned 1000 times and averaged. The background measurement was done to remove the noise floor. Fig. 5.12 shows the background subtracted spectral power in arbitrary units. All stronger lines show sidelobes (see Fig. 5.13) from the Hann apodization of the Fourier-Transform.

To assign the lines in the spectra, the positions of all maxima are compared with the line positions in the NIST Atomic Spectra Database (Standards and Technology 2009) for the nearest value. The line with the higher recorded intensity is chosen in the case of ambiguity. Fig. 5.14 shows the residuals between the maxima's positions of the lines and their nearest value from the database. For visibility, the uncertainties for the residuals are those from the database. The data follows a sawtooth pattern with a jump at ≈6000 Å. This pattern is reproducible with all low-pressure lamps. As the multimode fiber allows multiple slightly misaligned beams to be coupled into the spectrometer, each line appears unresolved. In

[5] Optical Spectrum Analyzer (OSA) 201C from Thorlabs

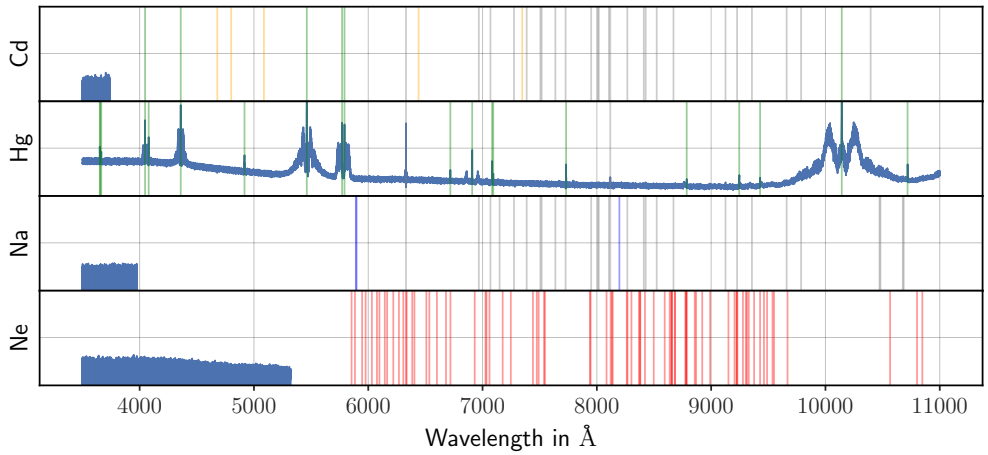


Figure 5.12: Spectra of the low-pressure gas lamps measured with the OSA201C from Thorlabs showing 138 identified lines from neon, sodium, mercury, cadmium, and argon. The labels on the left axis indicate the measured lamps. The identified lines are colored with corresponding elements (Green: Mercury, Yellow: Cadmium, Grey: Argon, Blue: Sodium, and Red: Neon). As argon is used as an ignition gas for sodium, mercury, and cadmium lamps, its lines occur in multiple spectra.

contrast, the intensity ratio between the lines varies over the wavelength range. The subplots in Fig. 5.14 show representative neon lines. From small wavelength to grander, the left peak increases while the other decreases. At ≈6000 Å, the lines alternate in being the maximal peak, which leads to a jump in the assigned line position explained above. As the residual is $\leq |4|$ pm over the complete range, it is comparable with zero within 2 σ compared with the spectrometer's spectral accuracy of 2 pm. This now provides a subset of spectral lines that occur in the lamps and allows an assignment of the lines in a spectrum measured with the double monochromator system by visual inspection.

In the second step, the low-pressure gas lamps are operated in the hollow integrating cylinder (compare Sec. 3.1.1). The spectral flux is recorded by an S1337-1010BQ photodiode from Hamamatsu positioned at the end of the double monochromator (Ref4, Fig. 3.1). The diode's signal is recorded with a Keithley 6485, while the double monochromator scans the spectrum from 3000 Å to 12 000 Å in 4 Å steps. Fig. 5.15 shows all measured spectra with the photocurrent[6] of the diode against the model wavelength λ_{Model} calculated from the encoder's measures (compare Sec. 3.1.2). A linear model $C_{\lambda,\text{Model}}$ is assumed to describe the calibration of λ_{Model}:

$$C_{\lambda,\text{Model}}(\lambda) = c_s\lambda + c_{\text{off}} \tag{5.11}$$

The bandpass shape of the double monochromator can be described as a rounded triangular function (Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020). In the range of Full-width at Half Maximum (FWHM), the shape can be approximated from a Gaussian function. For reasons of symmetry, the fitted line position is expected to match the value from a triangular fit. All lines in (see. Fig. 5.15) fitted and

[6] Each measurement point consists of an average of ≈ 100 samples of the amperemeter with a sampling rate of 12 Hz. At each measurement point, a background measurement with closed monochromator shutters is done with the same conditions. Its average is subtracted from the signal average.

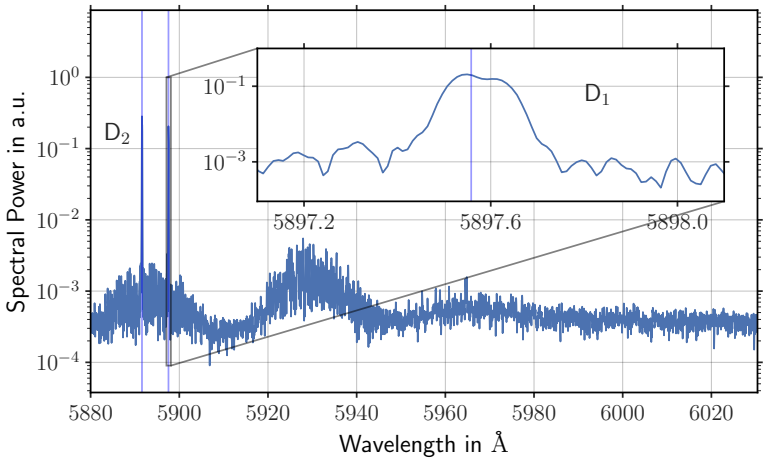


Figure 5.13: The prominent sodium doublet D_1 and D_2 at 589.755 814 7 Å and 5891.583 264 Å (vertical lines), respectively (Juncar et al. 1981). The sidelobes next to these strong lines caused by the finite scanning range from the OSA and the Hann apodization are well visible. The small plot shows a zoom into D_1. The line is visible as an unresolved doublet caused by the multimode fiber of 125 μm, allowing two modes to be coupled to the spectrometer with warped optical paths. The wavelength scale is in vacuum.

compared with the subset of spectral lines from the OSA measurements. The fitted positions are assigned to the lines by visual inspection. The mercury lamp spectrum allows the best first visual estimate for the start values of $C_{\lambda,\mathrm{Model}}$ as it provides less but good distinguishable strong lines equally distributed over the whole wavelength range. With this estimate for c_s and c_off, the lines of the other spectra are assigned visually (see Tab. A.2, A.3, A.4, and A.5 in the appendix). An intensity-weighted average of those lines in the subset is calculated that could not be resolved by the finite resolution of the double monochromator (18 Å). A multi-gauss fit is done for distinguishable but overlapping lines. In the case of neon, all lines below 5850 Å are directly taken from the database because those are below the OSA's sensitivity. In combination with the above calibration estimate from mercury, lines are assigned visually if there is no ambiguity within the 18 Å bandwidth.

Fig. 5.16 shows the difference Δ_Model between fit positions $\lambda_\mathrm{Model} - \lambda_\mathrm{NIST}$. All line deviations follow a linear trend. Eq. 5.11 is fitted onto the data. The fit has 91 degrees of freedom and a $\chi^2 = 91.34$. The residuals have a standard deviation of 0.28 Å. The calibration of the model wavelength is therefore $C_{\lambda,\mathrm{Model}}(\lambda) = 1.70(1)\lambda - 7.0(2) \times 10^{-4}$ Å. To achieve the NIST traceable wavelength:

$$\lambda_\mathrm{NIST} = \lambda_\mathrm{Model} - C_{\lambda,\mathrm{Model}}(\lambda_\mathrm{Model}) \tag{5.12}$$

Holmium-Didymium Filter

The above calibration of the model wavelength only applies to the band shape before the optical fiber. As the monochromator's exit slit's image is distorted (see Fig. 3.21) with dimensions of 200 μm × 1000 μm, only a fraction of the light can be coupled into the optical fiber with a core diameter of 400 μm. In that way, the fiber works as an aperture. The Holmium-Didymium (HoDi) filter imprints its transmission lines into the DUT's signal, allowing wavelength calibration by fitting the HoDi absorption lines measured before the optical fiber. If the spectrum before the fiber has a traceable calibration, the fit allows us to determine the shift introduced by the fiber concerning the calibrated wavelength scale.

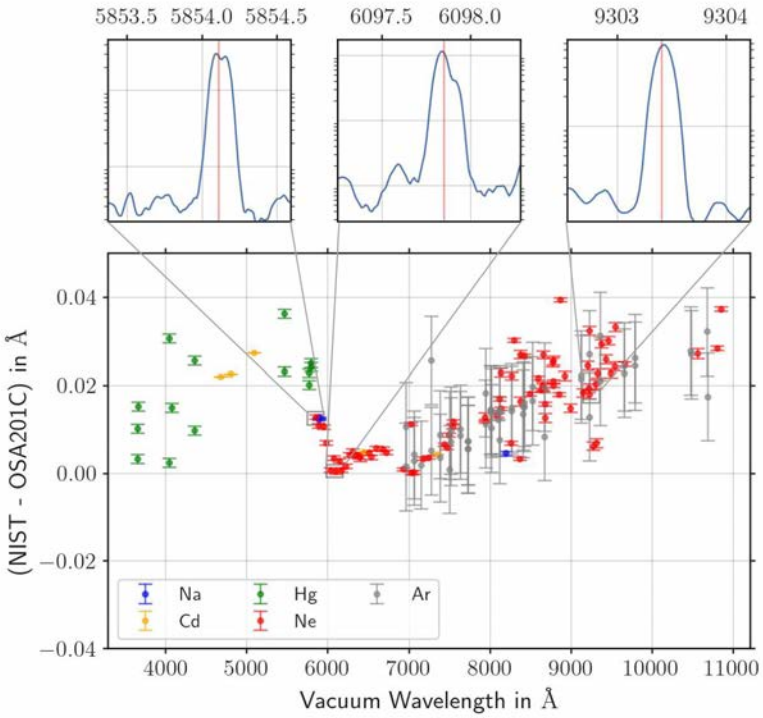


Figure 5.14: Residuals between the assigned spectral lines from the sodium, mercury, and cadmium lamps to their assigned truth values from NIST Spectral Lines Database (Standards and Technology 2009). The elements are distinguished by color: Green: Mercury, Yellow: Cadmium, Grey: Argon, Blue: Sodium, and Red: Neon. The error bars indicate the uncertainties from the NIST database. The subfigures above show three representative neon lines to visualize the misalignment error due to the multimode fiber.

To calibrate the Holmium-Didymium (HoDi) filter, the 150W-Xenon lamp is used as a broad-spectrum light source. It illuminates the integrating cylinder that contains the low-pressure gas lamps used in the previous λ_{Model} calibration. This setup ensures that the optical path remains consistent and allows for the transfer of traceable wavelength calibration. The transmission curve of the HoDi filter is scanned with the double monochromator setup. As above, the same photodiode Ref4 (see Fig. 3.1) records the monochromatic flux. The wavelength sweep is done twice, one with and a second without the HoDi filter in the beamline (see Sec. 3.19). Fig. 5.17 showcases the calibration of the HoDi. The parts of the transmission spectrum taken for line fits with the DUT's signal are shaded blueish.

DUT Wavelength Scale

When the DUT is calibrated, the imprinted transmission curve of the HoDi filter is calculated (see Fig. 5.18 red scatter, upper panel) by taking the ratio between the ratio of its recorded currents with and without a filter. The data from the DUT $T_{\text{DUT}}(\lambda_{\text{Model}})$ is stretched using two polynomials of first (Eq. 5.13a) and third order (Eq. 5.13b) in transmission and wavelength scale, respectively. As the bandwidth after the optical fiber is smaller than before, the transmission T_{DUT} is convoluted with a Gaussian kernel with a width σ_{fiber}. The parameter a, b, c, &, d are varied together with σ_{fiber} in a χ^2-fit. Instead of a global fit, each line is fitted individually onto the model data (blue scatter in Fig. 5.18), minimizing the χ^2 (Eq. 5.14):

$$C_{\text{T, HoDi}} = a + b \cdot T_{\text{DUT}}(\lambda_{\text{Model}} - C_{\lambda,\text{ HoDi}}) \circ G(\sigma_{\text{fiber}}) \tag{5.13a}$$

$$C_{\lambda,\text{ HoDi}} = c + d \cdot \lambda^2_{\text{Model}} \tag{5.13b}$$

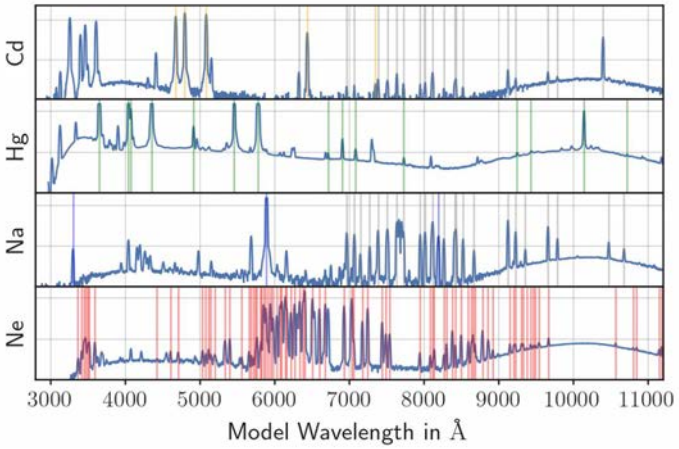


Figure 5.15: Spectra of the low-pressure gas lamps measured using the double monochromator setup, an S1337-1010BQ photodiode, and a one-channel Keithley 6485 amperemeter. The wavelength is given for the vacuum. Colored vertical lines mark the assigned spectral lines: Green: Mercury, Yellow: Cadmium, Grey: Argon, Blue: Sodium, and Red: Neon. A full list of all assigned spectral lines can be found in the appendix.

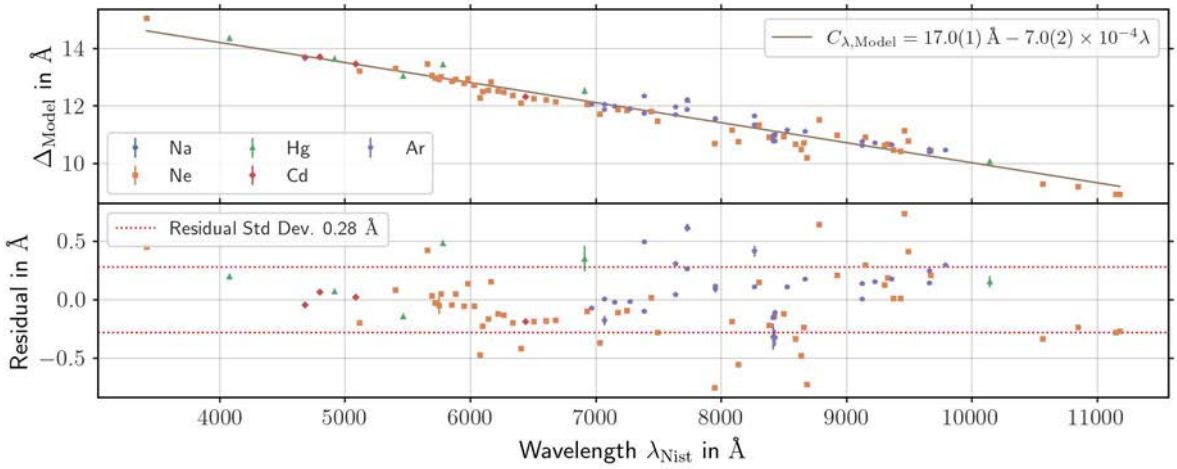


Figure 5.16: Upper plot: Calibration of the model wavelength $\Delta_{\mathrm{Model}} = \lambda_{\mathrm{Model}} - \lambda_{\mathrm{Nist}}$ in Å. The dots indicate Δ_{Model} for the 94 assigned spectral lines. The color shows the element (see legend). The straight line shows a linear fit for the calibration. The reduced $\chi^2_{\mathrm{red}} = 1.004$. Lower plot: The residual between fit results and model calibration. The dashed red line shows the standard deviation for the residuals. Note the uncertainties of the individual lines are too small to be visualized here.

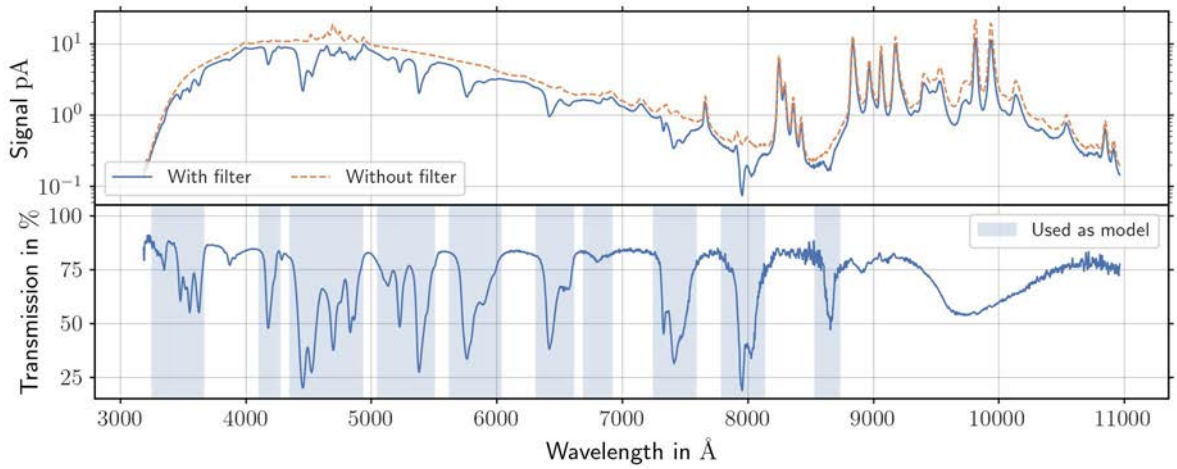


Figure 5.17: The signal of the reference diode after the HoDi filter with calibrated wavelength scale. The blue solid line depicts the signal with the filter in the beamline, while the orange dashed line shows the scan without the filter. The lower panel shows the retrieved transmission curve from the above signals. The blue-shaded regions visualize the data used as the calibrated model for fitting the transmission lines imprinted in the DUT signals.

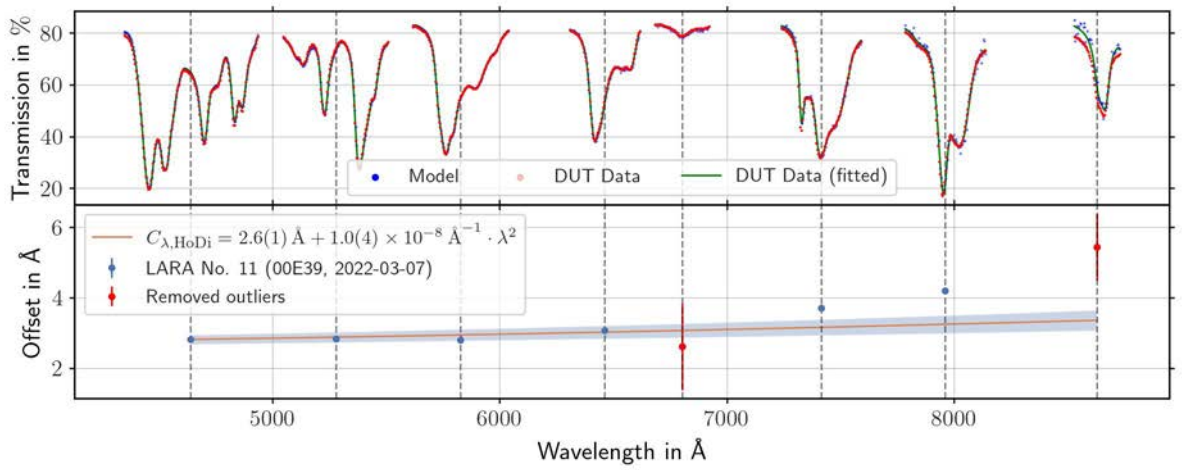


Figure 5.18: Examplary results for the line fits (upper panel) and HoDi wavelength calibration (lower panel) of the light after the optical fiber for the LARA No. 11. The blue scatter plot shows the calibrated reference data (described in Sec. 5.3.2) of the measured HoDi spectrum before the optical fiber. The red scatter shows the imprinted absorption lines in the DUT's photometric calibration data after the fiber, and the solid green line shows the line fit.

$$\chi^2 = \sum_i \frac{[T_{\mathrm{Model}}(\lambda_i) - C_{\mathrm{T,\,HoDi}}(\lambda_i)]^2}{\sigma^2_{\mathrm{Model,\,i}} + \sigma^2_{\mathrm{DUT,\,i}}} \tag{5.14}$$

With i the index of the measurement points and $\sigma_{\mathrm{Model,\,i}}$ & $\sigma_{\mathrm{DUT,\,i}}$ the respective statistical uncertainties from the measurements.

The green solid lines showcase the resulting fits, and the lower panel shows the values for $C_{\lambda,\,\mathrm{HoDi}}$ at their respective averaged wavelength. The last step is a second fit of $C_{\lambda,\,\mathrm{HoDi}}$ on the retrieved offsets blue scatter shown in the lower panel of Fig. 5.18 to get global values for C. Before fitting, outliers are removed as the signal-to-noise of some lines is insufficient to get reproducible results. To decide which points are removed, the ansatz of the Cooks distance[7] is used.

For the final wavelength scale after the optical fiber, Eq. 5.12 must be modified to:

$$\lambda_{\mathrm{NIST}} = \lambda_{\mathrm{Model}} - C_{\lambda,\,\mathrm{Model}}(\lambda_{\mathrm{Model}}) - C_{\lambda,\,\mathrm{HoDi}}(\lambda_{\mathrm{Model}}) \tag{5.15}$$

The uncertainties are discussed in Sec. 5.4.

5.3.3 Photometric Calibration

The scan involves recording the waveform signals from the LARA module under test (DUT) and reference LARA modules as a function of wavelength (λ). Four distinct signals are captured during this wavelength scan:

- $I_i^{s,b}(\lambda)$ This signal represents the waveform recorded from the LARA module under test (DUT) or the NIST-calibrated photodiode with the background at wavelength λ.

[7] A measure to judge the influence of a certain datapoint to determine the least square estimate (Cook 1977). The used code is adapted from Benjamin Bengfort, a contributor to the SciPy python package (Virtanen et al. 2020).

- $I_i^b(\lambda)$ The background for the same module is recorded subsequently without illumination (closed monochromator shutter).
- The reference module's signal, denoted as $I_{\text{ref}}^{s,b}(\lambda)$, is simultaneously recorded along with $I_{\text{ref}}^b(\lambda)$.

The scan is performed over a wavelength range of 2680 Å to 12 000 Å. Fig. 5.20 shows the average of the measured signals $\langle I_i^s(\lambda)\rangle$ and the ratio r_{DUT} The usual step size is 8 Å. As depicted in Fig. 5.20, an example of the recorded signal data from calibrating a usual LARA module is presented. In this graph, the orange curve represents the signal I_{DUT}^s obtained from LARA No. 11, while the blue curve represents the reference signal I_{ref}^s from the monitoring module. The data is collected across various wavelength ranges, with some regions scanned twice with a higher resolution of 4 Å, as indicated by the grey areas, and are listed in Tab. 5.1.

The observed discontinuities in the data (Fig. 5.20 top panel) result from adjustments made to control the flux intensity using neutral density filters in the beamline. These adjustments are necessary to avoid signal saturation and ensure accurate measurements are obtained throughout the scan.

The mid-plot shows the averaged ratios r_{11} and r_{NIST} after background subtraction, synchronization, and cosmic removal. The calibration C_{11} of LARA No. 11 is shown at the bottom but normalized at 5556 Å for better comparison.

C is exposed to various effects that must be corrected afterward. The NIST photodiodes are compared with the LARA modules that have different apertures (compare Fig. 5.1a and 5.5), which lead to different signals because of the tails of the fiber's image PSF-tails. The photodiodes have temperature-dependent internal quantum efficiencies in the infrared (Larason and Jeanne M. Houston 2008).

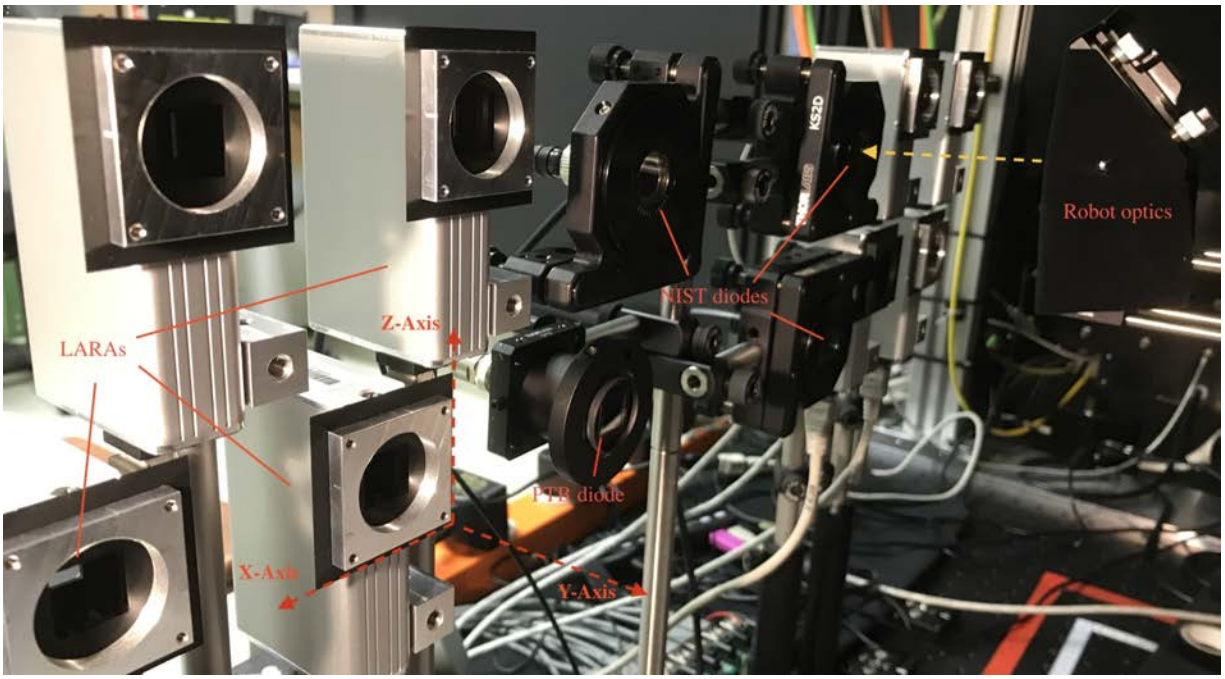


Figure 5.19: Eight LARAs during calibration in the dark box. The four diodes in the middle with the black mounting structure are the three NIST-provided S2261 absolute calibrated photodiodes and one S1337-1010BQ photodiode absolute calibrated in the UV from the Physikalisch Technischen Bundesanstalt (PTB). The output of the robot optics illuminates one of the NIST diodes (Yellow dashed arrow). Red dashed: The robot's axis width (X-axis), depth (Y-axis), and height (Z-axis).

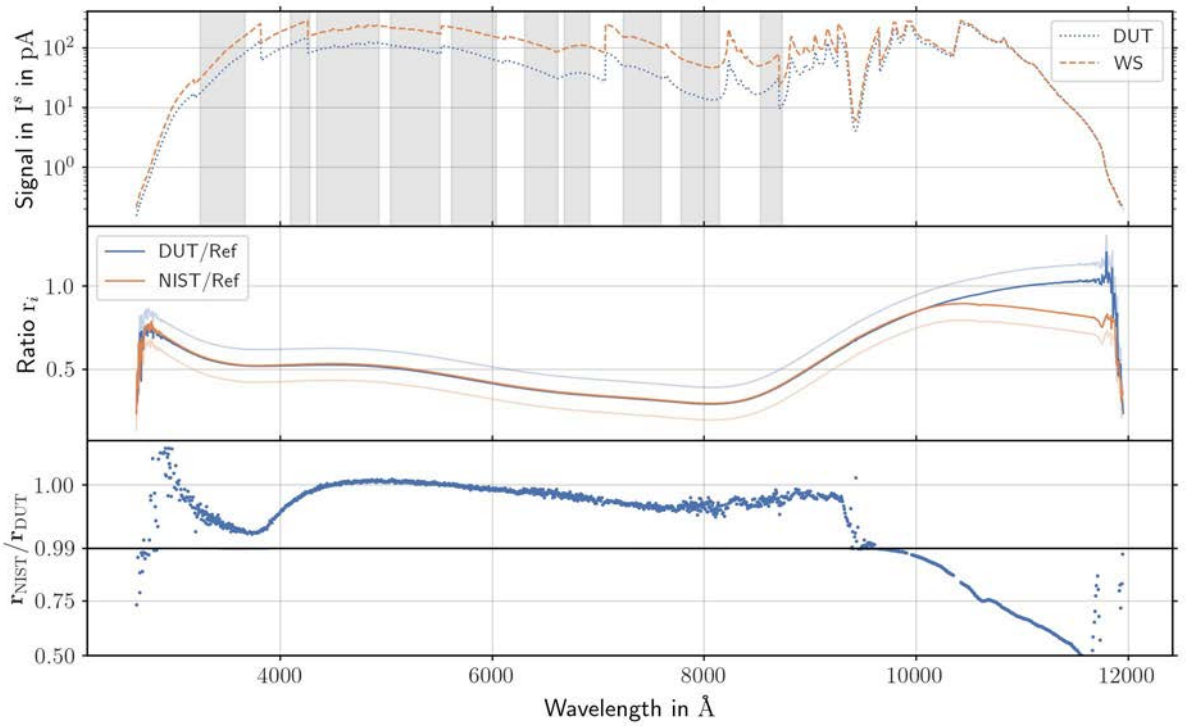


Figure 5.20: Top: Exemplary photocurrent I^s_{DUT} (orange) and I^s_{ref} (blue) from the calibration of LARA No. 11. The grey regions are scanned twice with and without HoDi-filter in the beamline to imprint its wavelength calibration in the data. Those regions are scanned with a resolution of 4 Å. These wavelength ranges are listed in Tab. 5.1. All other parts are scanned with 8 Å steps. The discontinuities are due to flux intensity reductions with neutral density filters in the beamline to avoid saturation. Middle: The ratio r_{DUT} between the above signals I^s_{11} and I^s_{ref}, the transparent lines are the same ratios but separated for better visuality. Bottom: The calibration $R_{11} = r_{\mathrm{NIST}}/r_{11}$ normalized at 5556 Å.

Start [Å]	Stop [Å]
2680	2990
3250	3380
3390	3700
4110	4300
4350	4970
5050	5540
5630	6070
6320	6650
6700	6950
7250	7630
7800	8170
8550	8770

Table 5.1: Wavelength ranges that are scanned twice with and without Hodi-filter and with a step-size of 4 Å.

Temperature Correction

The responsivity of a photodiode is temperature dependent as discussed in Sec. 3.2.1. Photodiodes are calibrated in a laboratory environment at 24.0(5) °C, while SNIFS is calibrated at the UH88 dome on Maunakea at 5 °C. This temperature difference requires an examination of the diode's temperature coefficient to achieve the target uncertainty and correct the discrepancy between the laboratory calibration and the measurements taken at the summit during commissioning.

As detailed in Sec. 3.2.2, a LARA plus S1337 photodiode is placed within a Thermal Vacuum Chamber (TVaC) to determine the temperature dependency of the feedback resistance, as illustrated in Fig. 3.29. Throughout these experiments, multiple wavelength scans ranging from 6500 Å to 11 000 Å were performed. The temperature was recorded with the internal thermometer of the 32bit-ADS1264 ADC. These measurements, alongside spectral response data from the identical monitoring photodiode on the robotic optics, are utilized to ascertain the relative change in spectral responsivity with changing temperature. This is achieved by dividing the ratio r of the diodes' photocurrents by the temperature difference ΔT at the time of readout between the monitoring diode at room temperature and the cooled diode inside the TVaC.

The temperature dependence is especially pronounced for wavelengths beyond ≈9000 Å. Therefore, the average data below this wavelength have been used to normalize the coefficient to 0 % K^{-1} since no temperature dependence is anticipated in that spectral region (Compare the data from NIST shown replotted in Fig. 5.21 (Larason and Jeanne M. Houston 2008).).

The upper graph in Fig. 5.21 illustrates the resulting temperature coefficient $\Delta S/(S\Delta T)$ of the S1337 photodiode. While the measured temperature coefficient agrees with the measurements reported by NIST (Larason and Jeanne M. Houston 2008) within one standard deviation of the statistical uncertainty up to ≈10 600 Å (indicated by the orange dashed line), the diode manufacturer has systematically provided a lower estimate of the coefficient, as indicated by the black dotted line.

The lower plot in Fig. 5.21 shows the residual scatter concerning a spline interpolation using the weighted averages of 32 Å[8] bins as nodes and the actual measurement data; in addition to the interpolation uncertainty (discussed in Sec. 5.4), the discrepancy between the spline interpolation and the NIST data for the S1337 is an estimate of the model uncertainty.

Finally, the temperature correction is applied to each calibration C to account for the temperature difference between the laboratory setup and the actual conditions during SCALA operations.

Aperture correction

As discussed in Sec. 5.3.1, aberration of the optics lead to errors in the optical fiber's image. Instead of a top-hat function, its PSF has tails that extend over the diode's active area. Therefore, the shape of a detector's photosensitive area influences its response during the illumination with

[8] Two times the bandwidth of the VASC.

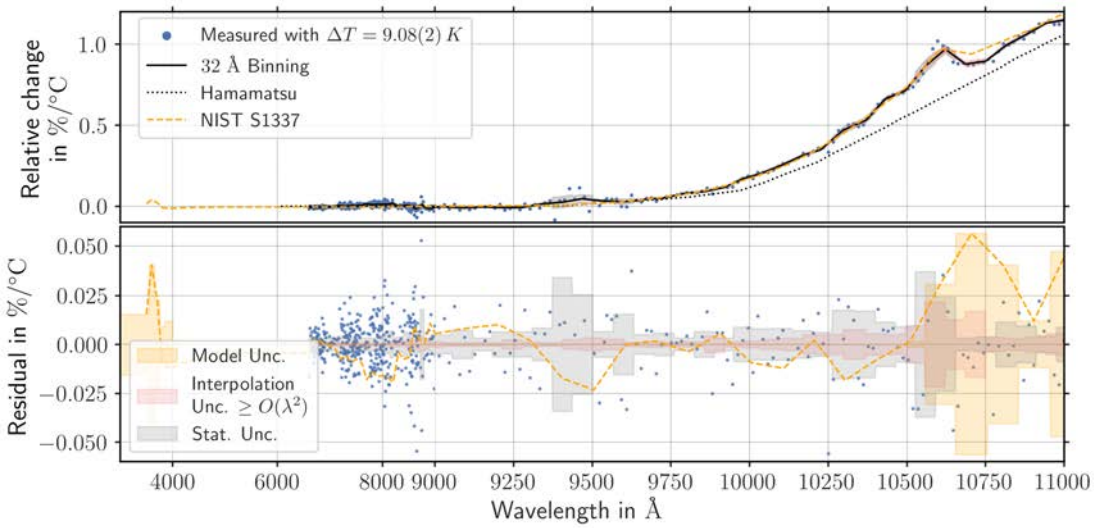


Figure 5.21: Relative change in sensitivity $\Delta S/(S\Delta T)$ for the S1337 photodiode measured with the VASC blue points, NIST published data orange dashed line, and manufacturer estimate black dotted line (top). The residual between the data and the 32 Å spline interpolation. Noteworthy, the increased scatter at 9450 Å is due to the absorption peak of the optical fiber (see Fig. 3.20b).

the optics. Fig. 5.22 shows the radially integrated fraction of the optics PSF in dependence on the distance to the optical axis. Even with a circular aperture of 10 mm diameter, more than 1 % of the flux is not collected. As a consequence, the collective area of a photosensor must have the same shape as the working standard to be calibrated with the substitution method. The S1337 photodiode, employed within the LARA systems, and the S2261, calibrated by NIST, feature different geometries. While the S1337 is squared, the S2261 has a circular active region.

To investigate the influence of the different shapes, the S2261 is measured consecutively with and without a squared aperture with (9.5 mm)[29] mimicking the shape of the S1337. Using Eq. 5.2 one gets the response ratio $S_\square/S_\bigcirc$ which is the response ratio between the aperatures of the S1337 ($S_\square$) and S2261 ($S_\bigcirc$). The upper plot in Fig. 5.23 depicts the measured response ratio (blue dots) with a 32 Å binning (black line). The region from 3600 Å to 4100 Å is a region in which at least one LARA has been saturated. As the curve is flat in this region, the uncertainty during interpolation is still sufficiently small.

The collective area of the S2261 photodiode has a diameter of 11.6 mm with an area of 105.7 mm^2, whereas the diagonal of the square aperture is 13.4 mm. Consequently, stray light on the corners of the aperture is not measured as measured by the LARAs. The missing area is 3.6 mm^2. Under the assumption that the 0.55 % stray light accumulated in average over all wavelengths by the full circle compared to the overlapping region is homogenous, one can estimate the influence of the missing 3.6 mm2 in the corners by scaling down the straylight with $0.5\,\text{mm}^2 \cdot (A_\square/A_{\text{Overlap}} - 1)/(A_\bigcirc/A_{\text{Overlap}} - 1) \approx 0.1\,\%$. Compared to the overall uncertainty budget of 0.5 %, this additional error is tolerable.

[9] Made out of PCB material painted in black color.

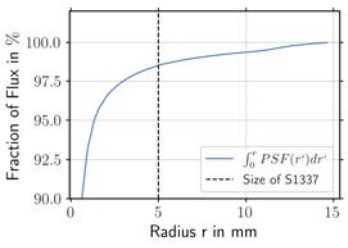


Figure 5.22: The solid blue curve represents the integrated Point Spread Function (PSF) from Fig. 5.6 as a function of the radius in millimeters. The dashed vertical line indicates the size of the photodiode S1337. This graph demonstrates that a significant portion of the stray light extends beyond the physical boundaries of the photodiode, highlighting the importance of the photodiode's shape in capturing the total flux.

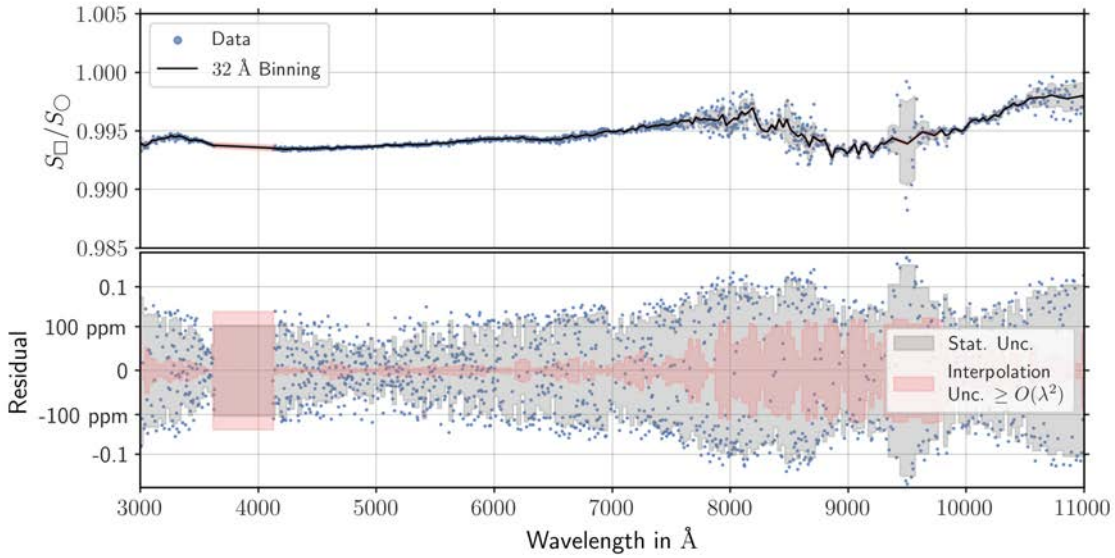


Figure 5.23: Upper Panel: The graph displays the spectral ratio ($S_\square/S_\bigcirc$) denoted by blue dots, with a spline interpolation represented by the black line, derived from 32 Å binning. Lower Panel: This section illustrates the residuals from the spline interpolation. Additionally, it includes an assessment of the statistical uncertainties (gray) and the interpolation uncertainties (red), the latter derived from quadratic spline interpolation.

Reproducibility

The splitting ratio, the ratio between the flux at the monitoring diode and the DUT, changes over time due to changes in the environmental conditions. The measurement against the working standard must be frequently repeated. It is redone every third to fifth day. When calculating the relative response of the monitoring diode (MD), the weighted spectral average of the MD comparison preceding and following the measurement is used. To quantify the uncertainty due to the splitting ratio, the change spectral change in the splitting ratio is calculated.

To do so, the ratio of the responses before $S_{\text{MD, Pre}}$ and after $S_{\text{MD, Post}}$ of two exemplary MD calibrations, separated by 4.9 days (blue dots), is shown in the upper panel of Fig. 5.24. The solid lines represent spline interpolations from 16 Å binning. The lower panel displays the residual concerning the interpolation; the scatter of the data around the trend is approximately 300 ppm, which is consistent with the statistical uncertainty estimated from the ratios of the individual waveforms originating from the amperemeter noise.

The drift between sequential MD calibrations, conducted before and after the measurement with the Device Under Test (DUT), is considered a systematic uncertainty. The maximum residual between the average and the 16 Å trend of the individual MD calibrations is taken as the 1σ uncertainty. It is not observed to be larger than 0.1 % over five days. The orange data points show the drift between two MD comparisons with the WS 44 days apart; the observed drift in the spectrally important region is also less than 0.2 %, and at most 0.3 % above 10 800 Å.

A possible explanation for this drift is the thermoreflectance of aluminum. At 8000 Å it has a thermoreflectance coefficient of $C_{\text{Al}} \approx 1.5 \times 10^{-4}\,\text{K}^{-1}$ (Favaloro, Bahk, and Shakouri 2015). The beam to the DUT

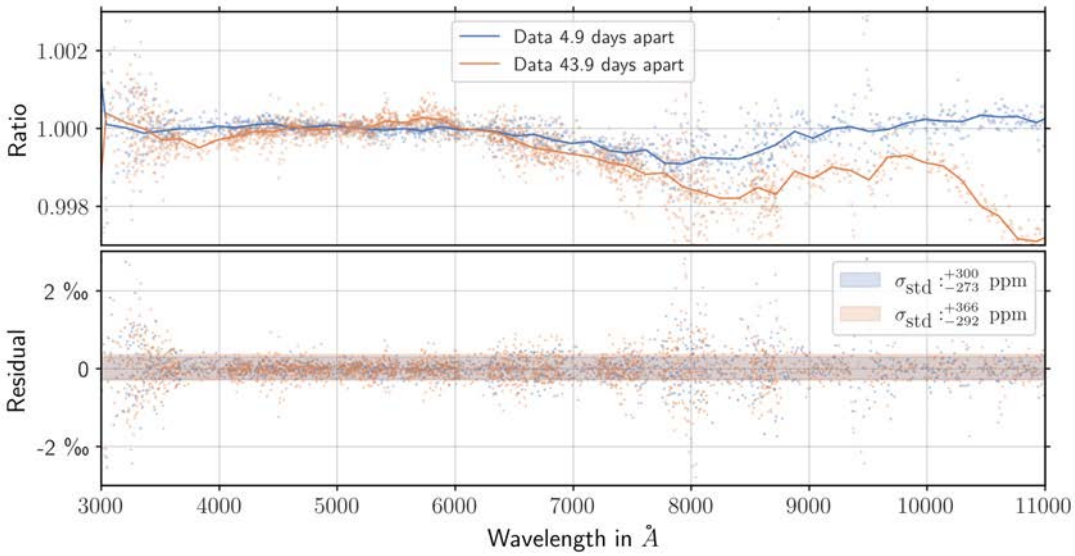


Figure 5.24: Illustration of the drift of the splitting ratio over time. Upper panel: Ratio of MD response before and after re-comparison (4.9 days apart) with spline interpolation (16 Å binning). Lower panel: Residuals indicating a typical scatter of about 300 ppm, aligning with amperemeter noise-derived statistical uncertainty. Systematic uncertainty remains below 0.1 % over five days, with a maximal observed drift below 0.3 % for longer intervals (44 days) above 10 800 Å.

in the SCALA relay optic (see Sec. 3.1.7) has four further reflections at UV-enhanced aluminum, which multiplies this effect. From Fig. 3.26 an approximation for the flux ratio $F_{\mathrm{MD}}/F_{\mathrm{DUT}}$ the MD and the DUT are exposed to is:

$$\frac{F_{\mathrm{MD}}}{F_{\mathrm{DUT}}}(\Delta T) = \frac{1 - r_{\mathrm{BS}}}{r_{\mathrm{BS}}} \cdot \frac{1}{r_{\mathrm{Al}}(1 + C_{\mathrm{Al}}\Delta T)^4} \tag{5.16}$$

with the reflectivity r_{BS} of the beam splitter, r_{Al} the reflectivity of aluminum, and ΔT the temperature difference to the nominal value r_{Al} is referred to. The change in the splitting ratio between two measurements at the temperatures T_0 and T_1 then:

$$\left(\frac{F_{\mathrm{MD}}}{F_{\mathrm{DUT}}}\right)_{T0} \Bigg/ \left(\frac{F_{\mathrm{MD}}}{F_{\mathrm{DUT}}}\right)_{T1=T0-\Delta T} = \frac{1}{(1 + C_{\mathrm{Al}}\Delta T)^4} \tag{5.17}$$

From Eq. 5.17, it follows that a change in the ratio is expected to be 600 ppm K^{-1}. 3.3 K are, therefore, suffice to explain a change of 2 ‰. These temperature changes are consistent with the observed temperature drifts in the laboratory. The decrease above 10 000 Å can be explained by the temperature dependence of the photodiodes, as the working standard NIST diode is of a different type than the monitoring diode.

5.4 Comparison of all LARAs

Fig. 5.25 shows the calibration results R_{LARA} for all eighteen LARAs that are used in the SCALA setup. The assignment of each LARA module to its respective beam is detailed in Tab. 3.10. The calibration data has been binned with a 16 Å interval using sigma clipping to present the solid lines, while the scatter plots illustrate individual sample points.

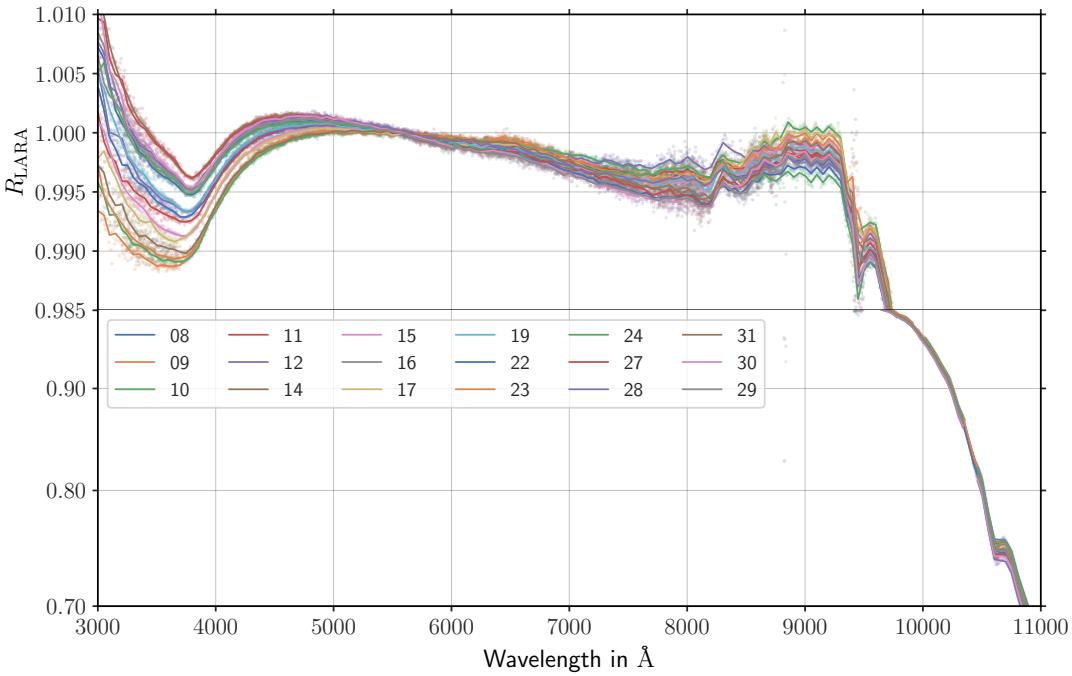


Figure 5.25: Calibration results $R_{\mathrm{LARA}} = QE_{\mathrm{LARA}}/QE_{\mathrm{NIST}}$ for all eighteen LARA modules used in SCALA. Which LARA is equipped into which beam can be taken from Tab. 3.10. The solid lines show a 16 Å-binning with sigma clipping. The scatter plot depicts the individual samples, that are shown to visualize the statistical scatter. The data is normalized at 5556 Å. The y-axis is divided into two different ranges for visibility.

The normalization point for the data is 5556 Å. The data is temperature corrected for 5 °C as expected at the summit. For comparison R_{LARA} is showcased as the ratio with the calibration results for LARA No. 11, already shown in Fig. 5.20 in Fig. 5.26. The responsivities vary with 0.2 % between 4500 Å to 9800 Å. Below 4500 Å the responses diverge up to 1 %. Above 9800 Å the responses diverge up to 1.5 %.

5.4.1 Calibration Summary and Uncertainty Discussion

To summarize the complete calibration, the corrections for the temperature, as well as the responsivity of the DUT adjusted for a circular aperture equivalent to that of the NIST diode, are applied to Eq. 5.7 and Eq. 5.6. This adjustment involves division by $S_{\bigcirc}/S_{\square}$ and applying a temperature coefficient $\delta_T^S(\lambda, 20\,^{\circ}\mathrm{C} - \langle T_{\mathrm{DUT}}\rangle)$ for the usage conditions, with T_{Cal} being the temperature during calibration, and T_{Usage}, the condition during usage (approximately 5 °C at the summit of Maunakea).

The calibration of the DUT is then represented by:

$$\hat{U}_i(\lambda, T) = U_{\mathrm{NIST}}(\lambda)[1 - \alpha(\langle T_{\mathrm{NIST}}\rangle_t - 20\,^{\circ}\mathrm{C})], \tag{5.18a}$$

$$R_{\mathrm{DUT}}(\lambda) = \frac{k_{\mathrm{DUT}}}{k_{\mathrm{NIST}}} \left\langle \frac{\hat{U}_{\mathrm{NIST}}(\lambda, T_{\mathrm{NIST}})}{\hat{U}_{\mathrm{Ref1}}(\lambda, T_{\mathrm{Ref1}})} \right\rangle_t \left\langle \frac{\hat{U}_{\mathrm{Ref2}}(\lambda, T_{\mathrm{Ref2}})}{\hat{U}_{\mathrm{DUT}}(\lambda, T_{\mathrm{DUT}})} \right\rangle_t \frac{S_{\bigcirc}}{S_{\square}}\, \delta_T^S(\lambda, T_{\mathrm{Cal}} - T_{\mathrm{Usage}}). \tag{5.18b}$$

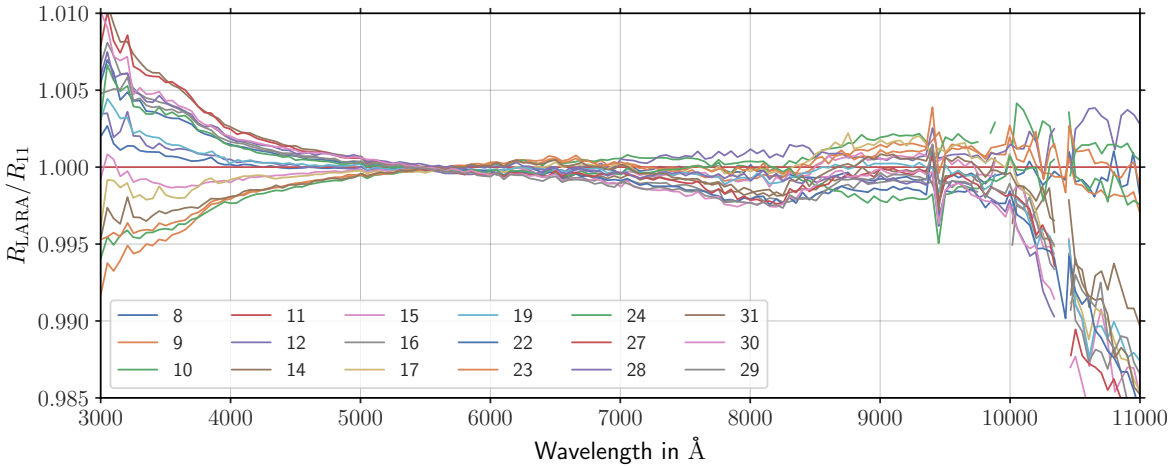


Figure 5.26: Comparision of the calibration results of the showcased in the former Fig. 5.25. The calibrations are normalized with LARA No. 11.

Here, U_{NIST}, U_{Ref1}, U_{Ref2}, and U_{DUT} are the voltage waveforms for the NIST, reference, monitoring, and DUT diodes, respectively, depending on time t. The temperature coefficient $\alpha = 2.1(2) \times 10^{-5}$ is the coefficient for the feedback resistor (Sec. 3.2.2), and $\langle T^i \rangle_t$ represents the average temperature during waveform recording. $\langle \cdots \rangle_t$ denotes the time average of the waveform ratios after synchronization and cosmic removal.

The term $k_{\mathrm{DUT}}/k_{\mathrm{NIST}}$ scales the flux but does not affect the color calibration or cosmological implications. Calibrations $R_{\mathrm{DUT}}(\lambda)$ are normalized such that $R_{\mathrm{DUT}}(5556\,\text{Å}) = 1$.

Various uncertainties must be considered. Fig. 5.28 summarizes all contributions detailed in the subsequent sections.

Statistical
Each measurement contributes statistical uncertainties determined by the ratio r_i of the standard deviation due to the noise in the LARAs. The upper plot of Fig. 5.28 shows the contributing uncertainties for LARA No. 11 as an example.

- Measurement of r_{NIST} (orange scatter).
- Measurement of r_{DUT} (blue scatter).
- Aperture correction measurements (dotted line).
- Temperature model measurements (dashed line).

Total statistical uncertainty is calculated as the quadratic sum of the above (black scatter). The statistical uncertainties, which depend on the signal-to-noise ratio that correlates with the light intensity, are approximately proportional to the inverse of the square root of the number of available photon fluxes (see Fig. 5.20).

Interpolation When comparing the responses of the instruments, the actual wavelengths differ between the measurements because the monochromators have limited reproducibility, and the wavelength hitting the DUT at a given target wavelength depends on the ambient conditions (see Sec. 3.1.2 and 5.3.2), the data must be interpolated. The data is binned to the resolution of the monochromator system (16 Å) and interpolated using a linear spline fit. The

NIST calibration data are interpolated directly. A quadratic interpolation is performed to estimate the interpolation uncertainty, and the maximum difference is taken as the uncertainty in bin i. $\max_i |y_i^{\text{linear}} - y_i^{\text{quadratic}}|$ where y_i is the interpolation value for the i-th bin. There are such interpolation uncertainties:

- NIST calibration (dashed-dotted line in the top panel of Fig. 5.28)
- Interpolation of r_{NIST} (dotted orange line)
- Interpolation of the aperture model (dashed green line)
- Temperature model interpolation (dashed-dotted line)

Since all components in Eq. 5.18 are multiplicative, the total interpolation uncertainty can be calculated as a quadratic sum (black solid line).

NIST The NIST calibration has a systematic uncertainty, shown as the blue solid line in the middle panel of Fig. 5.28.

Splitting Ration Drift As described in Sec. 5.3.3, the drift of the splitting ratio is considered as an additional uncertainty, shown as an orange dashed line. It is caused by the thermo-dependent reflectance of the four additional aluminum mirrors in front of the DUT and is proportional to $(1 + C_{\text{Al}} \cdot \Delta T)^{-1}$. Its effect is maximal at 8000 Å consistent with the maximum for C_{Al} (Favaloro, Bahk, and Shakouri 2015; Rosei and Lynch 1972).

Temperature Unc. The final resolution $\delta T = 0.5\,°\text{C}$ of the thermometer in the LARAs used to compare the temperatures at the time of calibration and their use in the UH88 gives an uncertainty for the temperature correction by multiplying this value with the spectral temperature coefficient model (green dotted line).

Temperature Model The difference between the temperature coefficient data published by NIST (Larason and Jeanne M. Houston 2008) and the model determined in this paper is taken as the uncertainty for the model itself (dashed orange line).

Angle of Incidence During the calibrations with the VASC, the comparison between the NIST diodes and the S1337 diodes of the LARA modules was performed with the same AoI. The main contribution of change in responsivity due to the AoI cancels out. The remaining part relies on the range of the reproducibility of the light beam's pointing vector that depend on the reproducibility of the HDR50 rotationary stages that adjust the relais optic. The manufacturer gives its reproducibility as 0.02 deg, which makes the error due to Eq. 5.10 negligible. In contrast, the calibrations done with SCALA are assumed to have an uncertainty in the AoI of 1 deg (see discussion in Sec. 6.2.2).

Wavelength HoDi/NIST The wavelength calibration eq. 5.15 on the wavelength scale calculated from the double monochromator model λ_{Model} (described in Sec. 3.1.2) results from two fits to the emission lines of a set of elements and to the absorption lines of holmium oxide, praseodymium, and neodymium. The residual scatter of the fits of the lines to the calibration models $C_{\lambda,\,\text{Model}}$ and $C_{\lambda,\,\text{HoDi}}$ is used as a measure of the uncertainty in the wavelength scale. In the case of model calibration, the standard deviation of the residual (shown in the lower panel of Fig. 5.16) is used as the global uncertainty. Even though the scatter is smaller in the visual than in the infrared, the low line density does not allow a statistical

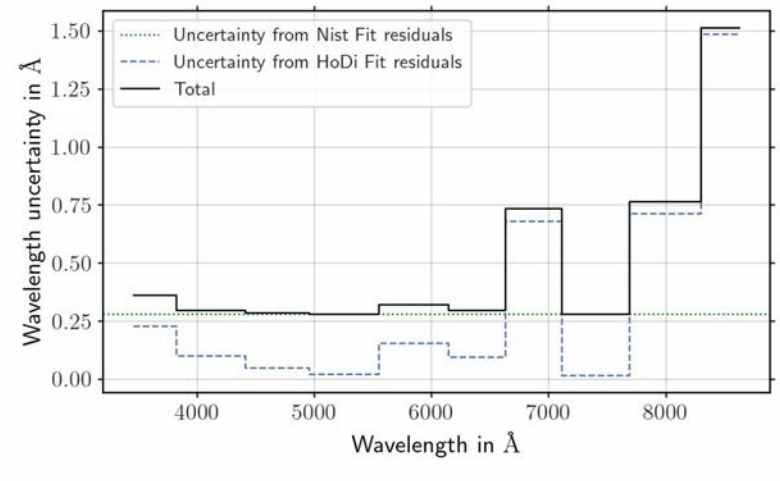


Figure 5.27: Unceratainty on the model calibration from fit residuals over the HoDi-filter and spectral lines.

approach to estimate the scatter for different wavelength regions. In addition, the distribution of the line residuals shows a structure that suggests misclassification of the lines in the database for those lines that do not follow the pattern of lines of the same element (see, e.g., neon between 6000 Å and 7000 Å). The global uncertainty for $C_{\lambda,\,\mathrm{Model}}$ is calculated to be 0.3 Å. For the HoDi fit, the line fit is performed several times with different devices (all LARAs and the NIST photodiode measurements; $\approx$ 30 measurements). This allows a statistical examination of the residuals for different wavelength regimes λ_{reg}. For each line fit, the median and 15.87th-percentile p_{16}, the median and 84.13th-percentile p_{84}, are calculated for the residual to $C_{\lambda,\,\mathrm{HoDi}}$, representing the 1-$\sigma$ region of the distribution. As a measure of the systematic uncertainty using $C_{\lambda,\,\mathrm{HoDi}}$, the maximum between the absolute values for the mentioned percentiles is used $\max_{\mathrm{reg}}(|p_{16}|, |p_{84}|)$. Fig. 5.27 shows the estimated uncertainties for the wavelength calibration. While the green dotted line shows the global systematic uncertainty for the model wavelength of 0.28 Å, the blue dashed line shows the above maximum between $|p_{16}|$ & $|p_{84}|$ of the residuals for each fit regime centered on its mean wavelength. The black solid line gives the wavelength calibration's final uncertainty as the former's quadratic sum. When converting the wavelength from air to vacuum, the uncertainties are due to the determination of the ambient conditions in the laboratory ($\sigma_{\mathrm{Temperature}} = 1\,^{\circ}\mathrm{C}$, $\sigma_{\mathrm{Pressure}} = 2\,\mathrm{hPa}$, $\sigma_{\mathrm{Humidity}} = 3\,\%$) leads to additional ≤ 0.01 Å uncertainty, which is neglected here.

The statistical uncertainty is directly related to the available flux and the actual sensitivity of the detectors. The VASC flux decreases at wavelengths less than 3600 Å due to the spectral limits of the laser-driven light source and greater than 10 500 Å the signal decreases because the photodiode's sensitivity decreases. At 9400 Å, the optical fiber has an absorption peak and limits the flux in this region. The region between 6500 Å and 9000 Å is a flux minimum of the LDLS spectrum. The statistical uncertainty for wavelengths greater than 9800 Å is limited by the scatter of the measurements for the temperature dependence. The statistical uncertainty derived from the scatter around the spline interpolation

is larger than the uncertainties derived from the noise of the LARA, suggesting that additional systematics (e.g. temperature stability during measurements or comparability between measurements) contribute to the scatter.

Spline interpolation works well in regions where the curvature (second derivative) of the determined curve is negligible. As the sensitivity of the photodiodes decreases at wavelengths below 4000 Å and above 9000 Å, the interpolation becomes less accurate, resulting in a larger uncertainty. Also, the sensitivity of the diodes has a local minimum at 3700 Å, a region of non-negligible curvature, which also leads to a higher interpolation error.

While the wavelength uncertainty is marginal, the reproducibility of the splitting ratio intercomparison dominates the uncertainties. It is also noteworthy that the temperature model uncertainty is dominant below 4000 Å and above 10 500 Å. More frequent intermediate measurements of the MD, making the relais optics symmetric, canceling out the thermo-dependent reflectance, and a better understanding of the temperature dependence of the diode can significantly improve the uncertainties. The next dominant contribution is the uncertainty given by the laboratory standard measurements done by NIST.

Fig. 5.29 shows the exemplary calibration $R_{\mathrm{DUT}} = QE_{\mathrm{DUT}}/QE_{\mathrm{NIST}}$ for LARA No. 11 from Eq. 5.18 with a binning of 16 Å. The residual shows a standard deviation of 300 ppm over the whole spectrum, which is consistent with the statistical uncertainties calculated from the scatter of the amperemeter noise (standard deviation of the waveform ratios $\sigma(r_i)$) for each sample. The bottom panel shows a summary of all uncertainties together. Statistical and splitting ratio variability are the dominant contributions of the VASC. Only in regions where the temperature model differs from the NIST-derived coefficient does the temperature dependence contribute more. For evaluation, it should be noted that the statistical uncertainties are suppressed by a factor of $\sqrt{18}^{-1} \approx 4^{-1}$ when the 18 LARAs are used simultaneously in SCALA. It is, therefore, the splitting ratio variability and temperature model below 4000 Å which, together with the NIST calibration, limits the uncertainties in the case of flux monitoring in SCALA in its range 3000 Å to 9500 Å.

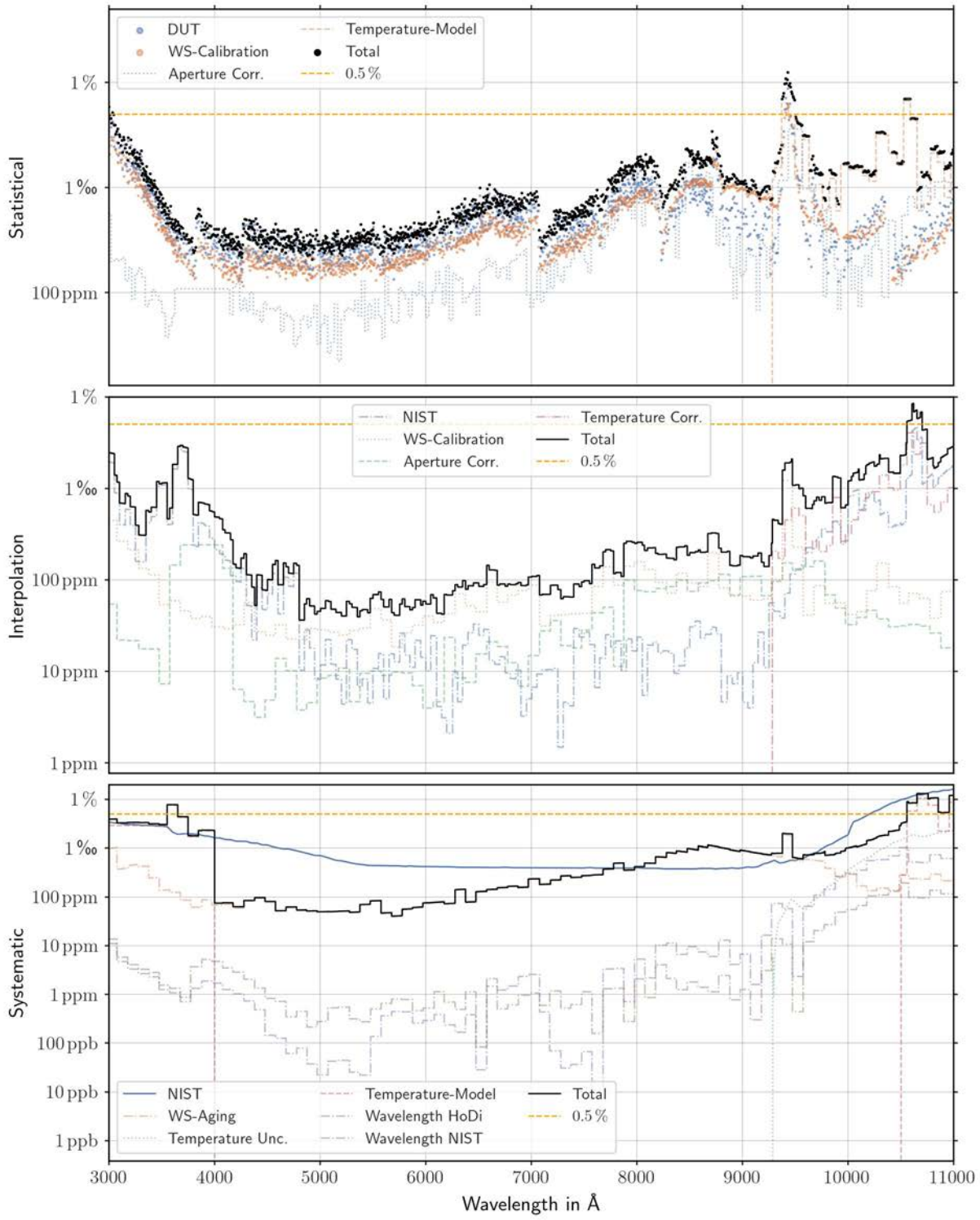


Figure 5.28: Comparative plot of photometric uncertainties at different wavelengths. The top panel (Interpolation) shows the uncertainties originating from interpolations, with distinct lines representing interpolations from NIST data (blue dashed-dotted line), the MD calibration (orange dotted line), aperture correction (green dashed line), and temperature correction (red dashed-dotted line), along with their cumulative effect (solid black line). The middle panel (Systematic) shows the systematic uncertainties, highlighting the contributions from NIST (solid line), splitting ratio variability (orange dashed-dotted line), temperature uncertainty (grey dotted line), temperature model (grey dashed line), and the two parts of the wavelength calibration (dashed-dotted lines), accompanied by the total uncertainty (solid black line). The bottom panel (Statistical) shows the statistical uncertainties, including DUT measurement (blue scatter points), MD calibration (orange scatter points), aperture correction (grey dotted line), temperature model (orange dashed line), and the total statistical uncertainty (solid black scatter points). Uncertainties are plotted on a logarithmic scale for clarity and ease of interpretation.

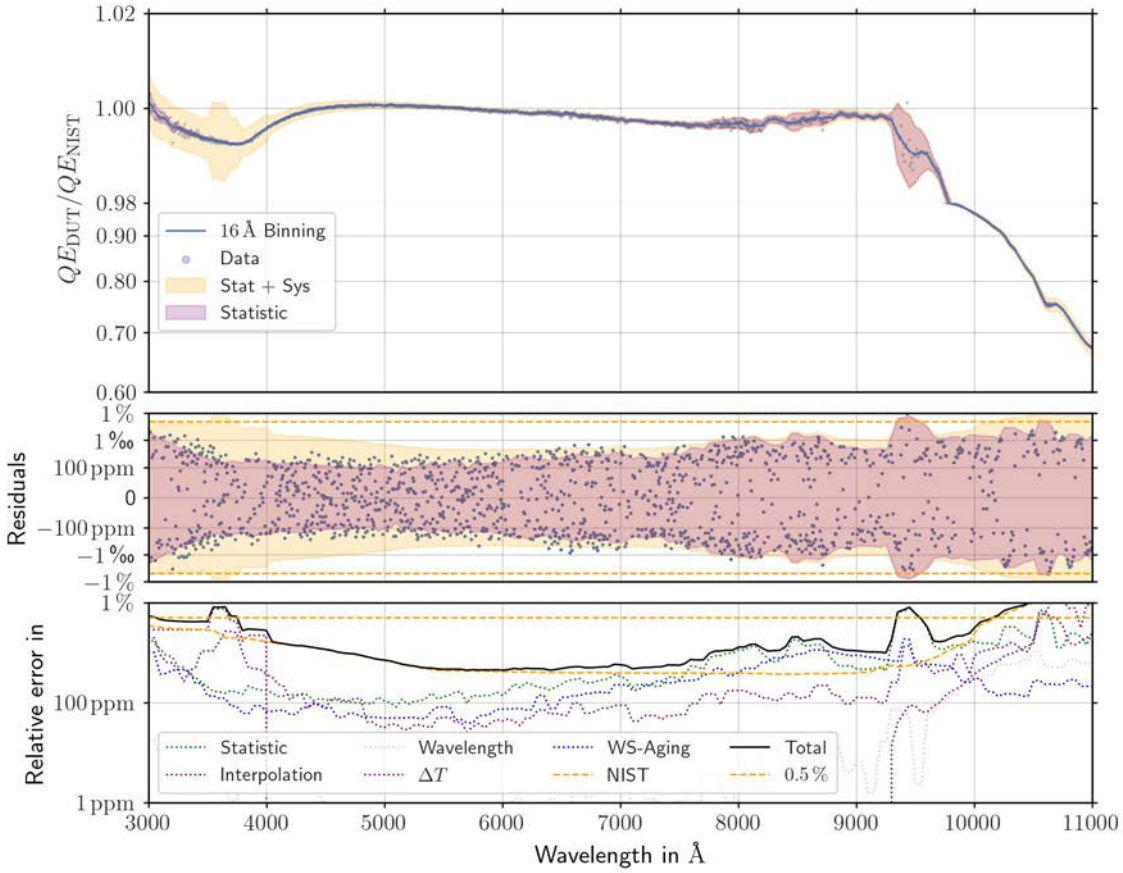


Figure 5.29: The upper plot $R_{\mathrm{DUT}} = QE_{\mathrm{DUT}}/QE_{\mathrm{NIST}}$ displays the calibration. It shows the calibration calculated after Eq. 5.18 (blue scatter plot) and a 16 Å Binning (blue solid line). An orange band indicates the combined statistical and systematic uncertainties, while the red band shows the statistical uncertainties. The residual plot (mid-pannel) shows the differences between the measured and a spline interpolated from the binning (blue scatter plot). The lower panel depicts the various uncertainties drawn in distinct colors: green for statistics, purple for interpolation, pink for wavelength, blue for splitting ratio variability, orange for NIST, and black solid for the total uncertainty.

5.5 Comparison with NIST calibration

Finally, we will discuss the capabilities of the VASC compared to the capabilities of the NIST calibration. To compare the capabilities of a relative calibration, it is sufficient to compare observables on which the absolute scale is canceled, such as the ratio r between two calibrations of two different photodiodes. When performing a comparative calibration between the two NIST-calibrated photodiodes 7J052 & 7J054, the sensitivity ratio from Eq. 5.2 $R_{\mathrm{VASC}} = r_{7\mathrm{J}054}/r_{7\mathrm{J}052}$ is equivalent to the ratio between the responses $R_{\mathrm{NIST}} = S_{7\mathrm{J}054}/S_{7\mathrm{J}052}$. Fig. 5.30 shows the ratio of the responsivities for the mentioned S2281 photodiodes measured by NIST (black solid line, upper panel) and measured with the VASC. The comparison is made several times with the VASC on different days. R_{VASC} is scaled by a factor of 1.2^{-1}, which is the ratio of the feedback resistance $k_{\mathrm{LARA07}}/k_{\mathrm{LARA06}}$ of the LARAs used to record the photocurrent[10].The lower plot shows the residual between the comparator setups. The yellow and red background colors show the combined systematic and statistical uncertainty and the statistical uncertainty, respectively. All residuals are in the 1-σ region.

The structure at 3600 Å can be explained by the different bandwidths of the comparator systems (NIST/SCF: 40 Å, VASC: 16 Å). The sensitivity of the diodes has a local minimum in this region, which is smoothed with a 40 Å kernel, while the VASC measurements resolve the minimum stepper. The remaining 100 ppm in the region above 7000 Å could be explained by the thermo-dependent reflectance of the aluminum mirrors in the relay optics.

In summary, the comparator systems are comparable in relative calibration within the uncertainties given by the systematic uncertainties of the VASC.

[10] The factor is determined by a least-squares fit of R_{VASC} to R_{NIST}. It is not important for color calibration as a global factor and will not be discussed further.

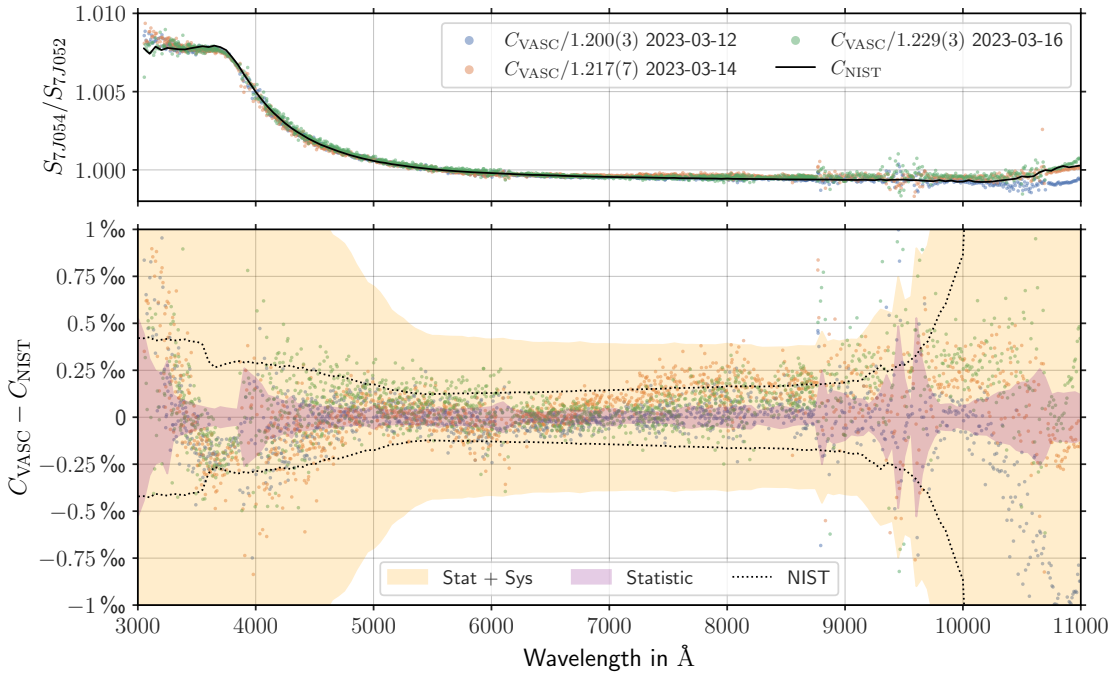


Figure 5.30: Comparison of the sensitivity ratios of S2281 photodiodes 7J052 & 7J054 as measured by NIST and VASC. The upper panel shows the NIST-measured ratio $R_{\mathrm{NIST}} = S_{7\mathrm{J}054}/S_{7\mathrm{J}052}$ as a solid black line, while the scaled R_{VASC} ratios measured on different days using the VASC system are shown with different colors. The lower panel shows the residual between R_{VASC} and R_{NIST}, with yellow and purple areas indicating the combined systematic and statistical and statistical uncertainties, respectively.

Calibration of Stars 6

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To transfer the calibration of the laboratory standard to CALSPEC, the responsivity of SNIFS is determined twice by observing CALSPEC stars and recording their response to the illumination with SCALA and the LARAs working as monitoring references. The standard analysis pipeline for SNIFS is used to extract its responsivity concerning CALSPEC. The typical use case involves multiple spectral measurements of CALSPEC stars during an observational run. During this work, stars are observed in four nights from 19th to 22nd June 2022 UTC[a].

The observation strategy: One SCALA calibration is performed on June 19, 2022. Each observation night n starts with the calibration k with SCALA on the day as in Lombardo, Küsters, et al. 2017. In contrast, the observations of stars were performed the complete night without a SCALA calibration in between[b]. At the end of the last night, the calibration with SCALA was repeated once. That gives four calibrations with CALSPEC and five with SCALA.

This work shows preliminary results for the comparison of CALSPEC and the laboratory standard. As will be discussed, the analysis of the SCALA-exposed SNIFS data is challenging using its pipeline. The focus is exclusively on observational data from the night of June 19, 2022, UTC, and its comparison with SCALA data. These results are used to determine the validity of the assumptions built into the SNIFS data pipeline when exposing it to SCALA and to develop appropriate solutions. The other nights are held blind for the final analysis that goes beyond this work.

The chapter starts by describing the calibration of SNIFS with CALSPEC (Sec. 6.1) to extract SNIFS' responsivity $S^{\star,n}_{\text{SNIFS}}$ against CALSPEC measured during the night $n \in [170, 171, 172, 173]$. Then the calibration with SCALA is described (Sec. 6.2) to get the deposited charge in each frame, which is needed to get the SNIFS responsivity against the NIST laboratory standard $S^{\text{NIST}}_{\text{SNIFS}}$. The chapter ends with the comparison of the two calibrations $S^{\star 170}_{\text{SNIFS}}$ and $S^{\text{NIST},k}_{\text{SNIFS}}$ is discussed in Sec. 6.3.

a: These days correspond to the 170th to the 173th day of the year 2022.

b: The tables A.7, A.8, A.9, and A.10 provide an overview of the observed stars, mostly secondary CALSPEC stars and the primary star GD151 (bold in the above tables). GD71 and G191B2B were not accessible in June from Hawaii.

6.1 Calibration with CALSPEC

Calibration of the SNIFS instrument with CALSPEC (denoted with a "⋆") standards is performed using the established analysis pipeline of SNIFS. If a specific calibration is referred to, the day of the year is added as a prefix (e.g., "⋆170" for the observation on 19th June 2022 UTC). A description of the instrument is given in section 3.3.1. The analysis under discussion was performed on the night of June 19, 2022. A short list of the observed stars is given in Tab. 6.2, and a comprehensive list is given in Tab. A.7 in the Appendix.

The processing of the SNIFS data takes several steps from "preprocessing" that removes artifacts of the CCD detectors, over the arrangement of the signal as a three-dimensional data cube $C(x, y, \lambda)$ (cube extraction)

(compare Sec. 3.3.1), the extraction of the spectrum with via PSF, to atmospheric corrections. The comparison between the observed spectra and the CALSPEC spectra is used to determine the conversion factor from ADUs to spectral flux $S^{\star}_{\mathrm{SNIFS}}$ of SNIFS relative to each standard star separately. As a final step, a polynomial fit is performed over all responsivities to obtain the responsivity $\langle S^{\star}_{\mathrm{SNIFS}} \rangle$ of SNIFS relative to CALSPEC.

Only the parts of the SNIFS pipeline that are necessary for the discussion of the comparison between laboratory standards and CALSPEC in Sec. 6.3 are highlighted here. For better understanding, the SNIFS pipeline is described in more detail in C. Buton et al. 2012; Clément Buton 2009; Yannick Copin 2013 and Rubin et al. 2022b.

6.1.1 Preprocessing and Cube Extraction

SNIFS uses CCD detectors that introduce their signatures into the measured signal, which need to be characterized and corrected in order to recover the true signal incident on the sensor surface. The "preprocessing" involves the following steps:

1. **Pedestal Subtraction:** Removal of the baseline analog signal from readout electronics.
2. **RoN and Gain Estimation:** Determining read-out noise (RoN) and conversion factor (gain) from electrons to ADUs.
3. **Signal Conversion and Variance:** Converting signal to electrons and estimating variance from Poisson and read noise.
4. **Cosmetic Defects Correction:** Addressing defects in detectors, such as "hot" or "cold" pixels, especially on CCD R.
5. **Bias Subtraction:** Eliminating the basic signal from readout electronics during zero exposure.
6. **Dark Current Subtraction:** Subtracting signal due to thermal effects during exposure and readout.
7. **Binary Offset Effect:** A CCD output voltage offset, influenced by the binary encoding of one pixel, affects subsequent pixels read one to three pixels later (Boone et al. 2018).

In the next step, the 225 spectra are extracted and arranged as three-dimensional data cubes $C(x, y, \lambda)$, where x and y are the pixels of the image of the star at the wavelength λ.

The star's image in the focal plane is imaged onto the Microlens array, which divides it into the 15×15 pixels. Each pixel has its optical path and is refracted by the Grism to form a spectrum at the CCD, which is called **spaxel** and is annotated as

$$C(x, y)_{\lambda} = \{C(x, y, \lambda) \mid \lambda \in [3300\,\text{Å}, 9700\,\text{Å}]\}$$

(compare Fig. 3.34).

Fig. 6.1 depicts the cube construction from the two-dimensional CCD plane. Each spaxel $C(x, y)_{\lambda}$ has its individual curvature. To illustrate the signal along the spaxel's trace ±4, pixels perpendicular to it (red line, upper pane) are summed up (lower pane). The middle pane shows

the straightened trace of an exemplary spaxel[1]. The pipeline uses multiple fits that consider the correlation with the background from the neighboring spaxels. For details, see Yannick Copin 2013. The last step involves mapping the pixels along the trace to the wavelength with the arc exposure involving line fitting, which is not discussed here. The wavelength calibration is described in Yannick Copin 2013.

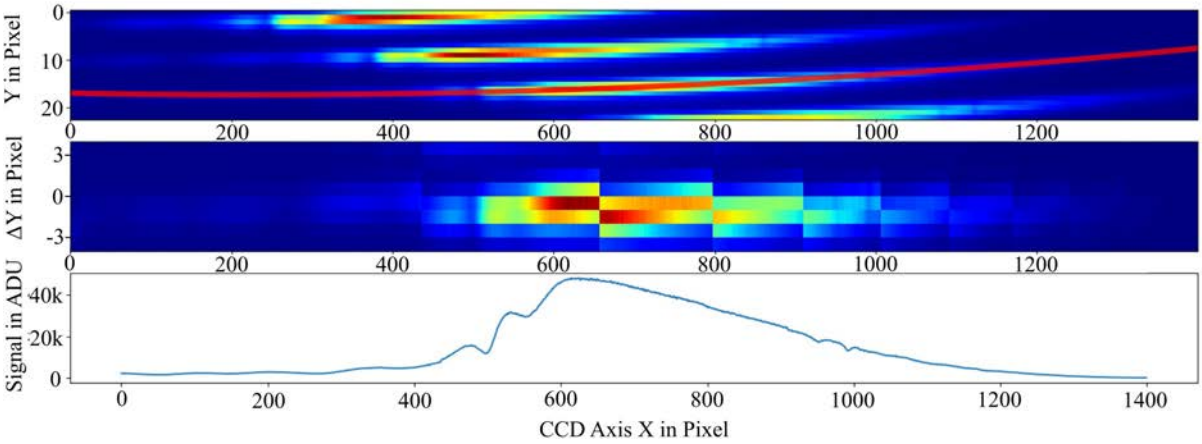


Figure 6.1: Ilustration of the Cube reconstruction from a SNIFS exposure. Upper pane: Image detail of a SNIFS continuums exposure. The red line shows the trace of the spaxel ID 28. Note that each spaxel has its curvature in the trace. Middle pane: Straightened trace of the spectrum of spaxel ID 28. The abscissa is the path along the trace in pixels, and the ordinate shows ±4 pixels perpendicular to the trace. Lower pane: For illustration: The sum of the eight pixels along the trace. The signal is shown in CCD ADUs. The plot is provided by Daniel Küsters 2023d via personal communication.

Poor pointing or a bad seeing[2] can lead to a non-negligible contribution of incident flux ending up outside the field of view of SNIFS, precluding simple aperture photometry[1] Instead, a model of the point spread function (PSF) $\mathscr{P}_{\text{Pipeline}}(r)$ is fitted to the data at each wavelength point to extract the spectrum from the cube. The PSF is an empirical model $\mathscr{P}$ based on a Gaussian $\mathscr{G}$ and a Moffat $\mathscr{M}$ function[2]

$$\mathscr{P}_{\text{Pipeline}}(r) = I \times [\eta \mathscr{G}(r) + \mathscr{M}(r)] \tag{6.1}$$

1: Like $F_{\text{aperture}}(\lambda) = \int_A dA' \, [C(x, y, \lambda) - Bkg(\lambda)]$ where F_{aperture} is the spatially integrated spectral flux at wavelength λ over the area A, and $Bkg(\lambda)$ is the wavelength-dependent sky background.

2: With $\mathscr{G}(r) = \exp\left(-\frac{r^2}{2\sigma^2}\right)$, and $\mathscr{M}(r) = \left[1 + \left(\frac{r}{\alpha}\right)^2\right]$, where $[\eta, I, \sigma, \alpha]$ is a set of possible free parameters.

6.1.2 Atmospheric Correction

The observed spectrum $F^{\star}_{\text{obs},i}(\lambda)$ of the i-th star can be described as the product of the star's spectrum from the database $F^{\star}(\lambda)$, the atmospheric transmission T_{atm}, the instrumental throughput $(T_{\text{tele}} T_{\text{SNIFS}})(\lambda)$, and the quantum efficiency of the CCD $QE(\lambda)$. Additionally, the atmospheric variability is denoted as $\delta T_i(\lambda, \hat{z}_i, t)$ depending on the line of sight $\hat{z}$ and the time t. The observed flux F_{obs} is given by the equation:

$$F^{\star}_{\text{obs},i}(\lambda) = F^{\star}_i(\lambda) \cdot S^{\star}_{\text{SNIFS}}(\lambda) \cdot 10^{-0.4 \times K(\lambda) \hat{X}(\hat{z}_i)} \cdot \delta T_i(\lambda, \hat{z}_i, t). \tag{6.2}$$

$S^{\star}_{\text{SNIFS}}(\lambda) = (T_{\text{tele}} T_{\text{SNIFS}} QE)(\lambda)$ summarizes the responsivity of SNIFS, including the telescope, instrument transmission, and the quantum efficiency of the CCD. It is common to express the atmospheric extinction

[1] The spaxel are numbered from 1 to 225. The spaxel serving as an example has the ID 28.
[2] The microscale fluctuation of the optical properties of air on a millisecond timescale.

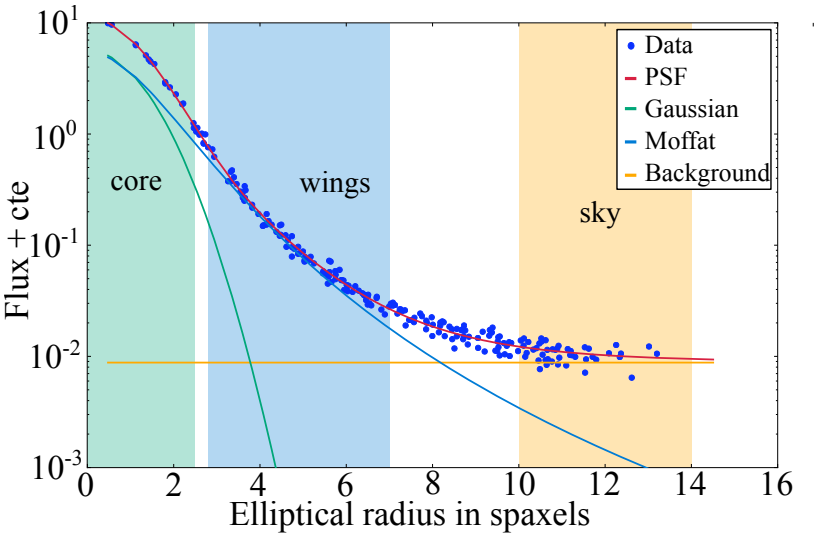


Figure 6.2: Details of the PSF components for an exposure of GD71. The blue graph shows the data at 4327 Å. The PSF is composed of three distinct parts: the core, which is described by a Gaussian function (green shaded); the wings, which are described by a Moffat function (blue shaded); and the sky background, which is described by a constant (orange shaded). A constant has been added to the PSF, to adjust the background. The plot is taken from Clément Buton 2009, and fonts are modified for readability.

as $T_{\text{atm}} = 10^{-0.4 \times K(\lambda)\hat{X}(\hat{z}_i)}$ (C. Buton et al. 2012). With the extinction coefficient $K(\lambda)$ that quantifies the spectral extinction per airmass $\hat{X}(\hat{z}_i)$ along $\hat{z}$. On photometric nights, the variability $\delta T_i(\lambda, \hat{z}_i, t)$ is assumed to be one within the measurement uncertainties (C. Buton et al. 2012).

The responsivity of SNIFS relative to CALSPEC, $S^{\star,n}_{\text{SNIFS}}$, is obtained by dividing the observed spectrum $F^{\star}_{\text{obs}}(\lambda)$ by the CALSPEC spectrum of the star $F^{\star}(\lambda)$ and correcting for the extinction $K(\lambda)\hat{X}(\hat{z}_i)$ while ignoring any variability in K.

With the assumption of a photometric night ($\delta T_i(\lambda, \hat{z}_i, t) = 1$) Eq. 6.2 simplifies and can be rewritten as:

$$\log_{10}\left(\frac{F^{\star}_{\text{obs},i}(\lambda)}{F^{\star}_{i}(\lambda)}\right) = \log_{10}\left(S^{\star,n}_{\text{SNIFS},i}(\lambda)\right) - 0.4 \times K_{\text{obs}}(\lambda)\hat{X}(\hat{z}_i) \qquad (6.3)$$

The observed extinction $K_{\text{obs}}(\lambda)\hat{X}(\hat{z}_i)$ can be identified as the slope of a linear relation between the airmass and the logarithmic ratio between the observed and "true" spectrum. Linear fits over the full set of standard stars i for each wavelength, allowing the extraction of the spectral extinction coefficient $K_{\text{obs}}(\lambda)$. The relation between airmass and relative flux is also called a "Langley plot". Its intersection with the y-axis is the logarithm of the responsivity $S^{\star,n}_{\text{SNIFS}}(\lambda)$.

To improve the accuracy of the atmospheric corrections, a physical model of the atmosphere, K_{model} is fitted to the observed extinction coefficient $K_{\text{obs}}(\lambda)$. The results of the fitted model, rather than the slopes, are used to correct the flux $F^{\star}_{\text{obs}}$.

The atmospheric extinction has several components k_j that can be summed to give the total model $K_{\text{model}}(\lambda) = \sum_j k_j(\lambda)$. k_j describes either absorption or scattering from:

a. Rayleigh scattering
b. Aerosol scattering
c. Ozone absorption
d. Telluric lines

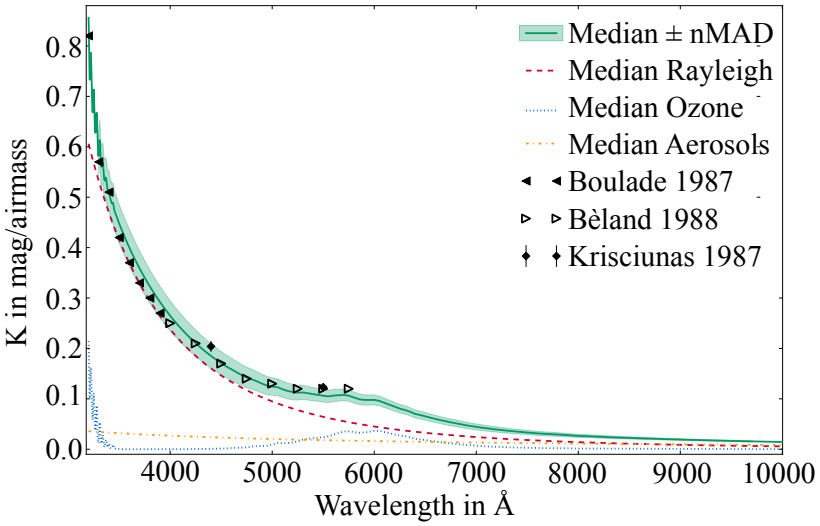


Figure 6.3: The average atmospheric extinction observed by SNfactory is shown as a solid line, with its constituent physical elements shown as dashed lines. For comparative analysis, historical extinction measurements from Maunakea as determined in previous studies by Boulade 1987 (diamonds), Bèland, Boulade, and Davidge 1988 (triangles), and Krisciunas et al. 1987 (stars) are also shown. Taken with modified fonts from C. Buton et al. 2012.

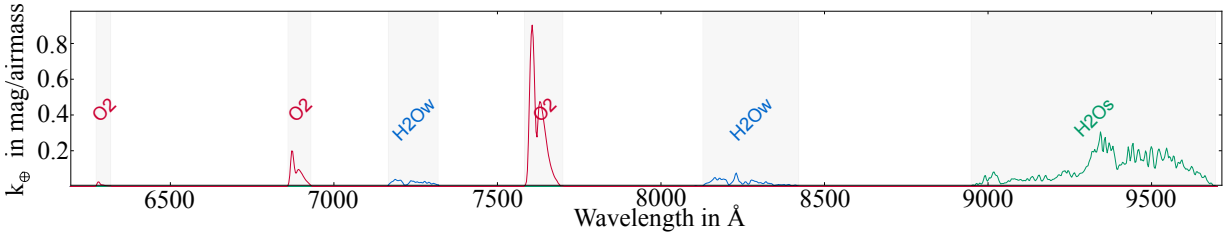


Figure 6.4: Telluric template derived by SNfactory. The spectral extinction $k_\oplus$ (d.) per airmass $X(\hat{z})$. The strong water lines are denoted as H2Os and H2Ow for the weak. From C. Buton et al. 2012 with modified fonts for better readability.

Fig. 6.3 illustrates the atmospheric extinction components a., b., and c.. This figure includes a comparison with data from former studies. The overall model is represented by a solid line, with its normalized median averaged deviation (nMAD) indicated by the shaded band. The individual components of the model are shown as dashed lines. It is noteworthy that the model agrees well with measurements made two decades earlier. The model $K_{\text{model}}(\lambda, \hat{z})$ is fited to the observed extinction $K_{\text{obs}}(\lambda, \hat{z}_i)$, with parameters detailed in C. Buton et al. 2012.

The absorption line templates for water and oxygen are shown in Fig. 6.4. These templates are derived from the high-resolution telluric spectral data from the Kitt Peak National Observatory (KPNO, Hinkle, Wallace, and Livingston 2003). The amplitude of these templates is adjusted to match the observed extinction data. For technical reasons, the telluric lines are fitted separately. This approach gives the SNIFS pipeline more flexibility for different scientific applications.

The model for the extinction above Maunakea, its variability, and the methodology of the fit method are described in C. Buton et al. 2012. The newest model is discussed in Rubin et al. 2022b.

After determination of the atmospheric extinction $K_{\text{Model}}(\lambda)$ the responsivity $S^{\star,n}_{\text{SNIFS},i}(\lambda)$ can be calculated from Eq. 6.3 for each star i. The average responsivity $\langle S^{\star,n}_{\text{SNIFS}}(\lambda)\rangle_i$ is obtained by fitting a polynomial over the complete set of individual responsivities $S^{\star,n}_{\text{SNIFS},i}(\lambda)$. The responsivity provided by the pipeline is a conversion factor from CCD ADUs to physical spectral flux, also referred to as "flux solution", implicitly including the CCD's gain.

Table 6.1: Summary of star observations for four consecutive nights. It details the total number of observed stars, the count of observations from CALSPEC sources, the frequency of GD153 observations, and the number of stars observed on multiple occasions. Additionally, the seeing conditions for each night are noted. In the final row, a cumulative total of these metrics across all nights is presented.

Date (UTC)	Total	CALSPEC	GD153	Multiple Obs	Seeing	Photometric?
2022-06-19	36	16	3	7	≈1.4″	No
2022-06-20	34	22	5	9	1.5″ to 2″	Yes
2022-06-21	36	25	7	10	1.5″ to 2″	Yes
2022-06-22	41	27	4	11	≈3″	No
Total	**147**	**90**	**19**	**37**	-	-

6.1.3 The Sample

The sample of stars observed on the nights of June 19-22, 2022 UTC, was determined using a Monte Carlo simulation. This simulation, developed by Dr. David Rubin, Assistant Professor at the University of Hawai'i at Mānoa, generates a random list of stars from the CALSPEC database and other stars commonly used for spectrophotometric calibrations. A list of standard stars used for SNIFS flux calibration purposes can be found in C. Buton et al. 2012 and in Rubin et al. 2022b.[3] The program simulates the expected airmasses of these stars at the time of their observation and the resulting Langley plot for atmospheric corrections (see Eq. 6.3). This procedure aims to find an observing schedule that minimizes uncertainties in the resulting atmospheric correction.

Tab. 6.1 summarizes the observations of the four nights. The table includes the total number of observations, the number of CALSPEC source observations, the number of observations of GD153, the number of stars observed multiple times, and the seeing conditions for each night. The last row provides the sum of these values across all nights. A full list of all observed stars is provided in Appendix Tab. A.7, Tab. A.8, Tab. A.9, and Tab. A.10.

Throughout four nights, the conditions varied. On June 19, the seeing was ≈1.4″, potentially influenced by the telescope's mirror being around 3 degrees warmer than the ambient temperature. Noteworthy is that the observation was performed without the dom's cooling system, which was non-operational during the observation. The following two nights, June 20 and 21, were classified as photometric, even though the seeing stayed between 1.5″ to 2″ for the reason above. However, on June 21, a notable issue arose with tracking errors for the stars HD116405 and HD180609 due to the use of an incorrect tracking mode, which can impact the precision of the observations. Finally, on June 22, the seeing conditions deteriorated to ≈3″, indicating poorer observational conditions compared to the earlier nights. The conditions were communicated by Greg Aldering in a private communication dated 2022-06-19.

Tab. 6.2 presents the list of stars used in the analysis discussed here, observed during the night of June 19th, in the order in which they were observed the first time. The selection includes stars from both CALSPEC and other sources, as determined by the Monte Carlo simulation mentioned above. Stars whose spectra are available in the CALSPEC database have links to their respective models.

[3] The ladder provides a discussion of the SNIFS standard star network and an illustration of the distribution of these stars in the sky.

Night of 19th June 2022 UTC			
BD+023375	BD+174708 (2)	BD+254655 (3)	BD+284211 (2)
BD+292091	**EG131 (6)**	**GD153 (3)**	HR4468
HR4963	HR8634	HR9087	**HD106252**
HD116405	**HD158485**	**HD159222**	**HD163466 (4)**
HD180609	**P041C**	**P177D (3)**	

Table 6.2: List of stellar observations conducted on the night of 19th June 2022, arranged in alphabetical order. The table is divided into four columns, with the number of observations in brackets next to stars observed multiple times. This compilation and the night's conditions were sourced from a communication with Greg Aldering dated 2022-06-19. For additional details, including airmass and corresponding data filenames, refer to Appendix Table A.7.

6.2 Calibration with SCALA

This section describes the measurement routine to calibrate SNIFS with the laboratory standard (Sec. 6.2.1) and how the flux into the telescope is calculated (Sec. 6.2.2).

6.2.1 Measurement Routine

The calibration procedure is adapted from the commissioning measurements of SCALA in 2015, as reported by Lombardo, Küsters, et al. 2017. The alignment of the telescope to SCALA (see Fig. 3.53d) is performed as described in Sec. 3.4.2. As mentioned in Sec. 3.4, a SNIFS frame is exposed to multiple wavelengths separated by 500 Å (see Fig. 3.42). Lombardo, Küsters, et al. 2017 have concluded that this spacing leads to negligible crosstalk between the wavelengths.

Fig. 6.5 visualizes the SNIFS exposures of the blue channel, the communication between the setups, and the algorithm used for the SNIFS exposure. The process starts with exposing SNIFS with the first wavelengths separated with the inner stepsize $\Delta\lambda_{\text{inner}}$ (array B00). The following wavelength arrays are shifted by $\Delta\lambda_{\text{outer}}$ for the subsequent exposure runs (B01 to R16). The steps are

$$\Delta\lambda_{\text{inner}} = 500\,\text{Å} \tag{6.4}$$

$$\Delta\lambda_{\text{outer}} = 30\,\text{Å} \tag{6.5}$$

achieving a 30 Å granularity. It is important to note that the wavelengths shown are target wavelengths, and the actual wavelength scale is derived from SNIFS wavelength calibration with arc exposures.

Communication between the setups is handled by Secure Shell Protocol (SSH) between the SNIFS and SCALA machines, with the main program running on the SNIFS machine, which executes SSH calls on the SCALA machine. This setup is shown in the lower left part of fig. 6.5, where the main program is represented by a red box and the SCALA machine by a black box with a gray background. The main program iterates through the wavelength arrays B0-R16 (see Tab. A.11), first screening the blue channel with B0-B16 and then exposing the red channel with R0-R16.

The exposure of an image begins with flat-fielding measurements and wavelength calibration with the SCU (described in section 3.3.1). After the arc and continuum exposures, the CCD is 'cleaned' with five consecutive readouts to ensure no residual charge remains. Then, the SNIFS shutter is opened, and the program makes an SSH call to request SCALA for

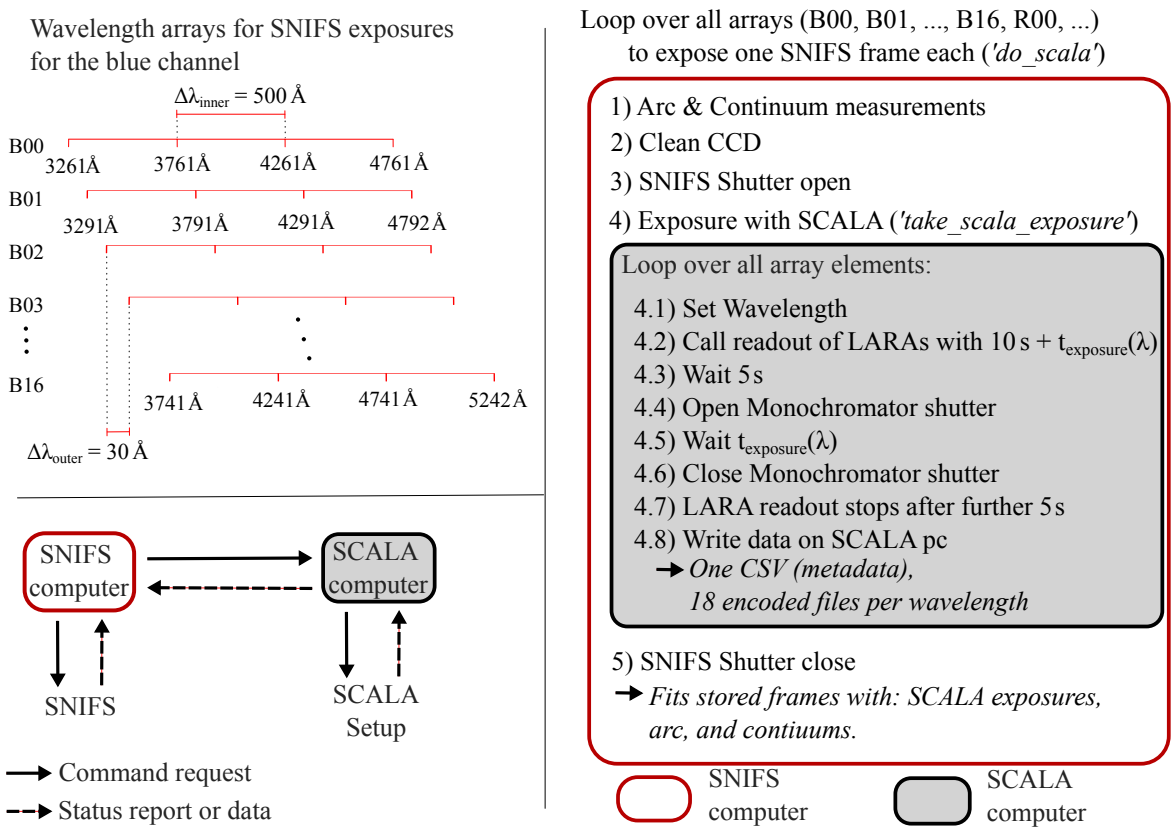


Figure 6.5: Upper left part: Schematic of the order of wavelengths at which SNIFS is exposed. A frame in SNIFS is exposed to multiple wavelengths (4 wavelengths in the blue channel and 13 wavelengths in the red channel). The inner spacing between each wavelength in a frame is $\Delta\lambda_{\text{inner}} = 5000$ Å. The outer distance of $\Delta\lambda_{\text{outer}} = 30$ Å shifts the next array. The arrays are labeled B for blue or R for red channel, together with incremental numbers. Lower left part: An overview of the data and command flow. The SNIFS computer runs the master program that controls SNIFS and the SCALA computer. The SCALA computer controls the monochromator and the LARA modules. Right part: An overview of the measurement algorithm.

exposure. The main program algorithm on the SNIFS machine follows the same procedure as Lombardo, Küsters, et al. 2017.

First, the SCALA monochromators set the wavelength, followed by initiating the readout of the LARAs with 10 s[4] added to the exposure time $t_{\text{exp}}(\lambda)$, which depends on the intensity of the xenon lamp at that wavelength and is chosen to ensure that the signal-to-noise ratio is greater than 100 per spaxel for each wavelength, twice[5] as suggested by Lombardo, Küsters, et al. 2017. After 5 s, the monochromator shutter (SCALA shutter) is opened, exposing the LARAs and SNIFS for the duration of $t_{\text{exp}}(\lambda)$, and then closed. The LARAs complete their readout after another 5 s, and the next wavelength is set after the data is written to disk. When the array is complete, the SNIFS shutter is closed, and the CCD is read out before proceeding to the next exposure. The software that controls the SCALA computer is hosted in a git repository (Bastian-Querner 2022), created by Rigault 2015.

[4] Long enough to ensure that all devices started readout. The additional data is also helpful for background determination.

[5] In contrast to Lombardo, Küsters, et al. 2017 the introduced chopper blocks light of half the time of exposure.

6.2.2 Calculation of the Radiant Energy

Determination of the radiant energy encompasses calculating the collected charge per LARA module and the radiant energy on the telescope combined with SNIFS. The impact of ionizing radiation on statistical uncertainty is also addressed, demonstrating that this contribution is insignificant.

Collected Charge

Fig. 6.6 illustrates the signal from one LARA of each SCALA module during a SCALA run. The signal square modulation is a result of the chopper and the monochromator's shutter. The red-shaded regions in the figure indicate periods when the chopper is closed (recorded by LARA No. 18), while the gray-shaded regions correspond to the monochromator's shutter, as identified from the recorded timestamps. Furthermore, the vertical dashed orange lines in the figure represent the positions of peaks from ionizing radiation in the background data, a topic further discussed in Sec. 6.2.2 Ionizing Radiation.

The red regions are used for correction via linear interpolation to account for varying background levels. The signal is then integrated over the entire waveform to calculate the collected charge per LARA, as described by the following equation:

$$C_{\mathrm{LARA}} = \frac{1}{R_f} \sum_{i=0}^{N_{\mathrm{samples}}} U_i \Delta t \tag{6.6}$$

where $R_f = 10\,\mathrm{G\Omega}$ represents the feedback resistance, and $\Delta t = (1200\,\mathrm{Hz})^{-1}$ is the sampling period.

Fig. 6.7 shows the collected charge C_{LARA} from Eq. 6.6 for LARAs 23 (in blue) and 30 (in orange), recorded on 2022-06-17. This figure also shows the photocurrent derived from the signal heights, assuming perfect square modulation and called effective. Visual inspection reveals that the red-circled outliers in the data are due to erroneous exposure time settings. These effects contribute to the statistical scatter with lower signal-to-noise. The photocurrent is found to vary between 1 pA and 30 pA, corresponding to a measured power between 2 pW and 100 pW. For an integration time varying between 20 s 300 s, this gives a deposited radiant energy of the order of 100 erg per sample.

The wavelength scale here is the monochromator wavelength, which has three different offsets depending on the configuration of gratings. The gratings are changed at 5500 Å and 7500 Å. The offsets are taken into account by the generation of the wavelength arrays. This results in an overlap at 5500 Å and a gap at 7500 Å, which is corrected after the wavelength calibration with the arc exposures.

Ionizing Radiation

The signal is modulated with a square function (see Fig. 6.6), and the study of an algorithm to remove the peaks before integration on the whole waveform is omitted because the contribution of the radiation to

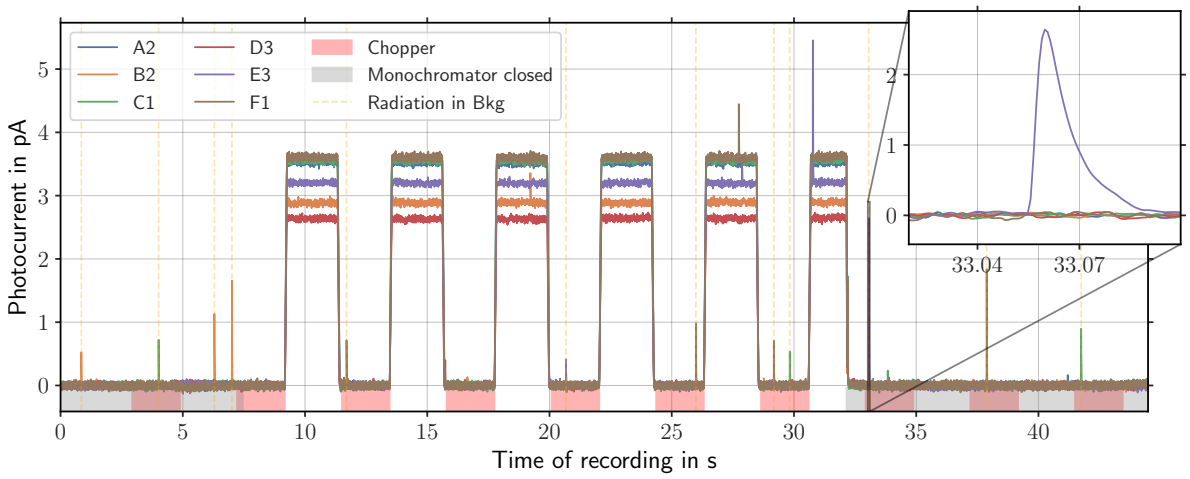


Figure 6.6: Signal from six LARAs, each representing one SCALA module at ≈5000 Å. The shaded regions indicate the closed chopper (red) and the shutter of the monochromator (gray) from the recorded time stamps. The vertical dashed orange lines show the positions of the ionizing radiation peaks in the background data. The small plot shows the integration window of a single peak (discussed in Sec. 6.2.2 Ionization Radiation).

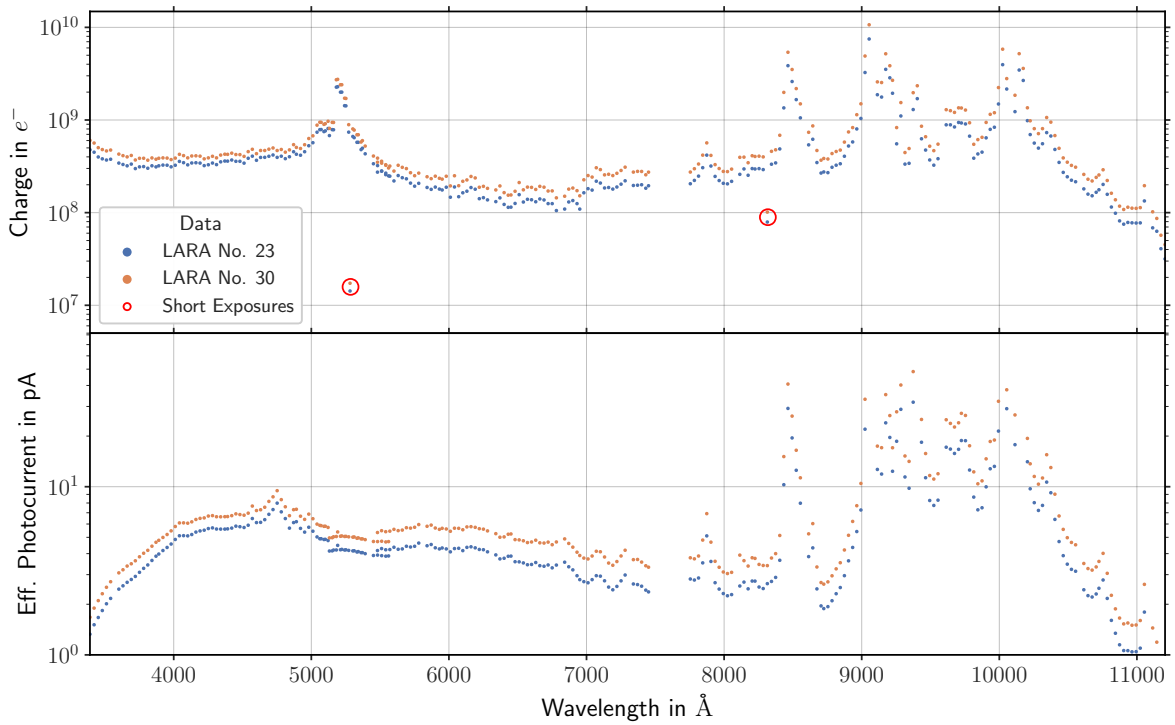


Figure 6.7: Collected charge and photocurrent for LARAs No. 23 and No. 30 from the run on 2022-06-17. Upper plot: Collected charge plotted as a scatter plot. Lower plot: Photocurrent derived from signal heights. Outliers are due to inaccurate background corrections from data losses. The overlap at 5500 Å and the gap at 7500 Å are due to the uncalibrated monochromator wavelength, which is discussed in the text.

the statistical uncertainty is seen to be less pronounced. Therefore, in contrast to the analysis of the VASC measurements, the ionizing radiation is not removed from the waveforms. Nevertheless, its contribution to the uncertainty must be quantified. The chopper information allows the use of standard algorithms from the programming libraries to find the peaks in the background data and compare the expected deposited charge per s in a waveform with the recorded photocurrent. The positions of the peaks in the background data (red and gray marked regions in Fig. 6.6) are determined using the 'find_peaks' function together with a Savitzky-Golay filter from the 'SciPy' Python package (Virtanen et al. 2020). A fixed empirical window of ±40 ms around the peak maximum is chosen to integrate[1] its charge (see the zoomed plot in Fig. 6.6). In the measurement of 2022-06-17, there are $N_{\text{peaks}} =\approx 10400$ peaks found over a recording time of 2.4 h times 18[6], corresponding to a count rate of $f_{\text{peak}} = 67.9(7)$ mHz. Fig. 6.8 shows the charge distribution $p(C)N_{\text{peaks}}$ on the left and its cumulative sum on the right. As seen in Fig. 6.7, the signal is mostly above 1 pA, which corresponds to $I_{\text{min}} = 6.2\times10^6\, e^-$. The expected value of the charge distribution for the deposited charge per peak is $\mu = \sum C_i p(C_i) = 9.9\times10^4\, e^-$, where 82 % of the peaks fall below that value. The averaged deposited charge per s is $\langle I_{\text{peaks}}\rangle = \mu f_{\text{peak}}$, which gives an upper limit max $\sigma_{\text{rel}}(I_{\text{peaks}})$ on the contribution to the statistical uncertainty if compared to I_{min}. $\sigma_{\text{rel}}(I_{\text{peaks}}) \leq \langle I_{\text{peaks}}\rangle / I_{\text{min}} = 750$ ppm, which is acceptable compared to the target of 0.5 %.

1: ($C = \sum_i U_i/R_f \cdot \Delta t$; with the voltage U_i recorded at the time i, the feedback resistance $R_f = 10\,\text{G}\Omega$ and the sampling period $\Delta t = (1200\,\text{Hz})^{-1}$)

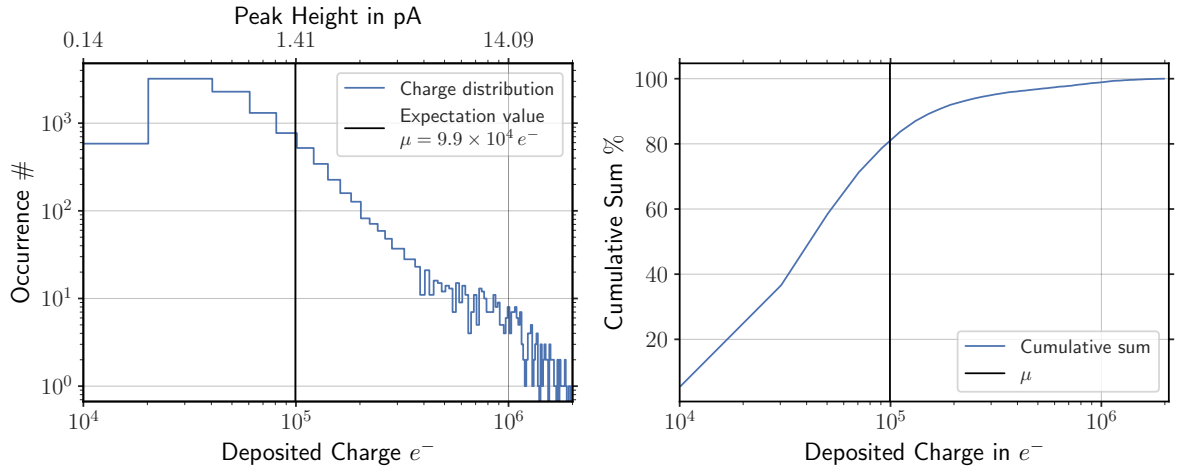


Figure 6.8: Left: Distribution of the deposited charge by cosmic radiation into the S1337 photodiodes in the LARA modules during the SCALA run in 2022-06-17. The weighted average ($\mu = 330\, e^-$) of the distribution is used for the upper limit of the comic's contribution to the statistical uncertainty. Right: Cumulative sum of the distribution.

Radiant Energy

The quantification of radiant energy that the telescope+SNIFS is exposed to by a specific beam involves computing the number of photons $N_{\gamma,\text{Beam}}$ based on the collected charge C_{LARA} obtained from Eq. 6.6 and the responsivity calibration R_{LARA} from Eq. 5.5. This is expressed in the equation:

[6] The 2.4 h is the time of the recorded background of only one LARA.

$$N_{\gamma,\text{Beam}} = k_{G,\text{LARA}} \frac{C_{\text{LARA}}(\lambda)}{QE_{\text{LARA}}(\lambda)} \tag{6.7}$$

Here, QE_{LARA} denotes the quantum efficiency for the individual LARA as detailed in Sec. 5.3.3, and $k_{G,\text{LARA}}$ is a geometrical factor accounting for the fact that each LARA only receives a fraction of the beam[2].

The value of the geometrical factor is $k_{G,\text{LARA}} = (A_{\text{Mirror}} - A_{\text{Shadow}})/A_{\text{LARA}} = 188(3)$[3]. This factor is used to scale the photon counts to a reasonable range and is assumed to be consistent across all beams, solely affecting the grayscale and not the color calibration. This calculation assumes that the LARA's position represents the entire mirror. This assumption is justified because the mirrors have new coatings, which are assumed to be homogenous.

The cumulative number of photons is the sum of photons from all individual beams:

$$N_{\gamma,\text{Total}} = \sum_{\text{Beam}}^{18} N_{\gamma,\text{Beam}}(\lambda) = k_G QE_{\text{NIST}} \sum_{\text{Beam}}^{18} \frac{C_{\text{LARA}}(\lambda)}{R_{\text{LARA}}(\lambda)} \tag{6.8}$$

In this equation, QE_{NIST} is the quantum efficiency of the NIST photodiode discussed in Sec. 5.3.3 and R_{LARA} is the responsivity calibration for each LARA from Eq. 5.5.

Figure 6.9a shows the radiant energy from SCALA into the telescope for each exposure and all runs. Note that the scatter in Fig. 6.9a does not represent statistical uncertainties because the exposure times varied. Figure 6.9b details the uncertainties associated with this energy.

The uncertainties for the interpolation (blue dashed line), the variability of the splitting ratio (red dotted line), the wavelength scale (orange dashed line) on the calibrations R_{LARA}, the statistics (green dashed line) and the contribution of the temperature correction (purple dashed line) between the laboratory conditions (≈24 °C) and the Maunakea summit (≈5 °C) are assumed to be uncorrelated and are therefore quadratically summed and divided by $\sqrt{18}$. The temperature model uncertainty (red dashed line) and the NIST calibration uncertainty (red solid line) contribute once. The total radiative energy uncertainty is the quadratic sum (black solid line).

There are four major contributions to the uncertainties in different regions:

Splitting Ratio Variability [7000 Å to 9500 Å] The asymmetry of the relay optics on the VASC setup results in a temperature-dependent splitting ratio between the monitor diode and the DUT. The thermo-reflectance of aluminum can be neutralized through the use of a symmetric relay.

Flux Limitation [7000 Å to 9500 Å] The flux limitation of the VASC (system throughput times LDLS flux) reaches several minima (see Fig. 5.20). Around 8000 Å, the output spectrum of the LDLS has a minimum, and at 8900 Å and 9800 Å, the fiber has absorption peaks. A change of optical components (e.g., degenerated mirrors due to high UV exposition, such as M1 and M2) in the VASC setup would improve the flux throughput.

2: It should be noted that the deviations $k_{\text{DUT}}/k_{\text{NIST}}$ from the nominal feedback resistance of 10 GΩ are omitted as they represent only an overall grayscale.

3: The area of the mirror is $A_{\text{Mirror}} = \pi(16\,\text{cm})^2/4$, and the area of the LARA's diode is $A_{\text{LARA}} = 1\,\text{cm}^2$. The shadow, created by the LARAs being mounted 1 cm away from the mirror's edge along with the dimensions of the LARA module and a conservative uncertainty of 1 cm, gives an area $A_{\text{Shadow}} \approx 13(4)\,\text{cm}^2$

Interpolation [below 4000 Å**]** The interpolation of the sensitivity calibration R_{LARA} is done with a linear spline interpolation and its uncertainty is estimated by the difference to a quadratic interpolation. Regions of large curvature in the response of the diode increase the uncertainty of the interpolation (compare the response of the working standard Fig. 3.6). Using a photodiode sensitivity model would eliminate the need for interpolation.

Temperature Model [below 4000 Å**]** Since the diodes have a temperature dependence below 4000 Å (Larason and Jeanne M. Houston 2008) a coefficient is not evaluated for wavelengths below 6000 Å. A full investigation of the temperature dependence of the diodes is required to obtain better uncertainties.

NIST Between **[**4000 Å **to** 7000 Å**]** the uncertainties are limited by those from the NIST calibration. Above this range, the uncertainties are limited by the flux limitation of the VASC and the thermo-reflection net effect due to the asymmetric relay optics on the gantry robot, which are described before.

and five minor contributions:

- Wavelength calibration of the VASC is below ten ppm below 900 Å and at least one order of magnitude smaller than the NIST uncertainties.
- Statistical uncertainties remains below NIST uncertainties below 7500 Å.
- $\Delta T_{\mathrm{VASC–SCALA}}$ only contributes above 9300 Å staying one order of magnitude below the NIST contribution.
- Temperature Model stays below ten ppm the whole range below 9000 Å and contributes similarly as $\Delta T_{\mathrm{VASC–SCALA}}$ above.
- Angle of Incidence 1° stays below 20 ppm the whole range.

In summary, in the region of 4000 Å to 7000 Å, the uncertainty is limited by the uncertainties given by the calibrations of the photodiodes traceable to the NIST laboratory standard. Above 8000 Å, the uncertainty is limited by the flux limitation of the VASC and the thermo-reflection net effect due to the asymmetric relay optics on the gantry robot. Improving the flux throughout the VASC system and using symmetric relay optics would improve uncertainty. Below 4000 Å, the usage of a TMM model for the photodiode's responsivity and a full investigation of the temperature dependence of the diodes would improve the uncertainty.

6.2.3 Charge Integration in SNIFS Spectra

The primary focus here is the extraction of the charge in the SCALA-exposed frames using the SNIFS pipeline (see Sec. 6.1.1). The resulting CCD counts are proportional to the number of produced photoelectrons per wavelength and spectral flux $F^{\mathrm{SCALA}}(\lambda)$. Fig. 6.10 shows the preprocessed spectrum of the SCALA exposure R11 taken on June 21, 2022. The spectrum is a combination of the blue and red channels. Each peak that corresponds to a SCALA exposure is labeled with its corresponding center wavelength λ_c (see Tab. A.11). The peaks from the first-order diffraction from the grism (compare Sec. 3.3.2 for a detailed explanation) are called SCALA peaks. The plot also shows parasitic signals from second-order diffraction from the grism (denoted as "2nd") and the

sidelobes from the diffraction of the square microlenses (denoted as "S").

As discussed in Daniel Küsters 2019, the square microlenses introduce a diffraction pattern, visible as crosses at the CCD (see arc illumination of Fig. 6.11). The red boxes show the pixels along the spaxel's trace that are summed up to construct the spectrum $C(\lambda, x_i, y_j)$ at the position (x_i, y_j) before PSF-fitting. The diffraction pattern of one spaxel leaks into its neighboring and creates a sidelobe approximately 326 Å aside from λ_c of the SCALA peak. Sidelobs from up to the fourth neighboring spaxel have been observed with decreasing amplitude by Daniel Küsters 2019. By adding multiple exposures of different wavelengths, the sidelobes add up to a discrete background and vary in amplitude. The use of different grating combinations (as outlined in Tab. 3.2) by the monochromators to maximize throughput introduces additional complexity. This results in varying offsets between the target wavelength (as requested by the monochromators) and the calibrated wavelength (established from arc calibrations). These offsets lead to changing periodicities in the sidelobes, with some merging with or appearing adjacent to the SCALA peaks, complicating straightforward spectral integration around each centered at λ_c SCALA peaks and its width Δ_λ as it was performed in Lombardo, Küsters, et al. 2017:

$$N(\lambda_c) = \int_{\lambda_c - \Delta_\lambda/2}^{\lambda_c - \Delta_\lambda/2} F^{\mathrm{SCALA}}(\lambda) d\lambda \tag{6.9}$$

As noted earlier, the diffraction also occurs during the usual SNIFS observations but results in a smooth background, which is easily removed.

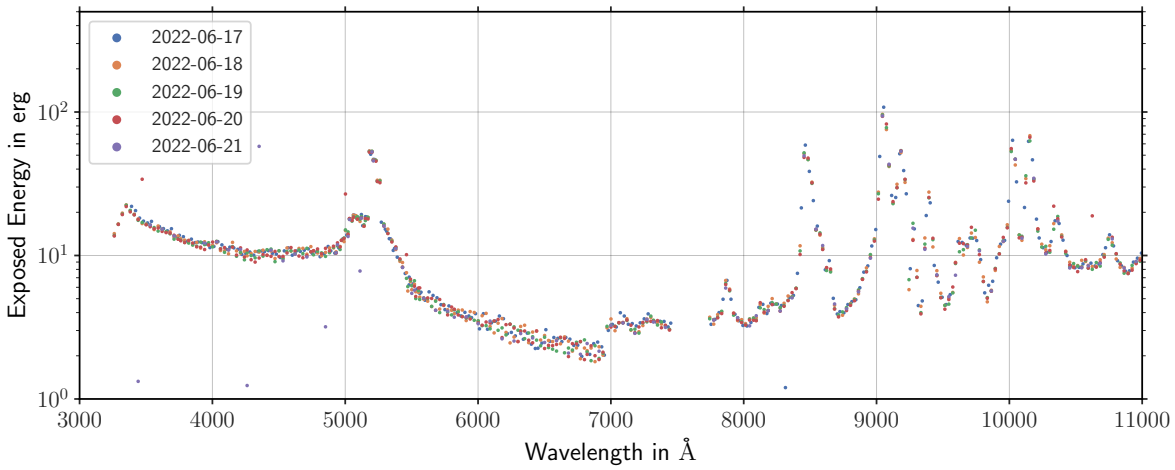
To minimize the contribution of the sidelobes, the integration window Δ_λ is changed dynamically depending on the amplitude of the SCALA peak. Only flux above a threshold of a fixed fraction f of the peak amplitude is integrated.

$$\Delta_{\lambda_c} = \{\lambda \mid F(\lambda) > f \cdot F(\lambda_c)\}$$

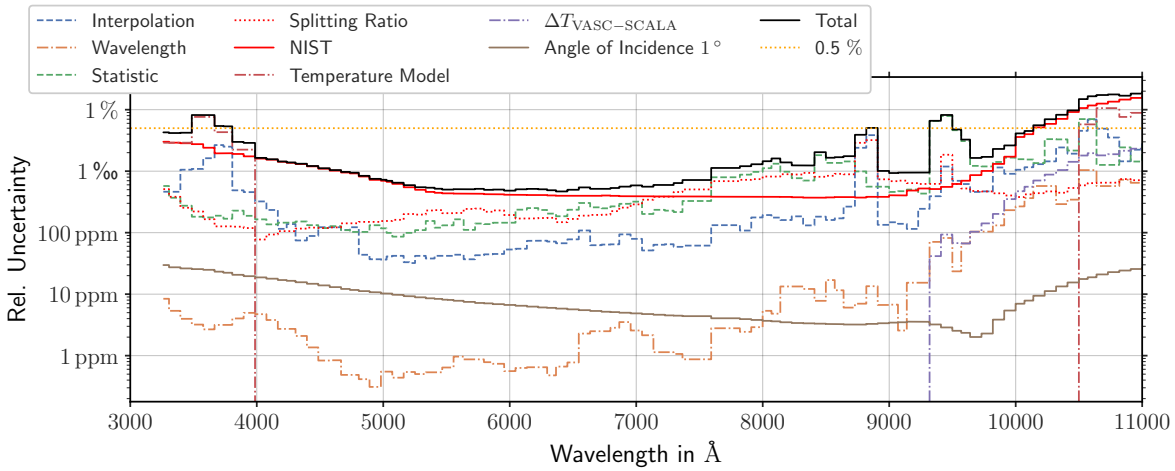
The resulting "aperture loss" was determined using SNIFS exposures containing only a single line. This was accomplished by the remote controllability of SCALA, which took advantage of several nights of unfavorable weather for these scans. Fig. 6.12 shows the "aperture loss" in two scenarios: one including measurements with flux greater than $f = 1/50$th (red line) of the peak amplitude and another with flux greater than $f = 1/10$th (blue line) and compares those losses with the xenon spectrum in yellow. Both curves show similarities in their small-scale structures. This small-scale structure is mainly due to the emission lines of the xenon lamp, with the uncalibrated SNIFS spectrum of SCALA's xenon lamp in orange. The double-monochromator cannot resolve these high-intense spectral lines, so they deform the SCALA peaks, affecting the dynamical charge summation. The xenon lines significantly impact the signal for smaller integration windows, but the overall trend is unaffected by the relative window width (private communication with Daniel Küsters 2023d).

The aperture loss uncovers a further artifact using SCALA. The artifact is a sidelobe of unknown origin, visible in Fig. 6.13. The figure shows the CCD exposure. The SCALA peak is in the center of the figure (green

dashed line), which is exposed to 6500 Å light. The sidelobe is highlighted with a black dashed line crossing the SCALA peak. This artifact results in additional signal for exposures with wavelengths between 6500 Å to 6800 Å. The aperture loss incorperates the SCALA sidelobe (compare Fig. 6.12).



(a) The aggregated energy of all beams for each exposure, represented across all calibrations. The energy is expressed in erg rather than the number of photons N_γ used to calculate SNIFS responsivity. The variation in the data points does not reflect statistical uncertainties due to differing exposure times. The overlap at 5500 Å and the gap at 7500 Å are due to the uncalibrated monochromator wavelength, discussed in the text in Sec. "6.2.2 Collected Charge".



(b) The summarized uncertainties of the deposited energy shown in Fig. 6.9a. The solid black line indicates the quadratic sum of all uncertainties, and the vertical dotted orange line indicates the targeted uncertainty of 0.5 %.

Figure 6.9: The radiant energy deposited into the telescope+SNIFS system for each exposure. The uncertainties are calculated as described in Sec. 6.2.2.

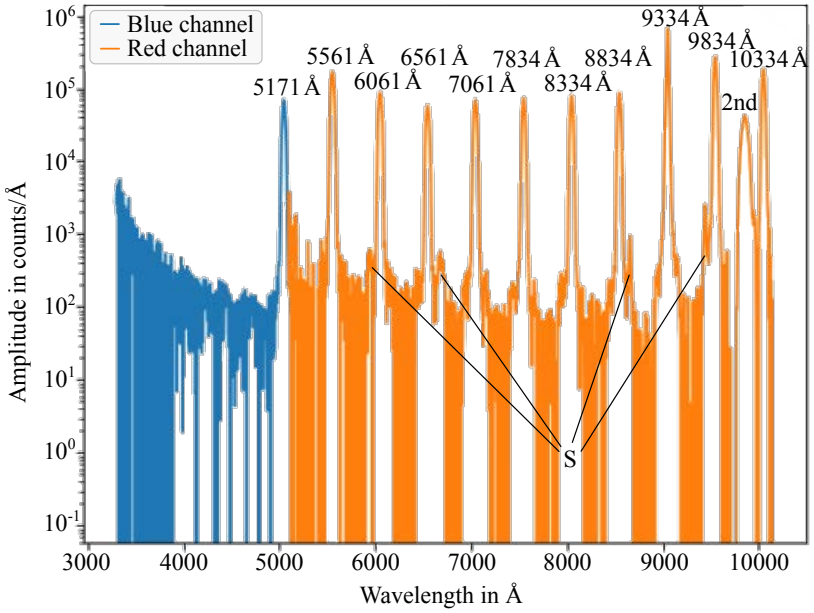


Figure 6.10: Preprocessed spectrum from the blue and red channel of the exposure R11 taken on June 21, 2022. The **"S"** depicts the sidelobes that arise from diffraction, and the **"2nd"** showcases the 2nd-order light from the bluest exposure in the blue channel (Daniel Küsters 2023d).

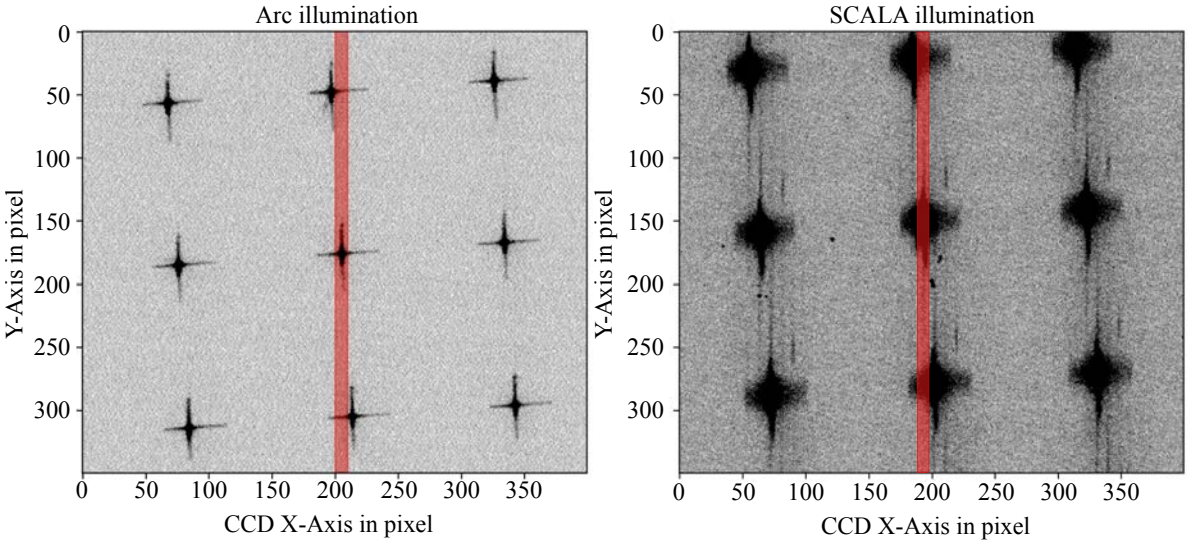


Figure 6.11: CCD exposures of the SNIFS blue channel to monochromatic light. The picture on the left shows an exposure with the blue channel's arc lamp, while the frame on the left is from SCALA. Note that the light from SCALA has a bandwidth of 52 ÅFWHM, while the width of the arc lines is limited by SNIFS' resolution of 5.23 Å. Each cross corresponds to one spaxel. The spectrum of the central spaxel is extracted from the area outlined by the red box. Taken and with modified labels from Daniel Küsters 2019.

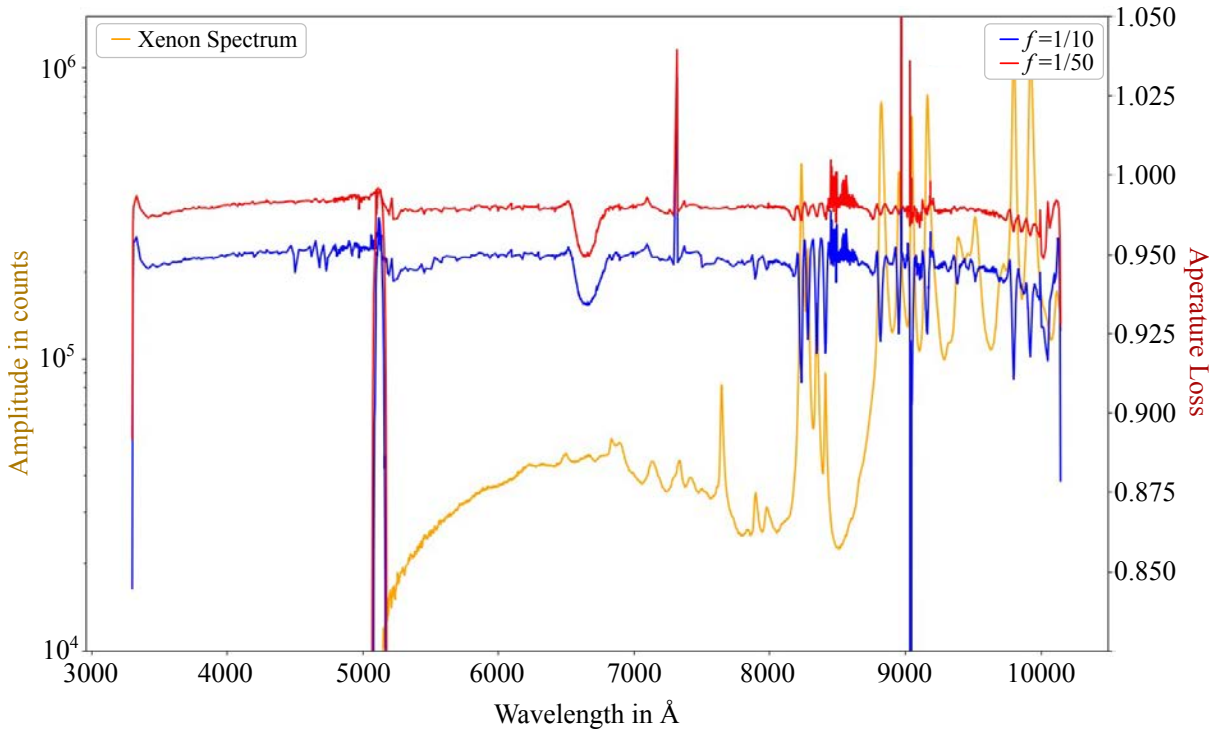


Figure 6.12: Resulting "aperture loss" for the dynamical summation window. The aperture loss is calculated by comparing the charge of the SCALA peaks dynamically integrated with its entire charge from single-line exposure. The blue curve shows the aperture loss for flux greater than $f = 1/10$th of the peak amplitude, while the red curve shows the correction for flux greater than $f = 1/50$th (right label). For comparison, the orange curve shows the entire xenon spectrum of the SCALA lamp. The 1/10th-aperture loss is applied to the spectra of the SNIFS exposures (Daniel Küsters 2023d).

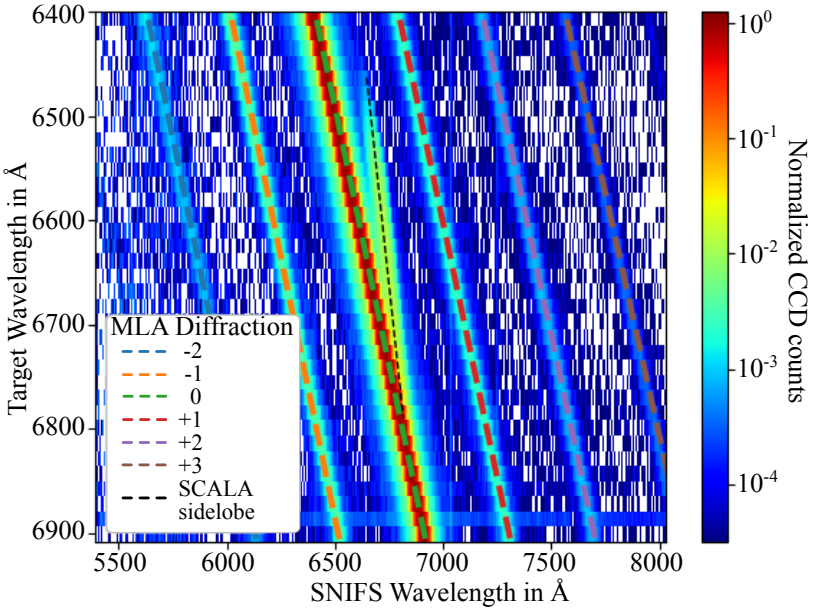


Figure 6.13: Artifact at the CCD using SCALA for wavelengths 6500 Å to 6800 Å. The figure shows an exposure of the red channel with SCALA at approximately 6500 Å for the SNIFS spectrum between 5500 Å and 8000 Å. The central green dashed line shows the SCALA peak crossing a sidelobe (black dashed line) of unknown origin. The spaxel's traces align with the abscissa labeled as SNIFS wavelength. The ordinate corresponds to SCALA's target wavelength. The other lines denoted as spaxels $\{-2, -1, +1, +2, +3\}$ are SNIFS sidelobes caused by the MLA diffraction. Note that the ordinate is stretched relative to the abscissa (Daniel Küsters 2023d).

6.3 Comparision Calibration with CALSPEC and SCALA

This section discusses the comparison between the calibration with SCALA and CALSPEC $S^{\star,n}_{\mathrm{SNIFS}}/S^{\mathrm{NIST}}_{\mathrm{SNIFS}}$. While processing the data from the SCALA measurements, it introduces artifacts, which originate from the diffraction of spaxels leaking one into the other, the sidelobe of unknown origin, and the intense spectral lines from the xenon lamp in the infrared. It was discussed that the aperture loss could resolve these effects.

The primary purpose of this section is to discuss the systematic effects that distort the calibration with SCALA. The comparison is made with the CALSPEC standard star observation ⋆170 from 2022-06-19. For this reason, the above-mentioned "aperture loss" is not applied to the comparison, allowing a complete discussion of all systematics.

It is essential to highlight that the conclusions and findings discussed in this section result from a close collaboration with Daniel Küsters 2023d. The author of this thesis performed maintenance of the SNIFS pipeline software in a computer that allows an examination of analysis on the SCALA data. At the same time, Daniel Küsters 2023d provided the expertise in the SNIFS instrument and performed the examination.

6.3.1 SNIFS Temperature Correction

A last step before comparison of the temperature dependency of the SNIFS instrument has to be taken into account. Rubin et al. 2022b mention a linear temperature dependency of SNIFS. Unfortunately, the dome's cooling system did not work from 19th to 22nd June 2022. Therefore, the instrument's temperature varied between the standard star observation and SCALA calibration. The temperature difference $\Delta T = \mathrm{SCALA} - T_\star$ between the instrument's temperature at the calibration with SCALA T_{SCALA} and the temperature of the standard star observation $T_\star$ the comparision $S^{\star,n}_{\mathrm{SNIFS}}/S^{\mathrm{NIST}}_{\mathrm{SNIFS}}$ is corrected with the temperature coefficient b_T from Rubin et al. 2022a. Fig. 6.14 shows b_T of SNIFS against the wavelength. The coefficient is displayed for both channels, and the spikes (highlighted in red-shaded regions) result from minor discrepancies in wavelength calibration or spectral resolution. These discrepancies become evident when comparing spectra that contain Balmer lines, where the lines do not perfectly align in center and width, leading to the observed spikes. The blue channel shows a strong wavelength-varying dependence of maximal approximately $2\,\%\,\mathrm{K}^{-1}$. In contrast, the red channel shows maximal dependence on the temperature of approximately $0.3\,\%\,\mathrm{K}^{-1}$; the coefficient is also flat, meaning the temperature does not introduce color in the calibration. Therefore, only the blue channel is corrected with b_T, with the red-shaded regions masked and linearly interpolated.

The correction is applied to the blue channel with:

$$S^{\mathrm{NIST}}_{\mathrm{SNIFS}}(\lambda, \Delta T) = \begin{cases} S^{\mathrm{NIST}}_{\mathrm{SNIFS}}(\lambda) \cdot (1 + b_T \Delta T) & \text{for blue channel} \\ S^{\mathrm{NIST}}_{\mathrm{SNIFS}}(\lambda) & \text{for red channel} \end{cases} \tag{6.10}$$

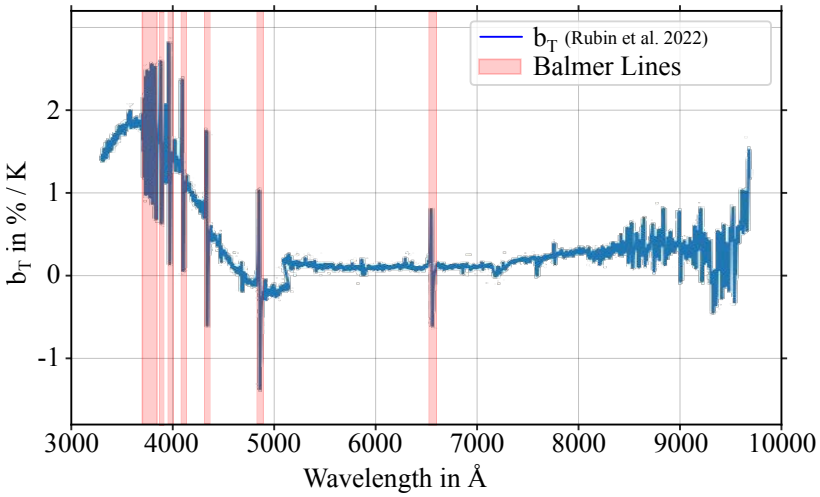


Figure 6.14: Temperature coefficient b_T affecting SNIFS's responsivity, derived from observations of standard stars over multiple nights. Data obtained from private correspondence as detailed in Rubin et al. 2022a.

Fig. 6.15 shows the effect of the temperature correction. The left plot shows the ratio $S^{\star,n}_{\mathrm{SNIFS}}/S^{\mathrm{NIST,\,i}}_{\mathrm{SNIFS}}$ without temperature correction with i being one of the five SCALA measurements between 17th and 21st June 2022, preluding the observation night. The corrected ratio is visualized in the mid-plot. The responsivities measured with SCALA are corrected by their difference to the instrument's temperature during the standard star observation $T_{\star 170} = 4.5\,^{\circ}\mathrm{C}$.

The ratio suggests a further temperature offset between the standard star observation and the SCALA calibration. If an instrument temperature of 2.5 °C is assumed, the ratio becomes flat (right plot), as discussed in the following.

6.3.2 Comparision

The preliminary comparison $\langle S^{\star,n}_{\mathrm{SNIFS}}(\lambda)\rangle_i/S^{\mathrm{NIST}}_{\mathrm{SNIFS}}$ of the responsivity calibrations is shown in Fig. 6.16.[7] The upper plot shows the ratio of the responsivity calibration $S^{\star 170}_{\mathrm{SNIFS}}/S^{\mathrm{NIST,\,k}}_{\mathrm{SNIFS}}$ of the five SCALA measurements k and the CALSPEC standard observation ⋆170. The responsivity $S^{\mathrm{NIST,k}}_{\mathrm{SNIFS}}$ of the blue channel is corrected to an instrument temperature $T_{\star 170} = 4.5\,^{\circ}\mathrm{C}$ during the standard star observation. The temperature correction overlaps all SCALA calibrations within a standard deviation of 1.1 %, but leaves a gradient of approximately 4.2 % per 1000 Å. The black solid line depicts the 35 Å-moving average[8] over all SCALA measurements k, denoted by their UTC dates in the legend. The grey-shaded region shows the standard deviation of the moving window. The lower plot shows the reproducibility between the SCALA exposures k. The graph shows several features that rely on systematic effects from the SNIFS instrument or assumptions made in the pipeline. These systematics are discussed in the following. The lower panel shows the reproducibility of the different SCALA calibrations k as the residual to the moving average. The shaded region also depicts the standard deviation of the moving window. The features denoted in the graphs and the outliers are discussed in the following section.

[7] The ratio of the blue channel is multiplied by 225/220 while the red channel is multiplied by 225/224. This scale factor considers that five spaxels in the blue and one in the red are

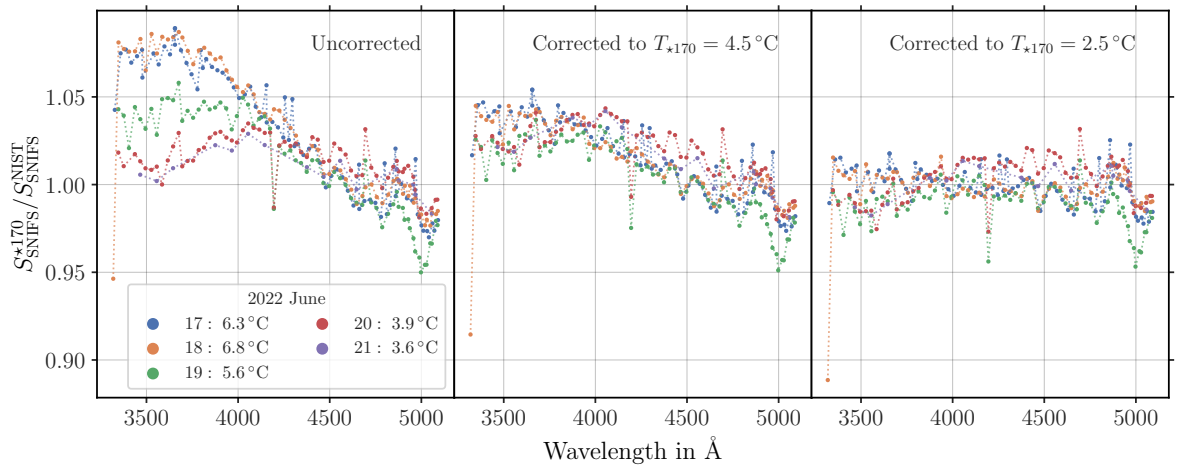


Figure 6.15: Influences of the temperature on the comparison $S^{\star,n}_{\mathrm{SNIFS}}/S^{\mathrm{NIST}}_{\mathrm{SNIFS}}$ in the blue channel. The left plot shows the comparison without temperature correction, while the mid plot shows the comparison with temperature correction to the actual instrument's temperature at the observation $T_{\star 170} = 4.5\,°C$. The right plot depicts the influence of a temperature correction with an offset of 2 K, corresponding to an instrumental temperature of 2.5 °C. The temperatures in the legend are the SNIFS temperatures during the SCALA calibrations.

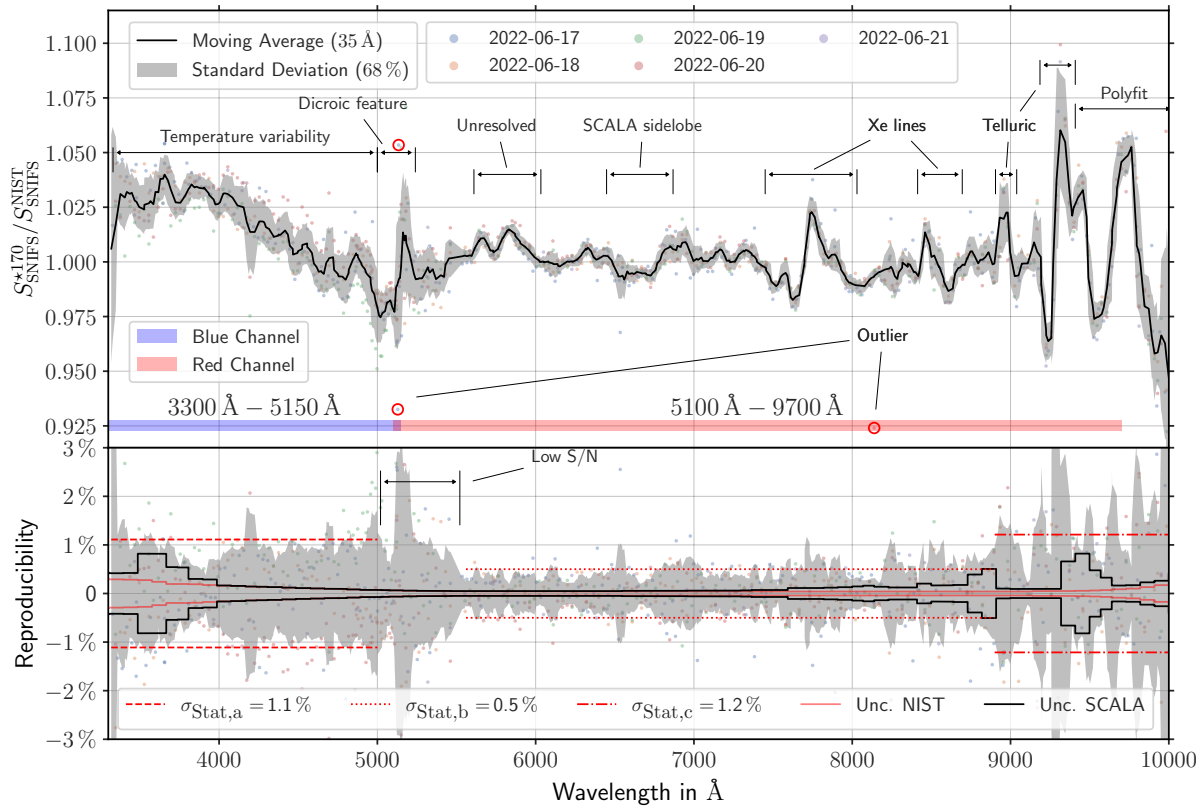
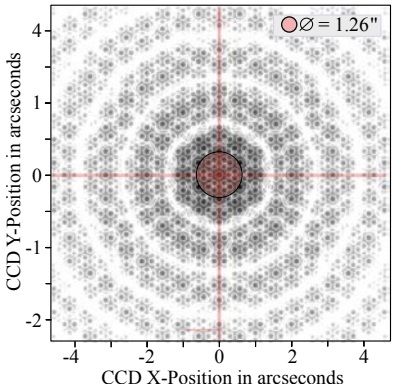
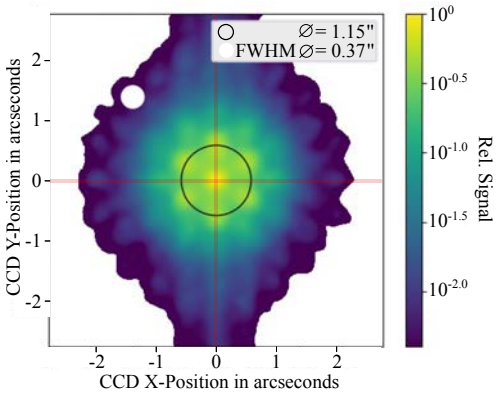


Figure 6.16: Comparision between SNIFS sensitivity calibrated using standard star observation ⋆170 and five calibrations with SCALA traced back to NIST laboratory standard. A 48 Å moving average illustrates the comparative trend (black line), and a gray shaded area indicates the standard deviation (68 %) of the data points within the moving window. The lower plot highlights the residuals to the trend, emphasizing the reproducibility between the five SCALA calibrations of 1.1 %, 0.5 %, and 1.2 % depending on the regions $a, b,$ or c. For comparison, the uncertainties on the reference measurements of SCALA are in red (NIST uncertainties) and black (NIST + SCALA). The adjustments for the blue channel reflect a correction to an instrument temperature of 2.5 °C compared to the actual measurement of 4.5 °C.

not working (Daniel Küsters 2023d).
[8] Compare with the SCALA bandwidth of 35 Å.



(a) Theoretical PSF from the Fourier transformation of the SCALA mask. Taken and added with labels from in Daniel Küsters 2019.



(b) Image from a star observation with short exposure time, showing the small scale SCALA structure as predicted in Daniel Küsters 2019 with a diameter of 0.37″. The signal is normalized by its maximum.

Figure 6.17: Comparision between the theoretical prediction of the telescope's PSF with SCALA mask, with a CCD image in SNIFS' photometric channel.

6.3.3 Systematic Effects on the SCALA Calibration

Several systematic effects have been identified in the calibration of SNIFS with SCALA. One significant issue is the sidelobes caused by diffraction at the MLA. These could be partially addressed by using a dynamical integration window, which is corrected through aperture loss, resulting in reproducibility in the red channel of 0.5 % between 5500 Å to 8800 Å and 1.2 % above. The blue channel shows a day-to-day reproducibility of 1.1 Å (see lower pannel in Fig. 6.16). Despite the MLA diffraction, effects related to the xenon lines, flaws in the telluric model, and the temperature-dependent focus in the blue channel's collimator and objective contribute to artifacts of this comparison. The following paragraphs summarize these effects.

Xe-Lines Fig. 6.12 shows the xenon spectrum. The high-intensity lines >7500 Å deform the triangular bandpass PSF of the double monochromator. This problem arose because it was decided to use the xenon lamp for the entire spectrum to save measurement time. Using the halogen lamp would solve this problem, but it would take more time.

Point Spread Function Another problem is the incorrect point spread function (PSF) assumed in the pipeline. The SCALA mask changes the PSF of the telescope. It generates a hexagonal small-scale structure with an F/1.25 airy disk as an enveloping function, leading to increased scatter in the PSF fits due to the model's inaccuracy. Theoretical and empirical observations of the PSF are shown in Figure 6.17. The left sub-figure shows the theoretical PSF from the Fourier transform of the SCALA mask, as computed in Daniel Küsters 2019, and the right sub-figure shows an image from a star observation[9] taken with the photometric channel that reveals the predicted small-scale "SCALA PSF" $\mathscr{P}_{\mathrm{SCALA}}(x, y)$. The SNIFS pipeline assumes $\mathscr{P}_{\mathrm{Pipeline}}(r)$ (see Eq. 6.1), which is monotonically decreasing in r and radially symmetric. The actual pattern is

[9] The image is one of multiple short-time exposures with exposure time that is smaller than the seeing's time scale. The image is chosen by visual inspection

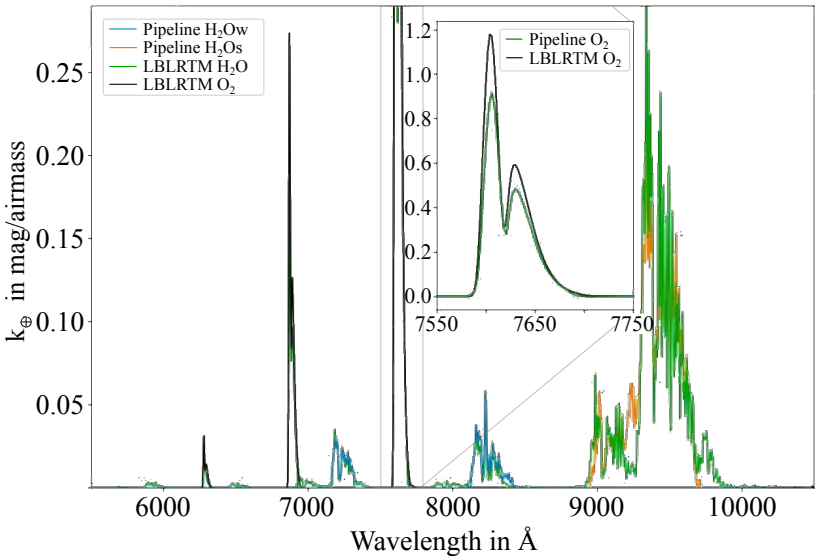


Figure 6.18: Comparison of the telluric template used in the SNIFS pipeline to a physics-based line-by-line radiative transfer model (LBLRTM) Rubin et al. 2022a

monotone and has a sixfold rotational symmetry instead of a continuous one. Even if $\mathscr{P}_{\text{Pipeline}}(r)$ could describe a smoothed SCALA PSF containing the same integral or a constant fraction κ of it, it is not guaranteed that either the PSF fit converges to a state where the integral over the area A satisfies $\int_A dA' \mathscr{P}_{\text{Pipeline}} = \kappa \int_A dA' \mathscr{P}_{\text{SCALA}}$, or κ is independent of wavelength. The result is an increased scatter in the flux solution of the standard star observation and the SCALA calibration.

Telluric Model and Polyfit The atmospheric correction in the red channel, especially the strong water vapor absorption in the telluric model, introduces oscillations above 8900 Å. Fig. 6.19 shows a comparison between the empirical telluric model from the KIT measurements (denoted as pipeline) and a physically based line-by-line radiative transfer model (LBLRTM) from Rubin et al. 2022a. The physics-based model differs in the strength of the oxygen lines and especially in the quality of the strong water absorption above 8900 Å. Compared to the LBLRTM (green line), the empirical model (orange line) overestimates the strong water absorption at 9250 Å, followed by an underestimation at 9350 Å. While the free parameter in the pipeline model fit can absorb a globally erroneous line intensity, the different shape of the water absorption introduces an artifact in the flux solutions of each standard star i.
Fig. 6.19 shows the conversion factors of all standard stars i (red lines), the standard star observation ⋆170, and the pipeline water model (blue line). The black line shows the final polynomial fit $\langle S^{\star,n}_{\text{SNIFS}}(\lambda)\rangle_i$ over the set of star calibrations i. Compared to the moving average (magenta line), the polynomial begins to oscillate at 9000 Å, unable to emulate the data. This oscillation is due to the overcorrection at 9250 Å and under-correction at 9350 Å by the empirical telluric model. As a result, these oscillations are visible in the comparison $\langle S^{\star,n}_{\text{SNIFS}}(\lambda)\rangle_i / S^{\text{NIST}}_{\text{SNIFS}}$ in Fig. 6.16.

SCALA sidelobe As depicted in Fig. 6.13 an artifact of unknown origin leads to an overestimation of the responsivity during the SCALA calibration between 6500 Å to 6800 Å. As discussed, this can be addressed by determining the "aperture loss".

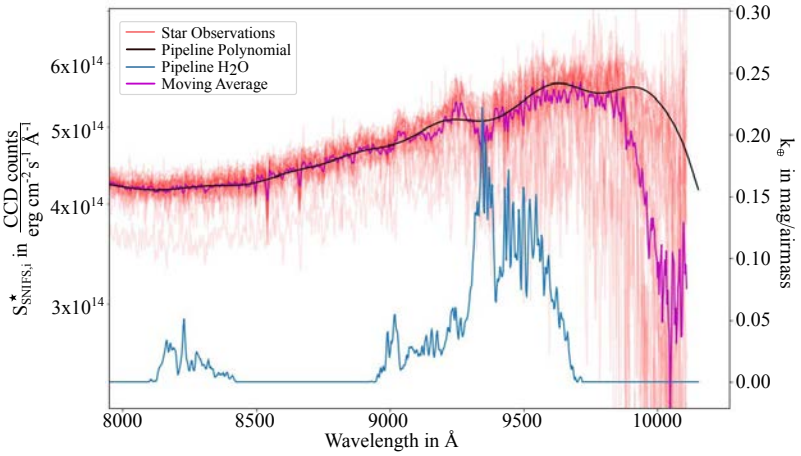


Figure 6.19: Comparison of the polynomial fit for correcting the water vapor absorption (black line) with the ratio between the observed spectra of the standard stars F_{SNIFS} and their model F_{Model}. The oscillations of the polynomial are clearly visible. Further, the overcorrection of the telluric model is also visible at 9250 Å (Daniel Küsters 2023d).

Focus and Temperature Variability The spectrographic system in the blue channel is based on refracting optics. Besides fused silica, calcium fluoride is used. The optic underlies thermal expansion, and the refractive index changes over temperature. As the effects do not compensate each other, the focus is temperature dependent. The temperature dependence of the focus is determined as ≈30 µm (Daniel Küsters 2023d). Each spaxel is a parabolic trace on the CCD. This trace's width depends on the PSF's width and the distance at which the image is out of focus. The focus also changes differently for different wavelengths because of the thermal dependence of the refractive index, and the index itself is wavelength-dependent. In the case of an exposure that is in focus, the spaxel trace may have an offset in position, and the pipeline is able to account for this. However, with a chromatic change in the refractive index and the focal length of the optics, the spaxel trace also changes in curvature. This results in the temperature-dependent sensitivity in the blue channel described in Rubin et al. 2022b. This problem can be addressed by modeling the SNIFS optics to determine the physical degrees of freedom for developing a better fit. The fact that the temperature correction flattens the ratio in the blue channel for 2.5 °C hints that the instrument is in focus at this temperature.

Low SN In the region between the channels 5000 Å to 5300 Å, the signal-to-noise is significantly lower because of low instrumental throughput. Lombardo, Küsters, et al. 2017 shows that the thought for both channel lies below 5 % in that region of the channel crossover. An increased statistical uncertainty is expected in this region.

Dicroic Crossover Yannick Copin 2013 highlights that the transmission of Dichroic beamsplitters oscillates in the range of 3900 Å to 4150 Å and 5000 Å to 5300 Å and varies with changing humidity. The effect is due to the hygroscopic nature of the layers used. In 2006, a system was installed to continuously blow dry nitrogen over the beamsplitter to stabilize the humidity. The residual fluctuations are corrected by an algorithm developed by É. Pécontal, Centre de Recherche Astrophysique de Lyon (CRAL). The humidity is determined by the relative position of the oscillations around 4000 Å at the centered spaxel (No. 113) from the continuum exposure of the SCU. Corrections are interpolated from a transmission correction library based on the observed position of this oscillation. This

correction assumes a continuous spectrum. A model for the optics is necessary to apply the correction, as described in "Focus and Temperature Variability". Further, the humidity stabilizer from 2006 is inoperable. Consequently, there is a possibility that the conditions experienced during the SCALA calibration and star observation fall outside the scope of Pécontal's model range.

Outlier Some outliers are due to faulty background subtraction in the signal of the monitoring LARAs. LARA module No. 18 records the chopper state needed for background subtraction. During the data transfer from the monitoring LARA modules to the SCALA computer, frequent data losses of the order of $\approx O(10)$ bits occurred. The data stream consists of a series of data packets containing one byte with the LARA ID, a running index of two bytes, the four-byte signal, and one byte for the shutter status. A recovery algorithm has been developed to trim the fractional data packets and NaNs by checking the data stream index to ensure data integrity. The algorithm is not able to recover every stream in minor cases. The removed outliers in Fig. 6.16 are identified by visual inspection. A more sophisticated algorithm is required to recover all data, but this effect is minor compared to the sidelobes.

In summary, the calibration of SNIFS with SCALA reveals several systematic problems. Diffraction effects, the xenon line, inaccuracies in the telluric model, and temperature-dependent focus in the collimator and camera objective of the blue channel contribute to increased measurement scatter and comparison artifacts. The high-intensity lines of the xenon spectrum above 7500 angstroms distort the triangular bandpass PSF, causing signal overestimation or underestimation.

In the red channel, atmospheric corrections, especially for water vapor absorption, cause oscillations and inaccuracies in the flux solutions. The temperature dependence of the blue channel focus due to refractive optical materials introduces variability in sensitivity. Statistical uncertainties increase in regions with low signal-to-noise ratios, such as between 5000 and 5300 angstroms. The behavior of the Dicroic corrections on the SCALA spectra needs to be further investigated. In addition, outlier problems arise from faulty background subtraction in the monitoring LARA signal.

A feature between 5610 Å to 6033 Å remains unresolved with a maximum deviation from CALSPEC of 1.5 %.

7 Summary and Outlook

Chapter 2 discusses supernova (SN) cosmology and motivates the need to calibrate the photometric system precisely. As described in Sec. 2.2, the zero-point calibration uncertainty $\sigma(ZP)$ of the different bands that measure SNIa lightcurves must surpass 0.1 % level in absolute flux to extract all statistical information for new generation instruments such as the 10-years Vera C. Rubin Observatory data. The current astrophysical standard is CALSPEC, which offers internal consistency at the 0.5 % level.

Lombardo, Greg Aldering, et al. 2014 suggested the "SNIFS CALibration Aperatus" (SCALA), a device to calibrate the "Supernova Integral Field Spectrograph" (SNIFS) on the University of Hawaii 2.2-meter telescope, that allows a relative flux calibration of standard stars that is traceable to the "National Institute of Standards and Technology" (NIST) laboratory standard. SNIFS serves as a comparator that compares the flux of the standard stars with the flux measured by the monitoring photodiode, that is calibrated by against the NIST laboratory standard. The work of Lombardo 2017 aimed at constructing SCALA and its first commissioning (Lombardo, Küsters, et al. 2017). Several issues with the setup were examined and summarized in Daniel Küsters 2019. The major findings are the crosstalk between SCALA beams and the reflection of the IS light shield. Next to these findings, the work suggested upgrading SCALA, allowing sub-percentage photometric comparison.

Motivated by these suggestions and the decision to equip each SCALA beam with a monitoring photodetector to improve the monitoring in the SCALA setup, the author of this work developed in close collaboration with Dr. Daniel Küsters (affiliated as Post Doc., Department of Physics. University of California, Berkeley) the "Variant Angle Spectral Comparator" (VASC) (see Chapter 3) as a comparator system that allows the transfer of the spectral-photometric calibration of one photodetector to another. To achieve calibration of 0.5 % in flux at SNIFS, the necessary uncertainty contribution due to the transfer shall be less than 0.1 %. The VASC is fully automatized and allows spectral and spatial screening of the photodetector. The gantry-robot enables the variation of different angles of incidence (AoI), contrasting to other competitive comparator systems used by the NIST or the Physikalisch Technischen Bundesanstalt. That allows a complete characterization of the detector's behavior, which is crucial for sub-percentage calibration transfers. The VASC can be calibrated with low-pressure gas lamps providing spectra with spectral lines spanning the range of 3500 Å to 11 000 Å with an uncertainty of below 1.5 Å. In the case of a silicon photodiode, this corresponds to < 10 ppm from 3500 Å to 9000 Å in the photometric scale. Photometric uncertainties on the calibrations of the VASC are dominated by the uncertainties below 8000 Å given by the comparative photodetector that is traced to the laboratory standard (in the case of this work, NIST) and above by the changing splitting ratio of the relay optics in the VASC due to its asymmetry.
It is noted that the author's contribution mainly relates to the develop-

ment of the automatization software for the VASC and its measurement routines. At the same time, Dr. Daniel Küsters provided his expertise in the optical design of the setup. The characterization of the VASC is published in Daniel Küsters, Bastian-Querner, Greg Aldering, Blot, et al. 2020; Daniel Küsters, Bastian-Querner, Greg Aldering, Karg, et al. 2022.

The VASC offers calibration possibilities that go beyond the SCALA project. An example is the QE measurement of CMOS test sensors ULTRASAT project, described in Chapter 4. Part of this work was the provision of automatization software for the use of the ULTRASAT test sensors and their implementation into the VASC's control software. The purpose of these measurements was the design decision of the Antireflection Coating for the final flight sensor of the satellite. The findings are published in Bastian-Querner, Kaipachery, et al. 2021a.

To equip each SCALA beam with a monitoring photodetector, it was necessary to deploy more than eighteen silicon photodiodes with traceable calibration. During the first commission in 2015, SCALA used the "Cooled Large Area Photodiode" (CLAP). The Paris Supernova Cosmology Group designed the CLAP, a photodetector module with a self-designed amperemeter plus commercial photodiode, for the SNDICE project (Juramy et al. 2008). SNDICE used its comparator setup to calibrate the CLAP against a NIST-provided photodiode, qouting a transfer uncertainty of 0.15 % in the range of 3500 Å to 9500 Å. This calibration was seven years old in 2015, leading to an aging uncertainty between 0.3 % and 1.5 % (Daniel Küsters 2019). After deliberating with the Paris Supernova Cosmology Group (Daniel Küsters 2023d), the decision was made not to manufacture additional CLAPs. This was primarily due to the departure of a key expert from the Paris Institute, whose specialized knowledge was crucial for reconstructing the CLAPs[1]. As a result, Dr. Daniel Küsters developed the "LArge Resistor Amperemeter" in collaboration with Marko Kossatz (Electronic Development Engineer, Deutsches Elektronen-Synchrotron (DESY)). Küsters characterized the amperemeter's noise and showed with the noise model (see Sec. 3.2.2) that the noise is limited by the thermal noise of the feedback resistor (Daniel Küsters 2023d). These amperemeters are equipped with a commercial photodiode S1337-1010BQ produced by Hamamatsu.

Chapter 3 ends with the description of the upgrade of SCALA, which was performed in June 2022. The upgrade includes the recoating of the mirrors, the implementation of a baffle for the integrating spheres, the implementation of monitoring diodes for each beam, the alignment of the beams, and the implementation of the second monochromator plus a chopper. The upgrade was carried out by Dr. Daniel Küsters and the author of this document, with support from Dr. Greg Aldering, a Senior Scientist at Lawrence Berkeley National Laboratory.

In Chapter 5, the VASC is used to perform a relative spectral calibration of the LARAs by using the comparison method with a monitor (see Chapter 2.3.1). It begins with the mathematical basis for photosensor calibration (Sec.5.1) and LARA data acquisition (Sec.5.2). The Chapter

[1] The CLAPs used custom-designed operational amplifiers (OPs) manufactured as application-specific integrated circuits (ASICs). This process requires specialized tuning expertise. For the interested reader, Barrelet, Lebbolo, and Sefri 2007 discusses the characterization of those "Low Current Amplifiers".

then discusses the LARA calibration processes, including wavelength scale calibration, alignment, and sensitivity determination (Sec.5.3). Finally, calibration results are presented, including LARA comparisons (see Sec. 5.4)), an uncertainty discussion (Sec. 5.5), and a comparison of VASC and NIST calibrations (see Sec. 5.4.1).
The uncertainty in the spectral calibration is determined as <0.3 Å for the wavelength range 3000 Å to 6500 Å, 0.75 Å for the wavelength range 6500 Å to 8500 Å, and 1.5 Å above. In the case of the used S1337 photodiodes, this uncertainty contributes to less than 10 ppm in the photometric scale for the wavelength range 3000 Å to 9000 Å and stays subordinate above. The resulting uncertainties of the LARAs are dominated by the uncertainties of the NIST calibration in the range of 4000 Å to 7500 Å and above 9500 Å. Below that range, the uncertainties are dominated by the temperature model and the interpolation uncertainties. Above that range, the uncertainties are dominated by the change in the splitting ratio due to the asymmetric relay optics and the flux limitation of the VASC. Above 10 000 Å, the temperature variability of the photodiodes takes the second leading role after the NIST uncertainties. The uncertainties are summarized in the Fig. 5.28.
Overall, aimed budget of 0.1 % for the VASC's error contribution is complied in nearly the complete SCALA interesting range of 3500 , to 10 000 , with two exceptions: the flux limitation at 9450 Å due to the optical fiber's absorption peak and the temperature model below 4000 Å. The comparison of the NIST-calibrated photodiodes shows that the VASC can reproduce the calibrations done by NIST within the uncertainties (see Figure 5.30).
A suggested improvement for the VASC is the implementation of symmetric relay optics to eliminate the temperature-dependent splitting ratio and a more sophisticated exploration of the temperature dependence of the photodiodes. The last improvement would be the use of a Transfer Matrix Model (TMM) for the photodiode's responsivity to eliminate the interpolation uncertainties.
This result ensures that the uncertainty contribution due to the calibration transfer **do not** limit the calibration of SNIFS to the 0.5 % level.

This work ends with the observation of standard stars and the comparison of the responsivity calibration of SNIFS with CALSPEC and SCALA. In this work, standard stars were observed over four nights from June 19–22, 2022 UTC, and SCALA calibrated SNIFS five times between these observations during the day. The observations were performed in collaboration with Dr. Greg Aldering and Dr. Daniel Küsters. The author of this work supervised the readout software and analyzed the SCALA data, while Drs. Aldering and Küsters contributed their expertise with the SNIFS instrument and the SCALA setup.
Chapter 6 begins with the description of the calibration of SNIFS with CALSPEC (Sec. 6.1) to extract SNIFS' responsivity $S^{\star,n}_{\mathrm{SNIFS}}$ against CALSPEC. Then the calibration with SCALA is described (Sec. 6.2) to get the deposited charge in each frame, which is needed to get the SNIFS responsivity against the NIST laboratory standard $S^{\mathrm{NIST}}_{\mathrm{SNIFS}}$. The uncertainties for the flux is exceeded the 0.1 % only due to the variability in the VASC's splitting ratio at 8700 Å and exceeds the 0.5 % at 9450 Å due to the flux limitation. The Chapter ends with the comparison of the two calibrations $S^{\star 170}_{\mathrm{SNIFS}}$ and $S^{\mathrm{NIST},k}_{\mathrm{SNIFS}}$ is discussed in Sec. 6.3.

The preliminary result is shown in the Figure 6.16. The night of June 19, 2022, serves as a "burning sample" for verification and examination of the assumption made in the SNIFS pipeline when using SCALA data. The other nights are left blind and not discussed in this work. The reproducibility of the SCALA calibrations is 1.1 %, 0.5 %, and 1.2 % for the blue channel, the red channel up to 8900 Å and above, respectively. The temperature correction coefficient is determined from observational data and suffers statistical uncertainty that contributes to the scatter in the blue channel. While the temperature correction is able to align the calibrations within 1.1 % standard deviation, a gradient of 4.2 % per 1000 Å is left. The gradient corresponds to a temperature offset of approximately 2 K, suggesting that the blue channel could be in focus with an instrumental temperature of 2.5 °C. The defect in the telluric model leads to an oscillating artifact in the polynomial fit over all standard star "flux solutions" above 9400 Å, which propagates to the result. The high curvature leads to an inaccurate determination of the trend with the performed moving average of the ratio, leading to an increased scatter in the residual. The examination uncloaked an artifact that is only visible in the use of SCALA but not observed during the arc exposures (SCALA sidelobe) and only one structure of 1.5 % at 5800 Å is left unresolved. The contribution of the author to these last findings is mainly the analysis of the SCALA data and the maintenance of the SNIFS pipeline software on an external computer for examination by Dr. Daniel Küsters.

A forward model for the SNIFS optics could be developed to address the temperature-dependent variations in the blue channel and the SCALA sidelobes, allowing their subtraction. To address the problem with xenon lines, the use of a halogen lamp instead of the xenon lamp could be a viable solution, albeit at the cost of increased measurement time. In addition, the telluric model could be improved by using a physics-based approach rather than relying on the current empirical model.

Towards the end of this work, the VASC was upgraded with a symmetric relay optic with a Schwarzschild design. This improvement significantly reduces optical aberrations, resulting in a refined point spread function (PSF). As a result, comparisons are less dependent on the photometer aperture, and the splitting ratio is stabilized against temperature changes. In addition, the parabolic mirrors M1 and M2, shown in Fig. 3.1, have been replaced to increase throughput. This was necessary due to their long-term degradation from high UV flux exposure. A recalibration of the LARAs would now limit the uncertainty primarily to the NIST calibration, with the only exception being the flux limitation at 9450 Å.

Marc Betoule et al. 2023 refers to this project as "the most advanced experiment" for bottom-up calibration of standard stars, a process last undertaken in the 1980s (Hayes 1985). Building on the results detailed in this work, SCALA emerges as a viable option for revisiting such calibration efforts.

8 Acknowledgments

This research was funded in part by a NIST Precision Measurement Grant awarded to the Department of Physics at the University of California, Berkeley, and supported by the Director of the Office of Science, Office of High Energy Physics of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231. Special thanks go to the engineering team at DESY for their invaluable assistance in the development of our flux calibration light source.

Further acknowledgments are due to the Summit crew of the UH88 for their unwavering technical support during both the upgrade and observing phases, sometimes even extending their support remotely. Special thanks go to Marko Kossatz for his technical contributions and the development of the LARAs and their readout hardware. Dr. Daniel Küsters deserves special mention for his sustained support throughout this thesis. Dr. Greg Aldering's expertise with SNIFS was instrumental during both the upgrade and observing phases, and his contributions are greatly appreciated.

I am also immensely grateful to Prof. Dr. Marek Kowalski for his mentorship and supervision, which have been instrumental in bringing this work to fruition. Finally, but most importantly, my deepest gratitude goes to my wife, Lea, and our parents, whose support and patience have been the cornerstone of my journey throughout the duration of this thesis.

A

Appendix

A.1 Inventory Laboratory Setup

The following table lists all devices used in the VASC. The devices are numbered and referenced in the text (see Sec. 3.1.1).

Table A.1: Summary of the laboratory inventory.

Device No.	Name	Remarks
S1	150 W Xenon arc lamp	Newport; Oriel Apex Monochromator Illuminator; MODEL 70812; SER. NO. 144; MFD. 11/2011
LDLS	Laser-Driven Light Source	EQ99-X from Energetiq Technology Inc.
S3	Low Pressure Gas Lamps	NaE (BGW/24/351); ZnE (BGW/16H/133); CdE (Narva/75); NeE (Narva/III/73); HgE (Narva/1/I/79)
S4	532 nm Laser	Thorlabs; CPS532-C2; 0.9 mW
M1	Parabolic Mirror	Thorlabs; MPD229 - F01; ⌀ = RFL = 2″, 90° OAP, UV Enhanced Aluminum
M2	Parabolic Mirror	Thorlabs; Customized Design; ⌀ = 2″ RFL = 8″, 90° OAP, UV Enhanced Aluminum
M3	Flat Mirror	Mounted on an HDR50/M rotation stage from Thorlabs to select between light sources.
M4	Spherical Mirror	Custom made from LASEROPTIK GmbH; RoC = 800 mm, ⌀ = 200 mm
F1	Motorized Filter Wheel	Lot Oriel Gruppe Europa; Model MSZ121 Ser.No.: 306; MFD 12/2005; Humboldt-Universität zu Berlin Inv.Nr. EEP00077
Mono1	Monochromator	Cornerstone Monochromator 260; Model 74100; Ser.No.: 715; MFD 10/2011; Physikal. Ins. Univ. Bonn: Inv. Nr.: 111150
Mono2	Monochromator	Cornerstone Monochromator 260; Model MSH301; Ser.No.: 393; MFD 12/2005; Humboldt-Universität zu Berlin Inv.Nr. EEP00077
	Encoder at Mono1	ERN 480 5000 27512-03; ID 385 480-33; W8; SN 69 423 449 B
	Encoder at Mono2	ERN 480 5000 27512-03; ID 385 480-33; W8; SN 69 423 452 B
M5	Off-Axis parabolic mirror	Thorlabs, $d = 1''$, $RFL = 2''$
M6	Concave cylindrical mirror	Thorlabs, $d = 1''$, $f = 2''$
M7	Flat mirror	Motorized in tilt with a GNL18-Z8 Goniometer from Thorlabs.
G1–4	UV fused silica windows	Used as beamsplitters to illuminate monitor detectors R1–4
Ref1	LARA01	Feedback Resistance 10 GΩ; Measurement range 0.1 pA to 250 pA (see Sec. 3.2)
Ref2	LARA13	Feedback Resistance 10 MΩ; Measurement range 0.1 nA to 250 nA (see Sec. 3.2)
HoDi	Holmium Didymium Liquid Filter	Mounted on linear motorized stage to move in and out of the beam. Type 667-UV45 from Hellma Analytics; 74977 D-K-18752-01-00 2021-05
Ref3	LARA32	Feedback Resistance 10 GΩ; Measurement range 0.1 pA to 250 pA (see Sec. 3.2)

A.2 Assigned Spectral Lines

To calibrate the calculated model wavelength from the grating's position of the monochromator (described in 3.1.2) spectral lines of four different low-pressure gas lamps are scanned as a function of the model wavelength. The lines are assigned to transition lines listed in the NIST Atomic Spectra Database (Standards and Technology 2009). The following tables list all assigned spectral lines used to calibrate the model wavelength concerning the NIST Database.

Table A.2: Assigned cadmium low-pressure gas lamp spectral lines with argon as ignition gas. The table shows the observed vacuum wavelength of the transitions from the database, its corresponding element, the energy levels (k: upper level, i: lower level), and the atom's electron configuration in its upper state.

	Element	Vaccum Wavelength in nm	E_i in cm^{-1}	E_k in cm^{-1}
0	Cd	467.94587	30113.9900	51483.9800
1	Cd	480.12540	30656.0870	51483.9800
2	Cd	508.72393	31826.9520	51483.9800
3	Ar	633.02240	194861.3529	210658.6020
4	Cd	644.02490	43692.3840	59219.7340
5	Ar	696.73520	93143.7600	107496.4166
6	Ar	706.91670	93143.7600	107289.7001
7	Cd	734.76941	51483.9800	65093.7020
8	Ar	738.60140	93750.5978	107289.7001
9	Ar	751.67210	93750.5978	107054.2720
10	Ar	763.72080	93143.7600	106237.5518
11	Ar	772.63330	94553.6652	107496.4166
12	Ar	795.03620	94553.6652	107131.7086
13	Ar	801.69900	93143.7600	105617.2700
14	Ar	811.75420	93143.7600	105462.7596
15	Ar	826.67940	95399.8276	107496.4166
16	Ar	841.05210	95399.8276	107289.7001
17	Ar	842.69630	93750.5978	105617.2700
18	Ar	852.37830	95399.8276	107131.7086
19	Ar	912.54710	93143.7600	104102.0990
20	Ar	922.70300	95399.8276	106237.5518
21	Ar	966.04350	93750.5978	104102.0990
22	Ar	978.71860	95399.8276	105617.2700
23	Ar	1039.54667	186470.3603	196089.9385

Table A.3: Assigned spectral lines of the mercury low-pressure gas lamp. The table shows the observed vacuum wavelength of the transitions from the database, its corresponding element, the energy levels (k: upper level, i: lower level), and the atom's electron configuration in its upper state. #: Intensity weighted average from lines in italics.

	Element	Vaccum Wavelength in nm	E_i in cm^{-1}	E_k in cm^{-1}
0	Hg	*365.11980*	*44042.9090*	*71431.1800*
		365.58833	*44042.9090*	*71396.0730*
		366.39303	*44042.909*	*71336.005*
		365.407762#		
1	Hg	404.770810	37644.9820	62350.325
2	Hg	407.898830	39412.2370	63928.120
3	Hg	435.956000	39412.2370	62350.325
4	Hg	491.744000	54068.6829	74404.560
5	Hg	546.226750	44042.9090	62350.325
6	Hg	*577.121010*	*54068.6829*	*71396.0730*
		579.227570	*54068.6829*	*71333.0530*
		578.118854#		
7	Hg	671.819000	63928.1200	78813.020
8	Hg	690.937000	62350.3250	76823.430
9	Hg	708.385400	62350.3250	76466.936
10	Hg	773.095200	63928.1200	76863.137
11	Hg	924.560000	68886.4300	79702.530
12	Hg	942.820000	69661.8030	80267.970
13	Hg	1014.253000	54068.6829	63928.120
14	Hg	1071.840000	68886.4300	78216.120

Table A.4: Assigned spectral lines of the sodium low-pressure gas lamp with argon as ignition gas. The table shows the observed vacuum wavelength of the transitions from the database, its corresponding element, the energy levels (k: upper level, i: lower level), and the atom's electron configuration in its upper state. *: Lines are assigned manually by visual inspection. #: Intensity weighted average from lines in italics.

	Element	Vaccum Wavelength in nm	E_i in cm^{-1}	E_k in cm^{-1}
0	Na	330.590800*	300387.82000	330636.75000
1	Na	*589.158326* (D_2)	*N/A*	*16973.36619*
	Na	*589.755815* (D_1)	*N/A*	*16956.17025*
	Na	589.357489#		
2	Ar	696.735200	93143.76000	107496.41660
3	Ar	706.916700	93143.76000	107289.70010
4	Ar	714.901200	93143.76000	107131.70860
5	Ar	727.494000	93750.59780	107496.41660
6	Ar	738.601400	93750.59780	107289.70010
7	Ar	*750.593500*	*95399.82760*	*108722.61940*
	Ar	*751.672100*	*93750.59780*	*107054.27200*
	Ar	750.593500#		
8	Ar	763.720800	93143.76000	106237.55180
9	Ar	772.633300	94553.66520	107496.41660
10	Ar	795.036200	94553.66520	107131.70860
11	Ar	*800.835900*	*93750.59780*	*106237.55180*
		801.699000	*93143.76000*	*105617.27000*
		801.315400#		
12	Ar	811.754200	93143.76000	105462.75960
13	Na	819.704340	16973.36619	29172.88700
14	Ar	826.679400	95399.82760	107496.41660
15	Ar	841.052100	95399.82760	107289.70010
16	Ar	842.696300	93750.59780	105617.27000
17	Ar	852.378300	95399.82760	107131.70860
18	Ar	867.032500	94553.66520	106087.25980
19	Ar	912.547100	93143.76000	104102.09900
20	Ar	922.703000	95399.82760	106237.55180
21	Ar	935.678700	95399.82760	106087.25980
22	Ar	966.043500	93750.59780	104102.09900
23	Ar	978.718600	95399.82760	105617.27000
24	Ar	1047.292300	94553.66520	104102.09900
25	Ar	1067.648900	104102.09900	113468.47300

Table A.5: Assigned spectral lines of the neon low-pressure gas lamp. The table shows the observed vacuum wavelength of the transitions from the database, its corresponding element, the energy levels (k: upper level, i: lower level), and the atom's electron configuration in its upper state. *: Lines are assigned manually by visual inspection.

	Element	Vaccum Wavelength in nm	E_i in cm^{-1}	E_k in cm^{-1}
0	Ne	337.08749*	134041.8400	163707.7261
1	Ne	341.88834*	134459.2871	163708.6029
2	Ne	344.86901*	134041.8400	163038.3544
3	Ne	347.35649*	134041.8400	162830.7073
4	Ne	350.22171*	134459.2871	163012.6247
5	Ne	352.14780*	135888.7173	164285.8872
6	Ne	359.46639*	135888.7173	163707.7261
7	Ne	442.66426*	148257.7898	170848.2740
8	Ne	461.12014*	150772.1118	172458.4270
9	Ne	471.13829*	148257.7898	169482.9862
10	Ne	503.91560*	149657.0392	169501.6353
11	Ne	508.17992*	149824.2215	169502.2893
12	Ne	511.79290*	148257.7898	167796.9475
13	Ne	514.64440*	150858.5079	170289.3243
14	Ne	520.53452*	150315.8612	169526.8869
15	Ne	534.25794*	148257.7898	166975.3424
16	Ne	540.20631*	134459.2871	152970.7328
17	Ne	556.39863*	149824.2215	167796.9475
18	Ne	565.82287*	150121.5922	167794.9709
19	Ne	569.13950*	148257.7898	165828.1766
20	Ne	572.08114*	150315.8612	167795.8966
21	Ne	574.98929*	149657.0392	167048.6694
22	Ne	576.60175*	149657.0392	167000.0317
23	Ne	581.30179*	149824.2215	167026.9923
24	Ne	582.17694*	149824.2215	167001.1327
25	Ne	585.41101	135888.7173	152970.7328
26	Ne	588.35252	134041.8400	151038.4524
27	Ne	591.52717*	150121.5922	167026.9923
28	Ne	594.64810	134041.8400	150858.5079
29	Ne	597.71895	134041.8400	150772.1118
30	Ne	603.16666	134459.2871	151038.4524
31	Ne	607.60193	134459.2871	150917.4307
32	Ne	609.78506	134459.2871	150858.5079
33	Ne	614.47629	134041.8400	150315.8612
34	Ne	616.52994	134818.6405	151038.4524
35	Ne	621.90013	134041.8400	150121.5922
36	Ne	626.82285	134818.6405	150772.1118
37	Ne	633.61791	134041.8400	149824.2215
38	Ne	638.47560	134459.2871	150121.5922
39	Ne	640.40177	134041.8400	149657.0392
40	Ne	650.83255	134459.2871	149824.2215
41	Ne	660.07754	135888.7173	151038.4524
42	Ne	668.01205	135888.7173	150858.5079
43	Ne	693.13787	135888.7173	150315.8612
44	Ne	703.43520	134041.8400	148257.7898
45	Ne	706.10543	148257.7898	162419.9818
46	Ne	717.59154	135888.7173	149824.2215
47	Ne	724.71631	134459.2871	148257.7898
48	Ne	744.09469	134818.6405	148257.7898
49	Ne	749.09335	148257.7898	161607.2609

Table A.6: Continueing Tab. A.5

	Element	Vaccum Wavelength in nm	E_i in cm^{-1}	E_k in cm^{-1}
50	Ne	753.78488	148257.7898	161524.1739
51	Ne	794.53654	149824.2215	162410.1736
52	Ne	808.46801	135888.7173	148257.7898
53	Ne	812.07818	150121.5922	162435.6780
54	Ne	813.86432	150121.5922	162408.6535
55	Ne	826.83491	150315.8612	162410.1736
56	Ne	830.26062	149657.0392	161701.4486
57	Ne	837.99093	149657.0392	161590.3412
58	Ne	842.07398	149824.2215	161699.6613
59	Ne	849.76932	149824.2215	161592.1200
60	Ne	859.36184	150772.1118	162408.6535
61	Ne	863.70190	150121.5922	161699.6613
62	Ne	865.67599	150858.5079	162410.1736
63	Ne	868.43062	150121.5922	161636.6175
64	Ne	878.30336	150315.8612	161701.4486
65	Ne	885.62980	150315.8612	161607.2609
66	Ne	892.19496	150315.8612	161524.1739
67	Ne	915.11829	150772.1118	161699.6613
68	Ne	920.42841	150772.1118	161636.6175
69	Ne	922.41108	150858.5079	161699.6613
70	Ne	930.34053	150858.5079	161607.2609
71	Ne	932.90663	150917.4307	161636.6175
72	Ne	937.58796	150858.5079	161524.1739
73	Ne	942.79655	150917.4307	161524.1739
74	Ne	946.18060	151038.4524	161607.2609
75	Ne	948.92849	148257.7898	158795.9924
76	Ne	953.67793	151038.4524	161524.1739
77	Ne	966.80709	148257.7898	158601.1152
78	Ne	1056.53029	152970.7328	162435.6780
79	Ne	1080.10009	150121.5922	159379.9935
80	Ne	1084.74479	150315.8612	159534.6196
81	Ne	1114.60715*	149824.2215	158795.9924
82	Ne	1118.05855*	149657.0392	158601.1152

A.3 Additional Data

The following tables show the list of observed stars. Four nights have been used from 2022-06-19 to 2022-06-22 to observe the stars. The particular star of interest is the primary CALSPEC star **GD153**, which is used to compare the laboratory standard with CALSPEC. The other two primary CALSPEC stars were not accessible in June from Hawaii. Please note that the reference date or time is the UTC, 12 hours before the local Hawaii time. This has the advantage that there is no date change during one observation. The stars listed at CALSPEC are in bold and hyperlinked to their corresponding model. The last table lists the targeted wavelengths of the SCALA exposures.

Table A.7: List of observed stars during the night 2022-06-19. The telescope's mirror was 3° warmer than the ambient temperature, leading to a seeing around ≈1.4″. The list was provided by Greg Aldering from personal communication from 2022-06-19.

No.	Object	Airmass	File
1	HR4468	1.272	P22_170_009_003_17_R.fits
2	HR4963	1.110	P22_170_011_003_17_R.fits
3	**GD153**	1.011	P22_170_013_003_17_R.fits
4	**HD106252**	1.107	P22_170_015_003_17_R.fits
5	**HD163466**	1.829	P22_170_017_003_17_R.fits
6	**HD158485**	1.629	P22_170_019_003_17_R.fits
7	BD+292091	1.537	P22_170_021_003_17_R.fits
8	BD+023375	1.388	P22_170_026_003_17_R.fits
9	BD+262606	1.008	P22_170_028_003_17_R.fits
10	**HD116405**	1.212	P22_170_030_003_17_R.fits
11	**P177D**	1.136	P22_170_032_003_17_R.fits
12	**GD153**	1.303	P22_170_034_003_17_R.fits
13	**EG131 (6)**	1.701	P22_170_036_003_17_R.fits
14	**HD180609**	1.630	P22_170_038_003_17_R.fits
15	**P041C**	1.698	P22_170_040_003_17_R.fits
16	**GD153**	1.845	P22_170_042_003_17_R.fits
17	**P177D**	1.188	P22_170_044_003_17_R.fits
18	**EG131 (6)**	1.236	P22_170_046_003_17_R.fits
19	BD+254655	1.730	P22_170_048_003_17_R.fits
20	**HD159222**	1.047	P22_170_050_003_17_R.fits
21	**EG131 (6)**	1.155	P22_170_053_003_17_R.fits
22	BD+174708	1.501	P22_170_055_003_17_R.fits
23	**EG131 (6)**	1.132	P22_170_057_003_17_R.fits
24	**P177D**	1.497	P22_170_059_003_17_R.fits
25	**EG131 (6)**	1.130	P22_170_061_003_17_R.fits
26	**EG131 (6)**	1.185	P22_170_063_003_17_R.fits
27	BD+284211	1.047	P22_170_065_003_17_R.fits
28	BD+254655	1.039	P22_170_067_003_17_R.fits
29	BD+284211	1.024	P22_170_070_003_17_R.fits
30	BD+254655	1.017	P22_170_072_003_17_R.fits
31	**HD163466**	1.808	P22_170_074_003_17_R.fits
32	**HD163466**	1.900	P22_170_077_003_17_R.fits
33	**HD163466**	1.969	P22_170_079_003_17_R.fits
34	BD+174708	1.001	P22_170_081_003_17_R.fits
35	HR8634	1.014	P22_170_083_003_17_R.fits
36	HR9087	1.169	P22_170_085_003_17_R.fits

Table A.8: List of observed stars during the night 2022-06-20. The telescope's mirror was 3° warmer than the ambient temperature, leading to larger seeing as in the night before. **The night is classified as photometric.** The list was provided by Greg Aldering from personal communication from 2022-06-20.

No.	Object	Airmass	File
1	HR4468	1.274	P22_171_016_003_17_R.fits
2	HR4963	1.110	P22_171_018_003_17_R.fits
3	**HD159222**	1.698	P22_171_020_003_17_R.fits
4	**HD163466**	1.943	P22_171_022_003_17_R.fits
5	**GD153**	1.029	P22_171_024_003_17_R.fits
6	BD+023375	1.631	P22_171_026_003_17_R.fits
7	**HD116405**	1.138	P22_171_028_003_17_R.fits
8	**GD153**	1.125	P22_171_031_003_17_R.fits
9	**HD93521**	1.959	P22_171_033_003_17_R.fits
10	**GD153**	1.215	P22_171_035_003_17_R.fits
11	BD+023375	1.184	P22_171_037_003_17_R.fits
12	**HD116405**	1.373	P22_171_041_003_17_R.fits
13	**GD153**	1.505	P22_171_043_003_17_R.fits
14	**HD180609**	1.570	P22_171_045_003_17_R.fits
15	**GD153**	1.824	P22_171_047_003_17_R.fits
16	BD+262606	1.226	P22_171_049_003_17_R.fits
17	**HD158485**	1.283	P22_171_052_003_17_R.fits
18	BD+023375	1.050	P22_171_054_003_17_R.fits
19	BD+262606	1.687	P22_171_067_003_17_R.fits
20	**HD180609**	1.398	P22_171_069_003_17_R.fits
21	**HD158485**	1.367	P22_171_071_003_17_R.fits
22	**HD159222**	1.126	P22_171_073_003_17_R.fits
23	**EG131 (6)**	1.128	P22_171_075_003_17_R.fits
24	BD+023375	1.228	P22_171_077_003_17_R.fits
25	**HD159222**	1.225	P22_171_079_003_17_R.fits
26	**HD163466**	1.465	P22_171_081_003_17_R.fits
27	**HD159222**	1.284	P22_171_083_003_17_R.fits
28	BD+174708	1.101	P22_171_085_003_17_R.fits
29	**HD158485**	1.650	P22_171_087_003_17_R.fits
30	**HD205905**	1.493	P22_171_089_003_17_R.fits
31	**EG131 (6)**	1.279	P22_171_091_003_17_R.fits
32	**HD158485**	1.899	P22_171_093_003_17_R.fits
33	**EG131 (6)**	1.392	P22_171_095_003_17_R.fits
34	**HD158485**	2.135	P22_171_097_003_17_R.fits

Table A.9: List of observed stars during the night 2022-06-21. Seeing remained poor between 1.5″ to 2″. **The night is classified as photometric.** For HD116405 and HD180609, the wrong tracking mode ('grabit'-mode, meaning without guiding) was used, leading to tracking errors. The list was provided by Greg Aldering from personal communication from 2022-06-21.

No.	Object	Airmass	File	UTC-Time
1	**HD116405**	1.101	22_172_009_003_17_R.fits	2022-06-21T05:53:32
2	**HD159222**	1.678	22_172_011_003_17_R.fits	2022-06-21T06:04:04
3	**HD163466**	1.929	22_172_013_003_17_R.fits	2022-06-21T06:12:47
4	**GD153**	1.031	22_172_015_003_17_R.fits	2022-06-21T06:33:46
5	BD+023375	1.616	22_172_017_003_17_R.fits	2022-06-21T06:49:24
6	**HD116405**	1.140	22_172_019_003_17_R.fits	2022-06-21T07:01:32
7	**GD153**	1.102	22_172_021_003_17_R.fits	2022-06-21T07:19:55
8	**HD93521**	1.847	22_172_023_003_17_R.fits	2022-06-21T07:29:16
9	**GD153**	1.192	22_172_025_003_17_R.fits	2022-06-21T07:55:02
10	BD+023375	1.204	22_172_027_003_17_R.fits	2022-06-21T08:10:29
11	**GD153**	1.334	22_172_029_003_17_R.fits	2022-06-21T08:31:31
12	**HD116405**	1.367	22_172_031_003_17_R.fits	2022-06-21T08:41:26
13	**GD153**	1.494	22_172_033_003_17_R.fits	2022-06-21T08:59:49
14	**HD180609**	1.575	22_172_035_003_17_R.fits	2022-06-21T09:14:23
15	**GD153**	1.806	22_172_037_003_17_R.fits	2022-06-21T09:36:23
16	**GD153**	2.022	22_172_039_003_17_R.fits	2022-06-21T09:53:51
17	**HD158485**	1.287	22_172_043_003_17_R.fits	2022-06-21T10:20:32
18	BD+023375	1.055	22_172_045_003_17_R.fits	2022-06-21T10:34:02
19	**HD158485**	1.303	22_172_047_003_17_R.fits	2022-06-21T10:44:29
20	BD+174708	1.631	22_172_049_003_17_R.fits	2022-06-21T10:58:32
21	BD+023375	1.089	22_172_051_003_17_R.fits	2022-06-21T11:12:21
22	**HD180609**	1.396	22_172_056_003_17_R.fits	2022-06-21T11:38:35
23	**HD159222**	1.142	22_172_059_003_17_R.fits	2022-06-21T11:51:20
24	**EG131 (6)**	1.130	22_172_061_003_17_R.fits	2022-06-21T12:11:08
25	BD+023375	1.276	22_172_064_003_17_R.fits	2022-06-21T12:28:50
26	**HD159222**	1.270	22_172_066_003_17_R.fits	2022-06-21T12:37:33
27	**HD163466**	1.500	22_172_068_003_17_R.fits	2022-06-21T12:46:20
28	**HD159222**	1.337	22_172_070_003_17_R.fits	2022-06-21T12:55:06
29	BD+174708	1.062	22_172_074_003_17_R.fits	2022-06-21T13:17:16
30	**HD158485**	1.733	22_172_076_003_17_R.fits	2022-06-21T13:28:22
31	**HD205905**	1.475	22_172_078_003_17_R.fits	2022-06-21T13:44:01
32	**EG131 (6)**	1.381	22_172_084_003_17_R.fits	2022-06-21T14:14:05
33	**HD158485**	2.112	22_172_086_003_17_R.fits	2022-06-21T14:23:51
34	**EG131 (6)**	1.573	22_172_088_003_17_R.fits	2022-06-21T14:48:37
35	**HD205905**	1.516	22_172_090_003_17_R.fits	2022-06-21T14:57:53
36	HR7596	1.427	22_172_092_003_17_R.fits	2022-06-21T15:08:26

Table A.10: List of observed stars during the night 2022-06-22. Seeing remained poor with ≈3″. The list was provided by Greg Aldering from personal communication from 2022-06-22.

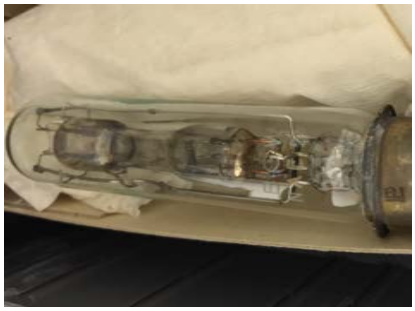
No.	Object	Airmass	File	UTC-Time
1	**HD158485**	1.844	22_173_041_003_17_R.fits	2022-06-22T05:51:57
2	**HD116405**	1.103	22_173_043_003_17_R.fits	2022-06-22T06:02:41
3	**HD163466**	1.898	22_173_045_003_17_R.fits	2022-06-22T06:13:43
4	**HD163466**	1.852	22_173_047_003_17_R.fits	2022-06-22T06:21:11
5	**GD153**	1.046	22_173_049_003_17_R.fits	2022-06-22T06:42:10
6	**GD153**	1.072	22_173_051_003_17_R.fits	2022-06-22T06:59:50
7	**HD93521**	1.701	22_173_053_003_17_R.fits	2022-06-22T07:09:07
8	**GD153**	1.130	22_173_055_003_17_R.fits	2022-06-22T07:28:20
9	**HD93521**	1.981	22_173_057_003_17_R.fits	2022-06-22T07:37:42
10	**HD93521**	2.075	22_173_059_003_17_R.fits	2022-06-22T07:45:20
11	**HD106252**	1.512	22_173_061_003_17_R.fits	2022-06-22T07:54:11
12	**HD158485**	1.366	22_173_063_003_17_R.fits	2022-06-22T08:05:30
13	**HD180609**	1.755	22_173_065_003_17_R.fits	2022-06-22T08:14:30
14	**HD180609**	1.725	22_173_067_003_17_R.fits	2022-06-22T08:22:01
15	BD+023375	1.132	22_173_069_003_17_R.fits	2022-06-22T08:35:38
16	BD+023375	1.106	22_173_071_003_17_R.fits	2022-06-22T08:50:26
17	BD+023375	1.090	22_173_073_003_17_R.fits	2022-06-22T09:00:56
18	**GD153**	1.759	22_173_077_003_17_R.fits	2022-06-22T09:28:01
19	**HD163466**	1.324	22_173_079_003_17_R.fits	2022-06-22T09:39:06
20	**EG131 (6)**	1.291	22_173_081_003_17_R.fits	2022-06-22T09:58:55
21	**HD159222**	1.034	22_173_083_003_17_R.fits	2022-06-22T10:09:11
22	BD+262606	1.377	22_173_085_003_17_R.fits	2022-06-22T10:21:58
23	**HD163466**	1.316	22_173_087_003_17_R.fits	2022-06-22T10:32:02
24	**P177D**	1.299	22_173_089_003_17_R.fits	2022-06-22T10:51:50
25	BD+023375	1.096	22_173_092_003_17_R.fits	2022-06-22T11:13:16
26	BD+262606	1.878	22_173_094_003_17_R.fits	2022-06-22T11:26:00
27	**HD158485**	1.375	22_173_096_003_17_R.fits	2022-06-22T11:35:40
28	**HD159222**	1.135	22_173_098_003_17_R.fits	2022-06-22T11:44:08
29	**EG131 (6)**	1.129	22_173_100_003_17_R.fits	2022-06-22T12:03:57
30	BD+023375	1.246	22_173_102_003_17_R.fits	2022-06-22T12:16:36
31	**HD158485**	1.497	22_173_104_003_17_R.fits	2022-06-22T12:26:15
32	**P177D**	1.925	22_173_108_003_17_R.fits	2022-06-22T12:50:53
33	**EG131 (6)**	1.200	22_173_110_003_17_R.fits	2022-06-22T13:10:58
34	BD+174708	1.046	22_173_112_003_17_R.fits	2022-06-22T13:23:56
35	BD+174708	1.033	22_173_114_003_17_R.fits	2022-06-22T13:34:53
36	**EG131 (6)**	1.334	22_173_116_003_17_R.fits	2022-06-22T13:58:30
37	**HD159222**	1.856	22_173_118_003_17_R.fits	2022-06-22T14:07:42
38	**EG131 (6)**	1.466	22_173_120_003_17_R.fits	2022-06-22T14:27:24
39	**HD159222**	2.225	22_173_122_003_17_R.fits	2022-06-22T14:36:26
40	**HD180609**	1.749	22_173_125_003_17_R.fits	2022-06-22T14:52:33
41	**HD205905**	1.545	22_173_127_003_17_R.fits	2022-06-22T15:07:04

Table A.11: Target wavelengths in one calibration of SNIFS. Each row is called 'run' and corresponds to one exposure of SNIFS. The wavelengths are adjusted in increasing order. The inner wavelengths step (left to right) is $\Delta\lambda_{\text{inner}} = 700$ Å between 6900 Å to 7400 Å and $\Delta\lambda_{\text{inner}} = 500$ Å else. The outer step is (up to down) $\Delta\lambda_{\text{outer}} = 30$ Å. The last digits deviate from that pattern due to rounding to full Å. The algorithm that generates these arrays is called *Read_wavelength_input* in the pySCALA package.

	Wavelength in Å												
Run	1	2	3	4	5	6	7	8	9	10	11	12	13
B00	3261	3762	4261	4761									
B01	3292	3791	4291	4792									
B02	3322	3821	4322	4822									
B03	3351	3852	4352	4852									
B04	3381	3882	4381	4882									
B05	3412	3911	4411	4912									
B06	3441	3941	4441	4942									
B07	3471	3971	4471	4972									
B08	3501	4002	4501	5002									
B09	3532	4032	4531	5032									
B10	3561	4061	4562	5062									
B11	3591	4091	4592	5092									
B12	3621	4121	4622	5122									
B13	3652	4151	4652	5152									
B14	3681	4182	4681	5182									
B15	3711	4212	4711	5212									
B16	3741	4241	4741	5242									
R00	4931	5431	5821	6321	6821	7321	8094	8595	9094	9594	10094	10594	11094
R01	4961	5461	5851	6351	6851	7351	8124	8624	9124	9624	10124	10624	11124
R02	4991	5491	5881	6381	6880	7380	8154	8654	9154	9654	10154	10654	11155
R03	5021	5521	5911	6411	6911	7411	8184	8684	9184	9684	10184	10684	11184
R04	5051	5552	5940	6441	6941	7441	8214	8714	9215	9714	10214	10714	11214
R05	5081	5471	5971	6471	6971	7744	8244	8744	9244	9744	10244	10744	11244
R06	5111	5501	6001	6501	7001	7774	8274	8774	9274	9774	10274	10774	11274
R07	5141	5531	6031	6531	7031	7804	8304	8804	9304	9804	10305	10804	11304
R08	5171	5561	6061	6561	7061	7834	8334	8834	9334	9834	10334	10834	11334
R09	5201	5591	6090	6591	7091	7864	8364	8864	9364	9864	10364	10864	11364
R10	5231	5621	6121	6621	7121	7894	8394	8894	9394	9894	10394	10894	11394
R11	5261	5651	6151	6651	7151	7924	8424	8924	9424	9924	10424	10924	11424
R12	5291	5681	6181	6681	7181	7954	8454	8954	9454	9954	10454	10954	11454
R13	5321	5711	6210	6710	7211	7985	8485	8984	9484	9985	10484	10984	11484
R14	5351	5741	6241	6741	7241	8014	8514	9014	9514	10014	10514	11014	11515
R15	5381	5771	6271	6771	7271	8044	8544	9045	9544	10045	10544	11044	11544
R16	5412	5801	6301	6801	7301	8074	8574	9074	9574	10074	10574	11074	11574

A.4 Technical Drawings and Specifications

The following pages contain technical drawings and specifications of the integrating sphere and the gas lamps used for spectral calibration of the VASC.



(a) Sodium gas lamp



(b) Two cadmium gas lamps

Figure A.1: Pictures of low-pressure gas lamps used to calibrate the Holmium-Didymium filter.



(a) Neon gas lamp



(b) Mercury gas lamp

Figure A.2: Pictures of low-pressure gas lamps used to calibrate the Holmium-Didymium filter.

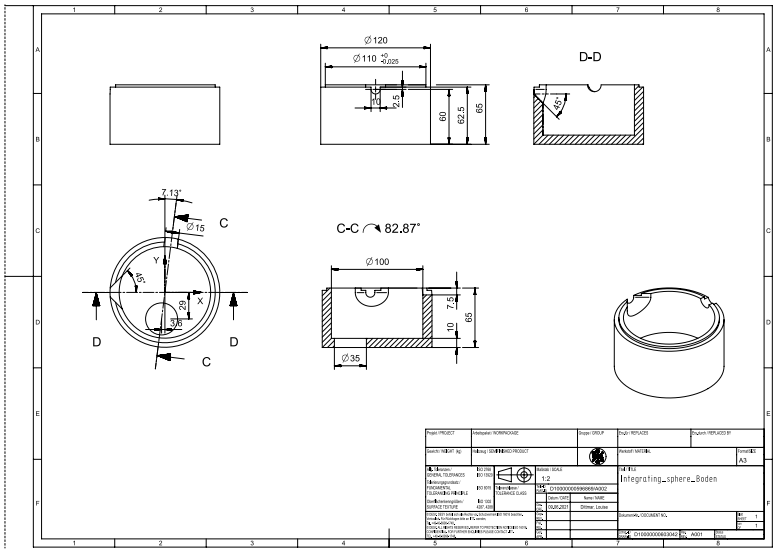


Figure A.3: Technical drawing of the lower part of the integrating cylinder (Daniel Küsters 2023b).

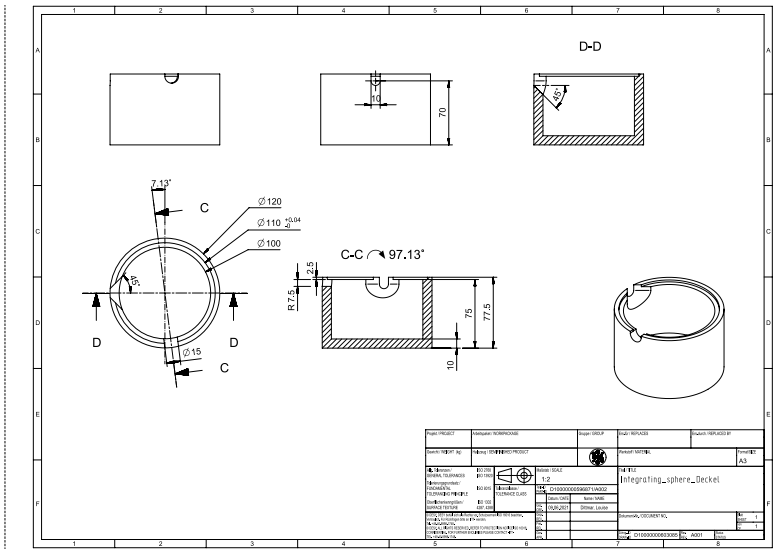


Figure A.4: Technical drawing of the upper part of the integrating cylinder (Daniel Küsters 2023b).

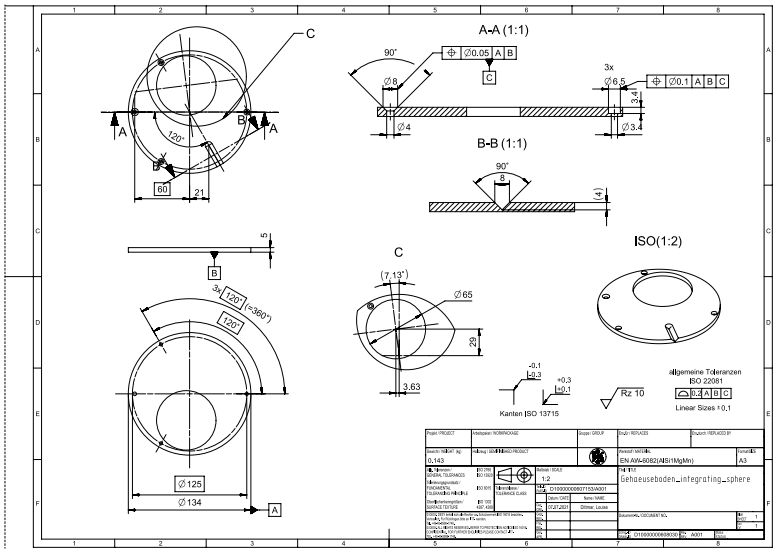


Figure A.5: Technical drawing of the bottom of the integrating cylinder's housing (Daniel Küsters 2023b).

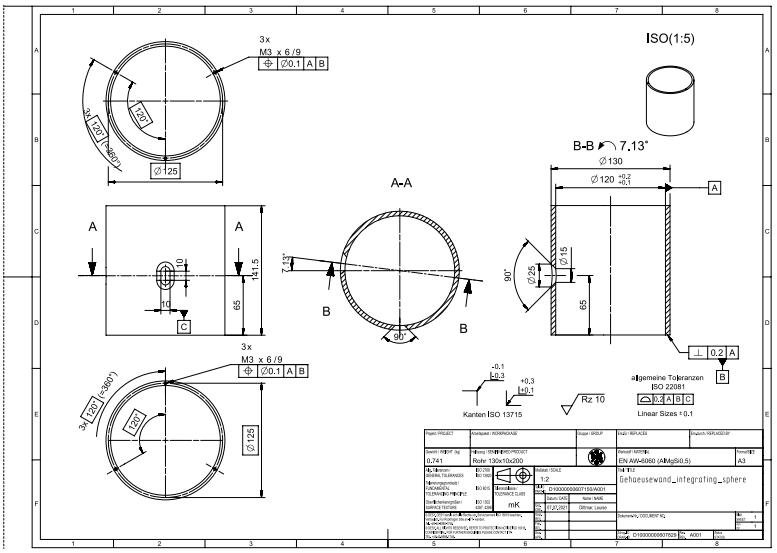


Figure A.6: Technical drawing of the walls of the integrating cylinder's housing (Daniel Küsters 2023b).

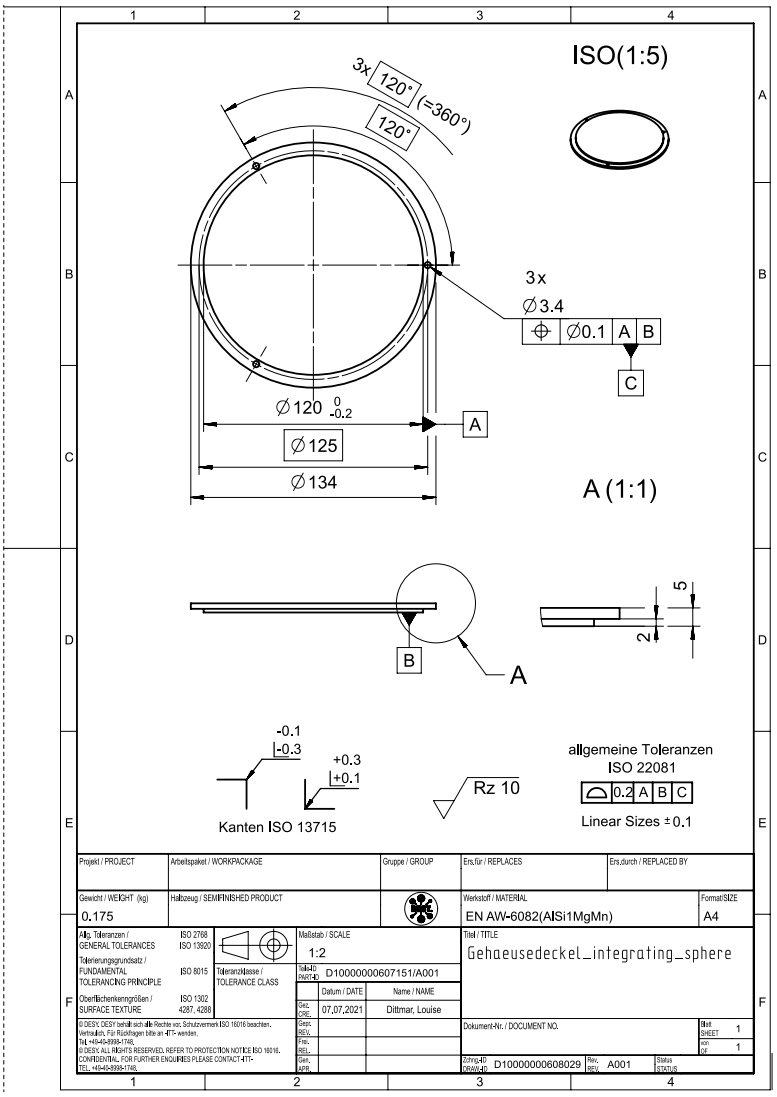


Figure A.7: Technical drawing of the cover plate of the integrating cylinder's housing (Daniel Küsters 2023b).

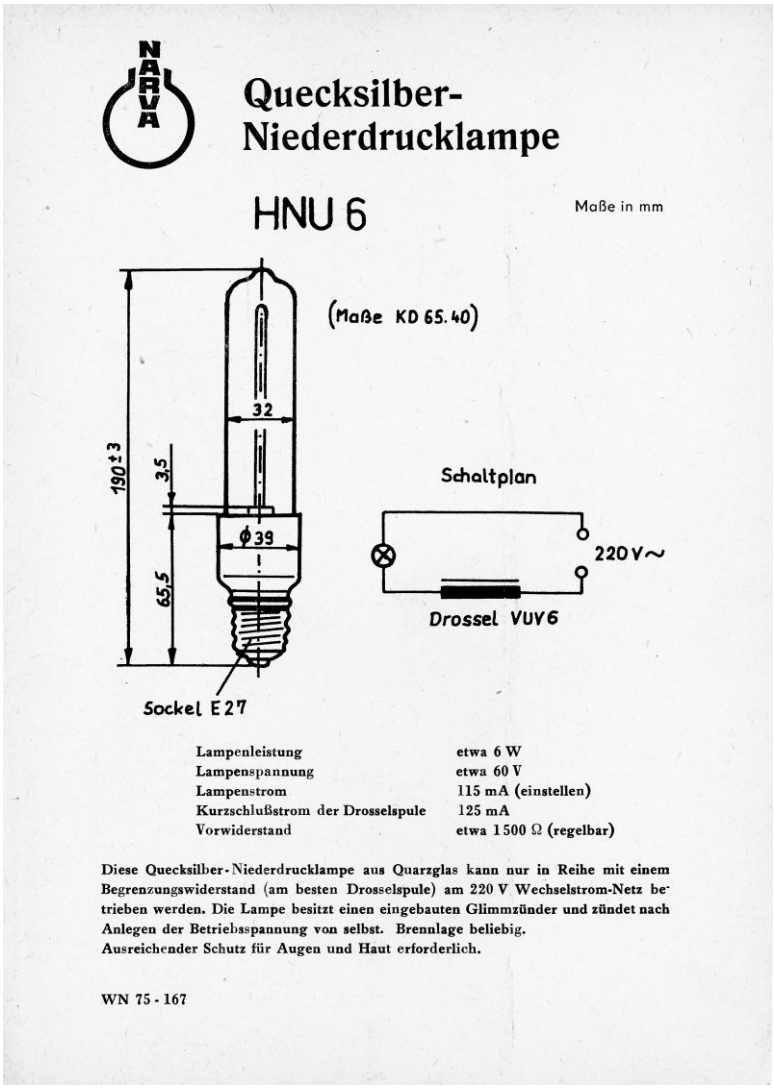


Figure A.8: Technical specification of the NARVA mercury low-pressure gas lamp.

A.5 Derivations

This section contains derivations of equations used in this thesis.

A.5.1 Relation between redshift and scale factor.

From the worldline of a light ray $ds^2 = 0$ it follows that the integral over the comoving time is the same regardless of a constant delay of emission Δt_e and receiving Δt_r:

$$\int_{t_e}^{t_0} \frac{dt}{a(t)} = \int_{t_e+\Delta t_e}^{t_0+\Delta t_0} \frac{dt}{a(t)} = \int_0^r \frac{dr'}{\sqrt{1-kr'^2}} \tag{A.1}$$

After a rearrangement of the integral limits, one finds:

$$\int_{t_e}^{t_e+\Delta t_e} \frac{dt}{a(t)} = \int_{t_r}^{t_r+\Delta t_r} \frac{dt}{a(t)} \tag{A.2}$$

As the delays are assumed to be small in the case of an oscillation period of an electromagnetic wave, the scale factor can be seen as constant and

be pulled out of the integral. After integration, we get:

$$\frac{\Delta t_e}{a(t_e)} = \frac{\Delta t_t}{a(t_r)} \tag{A.3}$$

As the periode Δt a light ray times c equals its wavelength:

$$\frac{\lambda_e}{\lambda_t} = \frac{a(t_e)}{a(t_r)} \tag{A.4}$$

With the definition of the redshift $z = \Delta\lambda/\lambda$, $\lambda_e = \lambda$, and $\lambda_r = \lambda + \Delta\lambda$:

$$\frac{a(t_r)}{a(t_e)} = 1 + z(t_e) \tag{A.5}$$

with $a(t_r)$ the scale factor at the time it is received on earth and $a(t_e)$ the scale factor at the time of emission. That leads us to the final relation between redshift and scale factor as $a(t_r)$ is defined to be one:

$$\boxed{a(t_e) = \frac{1}{z(t_e) + 1}} \tag{A.6}$$

A.5.2 The Horizon Problem

The Horizon Problem is the inability of the standard Big Bang model to predict the homogeneity of the Cosmic Microwave Background (CMB) radiation. Measurements from the Planck satellite show variations in the CMB of $< 10^{-4}$ (European Space Agency and Planck Collaboration 2013), which led to the conclusion that all matter content in the universe had to have causal contact in primordial times to achieve thermal equilibrium. The particle horizon $\Delta\tau = \int d\tau = \int da/H(a)$ is the maximal comoving distance a particle can travel since the initial singularity. To get the size $\Delta\tau_0^{CMB}$ of the regions with causal contact, one integrates from $a = 0$ to $a_{CMB} = 1/1088$. To get an angular distance, one needs the distance to the CMB $\Delta\tau_{CMB}^1$ with the integration boundaries from $a_{CMB} = 1/1088$ to $a_0 = 1$.

$$\Delta\tau_0^{CMB} = \int_0^{a_{CMB}=1/1088} \frac{da}{H_0\sqrt{\Omega_M/a^3 + \Omega_\Lambda}} \tag{A.7a}$$

$$\Delta\tau_{CMB}^1 = \int_{a_{CMB}=1/1088}^1 \frac{da}{H_0\sqrt{\Omega_M/a^3 + \Omega_\Lambda}} \tag{A.7b}$$

With the values of $\Omega_M \approx 0.3$ and $\Omega_\Lambda \approx 0.7$ we get distances of:

$$\begin{aligned} \Delta\tau_0^{CMB} &= \frac{0.06}{H_0} \\ \Delta\tau_{CMB}^1 &= \frac{3.15}{H_0} \end{aligned} \tag{A.8}$$

in units of the Hubble Time H_0^{-1}. To calculate the angular distance of regions that had causal contact since the time of singularity $\tau = 0$, one uses simple geometrics with $\Delta\tau_0^{CMB}$ being the radius at the CMB and $\Delta\tau_{CMB}^1 d_{CMB,1}$ the hypotenuse as it is the distance to the CMB, visualized

in Fig. A.9.

$$\boxed{\gamma \approx \sin\gamma = \frac{0.06}{3.15}\frac{360^\circ}{\pi} \approx 1.15^\circ} \tag{A.9}$$

This is approximately two times the size of the moon. From this, the standard Big Bang does not explain the homogeneity of the CMB as there are at least 30,000 different regions at the CMB that could not have been in causal contact. The result is plausible as the integrant in Eq. A.7 is dominated from later times. If one wishes to solve this problem, the integrant has to be dominated by early times (Baumann and McAllister 2015).

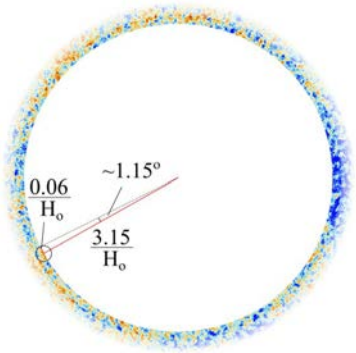


Figure A.9: An illustration of the horizon problem. The circles show the different particle horizons Eq. A.8. Both build a triangle that spans an angle of $1.15^\circ \approx 0.06/3.15$. Background CMB image credit: ESA and the Planck Collaboration (2013) (European Space Agency and Planck Collaboration 2013).

A.5.3 Approximated Model for Incidence Angle Dependency

The goal is to find an approximated version of the dependency on the incidence angle θ_i for the ratio between the responsivity of the detector $S(\theta_i, \lambda)$ and the responsivity at normal incidence $S(0, \lambda)$. If the different layers of a silicon photodiode are ignored and it is seen as a diode from silicon only, its transmission, described by the Fresnel equation, describes its external quantum efficiency at first-order (Chromey 2016).

The Fresnel equation describing the reflection R_s and transmission T_s of an s-polarized light ray at an interface between two media are:

$$R_\mathrm{s} = \left| \frac{n_1 \cos\theta_\mathrm{i} - n_2\sqrt{1-\left(\frac{n_1}{n_2}\sin\theta_\mathrm{i}\right)^2}}{n_1 \cos\theta_\mathrm{i} + n_2\sqrt{1-\left(\frac{n_1}{n_2}\sin\theta_\mathrm{i}\right)^2}} \right|^2 \tag{A.10}$$
$$T_\mathrm{s} = 1 - R_\mathrm{s}$$

with n_1 and n_2 the refractive indices of the medium the light is coming from and the silicon, respectively. Both refractive indices are wavelength-dependent. The wavelength dependency is now omitted in the equations. As only the change in responsibility $S(\theta_i)/S(0)$ concerning normal incidence is of interest, the respective change in the transmission $T_\mu(\theta_i)/T_\mu(0)$ is approximated to second order in θ_i with the polarization μ around 0°. The first and second-order derivative of $T_\mu(\theta_i)/T_\mu(0)$ involve complex calculations and is performed with the SymPy Python package (Meurer et al. 2017) that allows for symbolic calculations. Only the intermediate results are shown here.

First, we evaluate the normalization factor with the transmission at normal incidence which gives Eq. A.11:

$$T_\mu(0) = \frac{n_1 - n_2}{n_1 + n_2} \tag{A.11}$$

The approximation of $T_\mu(\theta_i)/T_\mu(0)$ is:

$$\frac{T_\mu(\theta_i)}{T_\mu(0)} \approx 1 + \left.\frac{dT_\mu(\theta_i)}{d\theta_i}\right|_{\theta_i=0} \frac{\theta_i}{T_\mu(0)} + \frac{1}{2}\left.\frac{d^2T_\mu(\theta_i)}{d\theta_i^2}\right|_{\theta_i=0} \frac{\theta_i^2}{T_\mu(0)} + O(\theta_i^3) \tag{A.12}$$

where the first term is one due to the normalization. Evaluating the first derivative at $\theta_i = 0$ shows that these vanish Eq. A.13:

$$\begin{aligned} \left.\frac{dT_s(\theta_i)}{d\theta_i}\right|_{\theta_i=0} &= 0 \\ \left.\frac{dT_p(\theta_i)}{d\theta_i}\right|_{\theta_i=0} &= 0 \end{aligned} \tag{A.13}$$

Note that the transmission has no linear dependence and the next nonvanishing term is the second derivative which we call curvature a. Even when we use this as a fit parameter in Sec. 5.3.1 it is shown for completeness in Eq. A.14:

$$\begin{aligned} \left.\frac{d^2T_s(\theta_i)}{d\theta_i^2}\right|_{\theta_i=0} \frac{1}{T_\mu(0)} &= -\frac{\frac{(n_1-n_2)^2\cdot\left(\frac{2n_1^2}{n_2}+2n_1\right)}{(n_1+n_2)^3} + \frac{(n_1-n_2)\left(\frac{2n_1^2}{n_2}-2n_1\right)}{(n_1+n_2)^2}}{1-\frac{(n_1-n_2)^2}{(n_1+n_2)^2}} \\ \left.\frac{d^2T_p(\theta_i)}{d\theta_i^2}\right|_{\theta_i=0} \frac{1}{T_\mu(0)} &= -\frac{\frac{(n_1-n_2)^2\cdot\left(\frac{2n_1^3}{n_2^2}+2n_2\right)}{(n_1+n_2)^3} + \frac{(n_1-n_2)\left(-\frac{2n_1^3}{n_2^2}+2n_2\right)}{(n_1+n_2)^2}}{1-\frac{(n_1-n_2)^2}{(n_1+n_2)^2}} \end{aligned} \tag{A.14}$$

The approximated model to describe the relative change in responsivity on the incident angle is:

$$\boxed{\frac{S(\theta_i)}{S(0)} \approx 1 + a\theta_i^2} \tag{A.15}$$

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